# JavaScript Object Reference

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## Object Methods and Properties

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Revised February 2026

| Name | Description |
|---|---|
| assign() | Copies properties from a source object to a target object |
| constructor | Returns the function that created an object's prototype |
| create() | Returns an object created from an existing object |
| defineProperties() | Adds or changes properties |
| defineProperty() | Adds or changes a property |
| delete | Deletes a property from an object: |
| entries() | Returns an array of the key/value pairs of an object |
| freeze() | Prevents any changes to an object |
| fromEntries() | Returns an object created from an iterable list of key/value pairs |
| getOwnPropertyDescriptor() | Returns an array of the keys of an object |
| getOwnPropertyDescriptors() | Returns an array of the keys of an object |
| getOwnPropertyNames() | Returns an array of the keys of an object |
| groupBy() | Groups object elements according to returned callback values |

| | |
|---|---|
| isExtensible() | Returns true if an object is extensible |
| isFrozen() | Returns true if an object is frozen |
| isSealed() | Returns true if an object is sealed |
| keys() | Returns an array of the keys of an object |
| preventExtensions() | Prevents adding new properties to an object |
| prototype | Lets you to add properties and methods to JavaScript objects |
| seal() | Prevents adding new or deleting existing object properties |
| toString() | Converts an object to a string and returns the result |
| valueOf() | Returns the primitive value of an object |
| values() | Returns an array of the property values of an object |

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JavaScript Object Advanced this

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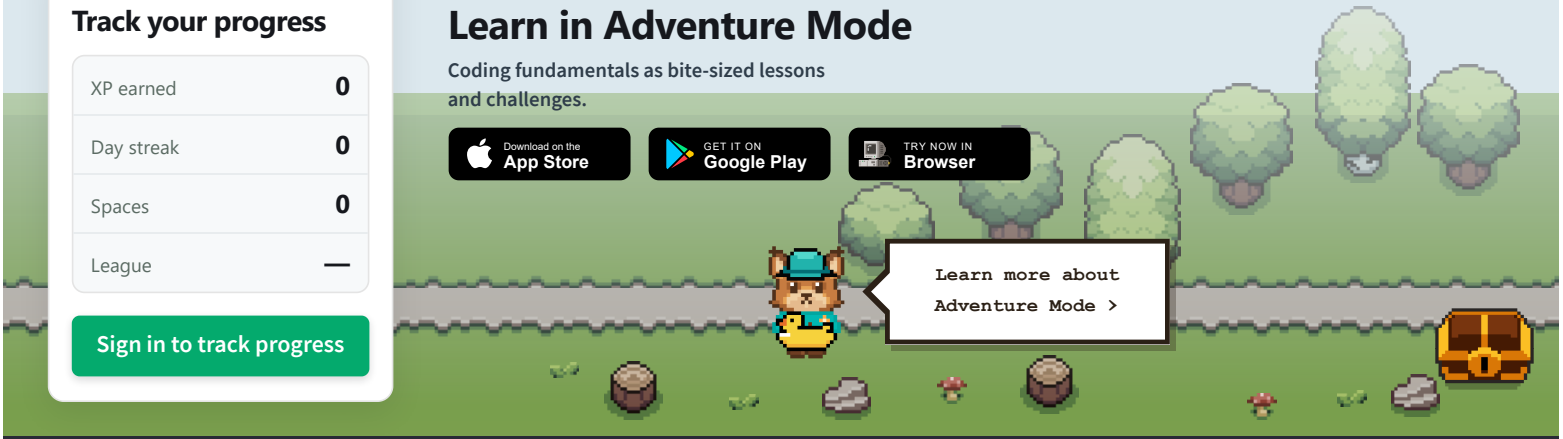
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# JavaScript Object Prototypes

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## Prototype Inheritance

All JavaScript objects inherit properties and methods from a prototype:

- `Date` objects inherit from `Date.prototype`
- `Array` objects inherit from `Array.prototype`

The `Object.prototype` is on the top of the prototype inheritance chain.

`Date` objects, `Array` objects, and all other objects inherit from `Object.prototype`.

## Adding Properties and Methods to Objects

Sometimes you want to add new properties (or methods) to all existing objects of a given type.

Sometimes you want to add new properties (or methods) to an object constructor.

Previously you have learned how to use an **object constructor**:

### Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
```

```
  this.age = age;
  this.eyeColor = eyecolor;
}

const myFather = new Person("John", "Doe", 50, "blue");
const myMother = new Person("Sally", "Rally", 48, "green");
```

Try it Yourself »

You **cannot** add a new property to an existing object constructor:

### Example

```
Person.nationality = "English";
```

Try it Yourself »

To add a new property to a constructor, you must add it to the constructor function:

### Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
  this.age = age;
  this.eyeColor = eyecolor;
  this.nationality = "English";
}
```

Try it Yourself »

## Using the **prototype** Property

The JavaScript `prototype` property allows you to add new properties to object constructors:

### Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
  this.age = age;
```

```
  this.eyeColor = eyecolor;
}

Person.prototype.nationality = "English";
```

Try it Yourself »

The JavaScript `prototype` property also allows you to add new methods to object constructors:

## Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
  this.age = age;
  this.eyeColor = eyecolor;
}

Person.prototype.name = function() {
  return this.firstName + " " + this.lastName;
};
```

Try it Yourself »

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## Warning

Only modify your **own** prototypes. Never modify the prototypes of standard JavaScript objects.

## Exercise

**Which prototype is on the top of the prototype inheritance chain?**

- Date.prototype
- Array.prototype

- Object.prototype
- Person.prototype

Submit Answer »

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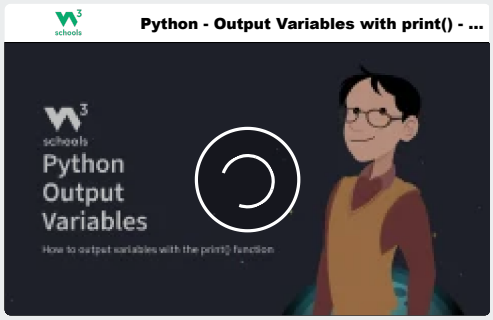
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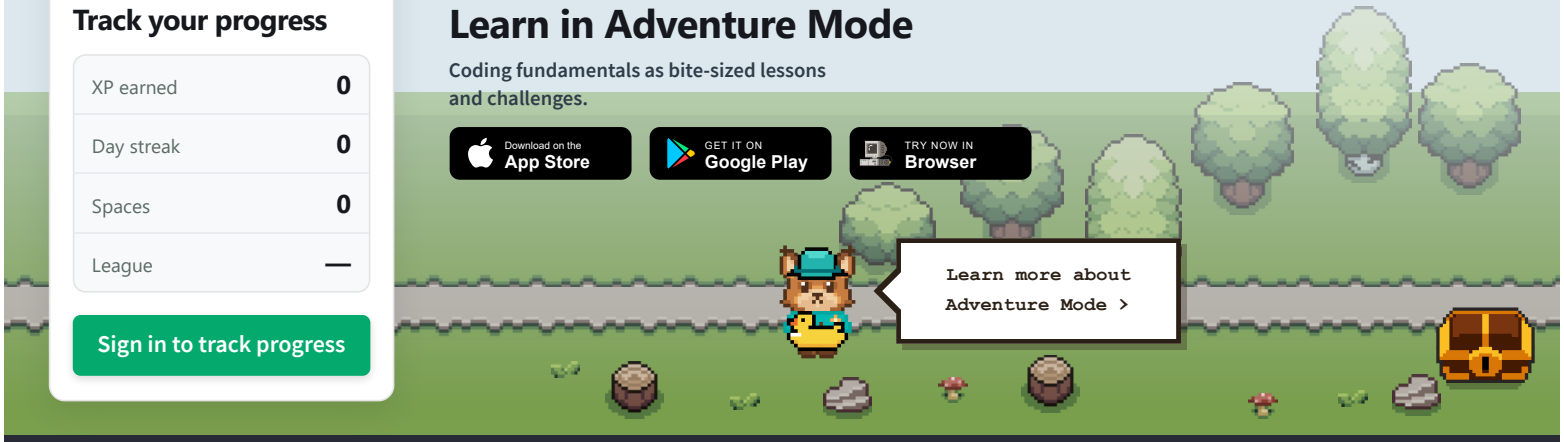
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# JavaScript Advanced Objects Code Challenge

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## Challenge: Advanced Objects

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Test your understanding of JavaScript by completing a small coding challenge.

**Instructions**

Inside the `<script>` element, complete the following steps:

1. Add a getter `description` to `car` that returns the brand and the model
2. Protect `car` with `Object.freeze()`, before the line that changes the year
3. Loop through `Object.entries(car)`. Add each key and value to `text`, with a `<br>` after each

The page should show:
**brand: Volvo**
**model: EX90**
**year: 2024**
**description: Volvo EX90**

Code Solution Check Code

```
<!DOCTYPE html>
<html>
<body>

<h2>Advanced Objects</h2>
<p id="demo"></p>

<script>
const car = {
```

```
  brand: "Volvo",
  model: "EX90",
  year: 2024,
  // Step 1: add a getter here
};

// Step 2: protect the object

// This line tries to change the year
car.year = 2030;

let text = "";
// Step 3: loop through the entries

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

Result

Click **Check Code** to see the result here.

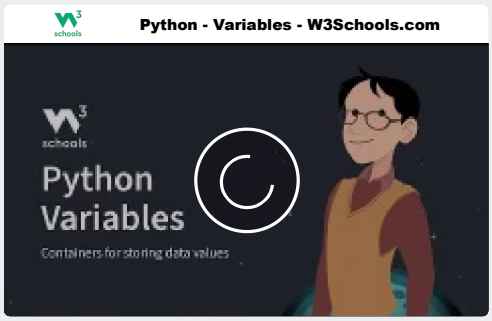
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# JavaScript Object.assign()

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## Example

```
// Create Target Object
const person1 = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Create Source Object
const person2 = {firstName: "Anne",lastName: "Smith"};

// Assign Source to Target
Object.assign(person1, person2);
```

Try it Yourself »

## Description

SHARE

The `Object.assign()` method copies properties from one or more source objects to a target object.

# Related Methods:

Object.assign() copies properties from a source object to a target object.

Object.create() creates an object from an existing object.

Object.fromEntries() creates an object from a list of keys/values.

# Syntax

```
Object.assign(target, source(s))
```

# Parameters

| Parameter | Description |
|---|---|
| *target* | Required.<br>An existing object. |
| *source* | Required.<br>One or more sources. |

# Return Value

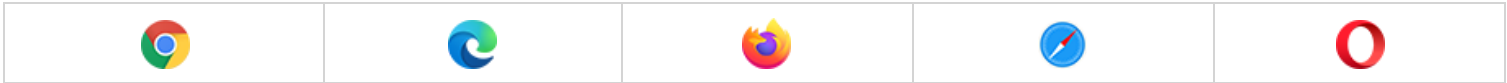
| Type | Description |
|---|---|
| Object | The target object. |

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# Browser Support

`Object.assign()` is an ECMAScript6 (ES6 2015) feature.

**JavaScript 2015** is supported in all browsers since **June 2017**:



| Chrome 51 | Edge 15 | Firefox 54 | Safari 10 | Opera 38 |
|---|---|---|---|---|
| May 2016 | Apr 2017 | Jun 2017 | Sep 2016 | Jun 2016 |

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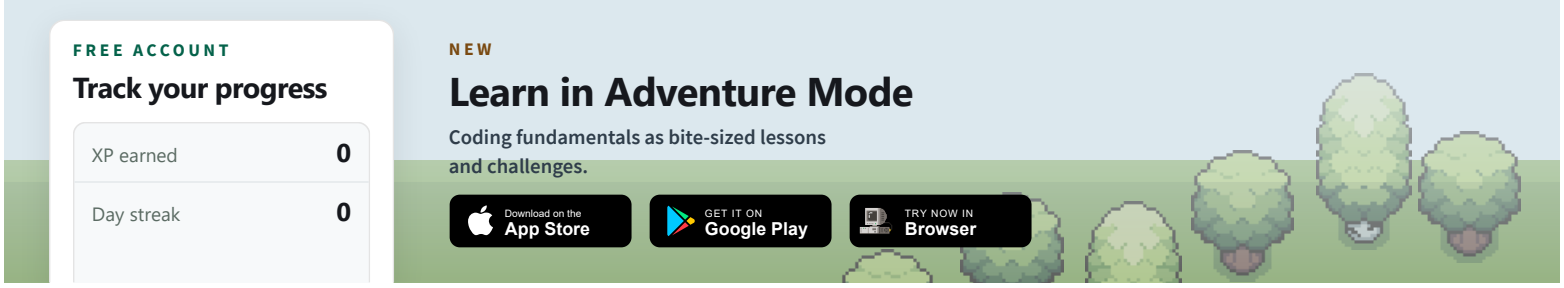
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# JavaScript Object constructor

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## Example

Get the object constructor:

```
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

let text = person.constructor;
```

Try it Yourself »

## Description

SHARE

The `constructor` property returns the function that created the Object prototype.

For JavaScript objects the `constructor` property returns:

**function Object() { [native code] }**

## Syntax

```
object.constructor
```

## Return Value

```
function Object() { [native code] }
```

## Browser Support

`constructor` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome | Edge | Firefox | Safari | Opera |

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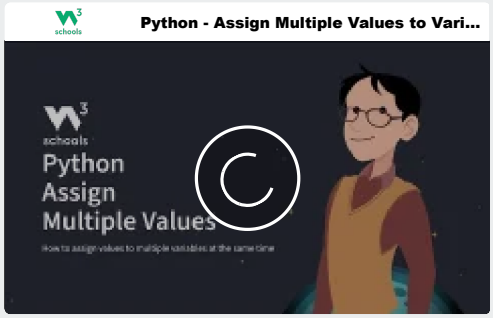
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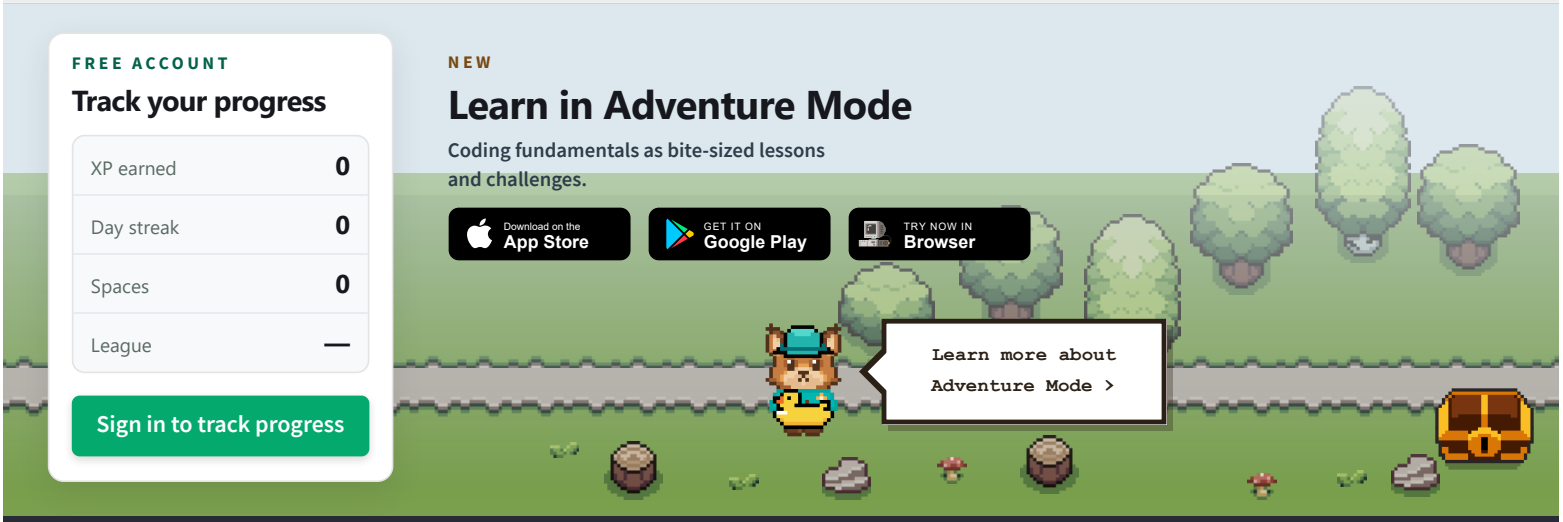
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# JavaScript Object.create()

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## Example

```
// Create an Object:
const person = {
  firstName: "John",
  lastName: "Doe"
};

// Create new Object
const man = Object.create(person);
man.firstName = "Peter";
```

Try it Yourself »

## Description

SHARE

The `Object.create()` method creates an object from an existing object.

## Related Methods:

Object.assign() copies properties from a source object to a target object.

Object.create() creates an object from an existing object.

Object.fromEntries() creates an object from a list of keys/values.

# Syntax

```
Object.create(object, properties)
```

# Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>An existing object. |
| *properties* | Optional.<br>A descriptor of properties to be added or changed:<br><br>value: *value*<br>writable : true\|false<br>enumerable : true\|false<br>configurable : true\|false<br>get : *function*<br>set : *function* |

# Return Value

| Type | Description |
|---|---|
| Object | The created object. |

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# Browser Support

`Object.create()` is an ECMAScript5 (ES5 2009) feature.

**JavaScript 2009** is supported in all browsers since **July 2013**:

| Chrome | IE/Edge | Firefox | Safari | Opera |
|---|---|---|---|---|
| 23 | 11 | 21 | 6 | 15 |

| Sep 2012 | Sep 2012 | Apr 2013 | Jul 2012 | Jul 2013 |
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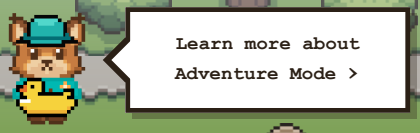
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# JavaScript Object.defineProperties()

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## Example

Add properties:

```
// Create an Object:
const person = {
  firstName: "John",
  lastName: "Doe"
};

// Add Properties
Object.defineProperties(person, {
  language: {value: "en"},
  year: {value: "Hello"}
});
```

Try it Yourself »

## Description

SHARE

The `Object.defineProperties()` method adds or changes object properties.

The `Object.defineProperties()` method lets you change property metadata.

The `Object.defineProperties()` method lets you add getters and setters.

See more example below.

## Related Methods:

Object.defineProperty() adds or changes one property.

Object.defineProperties() adds or changes many properties.

Object.getOwnPropertyNames() returns the property names of an object.

Object.getOwnPropertyDescriptor() returns the descriptor of a property.

Object.getOwnPropertyDescriptors() returns the descriptors of all properties.

## Syntax

```
Object.defineProperty(object, property, descriptor)
```

## Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>The object to edit. |
| *property* | Required.<br>The property to edit. |
| *descriptor* | Required.<br>A descriptor of the properties to be added or changed:<br><br>value: *value*<br>writable : true\|false<br>enumerable : true\|false<br>configurable : true\|false<br>get : *function*<br>set : *function* |

## Return Value

| Type | Description |
|---|---|

| | |
|---|---|
| Object | The object beeing passed, with the changes made. |

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## Browser Support

`Object.defineProperties()` is an ECMAScript5 (ES5 2009) feature.

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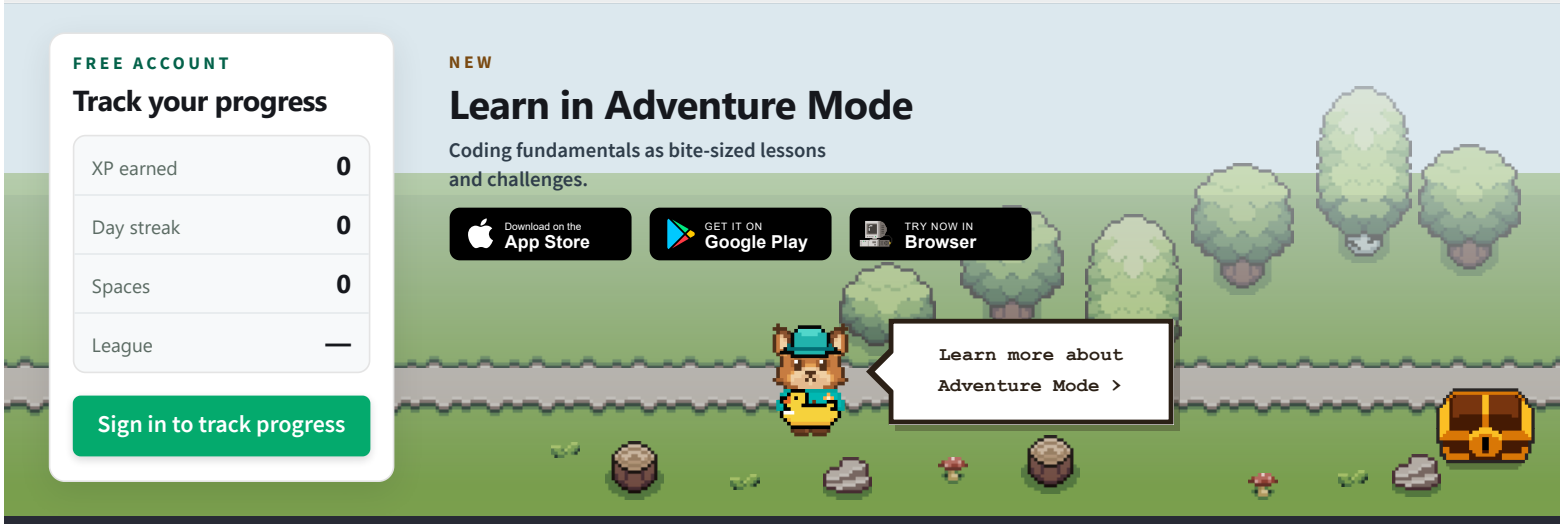
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# JavaScript Object.defineProperty()

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## Example

Add a property:

```
// Create an Object:
const person = {
  firstName: "John",
  lastName: "Doe",
  language: "EN"
};

// Add a new Property
Object.defineProperty(person, "year", {value:"2008"})
```

Try it Yourself »

Change a property:

```
// Create an object:
const person = {
  firstName: "John",
  lastName: "Doe",
  language: "EN"
};

// Change a property
Object.defineProperty(person, "language", {value:"NO"})
```

Try it Yourself »

More Examples Below !

# Description

SHARE

The `Object.defineProperty()` method adds or changes an object property.

The `Object.defineProperty()` method lets you change property metadata.

The `Object.defineProperty()` method lets you add getters and setters.

See more example below.

# Related Methods:

Object.defineProperty() adds or changes one property.

Object.defineProperties() adds or changes many properties.

Object.getOwnPropertyNames() returns the property names of an object.

Object.getOwnPropertyDescriptor() returns the descriptor of a property.

Object.getOwnPropertyDescriptors() returns the descriptors of all properties.

# Syntax

```
Object.defineProperty(object, property, descriptor)
```

# Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>The object. |
| *property* | Required.<br>The property name. |

| | |
|---|---|
| *descriptor* | Required.<br>A descriptor of the property to be added or changed:<br><br>value: *value*<br>writable : true\|false<br>enumerable : true\|false<br>configurable : true\|false<br>get : *function*<br>set : *function* |

# Return Value

| Type | Description |
|---|---|
| Object | The passed object with the changes made. |

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# More Examples

## Example

```
// Create an Object:
const person = {
  firstName: "John",
  lastName : "Doe",
  language : "NO",
};

// Change a Property:
Object.defineProperty(person, "language", {
  value: "EN",
  writable : true,
  enumerable : true,
  configurable : true
});

// Enumerate Properties
let txt = "";
for (let x in person) {
  txt += person[x] + "<br>";
}

// Display Properties
document.getElementById("demo").innerHTML = txt;
```

Try it Yourself »

Next example is the same code, except it hides the language property from enumeration:

## Example

```
// Create an Object:
const person = {
  firstName: "John",
  lastName : "Doe",
  language : "NO",
};

// Change a Property:
Object.defineProperty(person, "language", {
  value: "EN",
  writable : true,
  enumerable : false,
  configurable : true
});

// Enumerate Properties
let txt = "";
for (let x in person) {
  txt += person[x] + "<br>";
}
document.getElementById("demo").innerHTML = txt;
```

Try it Yourself »

# Setters and Getters

This example creates a setter and a getter to secure upper case updates of language:

## Example

```
// Create an Object:
const person = {
  firstName: "John",
  lastName : "Doe",
  language : "NO"
};

// Change a Property:
Object.defineProperty(person, "language", {
  get : function() { return language },
```

```
  set : function(value) { language = value.toUpperCase()}
});

// Change Language
person.language = "en";

// Display Language
document.getElementById("demo").innerHTML = person.language;
```

Try it Yourself »

This example uses a getter to join first name and last name:

## Example

```
// Create an Object:
const person = {
  firstName: "John",
  lastName : "Doe"
};

// Define a Getter
Object.defineProperty(person, "fullName", {
get: function () {return this.firstName + " " + this.lastName;}
});
```

Try it Yourself »

JavaScript Getters and Setters are perfect for creating counters:

## Example

```
// Define Setters and Getters
Object.defineProperty(obj, "reset", {
get : function () {this.counter = 0;}
});
Object.defineProperty(obj, "increment", {
get : function () {this.counter++;}
});
Object.defineProperty(obj, "decrement", {
get : function () {this.counter--;}
});
Object.defineProperty(obj, "add", {
set : function (value) {this.counter += value;}
});
Object.defineProperty(obj, "subtract", {
```

```
  set : function (value) {this.counter -= value;}
});
```

Try it Yourself »

## Browser Support

`Object.defineProperty()` is an ECMAScript5 (ES5 2009) feature.

**JavaScript 2009** is supported in all browsers since **July 2013**:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome<br>23 | IE/Edge<br>11 | Firefox<br>21 | Safari<br>6 | Opera<br>15 |
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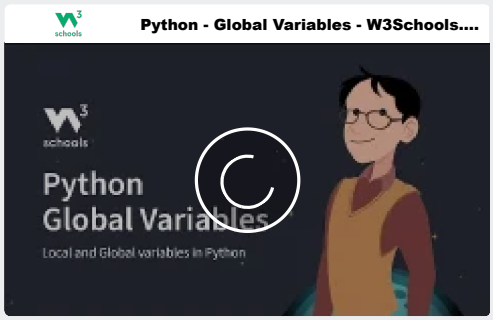
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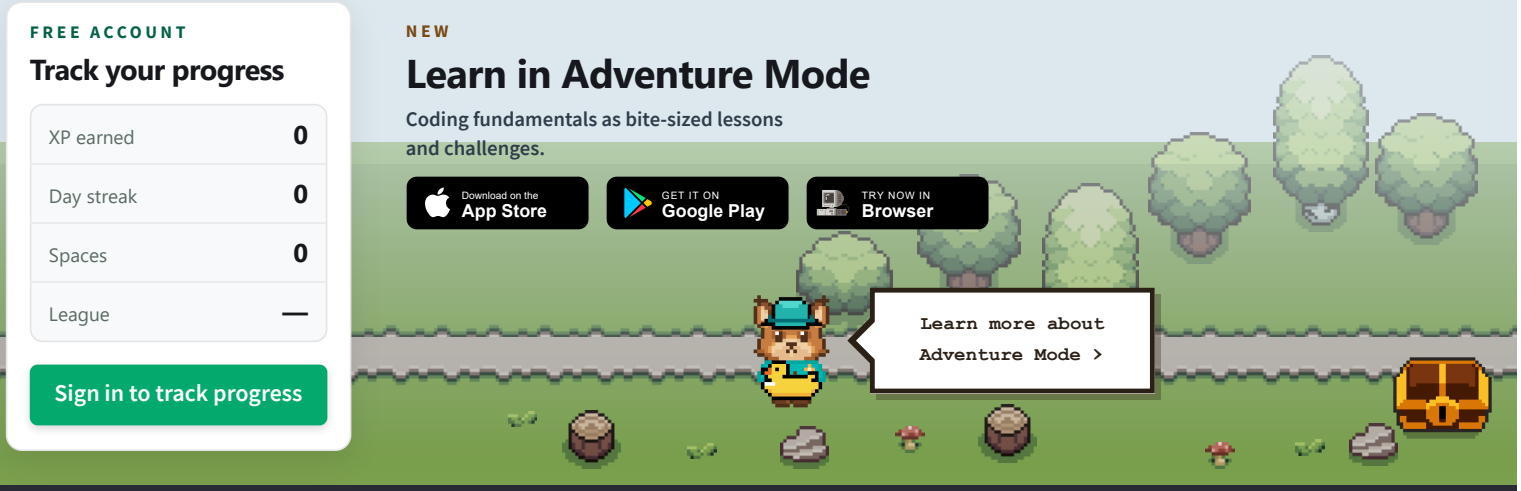
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# The delete Operator

‹ Previous JavaScript Operators Next ›

## Example

```
const person = {
  firstName:"John",
  lastName:"Doe",
  age:50,
  eyeColor:"blue"
};
delete person.age;
```

Try it Yourself »

## JavaScript delete

SHARE

The **delete** operator deletes a property from an object:

The delete operator deletes both the value of the property and the property itself.

After deletion, the property cannot be used before it is added back again.

The delete operator is designed to be used on object properties. It has no effect on variables or functions.

## Note

The delete operator should not be used on the properties of any predefined JavaScript objects (Array, Boolean, Date, Function, Math, Number, RegExp, and String).

This can crash your application.

## Browser Support

`delete` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome | Edge | Firefox | Safari | Opera |

‹ Previous JavaScript Operators Next ›

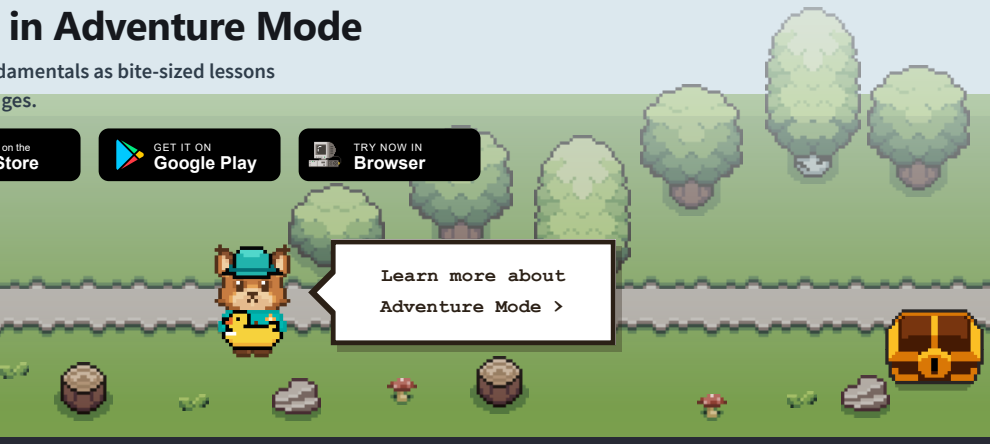
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# JavaScript Object.entries()

‹ Previous JavaScript Object Reference Next ›

## Examples

```
const person = {
  firstName : "John",
  lastName : "Doe",
  age : 50,
  eyeColor : "blue"
};

let text = Object.entries(person);
```

Try it Yourself »

Object.entries() makes it simpler to use objects in loops:

```
const fruits = {Bananas:300, Oranges:200, Apples:500};

let text = "";
for (let [fruit, value] of Object.entries(fruits)) {
  text += fruit + ": " + value + "<br>";
}
```

Try it Yourself »

More examples below.

# Description

SHARE

The `Object.entries()` method returns an array of the key/value pairs of an object.

The `Object.entries()` method does not change the original object.

## Related Methods:

Object.keys() returns the keys (properties) of any object type.

Object.values() returns the values of all object keys (properties).

Object.entries() returns the keys and values of any object types.

The methods above return an **Iterable** (enumerable array).

**Iterables** makes it simpler to use objects in loops and to convert objects into maps.

## Syntax

```
Object.values(object)
```

## Parameters

| Parameter | Description |
|---|---|
| *object* | Optional.<br>An object. |

## Return Value

| Type | Description |
|---|---|
| Array | An iterable array of the object's key/value pairs. |

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## More Examples

`Object.entries()` makes it simpler to convert objects to maps:

### Example

```
const fruits = {Bananas:300, Oranges:200, Apples:500};

const myMap = new Map(Object.entries(fruits));
```

Try it Yourself »

## Browser Support

`Object.entries()` is a JaveScript 2017 feature.

**ES 2017** is supported in all modern browsers since **September 2017**:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome 58 | Edge 15 | Firefox 52 | Safari 11 | Opera 45 |
| Apr 2017 | Apr 2017 | Mar 2017 | Sep 2017 | May 2017 |

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# JavaScript Object.freeze()

‹ Previous JavaScript Object Reference Next ›

## Example

```
"use strict"
// Create Object
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Freeze Object
Object.freeze(person)

// This will throw an error
person.age = 51;
```

Try it Yourself »

More Examples Below !

## Description

SHARE

The `Object.freeze()` method prevents any changes to an object.

The `Object.freeze()` method will fail silently in non-strict mode.

The `Object.freeze()` method will throw a TypeError in strict mode.

Frozen objects are read-only. No modification, addition or deletion of properties are allowed.

The `Object.isFrozen()` method can be used to check if an object is frozen.

## Related Methods:

Object.preventExtensions() allows modifications, but prevents addition of properties.

Object.seal() allows modifications, but prevents additions and deletions of properties.

Object.freeze() prevents modifications, additions and deletions of properties.

Object.isExtensible() returns true if an object is extensible.

Object.isSealed() returns true if an object is sealed.

Object.isFrozen() returns true if an object is frozen.

## Syntax

```
Object.freeze(object)
```

## Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>The object to freeze. |

## Return Value

| Type | Description |
|---|---|
| Object | The frozen object. |

## More Examples

## Example

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.freeze(fruits);

// This will trow an error:
fruits.push("Kiwi");
```

Try it Yourself »

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# Browser Support

`Object.freeze()` is an ECMAScript5 (ES5 2009) feature.

**JavaScript 2009** is supported in all browsers since **July 2013**:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome<br>23 | IE/Edge<br>11 | Firefox<br>21 | Safari<br>6 | Opera<br>15 |
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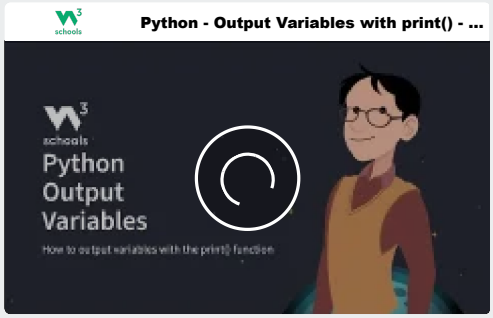
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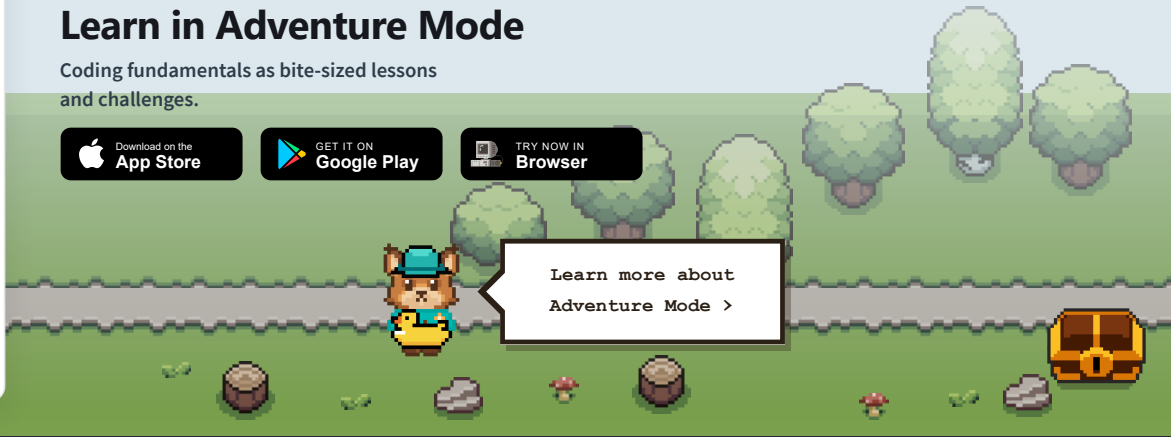
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# JavaScript Object.fromEntries()

‹ Previous JavaScript Object Reference Next ›

## Example

```
const fruits = [
  ["apples", 300],
  ["pears", 900],
  ["bananas", 500]
];

const myObj = Object.fromEntries(fruits);
```

Try it Yourself »

## Description

SHARE

The `fromEntries()` method creates an object from a list of key/value pairs.

## Related Methods:

Object.assign() copies properties from a source object to a target object.

Object.create() creates an object from an existing object.

Object.fromEntries() creates an object from a list of keys/values.

# Syntax

```
Object.fromEntries(iterable)
```

# Parameters

| Parameter | Description |
|---|---|
| *iterable* | Optional.<br>An iterable array or map. |

# Return Value

| Type | Description |
|---|---|
| Object | The object created from the key/value pairs. |

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# Browser Support

ES2019 added the Object method `fromEntries()` to JavaScript.

JavaScript Object `fromEntries()` is supported in all modern browsers since January 2020:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome 73 | Edge 79 | Firefox 63 | Safari 12.1 | Opera 60 |
| Mar 2019 | Jan 2020 | Oct 2018 | Mar 2019 | Apr 2019 |

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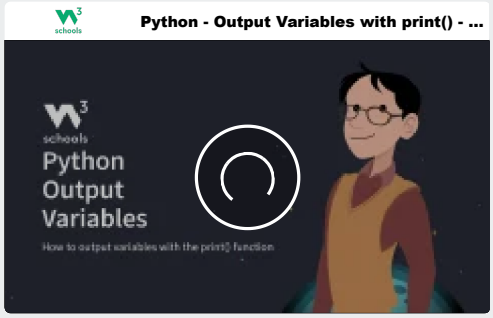
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# Object.getOwnPropertyDescriptor()

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## Example

```
// Create an Object
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Get Properties
let descriptor = Object.getOwnPropertyDescriptor(person);
```

Try it Yourself »

## Description

SHARE

The `Object.getOwnPropertyDescriptor()` method returns the property descriptors of an object.

The `Object.getOwnPropertyDescriptor()` method does not change the original object.

## Related Methods:

Object.defineProperty() adds or changes one property.

Object.defineProperties() adds or changes many properties.

Object.getOwnPropertyNames() returns the property names of an object.

Object.getOwnPropertyDescriptor() returns the descriptor of a property.

Object.getOwnPropertyDescriptors() returns the descriptors of all properties.

# Syntax

```
Object.getOwnPropertyDescriptor(object)
```

# Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>An iterable object. |

# Return Value

| Type | Description |
|---|---|
| Array | An interable array containing the keys of the object. |

# Browser Support

`Object.getOwnPropertyDescriptor()` is an ECMAScript5 (ES5 2009) feature.

**JavaScript 2009** is supported in all browsers since **July 2013**:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome<br>23 | IE/Edge<br>11 | Firefox<br>21 | Safari<br>6 | Opera<br>15 |
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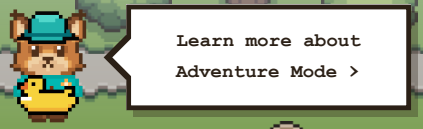
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# Object.getOwnPropertyDescriptors()

‹ Previous JavaScript Object Reference Next ›

## Example

```
// Create an Object
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Get Properties
let descriptors = Object.getOwnPropertyDescriptors(person);
```

Try it Yourself »

## Description

SHARE

The `Object.getOwnPropertyDescriptors()` method returns the property descriptors of an object.

The `Object.getOwnPropertyDescriptors()` method does not change the original object.

## Related Methods:

Object.defineProperty() adds or changes one property.

Object.defineProperties() adds or changes many properties.

Object.getOwnPropertyNames() returns the property names of an object.

Object.getOwnPropertyDescriptor() returns the descriptor of a property.

Object.getOwnPropertyDescriptors() returns the descriptors of all properties.

# Syntax

```
Object.getOwnPropertyDescriptors(object)
```

# Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>The object to get the property descriptors from. |

# Return Value

| Type | Description |
|---|---|
| Object | An object containing the property descriptors. |

# Browser Support

`Object.getOwnPropertyDescriptors()` is an ECMAScript5 (ES5 2009) feature.

**JavaScript 2009** is supported in all browsers since **July 2013**:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome<br>23 | IE/Edge<br>11 | Firefox<br>21 | Safari<br>6 | Opera<br>15 |
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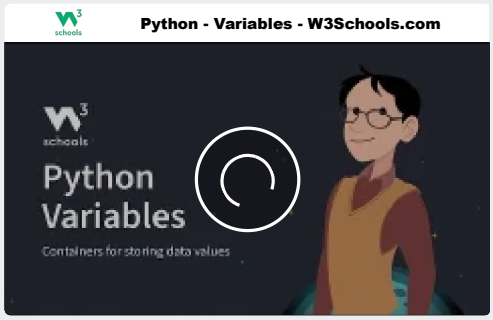
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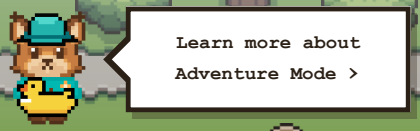
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# Object.getOwnPropertyNames()

‹ Previous JavaScript Object Reference Next ›

## Example

```
// Create an Object
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Get Properties
let props = Object.getOwnPropertyNames(person);
```

Try it Yourself »

## Description

SHARE

The `Object.getOwnPropertyNames()` method returns an array with the properties of an object.

The `Object.getOwnPropertyNames()` method does not change the original object.

## Related Methods:

Object.defineProperty() adds or changes one property.

Object.defineProperties() adds or changes many properties.

Object.getOwnPropertyNames() returns the property names of an object.

Object.getOwnPropertyDescriptor() returns the descriptor of a property.

Object.getOwnPropertyDescriptors() returns the descriptors of all properties.

# Syntax

```
Object.getOwnPropertyNames(object)
```

# Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>An iterable object. |

# Return Value

| Type | Description |
|---|---|
| Array | An array containing the properties of the object. |

# Browser Support

`Object.getOwnPropertyNames()` is an ECMAScript5 (ES5 2009) feature.

**JavaScript 2009** is supported in all browsers since **July 2013**:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome<br>23 | IE/Edge<br>11 | Firefox<br>21 | Safari<br>6 | Opera<br>15 |
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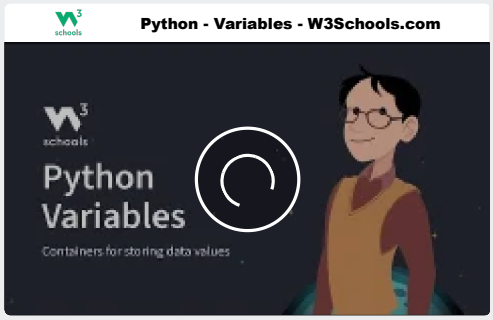
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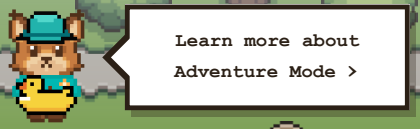



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# JavaScript Object.groupBy()

‹ Previous JavaScript Map Reference Next ›

## Example

```
// Create an Array
const fruits = [
  {name:"apples", quantity:300},
  {name:"bananas", quantity:500},
  {name:"oranges", quantity:200},
  {name:"kiwi", quantity:150}
];

// Callback function to Group Elements
function myCallback({ quantity }) {
  return quantity > 200 ? "ok" : "low";
}

// Group by Quantity
const result = Object.groupBy(fruits, myCallback);
```

Try it Yourself »

## Description

SHARE

The `Object.groupBy()` method groups elements of an object according to string values returned from a callback function.

The `Object.groupBy()` method does not change the original object.

## Note:

The elements in the original and in the returned object are the same.

Changes will be reflected in both the original and in the returned object.

## Object.groupBy() vs Map.groupBy()

The difference between Object.groupBy() and Map.groupBy() is:

Object.groupBy() groups elements into a JavaScript object.

Map.groupBy() groups elements into a Map object.

## Syntax

```
Object.groupBy(iterable, callback)
```

## Parameters

| Parameter | Description |
|---|---|
| *iterable* | Required.<br>An iterable array or map. |
| *callback* | Required.<br>A function to execute for each element.<br>The function should return a group name for the element. |

## Return Value

| Type | Description |
|---|---|
| Iterator | An iterable object with the grouped elements. |

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## Browser Support

`Object.groupby()` is a JavaScrip 2024 feature.

**ES 2024** is supported in all modern browsers since **March 2024**:

| Chrome | Edge | Firefox | Safari | Opera |
|---|---|---|---|---|
| 117 | 117 | 119 | 17.4 | 103 |
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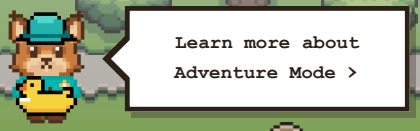
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# JavaScript Object Prototypes

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## Prototype Inheritance

All JavaScript objects inherit properties and methods from a prototype:

- `Date` objects inherit from `Date.prototype`
- `Array` objects inherit from `Array.prototype`

The `Object.prototype` is on the top of the prototype inheritance chain.

`Date` objects, `Array` objects, and all other objects inherit from `Object.prototype`.

## Adding Properties and Methods to Objects

Sometimes you want to add new properties (or methods) to all existing objects of a given type.

Sometimes you want to add new properties (or methods) to an object constructor.

Previously you have learned how to use an **object constructor**:

### Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
```

```
  this.age = age;
  this.eyeColor = eyecolor;
}

const myFather = new Person("John", "Doe", 50, "blue");
const myMother = new Person("Sally", "Rally", 48, "green");
```

Try it Yourself »

You **cannot** add a new property to an existing object constructor:

### Example

```
Person.nationality = "English";
```

Try it Yourself »

To add a new property to a constructor, you must add it to the constructor function:

### Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
  this.age = age;
  this.eyeColor = eyecolor;
  this.nationality = "English";
}
```

Try it Yourself »

## Using the **prototype** Property

The JavaScript `prototype` property allows you to add new properties to object constructors:

### Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
  this.age = age;
```

```
  this.eyeColor = eyecolor;
}

Person.prototype.nationality = "English";
```

Try it Yourself »

The JavaScript `prototype` property also allows you to add new methods to object constructors:

## Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
  this.age = age;
  this.eyeColor = eyecolor;
}

Person.prototype.name = function() {
  return this.firstName + " " + this.lastName;
};
```

Try it Yourself »

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## Warning

Only modify your **own** prototypes. Never modify the prototypes of standard JavaScript objects.

## Exercise

**Which prototype is on the top of the prototype inheritance chain?**

- Date.prototype
- Array.prototype

- Object.prototype
- Person.prototype

Submit Answer »

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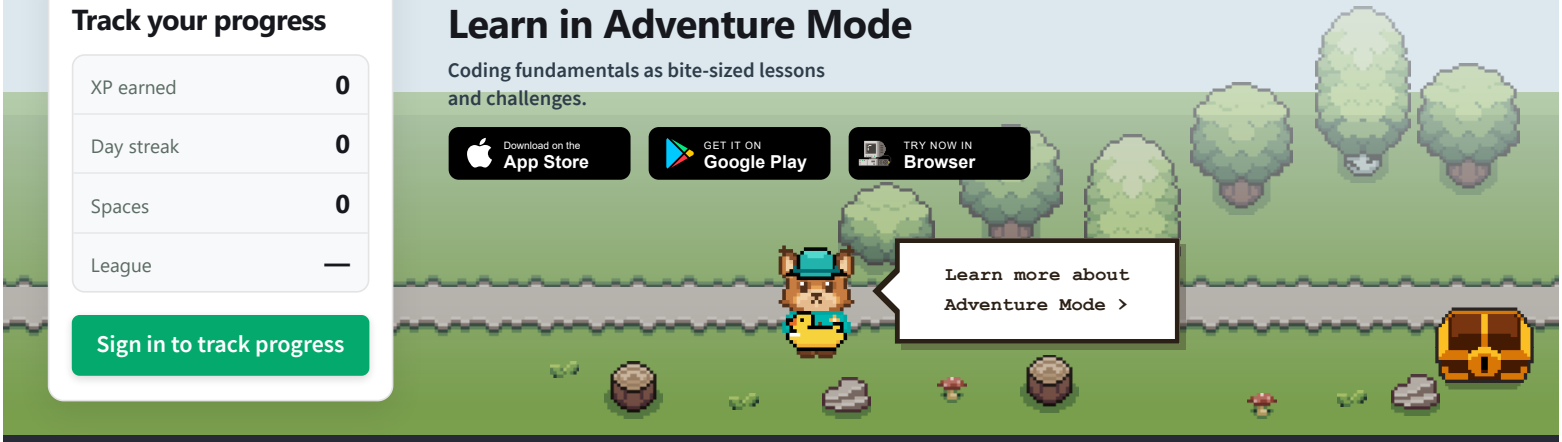
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# JavaScript Advanced Objects Code Challenge

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## Challenge: Advanced Objects

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Test your understanding of JavaScript by completing a small coding challenge.

**Instructions**

Inside the `<script>` element, complete the following steps:

1. Add a getter `description` to `car` that returns the brand and the model
2. Protect `car` with `Object.freeze()`, before the line that changes the year
3. Loop through `Object.entries(car)`. Add each key and value to `text`, with a `<br>` after each

The page should show:
**brand: Volvo**
**model: EX90**
**year: 2024**
**description: Volvo EX90**

Code Solution Check Code

```
<!DOCTYPE html>
<html>
<body>

<h2>Advanced Objects</h2>
<p id="demo"></p>

<script>
const car = {
```

```
  brand: "Volvo",
  model: "EX90",
  year: 2024,
  // Step 1: add a getter here
};

// Step 2: protect the object

// This line tries to change the year
car.year = 2030;

let text = "";
// Step 3: loop through the entries

document.getElementById("demo").innerHTML = text;
</script>

</body>
</html>
```

Result

Click **Check Code** to see the result here.

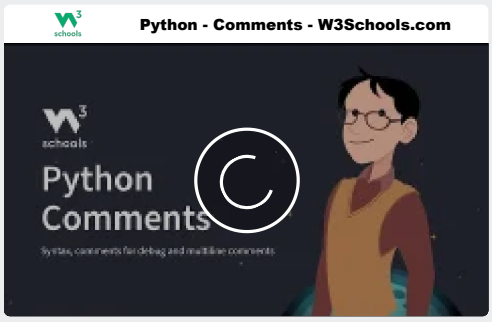
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# JavaScript Object.isExtensible()

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## Example

```
// Create Object
const person = {firstName:"John", lastName:"Doe"};

// Prevent Extensions
Object.preventExtensions(person);

// This will return false
let answer = Object.isExtensible(person);
```

Try it Yourself »

More Examples Below !

## Description

SHARE

The `Object.isExtensible()` returns true if an object is extensible.

## Related Methods:

Object.preventExtensions() allows modifications, but prevents addition of properties.

Object.seal() allows modifications, but prevents additions and deletions of properties.

Object.freeze() prevents modifications, additions and deletions of properties.

Object.isExtensible() returns true if an object is extensible.

Object.isSealed() returns true if an object is sealed.

Object.isFrozen() returns true if an object is frozen.

# Syntax

```
Object.isExtensible(object)
```

# Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>The object to check. |

# Return Value

| Type | Description |
|---|---|
| Boolean | `true` if the object is extensible.<br>`false` if not. |

# More Examples

## Example

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Prevent Extensions
Object.preventExtensions(fruits);

// This will return false
let answer = Object.isExtensible(fruits);
```

Try it Yourself »

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# Browser Support

`Object.isExtensible()` is an ECMAScript5 (ES5 2009) feature.

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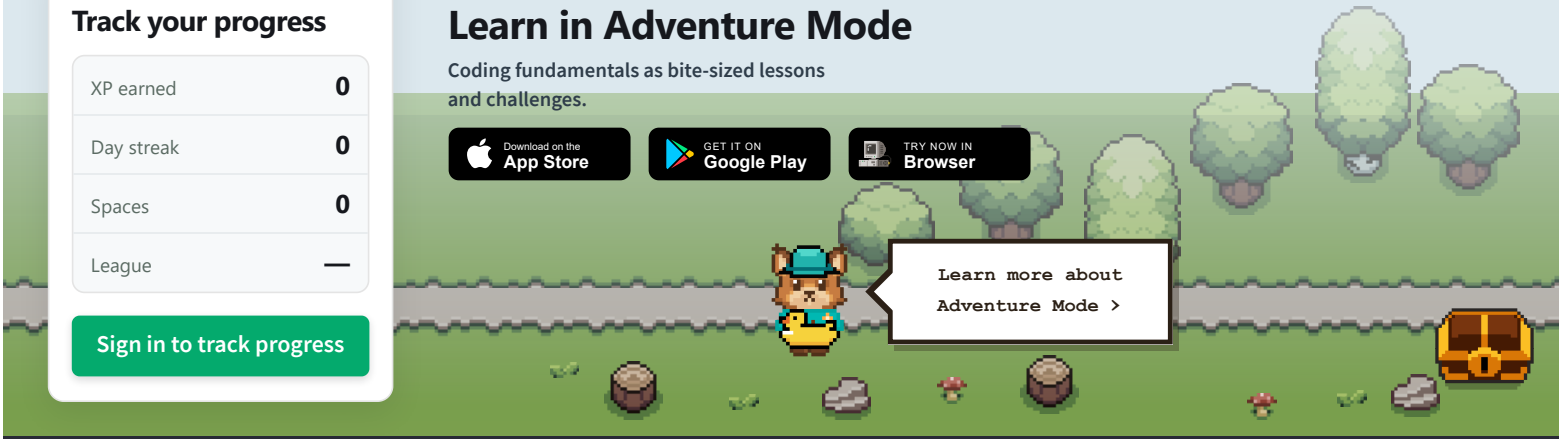
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# JavaScript Object.isFrozen()

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## Example

```
// Create Object
const person = {firstName:"John", lastName:"Doe"};

// Freeze Object
Object.freeze(person);

// This will return true
let text = Object.isFrozen(person);
```

Try it Yourself »

More Examples Below !

## Description

SHARE

The `Object.isFrozen()` returns true if an object is frozen.

## Related Methods:

Object.preventExtensions() allows modifications, but prevents addition of properties.

Object.seal() allows modifications, but prevents additions and deletions of properties.

Object.freeze() prevents modifications, additions and deletions of properties.

Object.isExtensible() returns true if an object is extensible.

Object.isSealed() returns true if an object is sealed.

Object.isFrozen() returns true if an object is frozen.

# Syntax

```
Object.isFrozen(object)
```

# Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>The object to check. |

# Return Value

| Type | Description |
|---|---|
| Boolean | `true` if the object is frozen.<br>`false` if not. |

# More Examples

## Example

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.freeze(fruits);

// This will return true:
let answer = Object.isFrozen(fruits);
```

Try it Yourself »

# Browser Support

`Object.isFrozen()` is an ECMAScript5 (ES5 2009) feature.

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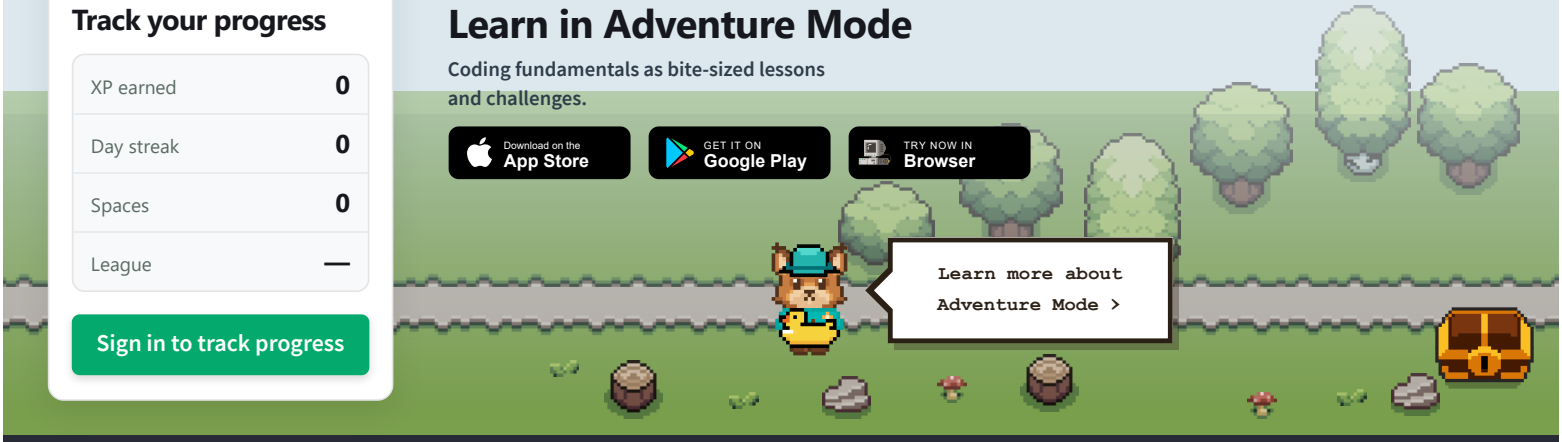



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# JavaScript Object.isSealed()

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## Example

```
// Create Object
const person = {firstName:"John", lastName:"Doe"};

// Seal Object
Object.seal(person);

// This will return true
let answer = Object.isSealed(person);
```

Try it Yourself »

More Examples Below !

## Description

SHARE

The `Object.isSealed()` returns true if an object is sealed.

## Related Methods:

Object.preventExtensions() allows modifications, but prevents addition of properties.

Object.seal() allows modifications, but prevents additions and deletions of properties.

Object.freeze() prevents modifications, additions and deletions of properties.

Object.isExtensible() returns true if an object is extensible.

Object.isSealed() returns true if an object is sealed.

Object.isFrozen() returns true if an object is frozen.

# Syntax

```
Object.isSealed(object)
```

# Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>The object to check. |

# Return Value

| Type | Description |
|---|---|
| Boolean | `true` if the object is sealed.<br>`false` if not. |

# More Examples

## Example

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Seal Array
Object.seal(fruits);

// This will return true
let answer = Object.isSealed(fruits);
```

Try it Yourself »

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## Browser Support

`Object.isSealed()` is an ECMAScript5 (ES5 2009) feature.

**JavaScript 2009** is supported in all browsers since **July 2013**:

| Chrome | IE/Edge | Firefox | Safari | Opera |
|---|---|---|---|---|
| 23 | 11 | 21 | 6 | 15 |
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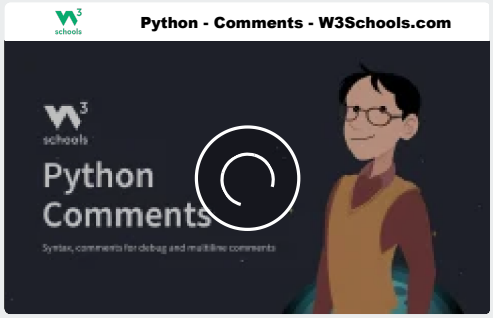
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# JavaScript Object.keys()

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## Example

```
// Create an Object
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Get the Keys
const keys = Object.keys(person);
```

Try it Yourself »

More Examples Below !

## Description

SHARE

The `Object.keys()` method returns an array with the keys of an object.

The `Object.keys()` method does not change the original object.

# Related Methods:

Object.keys() returns the keys (properties) of any object type.

Object.values() returns the values of all object keys (properties).

Object.entries() returns the keys and values of any object types.

The methods above return an **Iterable** (enumerable array).

**Iterables** makes it simpler to use objects in loops and to convert objects into maps.

# Syntax

```
Object.keys(object)
```

# Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>An iterable object. |

# Return Value

| Type | Description |
|---|---|
| Array | An array containing the keys of the object. |

# More Examples

## Examples

This example list only the enumerable properties of an object:

```
// Create an Object
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};
```

```
// Change Property
Object.defineProperty(person, "age", {enumerable:false});

// Get the Keys
const keys = Object.keys(person);
```

Try it Yourself »

Use Object.keys() on an array:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
const keys = Object.keys(fruits);
```

Try it Yourself »

Use Object.keys() on a string:

```
const fruits = "Banana";
const keys = Object.keys(fruits);
```

Try it Yourself »

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# Browser Support

`Object.keys()` is an ECMAScript5 (ES5 2009) feature.

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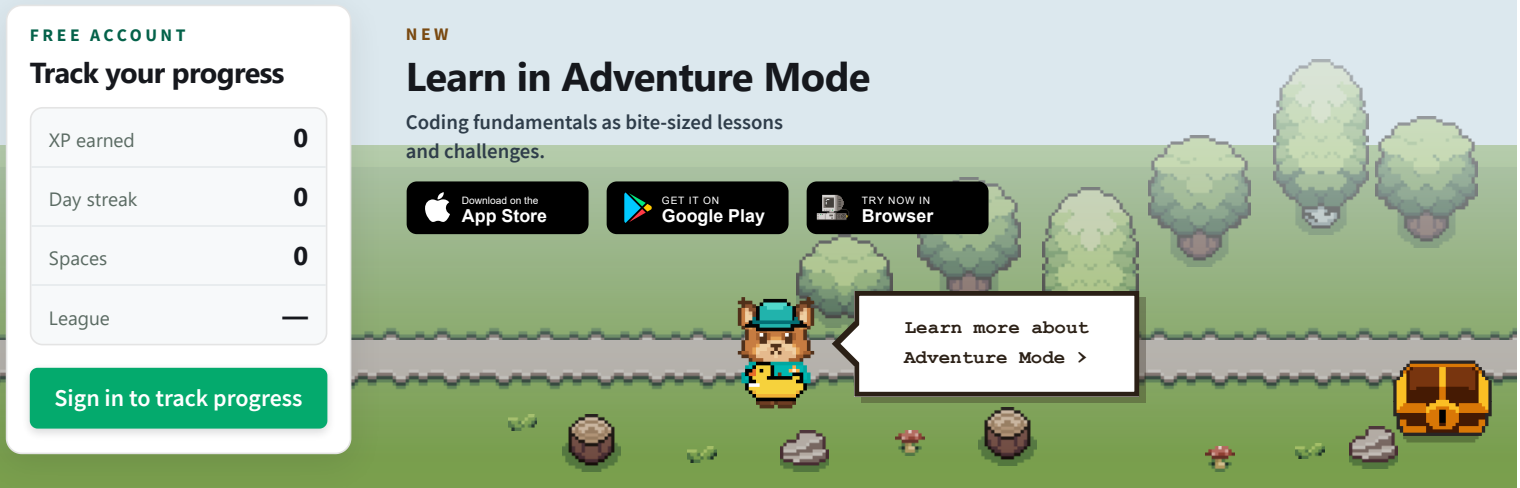
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# JavaScript Object.preventExtensions()

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## Example

```
// Create Object
const person = {firstName:"John", lastName:"Doe"};

// Prevent Extensions
Object.preventExtensions(person);

// This will throw an error
person.nationality = "English";
```

Try it Yourself »

More Examples Below !

## Description

SHARE

The `Object.preventExtensions()` method prevents adding properties to an object.

You can use `Object.isExtensible()` to check if an object is extensible.

## Related Methods:

Object.preventExtensions() allows modifications, but prevents addition of properties.

Object.seal() allows modifications, but prevents additions and deletions of properties.

Object.freeze() prevents modifications, additions and deletions of properties.

Object.isExtensible() returns true if an object is extensible.

Object.isSealed() returns true if an object is sealed.

Object.isFrozen() returns true if an object is frozen.

# Syntax

```
Object.preventExtensions(object)
```

# Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>The object to prevent extensions to. |

# Return Value

| Type | Description |
|---|---|
| Object | The object that was modified. |

# More Examples

## Example

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.preventExtensions(fruits);

// This will throw an error:
fruits.push("Kiwi");
```

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# Browser Support

`Object.preventExtensions()` is an ECMAScript5 (ES5 2009) feature.

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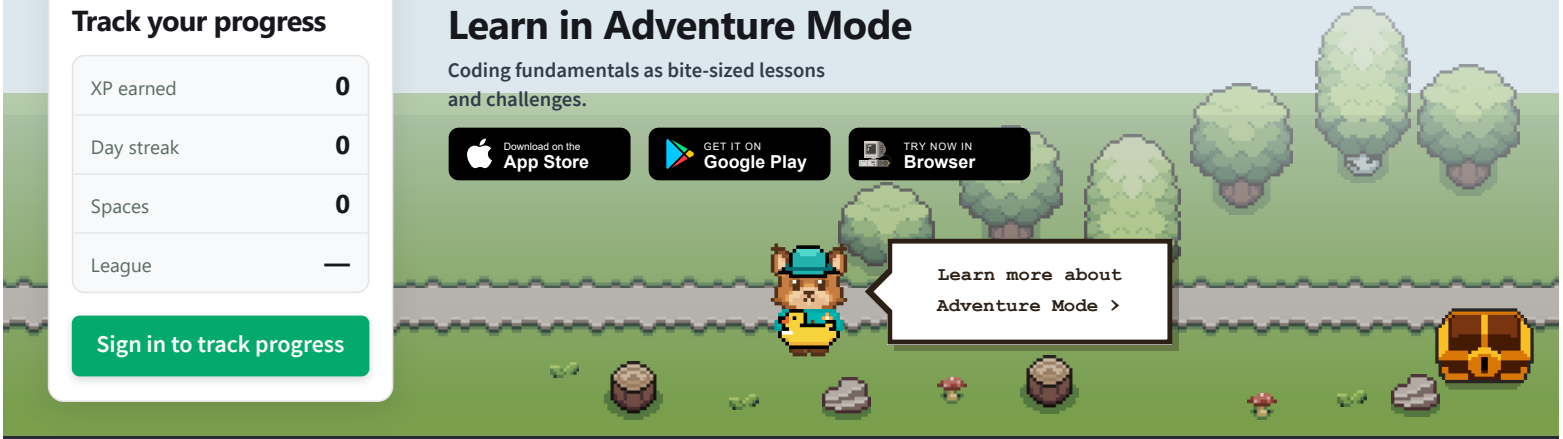
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# JavaScript Object prototype

‹ Previous JavaScript Object Reference Next ›

## Example

Use the prototype property to add a new property to all objects of a given type:

```
function employee(name, jobtitle, born) {
  this.name = name;
  this.jobtitle = jobtitle;
  this.born = born;
}
employee.prototype.salary = 2000;

const fred = new employee("Fred Flintstone", "Caveman", 1970);
```

Try it Yourself »

## Description

SHARE

The `prototype` is a global property available with all JavaScript objects.

The `prototype` property allows you to add new properties and methods to objects.

## Syntax

```
object.prototype.name = value
```

# Browser Support

`prototype` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

| Chrome | Edge | Firefox | Safari | Opera |
|---|---|---|---|---|

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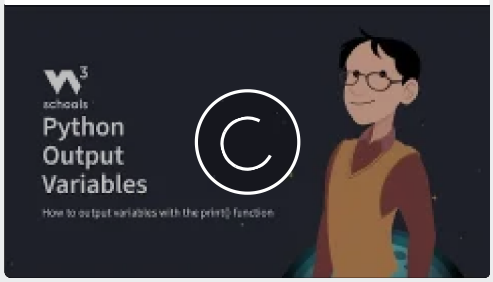
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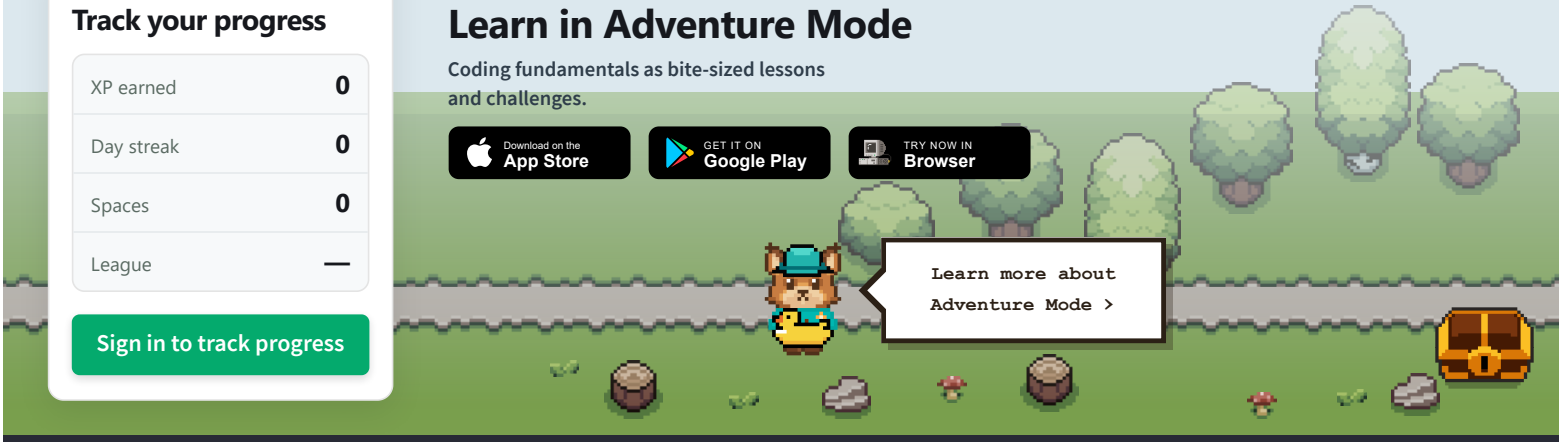
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# JavaScript Object.seal()

‹ Previous JavaScript Object Reference Next ›

## Example

```
"use strict"
// Create Object
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Seal Object
Object.seal(person)

// This will throw an error
delete person.age;
```

Try it Yourself »

More Examples Below !

## Description

SHARE

The `Object.seal()` method prevents additions or deletions of new properties.

The `Object.seal()` method makes existing properties non-configurable.

The `Object.seal()` method will fail silently in non-strict mode.

The `Object.seal()` method will throw a TypeError in strict mode.

The `Object.isSealed()` method can be used to check if an object is sealed.

## Related Methods:

Object.preventExtensions() allows modifications, but prevents addition of properties.

Object.seal() allows modifications, but prevents additions and deletions of properties.

Object.freeze() prevents modifications, additions and deletions of properties.

Object.isExtensible() returns true if an object is extensible.

Object.isSealed() returns true if an object is sealed.

Object.isFrozen() returns true if an object is frozen.

## Syntax

```
Object.seal(object)
```

## Parameters

| Parameter | Description |
|---|---|
| *object* | Required.<br>The object to seal. |

## Return Value

| Type | Description |
|---|---|
| Object | The sealed object. |

## More Examples

## Example

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.seal(fruits);

// This will throw an error:
fruits.push("Kiwi");
```

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# Browser Support

`Object.seal()` is an ECMAScript5 (ES5 2009) feature.

**JavaScript 2009** is supported in all browsers since **July 2013**:

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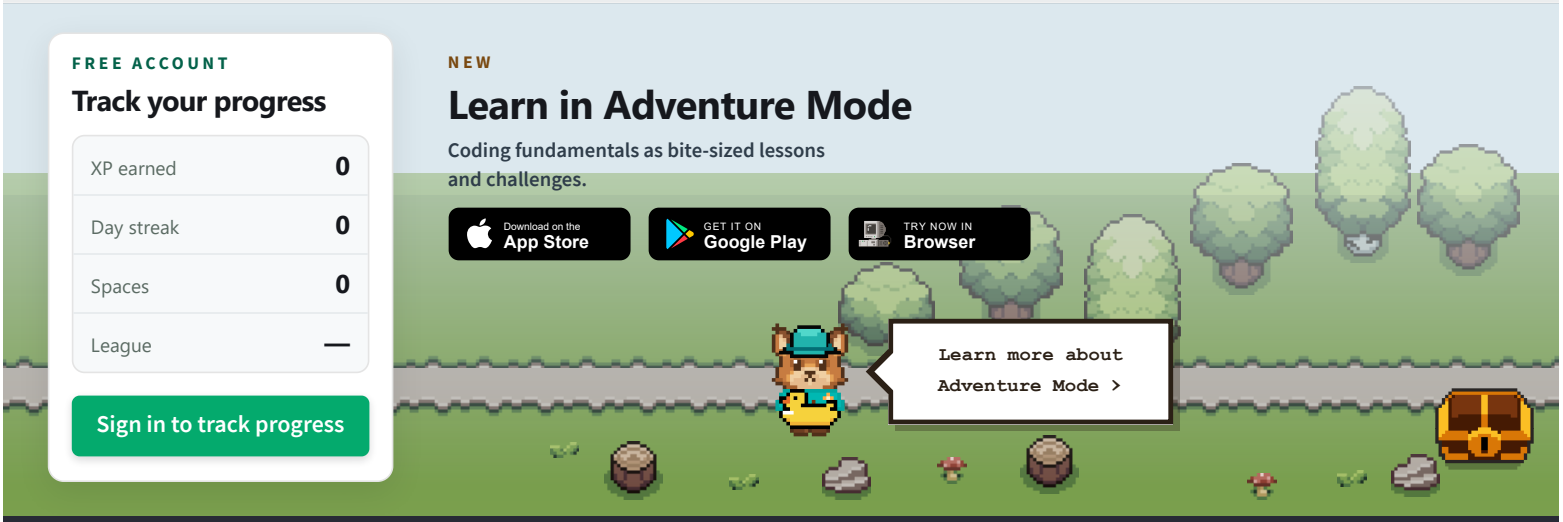
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# JavaScript Object toString()

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## Examples

Using toString() on an array:

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
let text = fruits.toString();
```

Try it Yourself »

Using toString() on an object:

```
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};
const keys = person.toString();
```

Try it Yourself »

Using Object.toString() on an object:

```
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
```

```
};
const keys = Object.toString(person);
```

Try it Yourself »

# Description

SHARE

The `toString()` method returns an object as a string.

The `toString()` method returns "[object Object]" if it cannot return a string.

`Object.toString()` always returns the object constructor.

The `toString()` method does not change the original object.

# Note

Every JavaScript object has a `toString()` method.

The `toString()` method is used by JavaScript when an object needs to be displayed as a text (like in HTML), or when an object needs to be used as a string.

# Browser Support

`toString()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

| Chrome | Edge | Firefox | Safari | Opera |
|---|---|---|---|---|

# Syntax

```
object.toString()
```

# Parameters

| NONE |
|---|

# Return Value

A string representing the object.

Or "[object type]" if it cannot return a string.

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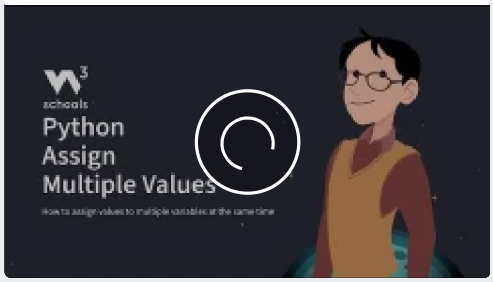
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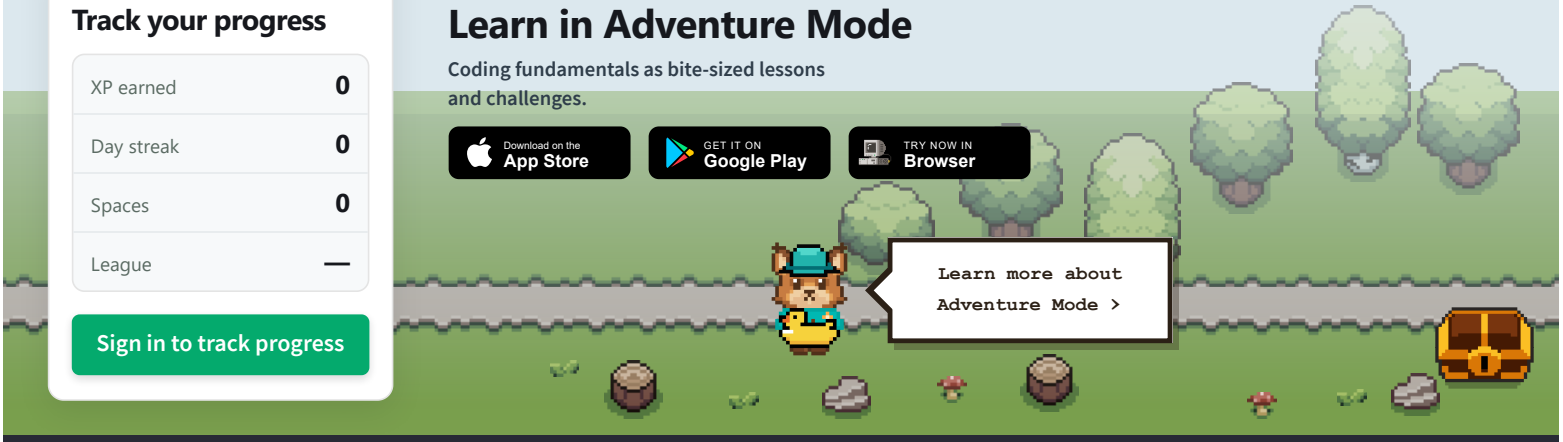
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# JavaScript Object valueOf()

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## Examples

```
// Create a person object
const person = {
  firstName:"John",
  lastName:"Doe",
  age:50,
  eyeColor:"blue"
}
let text = person.valueOf();
```

Try it Yourself »

## Description

SHARE

The `valueOf()` method returns the primitive value of an object.

If the object has no primitive value, `valueOf()` returns the object itself.

The `valueOf()` method does not change the original object.

## Syntax

```
object.valueOf()
```

## Parameters

NONE

## Return Value

The primitive value of an object.
Normally, it will return the object itself.
If the object is an array, it will return the primitive values.

## Browser Support

`valueOf()` is an ECMAScript1 (JavaScript 1997) feature.

It is supported in all browsers:

| Chrome | Edge | Firefox | Safari | Opera |
|---|---|---|---|---|

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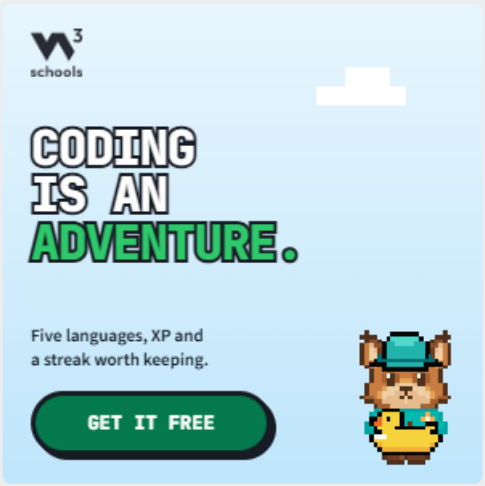
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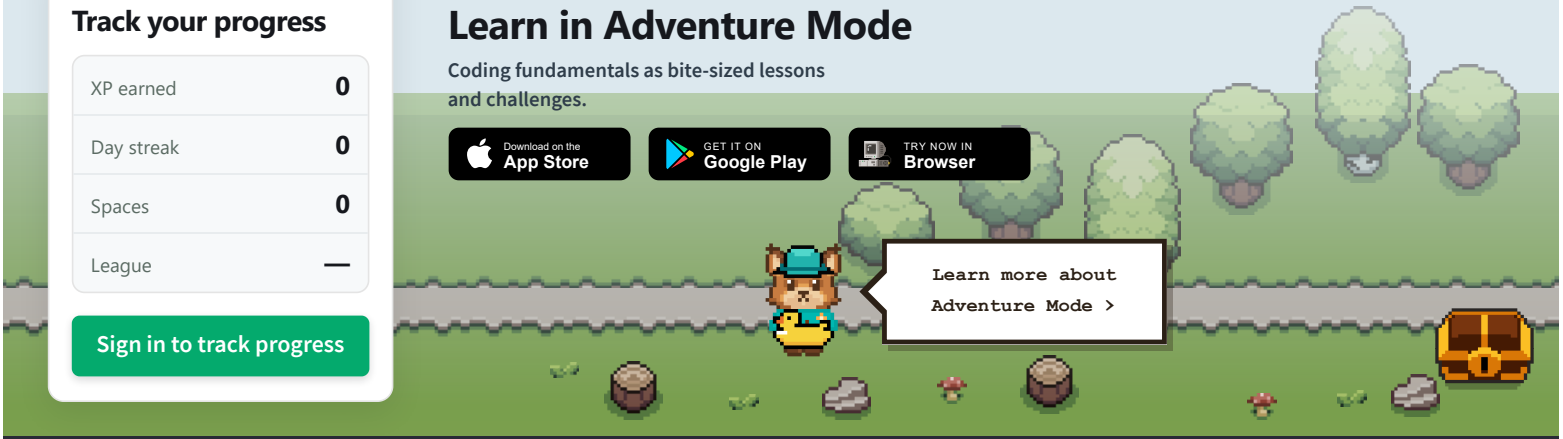
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# JavaScript Object.values()

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## Example

```
const person = {
  firstName : "John",
  lastName : "Doe",
  age : 50,
  eyeColor : "blue"
};

let text = Object.values(person);
```

Try it Yourself »

## Description

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The `Object.values()` method returns an array of the property values of an object.

The `Object.values()` method does not change the original object.

## Related Methods:

Object.keys() returns the keys (properties) of any object type.

Object.values() returns the values of all object keys (properties).

Object.entries() returns the keys and values of any object types.

The methods above return an **Iterable** (enumerable array).

**Iterables** makes it simpler to use objects in loops and to convert objects into maps.

# Syntax

```
Object.values(object)
```

# Parameters

| Parameter | Description |
|---|---|
| *object* | Optional.<br>An object. |

# Return Value

| Type | Description |
|---|---|
| Array | An iterable array of the object's property values. |

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# Browser Support

ECMAScript 2017 added the `Object.values()` method to objects.

`Object.values()` is supported in all modern browsers since March 2017:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome 54 | Edge 14 | Firefox 47 | Safari 10.1 | Opera 41 |
| Oct 2016 | Aug 2016 | Jun 2016 | Mar 2017 | Oct 2016 |

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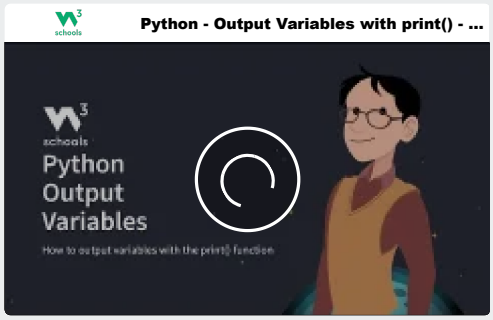
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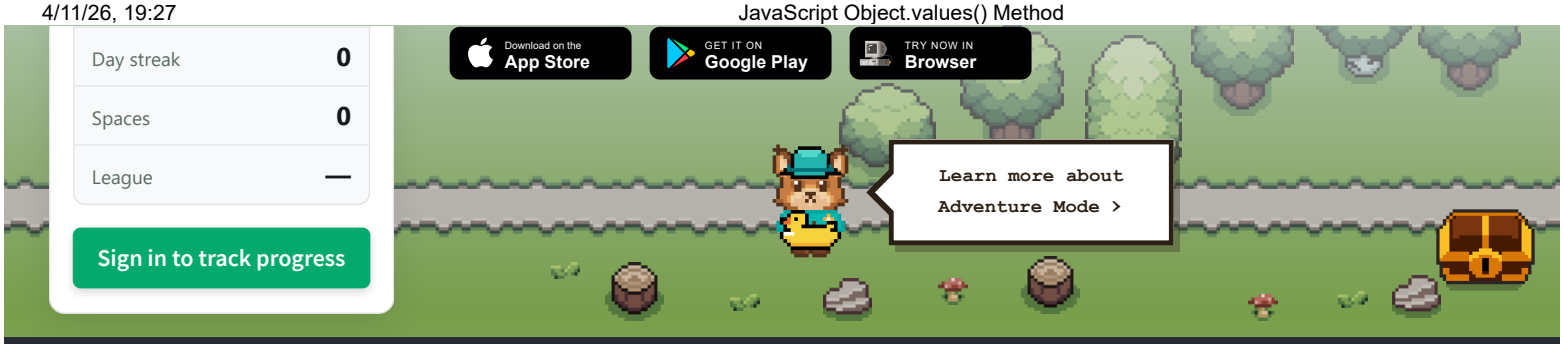

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# JavaScript Object Definitions

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## Methods for Defining JavaScript Objects

- Using an Object Literal
- Using the `new` Keyword
- Using `Object.assign()`
- Using `Object.create()`
- Using `Object.fromEntries()`

## Using an Object Literal

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An object literal is a list of property **key:values** inside curly braces **{ }**.

```
{firstName:"John", lastName:"Doe", age:50, eyeColor:"blue"};
```

### Example

```
// Create an Object
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
```

```
  eyeColor: "blue"
};
```

Try it Yourself »

# Using the **new** Keyword

## Example

```
// Create an Object
const person = new Object({
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
});
```

Try it Yourself »

The examples above do exactly the same.

But, there is no need to use `new Object()`.

For readability, simplicity and execution speed, use the **object literal** method.

Objects written as name value pairs are similar to:

- Associative arrays in PHP
- Dictionaries in Python
- Hash tables in C
- Hash maps in Java
- Hashes in Ruby and Perl

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# JavaScript Object.create()

The `Object.create()` method creates an object from an existing object.

### Example

```
// Create an Object:
const person = {
  firstName: "John",
  lastName: "Doe"
};

// Create new Object
const man = Object.create(person);
man.firstName = "Peter";
```

Try it Yourself »

## JavaScript Object fromEntries()

ES2019 added the Object method `fromEntries()` to JavaScript.

The `fromEntries()` method creates an object from iterable key / value pairs.

### Example

```
const fruits = [
["apples", 300],
["pears", 900],
["bananas", 500]
];

const myObj = Object.fromEntries(fruits);
```

Try it Yourself »

## Browser Support

`fromEntries()` is an ECMAScript 2019 feature.

**ES2019** is supported in all modern browsers since **January 2020**:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome<br>66 | Edge<br>79 | Firefox<br>61 | Safari<br>12 | Opera<br>50 |

| Apr 2018 | Jan 2020 | Jun 2018 | Sep 2018 | May 2018 |
|---|---|---|---|---|

# JavaScript Object.assign()

The `Object.assign()` method copies properties from one or more source objects to a target object.

## Example

```
// Create Target Object
const person1 = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Create Source Object
const person2 = {firstName: "Anne",lastName: "Smith"};

// Assign Source to Target
Object.assign(person1, person2);
```

Try it Yourself »

# In JavaScript, Objects are King.

## If you Understand Objects, you Understand JavaScript.

In JavaScript, almost "everything" is an object.

- Objects are objects
- Maths are objects
- Functions are objects
- Dates are objects
- Arrays are objects
- Maps are objects
- Sets are objects

All JavaScript values, except primitives, are objects.

# JavaScript Primitives

A **primitive data type** is a data type that can only store a single primitive value.

JavaScript defines 7 types of primitive data types:

| Type | Example value |
|---|---|
| `string` | "Hello" |
| `number` | 3.14 |
| `boolean` | true |
| `bigint` | 12345678901234n |
| `null` | null |
| `undefined` | undefined |
| `symbol` | Symbol("id") |

# Immutable

Primitive values are immutable (they are hardcoded and cannot be changed).

if x = 3.14, you can change the value of x, but you cannot change the value of 3.14.

| Value | Type | Comment |
|---|---|---|
| "Hello" | string | "Hello" is always "Hello" |
| 3.14 | number | 3.14 is always 3.14 |
| true | boolean | true is always true |
| false | boolean | false is always false |
| null | null | null is always null |
| undefined | undefined | undefined is always undefined |

# JavaScript Objects are Mutable

Objects are mutable: They are addressed by reference, not by value.

If person is an object, the following statement will not create a copy of person:

```
const x = person;
```

The object x is **not a copy** of person. The object x **is** person.

The object x and the object person share the same memory address.

Any changes to x will also change person:

## Example

```
//Create an Object
const person = {
  firstName:"John",
  lastName:"Doe",
  age:50, eyeColor:"blue"
}

// Try to create a copy
const x = person;

// This will change age in person:
x.age = 10;
```

Try it Yourself »

## Exercise

What is the value of `person.age` after this code runs?

```
const person = {firstName:"John", age:50};
const x = person;
x.age = 10;
```

- 50
- undefined
- 10

Submit Answer »

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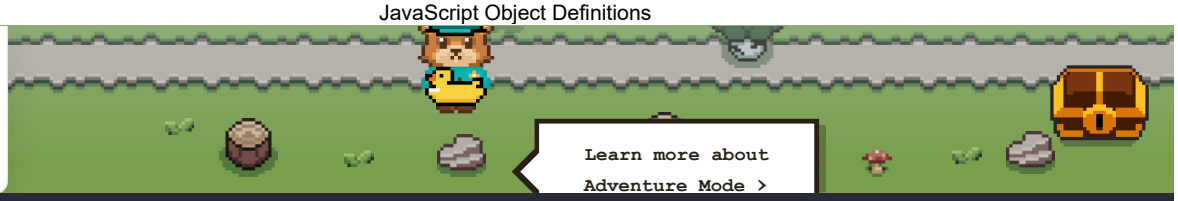
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# **this** in JavaScript Objects

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## **this** in an Object Method

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### Example

```
const person = {
  firstName: "John",
  lastName : "Doe",
  id       : 5566,
  fullName : function() {
    return this.firstName + " " + this.lastName;
  }
};
```

Try it Yourself »

When used in an object method, `this` refers to **the object**.

In the example above, `this` refers to **the person object**.

Because the **fullName** method is a method of the **person** object.

## Object Method Binding

In these examples, `this` is the **person object**:

### Example

```
const person = {
  firstName  : "John",
  lastName   : "Doe",
  id         : 5566,
  myFunction : function() {
    return this;
  }
};
```

Try it Yourself »

### Example

```
const person = {
  firstName: "John",
  lastName : "Doe",
  id       : 5566,
  fullName : function() {
    return this.firstName + " " + this.lastName;
  }
};
```

Try it Yourself »

i.e. **this.firstName** is the **firstName** property of **this** (the person object).

## Explicit Function Binding

The `call()` and `apply()` methods are predefined JavaScript methods.

They can both be used to call an object method with another object as argument.

## See Also:

The Function call() Method

The Function apply() Method

The Function bind() Method

The example below calls person1.fullName with person2 as an argument, **this** refers to person2, even if fullName is a method of person1:

## Example

```
const person1 = {
  fullName: function() {
    return this.firstName + " " + this.lastName;
  }
}

const person2 = {
  firstName:"John",
  lastName: "Doe",
}

// Return "John Doe":
person1.fullName.call(person2);
```

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# Function Borrowing

With the `bind()` method, an object can borrow a method from another object.

This example creates 2 objects (person and member).

The member object borrows the fullname method from the person object:

## Example

```
const person = {
  firstName:"John",
  lastName: "Doe",
  fullName: function () {
    return this.firstName + " " + this.lastName;
  }
}

const member = {
  firstName:"Hege",
  lastName: "Nilsen",
}
```

```
let fullName = person.fullName.bind(member);
```

Try it Yourself »

## Exercise ?

**Which two methods can call an object method with another object as argument?**

- get() and set()
- push() and pop()
- call() and apply()
- keys() and values()

Submit Answer »

# What is **this**?

In JavaScript, the `this` keyword refers to an **object**.

The `this` keyword refers to **different objects** depending on how it is used:

| |
|---|
| Alone, `this` refers to the **global object**. |
| In a function, `this` refers to the **global object**. |
| In a function, in strict mode, `this` is `undefined`. |
| In an object method, `this` refers to the **object**. |
| In an event, `this` refers to the **element** that received the event. |
| In methods like `call()`, `apply()` and `bind()`, `this` can refer to **any object**. |

`this` is not a variable.

`this` is a keyword.

You cannot change the value of `this`.

## **This** Precedence

Use the following precedence of order to determine which object `this` refers to:

| Order | Object | **Because** |
|---|---|---|
| 1 | bind() | `this` is in a function being called using bind() |
| 2 | apply() | `this` is in a function being called using apply() |
| 2 | call() | `this` is in a function being called using call() |
| 3 | Object method | `this` is in an object function (method) |
| 4 | Global scope | `this` is in a function in the global scope |

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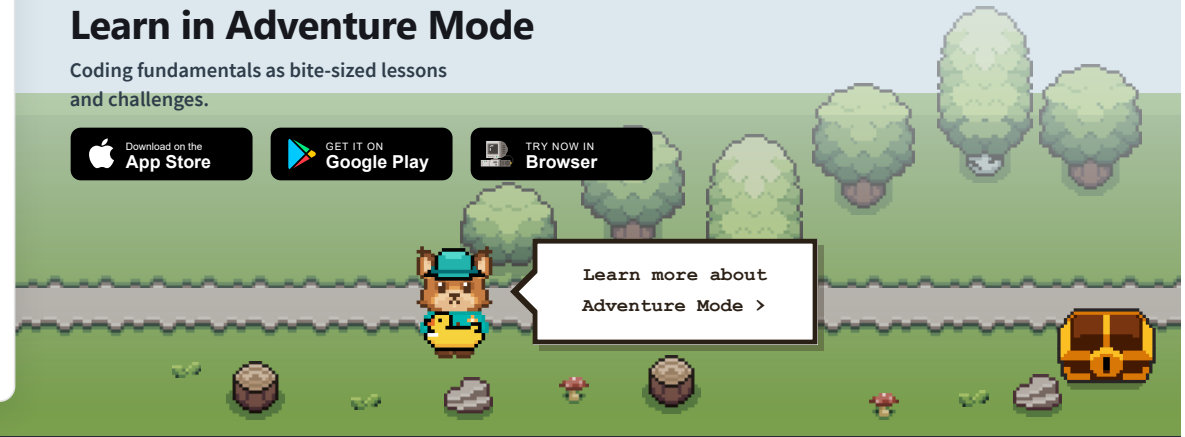
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# JavaScript Object Iterations

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## General Methods

```
// Copies properties from a source object to a target object
Object.assign(target, source)

// Creates an object from an existing object
Object.create(object)

// Returns an array of the key/value pairs of an object
Object.entries(object)

// Creates an object from a list of keys/values
Object.fromEntries()

// Returns an array of the keys of an object
Object.keys(object)

// Returns an array of the property values of an object
Object.values(object)

// Groups object elements according to a function
Object.groupBy(object, callback)
```

## JavaScript Object.entries()

SHARE

ECMAScript 2017 added the `Object.entries()` method to objects.

`Object.entries()` returns an array of the key/value pairs in an object:

## Example

```
const person = {
  firstName : "John",
  lastName : "Doe",
  age : 50,
  eyeColor : "blue"
};

let text = Object.entries(person);
```

Try it Yourself »

`Object.entries()` makes it simple to use objects in loops:

## Example

```
const fruits = {Bananas:300, Oranges:200, Apples:500};

let text = "";
for (let [fruit, value] of Object.entries(fruits)) {
  text += fruit + ": " + value + "<br>";
}
```

Try it Yourself »

`Object.entries()` also makes it simple to convert objects to maps:

## Example

```
const fruits = {Bananas:300, Oranges:200, Apples:500};

const myMap = new Map(Object.entries(fruits));
```

Try it Yourself »

`Object.entries()` is supported in all modern browsers since March 2017.

# JavaScript Object.fromEntries()

The `fromEntries()` method creates an object from a list of key/value pairs.

## Example

```
const fruits = [
  ["apples", 300],
  ["pears", 900],
  ["bananas", 500]
];

const myObj = Object.fromEntries(fruits);
```

Try it Yourself »

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# JavaScript Object.values()

`Object.values()` is similar to `Object.entries()`, but returns a single dimension array of the object values:

## Example

```
const person = {
  firstName : "John",
  lastName : "Doe",
  age : 50,
  eyeColor : "blue"
};

let text = Object.values(person);
```

Try it Yourself »

`Object.values()` is supported in all modern browsers since March 2017.

# JavaScript Object.groupBy()

ES2024 added the `Object.groupBy()` method to JavaScript.

The `Object.groupBy()` method groups elements of an array (or other iterable) according to string values returned from a callback function.

The `Object.groupBy()` method does not change the original array.

## Example

```
// Create an Array
const fruits = [
  {name:"apples", quantity:300},
  {name:"bananas", quantity:500},
  {name:"oranges", quantity:200},
  {name:"kiwi", quantity:150}
];

// Callback function to Group Elements
function myCallback({ quantity }) {
  return quantity > 200 ? "ok" : "low";
}

// Group by Quantity
const result = Object.groupBy(fruits, myCallback);
```

Try it Yourself »

# Browser Support

`Object.groupBy()` is an ECMAScript 2024 feature.

JavaScript 2024 is supported in new browsers since **March 2024**:

|  |  |  |  |  |
|---|---|---|---|---|
| Chrome 117 | Edge 117 | Firefox 119 | Safari 17.4 | Opera 103 |
| Sep 2023 | Sep 2023 | Oct 2023 | Okt 2024 | May 2023 |

# Warning

ES2024 features are relatively new.

Older browsers may need an alternative code (Polyfill)

# Object.groupBy() vs Map.groupBy()

The difference between Object.groupBy() and Map.groupBy() is:

Object.groupBy() groups elements into a JavaScript object.

Map.groupBy() groups elements into a Map object.

# JavaScript Object.keys()

The `Object.keys()` method returns an array with the keys of an object.

## Example

```
// Create an Object
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Get the Keys
const keys = Object.keys(person);
```

Try it Yourself »

# JavaScript for...in Loop

The JavaScript `for...in` statement loops through the properties of an object.

## Syntax

```
for (let variable in object) {
  // code to be executed
}
```

The block of code inside of the `for...in` loop will be executed once for each property.

Looping through the properties of an object:

## Example

```
const person = {
  fname:" John",
  lname:" Doe",
  age: 25
};

let txt = "";
for (let x in person) {
  txt += person[x];
}
```

Try it Yourself »

## Exercise

What is the value of `keys` after this code runs?

```
const person = {firstName:"John", age:50};
const keys = Object.keys(person);
```

- ["John", 50]
- 2
- ["firstName", "age"]
- [["firstName", "John"], ["age", 50]]

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# JavaScript Object Accessors

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## JavaScript Getters and Setters

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Getters and setters allow you to define Object Accessors (Computed Properties).

## JavaScript Getter (The get Keyword)

This example uses a `lang` property to `get` the value of the `language` property.

### Example

```
// Create an object:
const person = {
  firstName: "John",
  lastName: "Doe",
  language: "en",
  get lang() {
    return this.language;
  }
};

// Display data from the object using a getter:
document.getElementById("demo").innerHTML = person.lang;
```

Try it Yourself »

# JavaScript Setter (The set Keyword)

This example uses a `lang` property to `set` the value of the `language` property.

## Example

```
const person = {
  firstName: "John",
  lastName: "Doe",
  language: "",
  set lang(lang) {
    this.language = lang;
  }
};

// Set an object property using a setter:
person.lang = "en";

// Display data from the object:
document.getElementById("demo").innerHTML = person.language;
```

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# JavaScript Function or Getter?

What is the difference between these two examples?

## Example 1

```
const person = {
  firstName: "John",
  lastName: "Doe",
  fullName: function() {
    return this.firstName + " " + this.lastName;
  }
};

// Display data from the object using a method:
document.getElementById("demo").innerHTML = person.fullName();
```

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## Example 2

```
const person = {
  firstName: "John",
  lastName: "Doe",
  get fullName() {
    return this.firstName + " " + this.lastName;
  }
};

// Display data from the object using a getter:
document.getElementById("demo").innerHTML = person.fullName;
```

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Example 1 accesses fullName as a function: person.fullName().

Example 2 accesses fullName as a property: person.fullName.

The second example provides a simpler syntax.

# Data Quality

JavaScript can secure better data quality when using getters and setters.

Using the `lang` property, in this example, returns the value of the `language` property in upper case:

## Example

```
// Create an object:
const person = {
  firstName: "John",
  lastName: "Doe",
  language: "en",
  get lang() {
    return this.language.toUpperCase();
  }
};

// Display data from the object using a getter:
document.getElementById("demo").innerHTML = person.lang;
```

Try it Yourself »

Using the `lang` property, in this example, stores an upper case value in the `language` property:

### Example

```
const person = {
  firstName: "John",
  lastName: "Doe",
  language: "",
  set lang(lang) {
    this.language = lang.toUpperCase();
  }
};

// Set an object property using a setter:
person.lang = "en";

// Display data from the object:
document.getElementById("demo").innerHTML = person.language;
```

Try it Yourself »

## Why Use Getters and Setters?

- It gives simpler syntax
- It allows equal syntax for properties and methods
- It can secure better data quality
- It is useful for doing things behind-the-scenes

Avoid getters or setters if you only need simple values (prevents overuse).

## Object.defineProperty()

The `Object.defineProperty()` method can also be used to add Getters and Setters:

### A Counter Example

The getter only reads the value. The setter changes it, and ignores negative numbers.

```
// Define object
const obj = {_count : 0};

// Define a getter and a setter
Object.defineProperty(obj, "count", {
  get : function () {return this._count;},
  set : function (value) {
    if (value >= 0) {this._count = value;}
  }
});

// Play with the counter:
obj.count = 5;
obj.count = obj.count + 1;
obj.count = -3;
let text = obj.count;
```

Try it Yourself »

## Exercise ?

**Which code displays "John Doe" with the getter in this object?**

```
const person = {
  firstName: "John",
  lastName: "Doe",
  get fullName() {
    return this.firstName + " " + this.lastName;
  }
};
```

- person.fullName()
- person.fullName
- person.get.fullName
- person.get(fullName)

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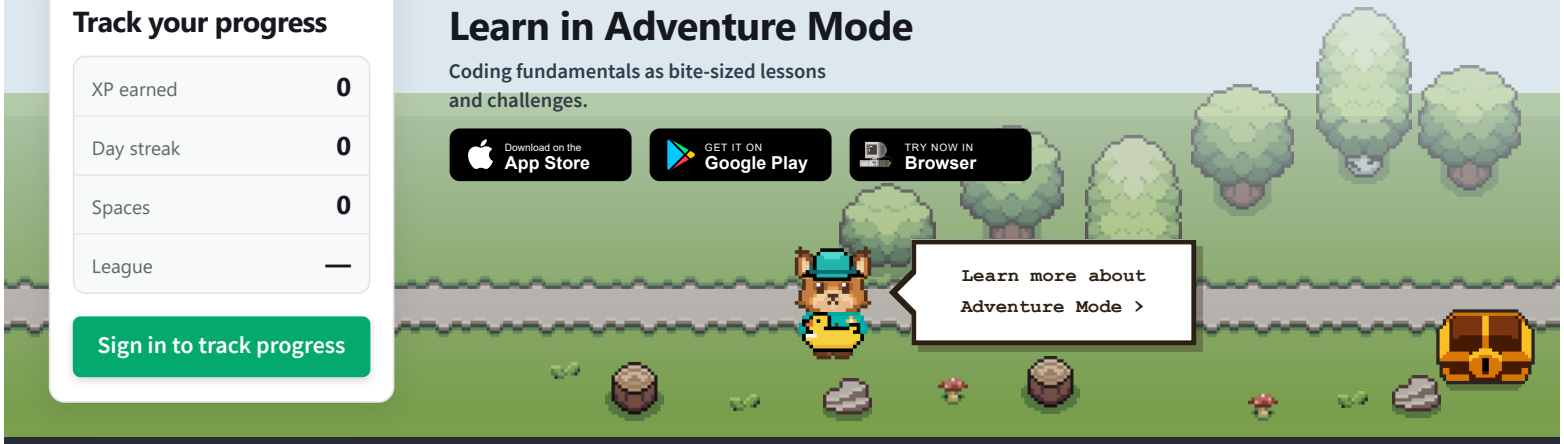



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# JavaScript Object Management

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## Property Management Methods

```
// Adding or changing an object property
Object.defineProperty(object, property, descriptor)

// Adding or changing object properties
Object.defineProperties(object, descriptors)

// Accessing a Property
Object.getOwnPropertyDescriptor(object, property)

// Accessing Properties
Object.getOwnPropertyDescriptors(object)

// Returns all properties as an array
Object.getOwnPropertyNames(object)

// Accessing the prototype
Object.getPrototypeOf(object)
```

## JavaScript Object.defineProperty()

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The `Object.defineProperty()` method can be used to:

- Add a new property to an object
- Change property values
- Change property metadata
- Change object getters and setters

Syntax:

```
Object.defineProperty(object, property, descriptor)
```

# Adding a new Property

This example adds a new property to an object:

## Example

```
// Create an Object:
const person = {
  firstName: "John",
  lastName : "Doe",
  language : "EN"
};

// Add a Property
Object.defineProperty(person, "year", {value:"2008"});
```

Try it Yourself »

# Changing a Property Value

This example changes a property value:

## Example

```
// Create an Object:
const person = {
  firstName: "John",
  lastName : "Doe",
  language : "EN"
};

// Change a Property
Object.defineProperty(person, "language", {value : "NO"});
```

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# Property Attributes

All properties have a name. In addition they also have a value.

The value is one of the property's attributes.

Other attributes are: enumerable, configurable, and writable.

These attributes define how the property can be accessed (is it readable?, is it writable?)

In JavaScript, all attributes can be read, but an assignment can only change the value attribute (and only if the property is writable).

The `Object.defineProperty()` method can change the other attributes (if the property is configurable).

# Changing Meta Data

The following property meta data can be changed:

```
writable : true      // Property value can be changed
enumerable : true    // Property can be enumerated
configurable : true  // Property can be reconfigured
```

```
writable : false     // Property value can not be changed
enumerable : false   // Property cannot be enumerated
configurable : false // Property cannot be reconfigured
```

Getters and setters can also be changed:

```
// Defining a getter
get: function() { return language }
// Defining a setter
set: function(value) { language = value }
```

This example makes language read-only:

```
Object.defineProperty(person, "language", {writable:false});
```

This example makes language not enumerable:

```
Object.defineProperty(person, "language", {enumerable:false});
```

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# JavaScript getOwnPropertyNames()

The `Object.getOwnPropertyNames()` method can:

- List object properties

## Syntax

```
Object.getOwnPropertyNames(object)
```

# List all Object Properties

This example gets all properties of an object:

## Example

```
// Create an Object
const person = {
  firstName: "John",
  lastName : "Doe",
  language : "EN"
};

// Get all Properties
Object.getOwnPropertyNames(person);
```

Try it Yourself »

`Object.getOwnPropertyNames()` will also list properties that are not enumerable:

### Example

```
// Create an Object
const person = {
  firstName: "John",
  lastName : "Doe",
  language : "EN"
};

// Set the language Property not enumerable
Object.defineProperty(person, "language", {enumerable:false});

// Get all Properties
Object.getOwnPropertyNames(person);
```

Try it Yourself »

# JavaScript Object.keys()

The `Object.keys()` method can:

- List enumerable object properties

### Syntax

```
Object.keys(object)
```

# List Enumerable Object Properties

This example uses `Object.keys()` instead of `Object.getOwnPropertyNames()`:

### Example

```
// Create an Object
const person = {
  firstName: "John",
  lastName : "Doe",
  language : "EN"
};

// Change the "language" Property
Object.defineProperty(person, "language", {enumerable:false});
```

```
// Get all Enumerable Properties
Object.keys(person);
```

Try it Yourself »

The getOwnPropertyNames() method returns all properties.

The Object.keys() method returns all enumerable properties.

If you define object properties without **enumerable:false**, the two methods will return the same.

# Adding Getters and Setters

The `Object.defineProperty()` method can also be used to add Getters and Setters:

## Example

```
//Create an object
const person = {firstName:"John", lastName:"Doe"};

// Define a getter
Object.defineProperty(person, "fullName", {
  get: function () {return this.firstName + " " + this.lastName;}
});
```

Try it Yourself »

# A Counter Example

The getter only reads the value. The setter changes it, and ignores negative numbers.

## Example

```
// Define object
const obj = {_count : 0};

// Define a getter and a setter
Object.defineProperty(obj, "count", {
  get : function () {return this._count;},
  set : function (value) {
```

```
    if (value >= 0) {this._count = value;}
  }
});

// Play with the counter:
obj.count = 5;
obj.count = obj.count + 1;
obj.count = -3;
let text = obj.count;
```

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## Prototype Properties

JavaScript objects inherit the properties of their prototype.

The `delete` keyword does not delete inherited properties, but if you delete a prototype property, it will affect all objects inherited from the prototype.

### Exercise ?

**What does this code do?**

```
const person = {firstName:"John", lastName:"Doe"};

Object.defineProperty(person, "year", {value:"2008"});
```

- It creates a new object named `year`
- It deletes the `year` property
- It adds a new property `year` to `person`
- It changes the `firstName` property to `"2008"`

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# JavaScript Object Protection

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## Object Protection Methods

```
// Prevents re-assignment
const car = {type:"Fiat", model:"500", color:"white"};

// Prevents adding object properties
Object.preventExtensions(object)

// Returns true if properties can be added to an object
Object.isExtensible(object)

// Prevents adding and deleting object properties
Object.seal(object)

// Returns true if object is sealed
Object.isSealed(object)

// Prevents any changes to an object
Object.freeze(object)

// Returns true if object is frozen
Object.isFrozen(object)
```

## Using const

The most common way to protect an object from being changed is by using the `const` keyword.

With `const` you **cannot** re-assign the object, but you **can** change the value of a property, delete a property or create a new property.

# JavaScript Object.preventExtensions()

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The `Object.preventExtensions()` method prevents adding properties to an object.

## Example

```
"use strict"
// Create Object
const person = {firstName:"John", lastName:"Doe"};

// Prevent Extensions
Object.preventExtensions(person);

// This will throw an error
person.nationality = "English";
```

Try it Yourself »

Since arrays are objects, arrays can be prevented from extensions too:

## Example

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.preventExtensions(fruits);

// This will throw an error:
fruits.push("Kiwi");
```

Try it Yourself »

The `Object.preventExtensions()` method will fail silently in non-strict mode and throw a TypeError in strict mode.

# JavaScript Object.isExtensible()

You can use `Object.isExtensible()` to check if an object is extensible.

The `Object.isExtensible()` returns true if an object is extensible.

## Examples

```
// Create Object
const person = {firstName:"John", lastName:"Doe"};

// Prevent Extensions
Object.preventExtensions(person);

// This will return false
let answer = Object.isExtensible(person);
```

Try it Yourself »

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Prevent Extensions
Object.preventExtensions(fruits);

// This will return false
let answer = Object.isExtensible(fruits);
```

Try it Yourself »

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# JavaScript Object.seal()

The `Object.seal()` method prevents additions or deletions of new properties.

The `Object.seal()` method makes existing properties non-configurable.

The `Object.isSealed()` method can be used to check if an object is sealed.

## Example

```
"use strict"
// Create Object
const person = {
  firstName: "John",
```

```
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Seal Object
Object.seal(person)

// This will throw an error
delete person.age;
```

Try it Yourself »

Since arrays are objects, arrays can be sealed too:

### Example

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.seal(fruits);

// This will throw an error:
fruits.push("Kiwi");
```

Try it Yourself »

The `Object.seal()` method will fail silently in non-strict mode and throw a TypeError in strict mode.

## JavaScript Object.isSealed()

The `Object.isSealed()` method can be used to check if an object is sealed.

The `Object.isSealed()` returns true if an object is sealed.

### Examples

```
// Create Object
const person = {firstName:"John", lastName:"Doe"};

// Seal Object
Object.seal(person);
```

```
// This will return true
let answer = Object.isSealed(person);
```

Try it Yourself »

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];

// Seal Array
Object.seal(fruits);

// This will return true
let answer = Object.isSealed(fruits);
```

Try it Yourself »

# JavaScript Object.freeze()

The `Object.freeze()` method prevents any changes to an object.

Frozen objects are read-only.

No modification, addition or deletion of properties are allowed.

## Example

```
"use strict"
// Create Object
const person = {
  firstName: "John",
  lastName: "Doe",
  age: 50,
  eyeColor: "blue"
};

// Freeze Object
Object.freeze(person)

// This will throw an error
person.age = 51;
```

Try it Yourself »

Since arrays are objects, arrays can be frozen too:

### Example

```
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.freeze(fruits);

// This will throw an error:
fruits.push("Kiwi");
```

Try it Yourself »

The `Object.freeze()` method will fail silently in non-strict mode and throw a TypeError in strict mode.

## JavaScript Object.isFrozen()

The `Object.isFrozen()` method can be used to check if an object is frozen.

The `Object.isFrozen()` returns true if an object is frozen.

### Examples

```
// Create Object
const person = {firstName:"John", lastName:"Doe"};

// Freeze Object
Object.freeze(person);

// This will return true
let answer = Object.isFrozen(person);
```

Try it Yourself »

```
// Create Array
const fruits = ["Banana", "Orange", "Apple", "Mango"];
Object.freeze(fruits);

// This will return true:
let answer = Object.isFrozen(fruits);
```

Try it Yourself »

## Exercise ?

**Which method prevents any changes to an object (no modification, addition or deletion of properties)?**

- `Object.seal()`
- `Object.preventExtensions()`
- `Object.isFrozen()`
- `Object.freeze()`

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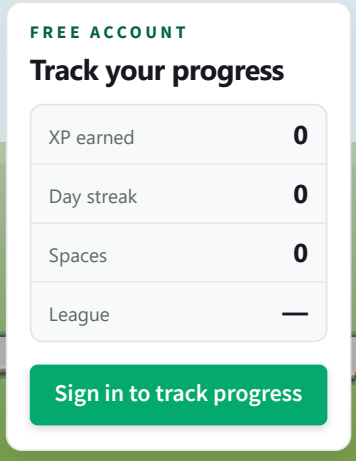
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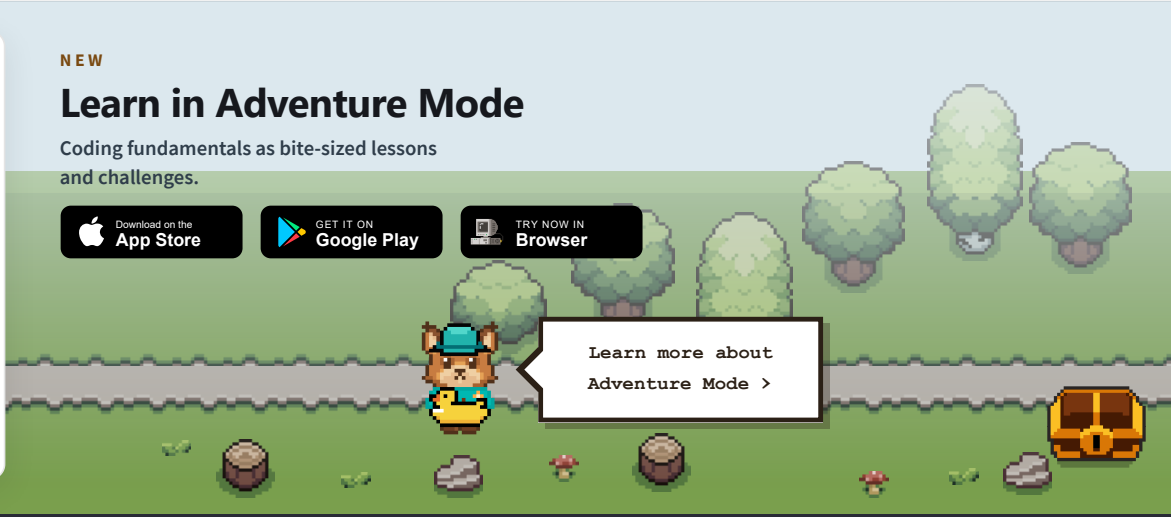
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# JavaScript Object Prototypes

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## Prototype Inheritance

All JavaScript objects inherit properties and methods from a prototype:

- `Date` objects inherit from `Date.prototype`
- `Array` objects inherit from `Array.prototype`

The `Object.prototype` is on the top of the prototype inheritance chain.

`Date` objects, `Array` objects, and all other objects inherit from `Object.prototype`.

## Adding Properties and Methods to Objects

Sometimes you want to add new properties (or methods) to all existing objects of a given type.

Sometimes you want to add new properties (or methods) to an object constructor.

Previously you have learned how to use an **object constructor**:

### Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
```

```
  this.age = age;
  this.eyeColor = eyecolor;
}

const myFather = new Person("John", "Doe", 50, "blue");
const myMother = new Person("Sally", "Rally", 48, "green");
```

Try it Yourself »

You **cannot** add a new property to an existing object constructor:

### Example

```
Person.nationality = "English";
```

Try it Yourself »

To add a new property to a constructor, you must add it to the constructor function:

### Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
  this.age = age;
  this.eyeColor = eyecolor;
  this.nationality = "English";
}
```

Try it Yourself »

## Using the **prototype** Property

The JavaScript `prototype` property allows you to add new properties to object constructors:

### Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
  this.age = age;
```

```
  this.eyeColor = eyecolor;
}

Person.prototype.nationality = "English";
```

Try it Yourself »

The JavaScript `prototype` property also allows you to add new methods to object constructors:

## Example

```
function Person(first, last, age, eyecolor) {
  this.firstName = first;
  this.lastName = last;
  this.age = age;
  this.eyeColor = eyecolor;
}

Person.prototype.name = function() {
  return this.firstName + " " + this.lastName;
};
```

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## Warning

Only modify your **own** prototypes. Never modify the prototypes of standard JavaScript objects.

## Exercise ?

**Which prototype is on the top of the prototype inheritance chain?**

- Date.prototype
- Array.prototype

- Object.prototype
- Person.prototype

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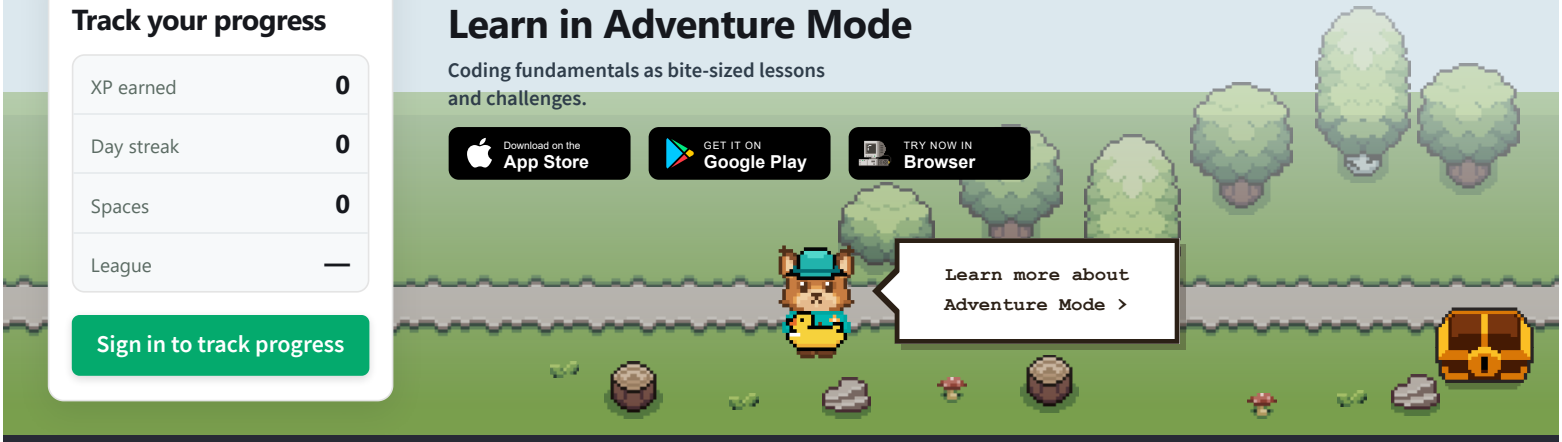
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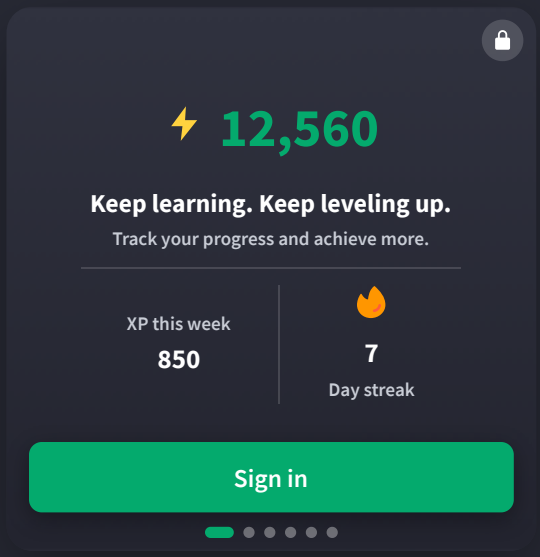
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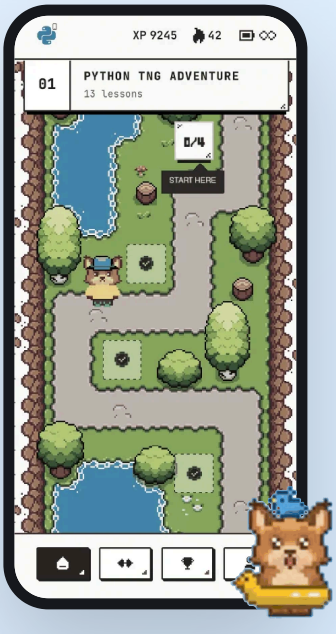
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</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>
```

```
</body>
</html>
```

Try it Yourself »

Learn HTML | HTML Reference | HTML Certificate

## All languages

**HTML**
The building blocks of every webpage

**CSS**
Style and layout for web pages

**JavaScript**
Interactive and dynamic web pages

**SQL**
Query and manage databases

**Python**
Popular language for web, data & AI

**Java**
Object-oriented programming language

**C**
A powerful general-purpose language

**C++**
The language behind games & engines

**C#**
Build apps, games and web services

**PHP**
Server-side scripting language

**React**
Build interactive user interfaces

**MySQL**
Store and manage relational data

**Excel**

**DSA**

Spreadsheets and data

Data Structures and Algorithms

### And 40+ more

W3.CSS, TypeScript, Git, Django, NumPy, jQuery, Bootstrap, R, AI, Data Science and many more

**Browse all tutorials**

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## Save your progress. Everywhere.

Learning is free with or without an account. Sign up to keep your XP, streaks and progress, on w3schools.com and in the Adventure app.

- ✓ Progress tracking
- ✓ XP & streaks
- ✓ Ad-free learning
- ✓ Access to App
- ✓ Challenges
- ✓ Practice problems
- ✓ Leagues
- ✓ Build & host websites



### Start in one click

Create your free account and pick up where you left off, anywhere.

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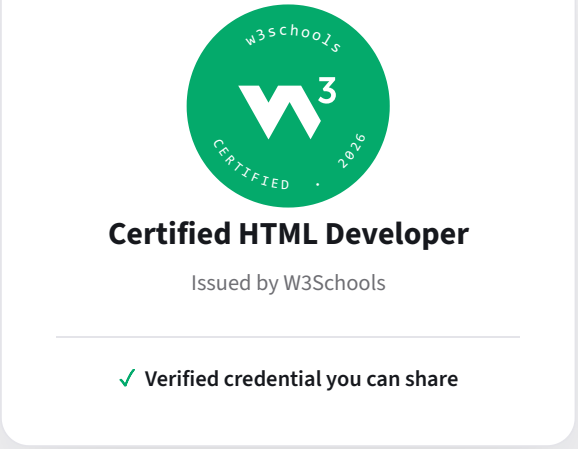
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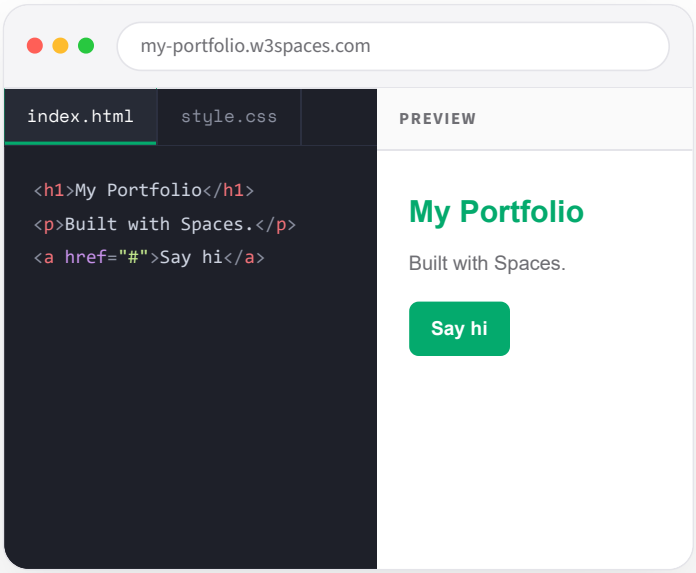
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## Kickstart your career.

Get certified by completing a course. Document your skills with a credential trusted by employers since 1999.

Get certified | See all certificates




W3SCHOOLS SPACES

## Build and host websites.

Code directly in your browser, nothing to install. When you are ready, publish your site with one click.

✓ No setup ✓ Templates to start from

✓ Hosting included

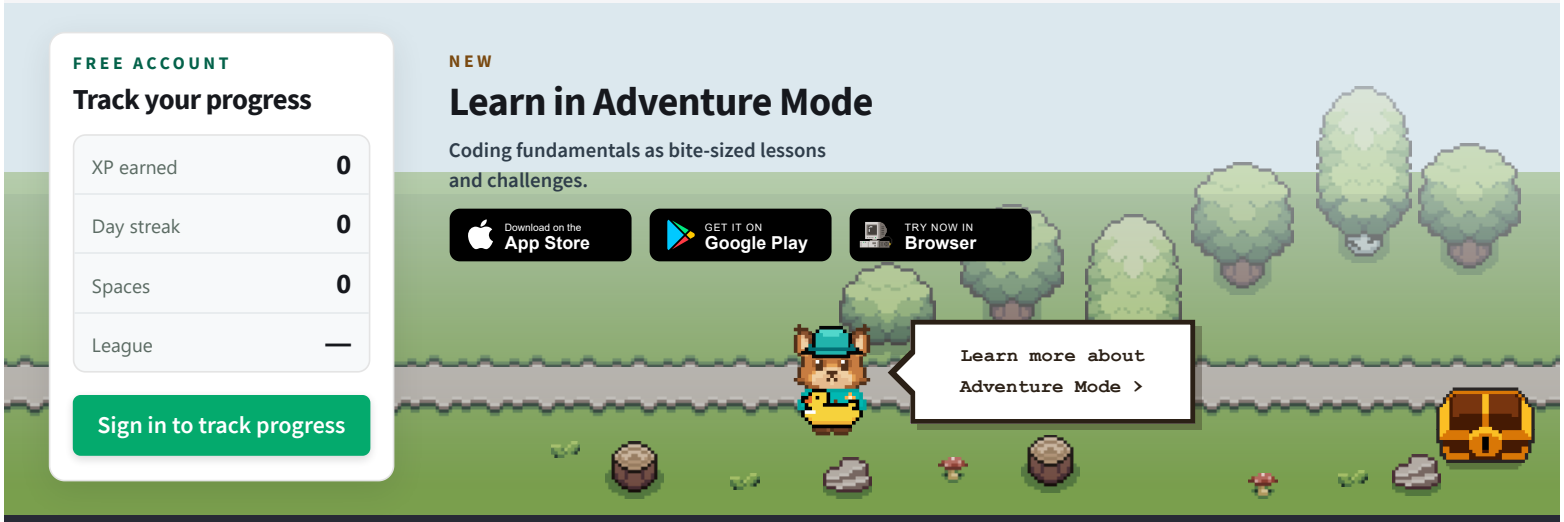
Try Spaces | Learn more

TEST YOUR SKILLS

## Practice with W3Schools

Exercises, quizzes and a new coding challenge every week.

Exercises | Quizzes | Weekly problem




W3Schools

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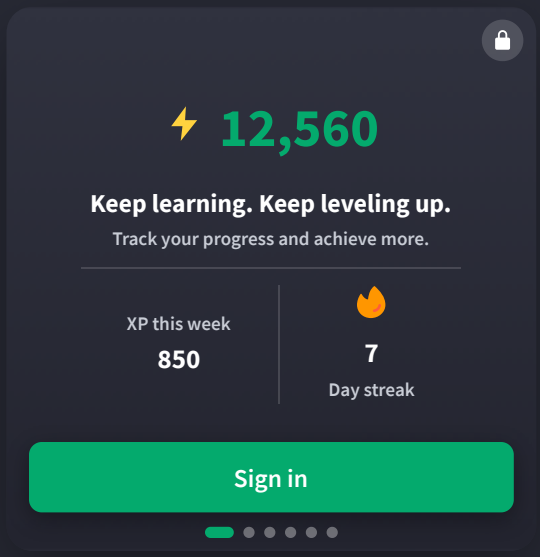
# Learn to Code

Free tutorials, examples and references.
**No sign-up needed, just start learning.**

Or sign up to unlock premium features:

- ✓ Progress tracking
- ✓ Ad-free learning
- ✓ Practice problems
- ✓ Challenges
- ✓ Build & host websites
- ✓ XP & Streaks
- ✓ Leagues
- ✓ Avatar

Already a member? Sign in



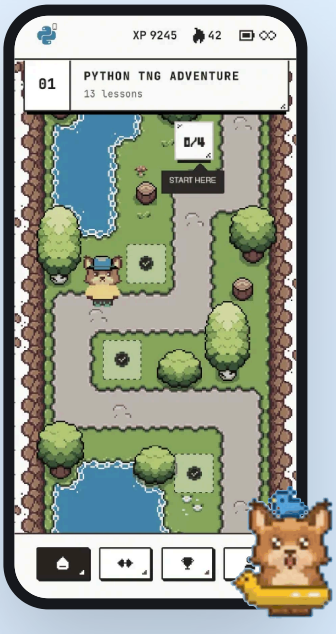
NEW

# W3Schools Adventure App

The W3Schools Adventure app is here! Coding fundamentals as a game. Bite-sized lessons, streaks and XP.
Free on iOS and Android.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser

Learn more about the app ›




TUTORIALS

## What do you want to learn?

Pick a language and start right away. Every tutorial is packed with examples you can run and edit in your browser:

HTML | CSS | JavaScript | Python | SQL

HTML example

```
<!DOCTYPE html>
<html>
<head>
<title>HTML Tutorial</title>
</head>
<body>

<h1>This is a heading</h1>
<p>This is a paragraph.</p>
```

```
</body>
</html>
```

Try it Yourself »

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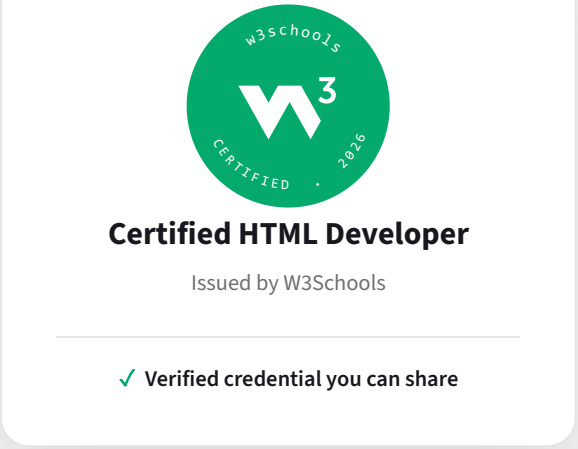
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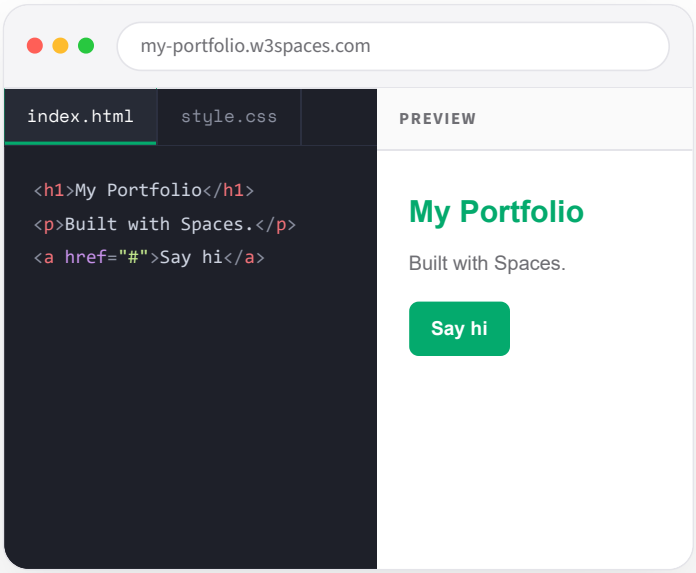
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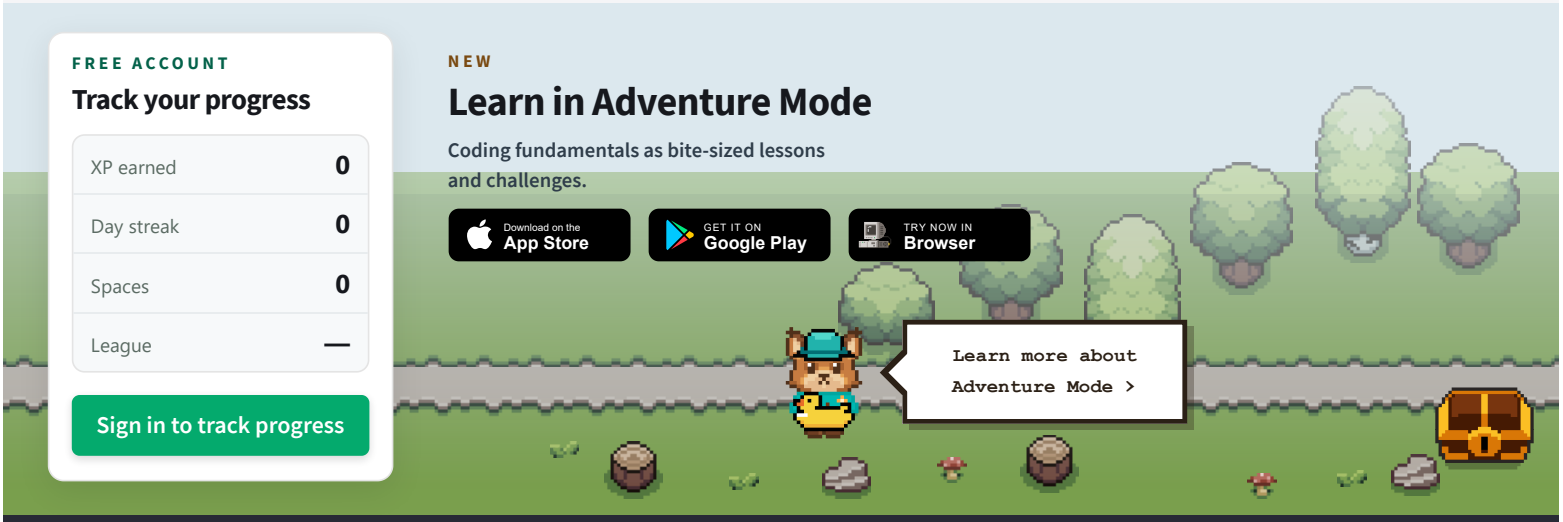
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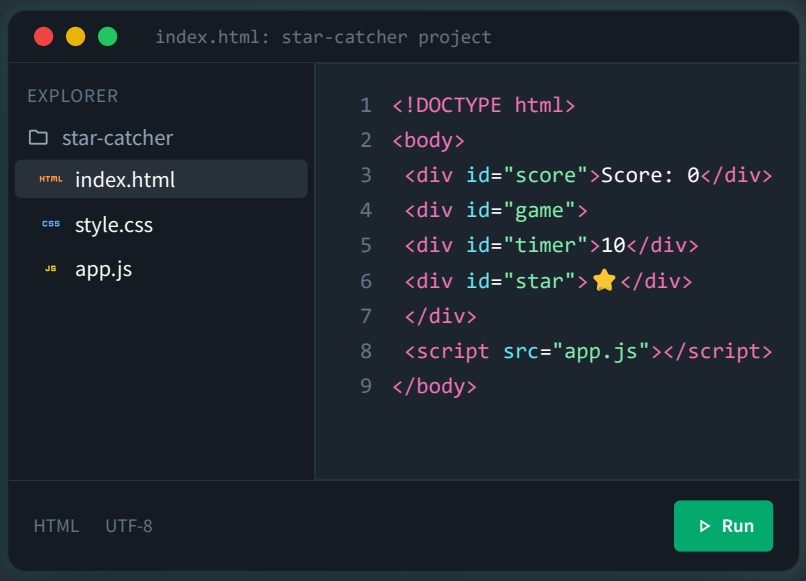
BROWSER-BASED IDE

# Create a Website

Code websites directly in the browser with W3Schools Spaces. Practice, build, and host - all in one place.

Practice kAI Tutor Build Host

Get Started for Free See Features




**2M+** Spaces Created

**70M+** Monthly Visitors

**25+** Years Teaching Code

# Built for Everyone

Whether you're just starting or building your portfolio, Spaces grows with you.

### Beginners

No setup, no confusion. Just open your browser and start coding. Perfect for your first website.

- ✓ Zero installation required
- ✓ Beginner-friendly templates
- ✓ kAI helps when you're stuck

### Students & Hobbyists

Practice what you learn in tutorials. Build real projects and create a portfolio that gets noticed.

- ✓ Host projects
- ✓ Practice any language
- ✓ Share with a live link

### Teachers & Schools

Give your students a coding environment that works instantly. No IT headaches.

- ✓ Works on any device
- ✓ Team management
- ✓ Integrates with W3Schools tutorials

Trusted by developers at companies worldwide

Google Microsoft OpenAI Amazon Web Services shopify

ONLY AT W3SCHOOLS

## Learn → Practice → Build

The only browser IDE connected to the world's largest web developer learning platform.

1

### Learn from Tutorials

2

### Practice in Spaces

3

### Build Real Projects

Free tutorials for HTML, CSS, JavaScript, Python, and more. Trusted by millions.

Apply what you learn immediately. No context switching between learning and coding.

Create portfolio-worthy websites. Host and share them with the world.

Explore W3Schools Tutorials

# Simple Pricing

Start a smaller plan and upgrade if you need more.

Monthly  Save by paying annually

### Hero

**$2.99** /Month

Billed annually ($4.99 if billed monthly)

Build and host your frontend projects

- Track your progress
- Ad free
- Unlimited energy in W3Schools Adventure app
- Code challenges
- Assessments
- Your personal avatar
- Compete in leaderboard
- Build websites
- Up to 10 static Spaces
- Reliable hosting with included monthly bandwidth
- Cancel Anytime

MOST POPULAR

### Plus

**$4.99** /Month

Billed annually ($14.99 if billed monthly)

This is for you that want to learn and reach your goals faster! Work with 18 different programming languages, ad free experience.

- Everything in Hero
- 1 full-stack Space for dynamic applications
- Up to 15 static Spaces
- Secure and high-capacity hosting for real-world traffic
- 10 GB total storage
- Upload files up to 50 MB
- 50% XP boost
- Priority support
- Cancel Anytime

Get Hero

Get Plus

BEST VALUE

**Full Access**

**$499** / one time

Get everything W3Schools has to offer with a single one-time payment.

- All Plus features
- Unlimited courses
- All Certification exams
- All Career Paths
- Get exclusive Full Access skin
- Lifetime access: stays active for the lifetime of the platform

Get Full Access

Are you a teacher? Check out W3Schools Campus (previously Academy)

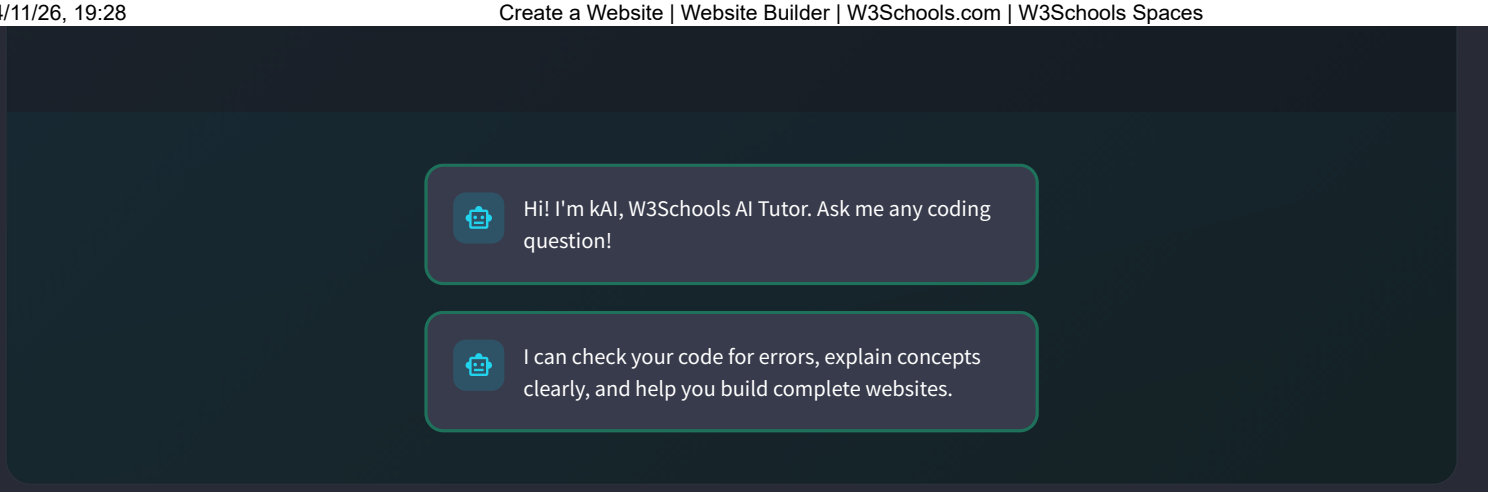
AI-POWERED

# Meet kAI

Your AI coding tutor, built into Spaces

Learn to code more effectively with kAI. Get instant help debugging, improving code structure, understanding concepts, or even generating complete websites from your description.

- Debug your code and explain errors
- Improve code structure and best practices
- Generate websites from descriptions




# Powerful Code Editor

Front-end, back-end, or full-stack - everything you need in one place.

### Cloud-Based

No installation. Access from anywhere, any device.

### Terminal & Log

Debug and troubleshoot your code in real-time.

### File Navigator

All your files in one place. Easy to manage.

### Database

Built-in database. View and manage your data.

### Package Manager

Install frameworks and libraries in one click.

### Analytics

Track visitors and understand your traffic.

### Custom Domains

Connect your own domain or use ours free.

### SSL Included

Free SSL certificate. Your site is secure.

Create Your Space Today

## Practice Any Language

Master web development with these technologies.

HTML

CSS

JavaScript

Python

PHP

React

Vue

Node.js

Start Practicing Today

## Start with Templates

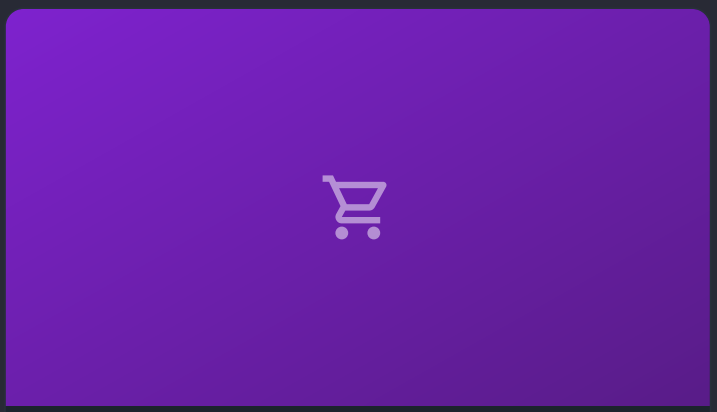
Build powerful websites in just a few clicks.



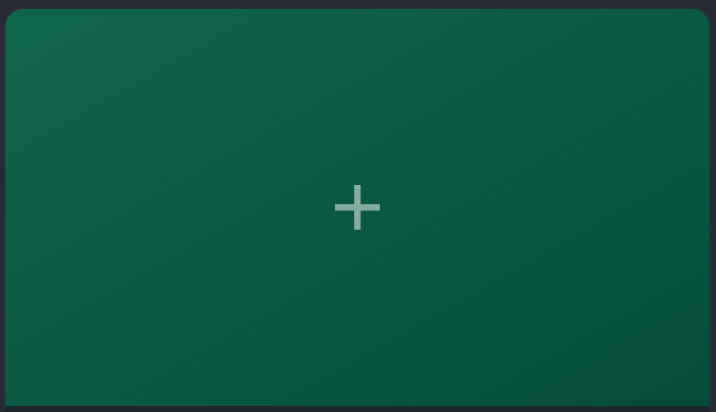
Portfolio



Blog



Webshop



View All Templates

Build Your Own Project or explore more projects

# How It Works

From idea to live website in minutes.

1

**Sign Up**

Create your W3Schools account in seconds.

2

**Choose a Template**

Or start from scratch with a blank project.

3

**Edit Your Code**

Write code directly in your browser.

4

**Share with the World**

Your site is live instantly.

# What Developers Say

Join millions who learned to code with W3Schools.

★★★★★

"Finally a browser IDE that's actually simple. Spaces just works - open it and start coding."

M Marcus T.
Learning Web Development

★★★★★

"kAI is like having a patient teacher. It explains my errors without making me feel stupid. Perfect for beginners like me."

S Sarah K.
Computer Science Student

★★★★★

"I teach coding to 30 students. Spaces means no more 'it doesn't work on my laptop' excuses. Everyone has the same environment."

J James R.
High School Teacher

Join them. Start building today.

Sign Up Today

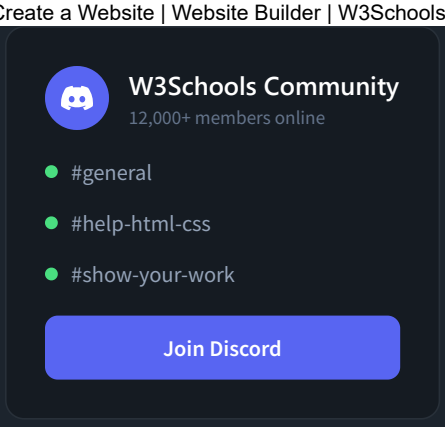
# You're Never Alone

Learning to code can be hard. We make it easier with support at every step.

**kAI Assistant**
Get instant help debugging, understanding concepts, or generating code.

**Help Center**
Searchable knowledge base with guides, tutorials, and FAQs.

**Discord Community**
Connect with other learners. Ask questions. Share your projects.




FOR EDUCATORS

# Spaces in Your Classroom

Give your students a professional coding environment without the IT headaches. No installations, no configuration - just learning.

**Project-Based Learning**
Students build real websites and apps, not just run exercises.

**Secure & Safe**
Sandboxed environment. Students can't break anything or access harmful content.

**Privacy Compliant**
GDPR compliant. No tracking of student data for advertising.

**Teacher Dashboard**
Manage students, track progress, and create custom learning paths.

Explore W3Schools Campus

**Mrs. Coleman**
Computer Science Teacher, Georgia State, US

*"Before Spaces, I spent half of every class helping students install software. Now they just open their browser and code. It's transformed how I teach."*

★★★★★ Teaching 150+ students

| 50% | 0 | 100% |
| --- | --- | --- |
| Less setup time | IT tickets | Browser-based |

# Frequently Asked Questions

What's the difference between Basic and Full-Stack Spaces?

Can I use my own domain?

What happens if I cancel my subscription?

Can I subscribe for multiple users?

# Ready to Build?

Create your first website in minutes.

Get Started for Free

Cancel anytime.

FREE ACCOUNT

Track your progress

XP earned 0

Day streak 0

Spaces 0

League —

Sign in to track progress

NEW

Learn in Adventure Mode

Coding fundamentals as bite-sized lessons and challenges.

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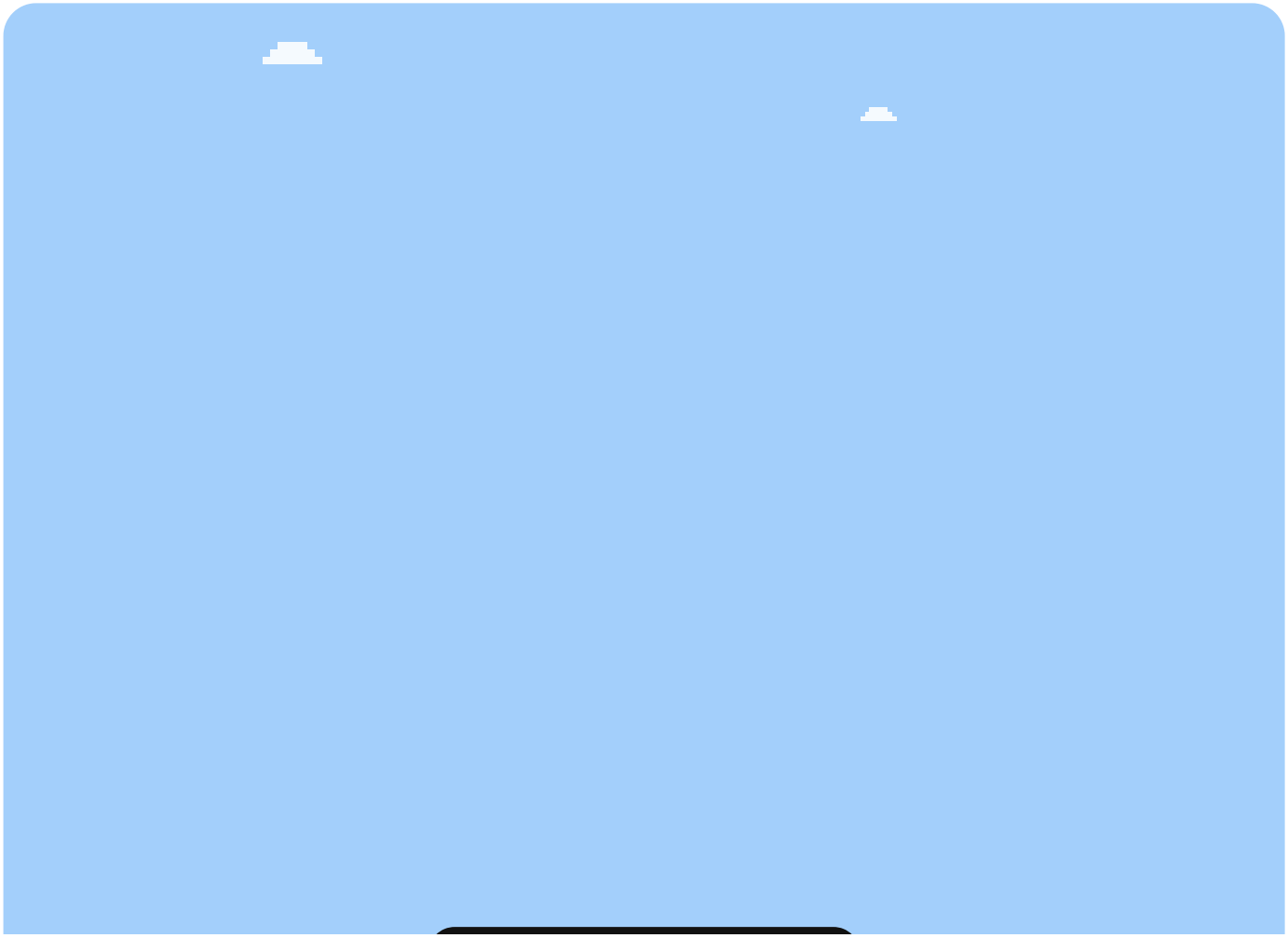
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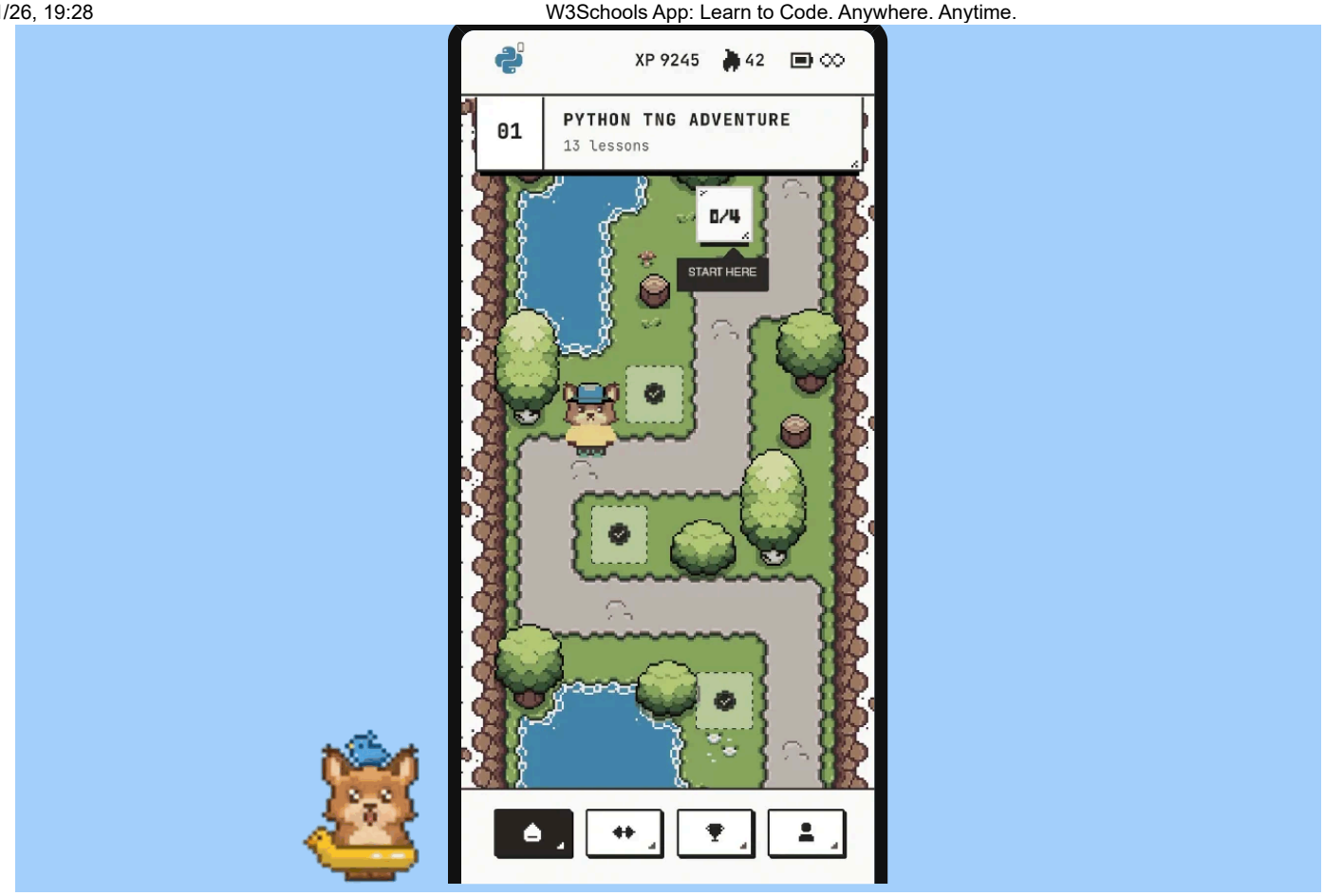
THE W3SCHOOLS APP

# LEARN TO CODE. ANYWHERE. ANYTIME.

Coding fundamentals as a game. Bite-sized lessons, streaks and XP. Free, and your first line of code is two minutes away.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser






HTML · CSS · JavaScript · Python · Java · SQL

INTERACTIVE

# TRY YOUR FIRST LESSON

One of the lines below has an error. Tap the line that is wrong.



Hi, I am your tutor. Find the bug and the XP is yours.

```
1  <h1>World</h2>
```

```
<p>Welcome!</p>

<p>World!</p>
```

THE METHOD

# HOW ADVENTURE WORKS

We've turned the complex journey of learning to code into a simple, engaging adventure.



### FOLLOW THE MAP

Just follow the structured path and learn wherever you are.



### PRACTICE AS YOU GO

Write real code in bite-sized interactive challenges.

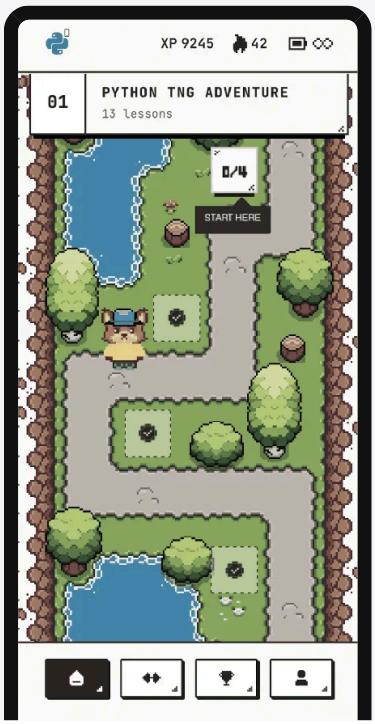


### STAY CONSISTENT

Build a streak and unlock special rewards.

SCREENSHOTS

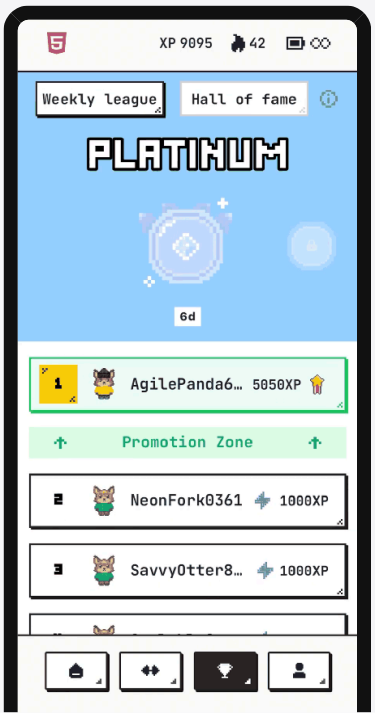
## A PEEK INSIDE




Adventure map




Lessons with your tutor




Weekly league

WHY IT STICKS

# A BETTER WAY TO LEARN CODING

Built with learning science and game design principles to make coding stick.

1

### BITE-SIZED LEARNING

Learn through quick coding activities that fit easily into your daily routine.

2

### WEEKLY QUESTS

Complete weekly challenges, practice your skills, and unlock special rewards.

3

### GROW & COMPETE

Track your progress, climb the leaderboard, and stay motivated while learning.

4

### CODE, LEARN, UNLOCK

Unlock skins and customize your Lynx as you complete activities and grow your skills.

W3SCHOOLS

# FROM THE SITE THAT TAUGHT THE WEB

| 1999 | 6 | 5 min | 0 |
|---|---|---|---|
| teaching code since | languages to learn | a day is all it takes | price. it is free |

FREE ON IOS AND ANDROID

## START YOUR STREAK TODAY

Day one is how every streak starts.

Download on the App Store | GET IT ON Google Play | TRY NOW IN Browser

iPhone Android

scan with your phone

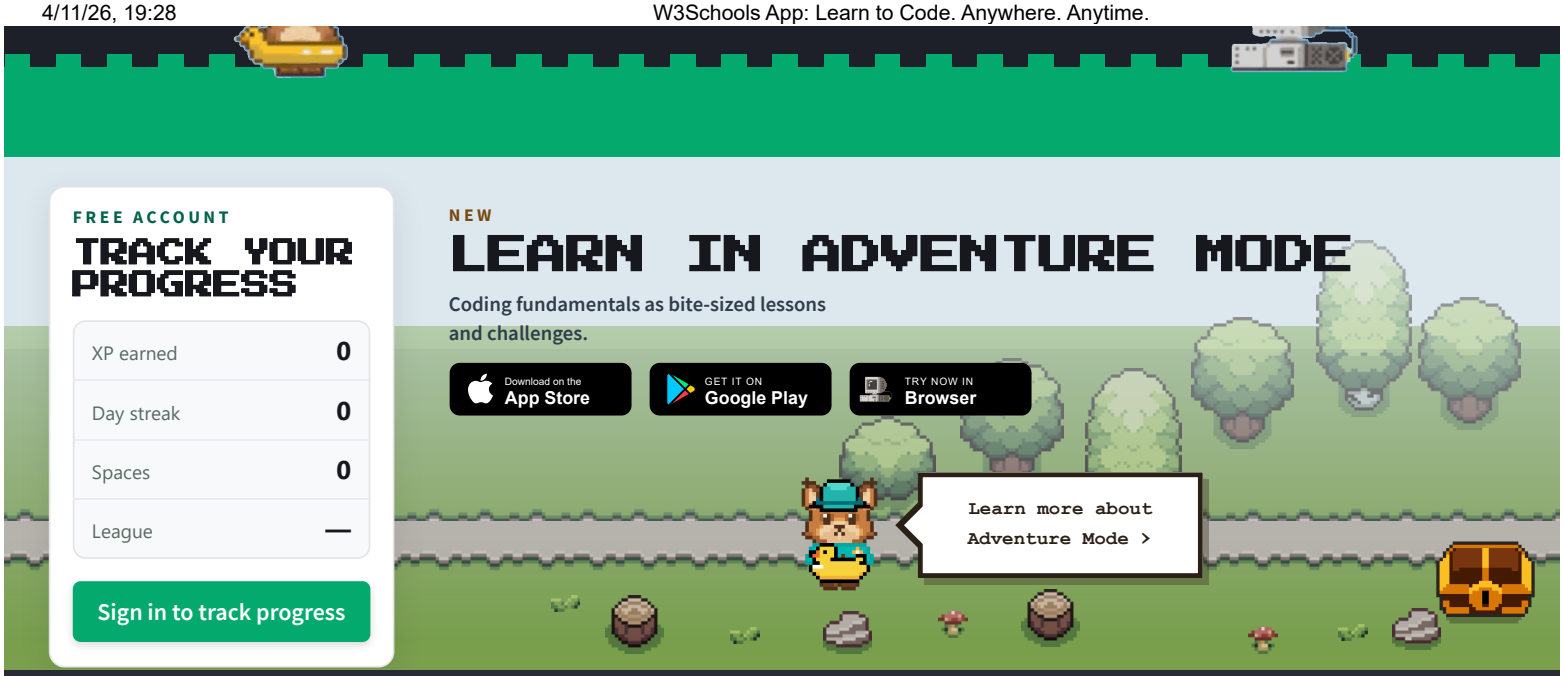
Free to download and free to start. Some features require a subscription, handled safely by the App Store or Google Play. Cancel anytime in your store account settings.



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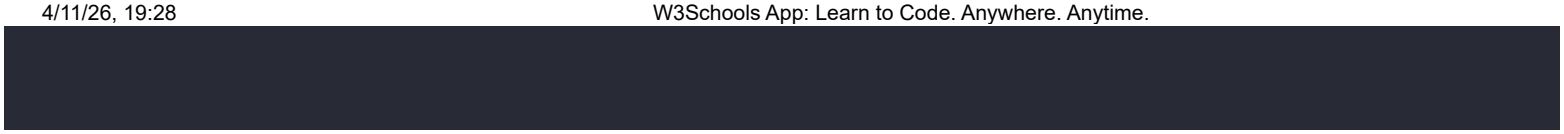
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# Practice Coding Problems

Improve your skills by solving real coding problems.

Write code, submit, and get instant feedback.

**130** practice problems per language · **12** languages




## Subscribe to Weekly Practice Problem

Get the latest coding practice problem delivered to your inbox every week.

Subscribe

Sign in to track your progress

## Select a Language

### Python
General-purpose, beginner-friendly language.

130 practice problems »

### Java
Object-oriented, widely used in enterprise.

130 practice problems »

### C
Low-level systems programming language.

130 practice problems »

### C++
Powerful language for performance-critical apps.

130 practice problems »

### C#
Modern language for .NET and game development.

130 practice problems »

### PHP
Server-side scripting for web development.

130 practice problems »

### Kotlin
Modern language for Android and JVM.

130 practice problems »

### JavaScript
The language of the web. Runs in Node.js.

130 practice problems »

### TypeScript
Typed JavaScript for larger applications.

130 practice problems »

### Swift
Apple's language for iOS and macOS apps.

130 practice problems »

### Rust
Safe, fast systems programming language.

130 practice problems »

### R
Statistical computing and data analysis.

130 practice problems »

## How It Works

Each challenge gives you a **problem description** and **starter code** in an online editor.

Your program reads input and prints the correct output. When you submit, your code is tested against **multiple hidden test cases** with different inputs.

Your score is the percentage of tests passed. Pass all tests to earn full XP. Partial credit is given for partially correct solutions.

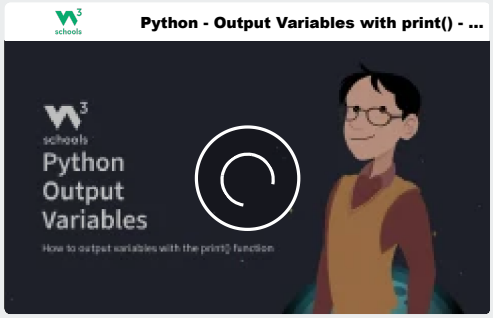
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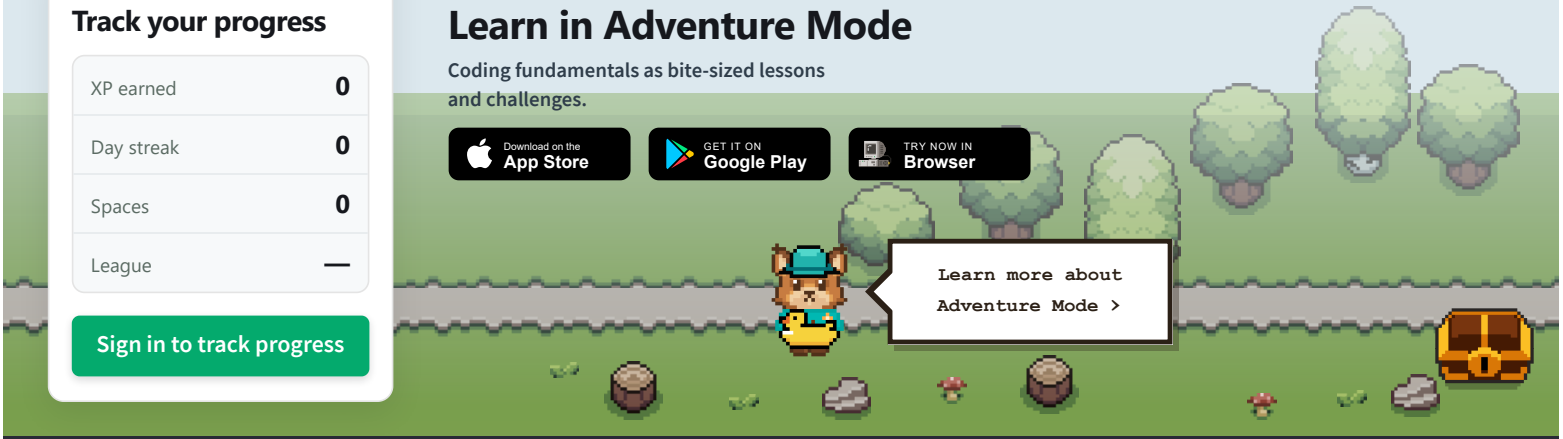
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# HTML Tutorial

‹ Home

TRY ADVENTURE MODE

Next ›

## Learn HTML

HTML is the standard markup language for Web pages.

With HTML you can create your own Website.

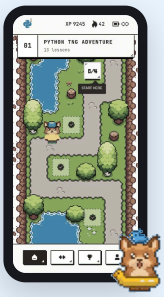
HTML is easy to learn - You will enjoy it!

Learn HTML now ›

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Learn HTML in bite-sized quests with XP and streaks. Your progress grows for each new adventure! Free on iOS and Android.



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## Learning by Examples

SHARE

With our "Try it Yourself" editor, you can edit the HTML code and view the result in the browser:

### Example

```
<!DOCTYPE html>
<html>
<head>
```

```
<title>Page Title</title>
</head>
<body>

<h1>This is a Heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

Try it Yourself »

**Click on the "Try it Yourself" button to see how it works.**

# HTML Examples

This tutorial supplements all explanations with clarifying examples.

See all HTML Examples

REMOVE ADS

# HTML Exercises

Many chapters in this tutorial end with an exercise where you can check your level of knowledge.

## Exercise ?

**What is a correct syntax for an HTML hyperlink?**

- `<a href='/home.htm'>Visit W3Schools.com!</a>`
- `<link href='/home.htm'>Visit W3Schools.com!</link>`
- `<alink href='/home.htm'>Visit W3Schools.com!</alink>`

Submit Answer »

See all HTML Exercises

## HTML Quiz Test

Test your HTML skills with our HTML Quiz!

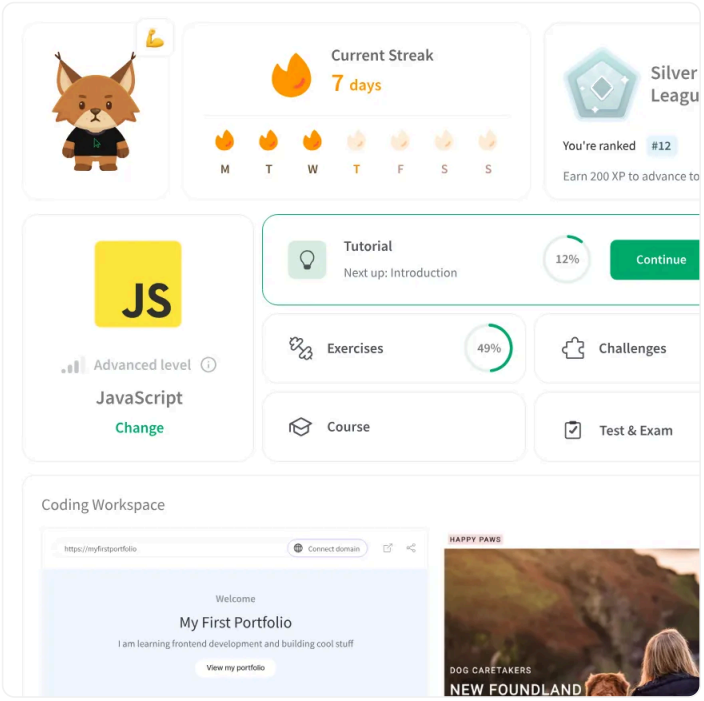
Start HTML Quiz!

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### Note

This is an optional feature. You can study at W3Schools without creating an account.

## HTML References

At W3Schools you will find complete references about HTML elements, attributes, events, color names, entities, character-sets, URL encoding, language codes, HTTP messages, browser support, and more:

| | | |
|---|---|---|
| HTML Elements | Browser Support | Attributes |

| | | |
|---|---|---|
| Global Attributes | Event Attributes | Color Names |
| Canvas | Audio/Video DOM | Character Sets |
| URL Encoding | Language Codes | Country Codes |
| HTTP Messages | Px to Em Converter | Keyboard Shortcuts |

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## Video: HTML for Beginners




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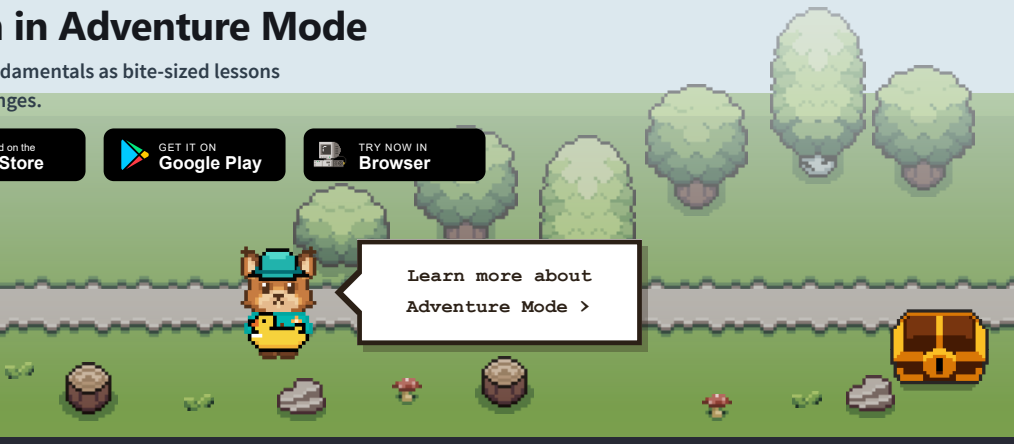
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# CSS Tutorial

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Next ›

## Learn CSS

CSS is the language we use to style an HTML document.

CSS describes how HTML elements should be displayed.

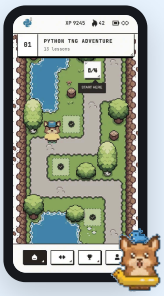
This tutorial will teach you CSS from basic to advanced.

Learn CSS now ›

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## Examples in Each Chapter

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This CSS tutorial contains over 700 CSS examples.

With our online editor, you can edit the CSS, and click on a button to view the result.

## CSS Example

```
body {
  background-color: lightblue;
}

h1 {
  color: white;
  text-align: center;
}

p {
  font-family: verdana;
  font-size: 20px;
}
```

Try it Yourself »

**Click on the "Try it Yourself" button to see how it works.**

# CSS Examples

Learn from over 700 examples! With our editor, you can edit the CSS, and click on a button to view the result.

Go to CSS Examples!

# Use the Menu

We recommend reading this tutorial, in the sequence listed in the menu.

If you have a large screen, the menu will always be present on the left.

If you have a small screen, open the menu by clicking the top menu sign ☰.

# CSS Templates

We have created some responsive W3.CSS templates for you to use.

You are free to modify, save, share, and use them in all your projects.

Free CSS Templates!

REMOVE ADS

# CSS Exercises

Many chapters in this tutorial end with an exercise where you can check your level of knowledge.

## Exercise ?

**Drag and drop the correct property to set a green background color for a `<div>` element.**

```
div {
  [          ]: green;
}
```

bg-color | color | opacity | background-color

Submit Answer »

See all CSS Exercises

# CSS Quiz

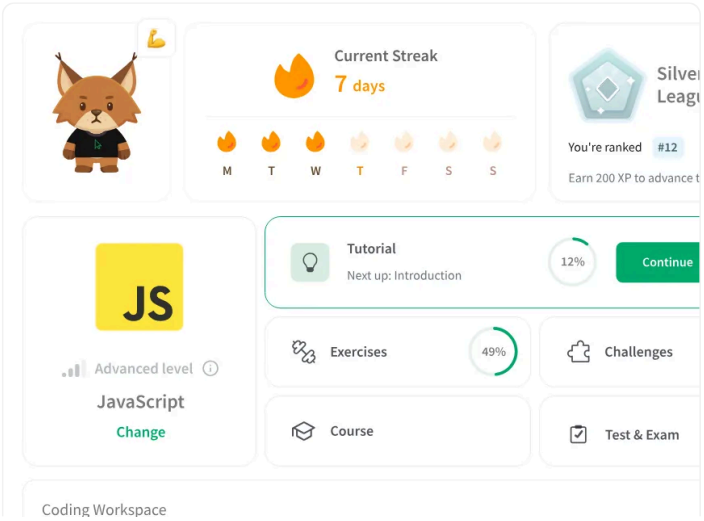
Test your CSS skills with a quiz.

Start CSS Quiz!

# Track Your Progress

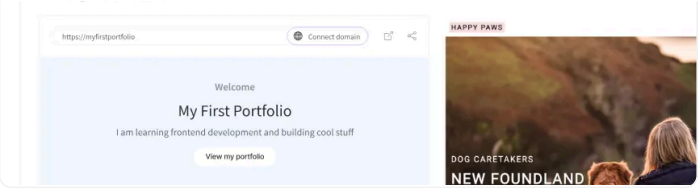
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✓ Create your own personal website

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## Note

This is an optional feature. You can study at W3Schools without creating an account.

# CSS References

At W3Schools you will find complete CSS references of all properties and selectors with syntax, examples, browser support, and more.

| | | |
|---|---|---|
| CSS Properties | CSS Browser Support | CSS Selectors |
| CSS Combinators | CSS Pseudo-classes | CSS Pseudo-elements |
| CSS At-rules | CSS Functions | CSS Animatable |
| CSS Web Safe Fonts | CSS Units | Px to Em Conversion |
| CSS Colors | CSS Default Values | CSS Entities |
| CSS Aural | | |

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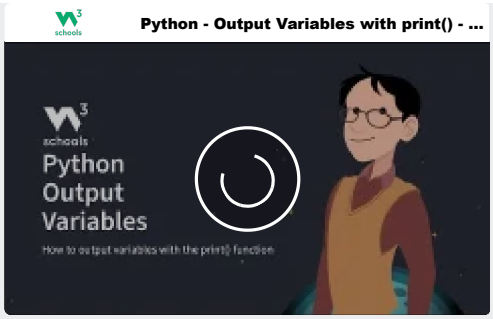



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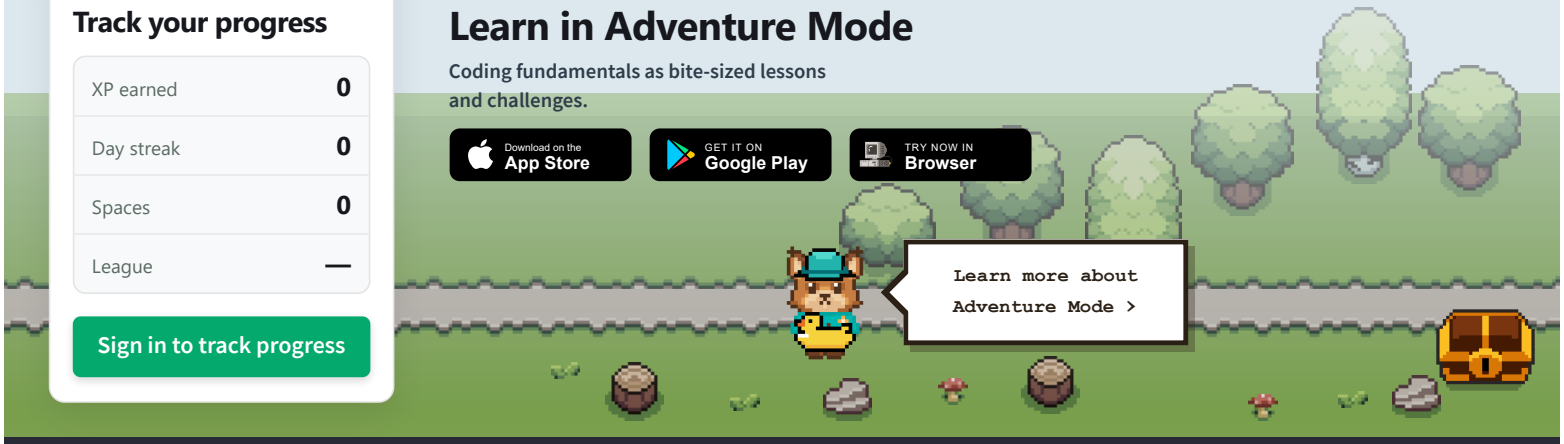
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# JavaScript Tutorial

‹ Home

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## Learn JavaScript

**The Programming Language of the Web**

JavaScript is easy to learn.

This tutorial covers everything from basic JavaScript up to the latest 2026 version.

Start learning JavaScript now »

## Examples in Each Chapter

SHARE

With our **Try it Yourself** editor, you can edit the source code and view the result.

### Example

#### My First JavaScript

Click me to display Date and Time

Try it Yourself »

## Use the Menu

We recommend reading this tutorial, **in the sequence listed in the menu**.

If you have a large screen, the menu will always be present on the left.

If you have a small screen, open the menu by clicking the top menu sign ☰.

## Learn by Examples

Examples are better than 1000 words.

Examples are often easier to understand than text explanations.

This tutorial supplements all explanations with clarifying **Try it Yourself** examples.

If you try all the examples, you will learn a lot about JavaScript, in a very short time!

JavaScript Examples »

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## Why Study JavaScript?

JavaScript is one of the **3 languages** all web developers **must** learn:

1. **HTML** to define the content of web pages
2. **CSS** to specify the layout of web pages
3. **JavaScript** to program the behavior of web pages

## Learning Speed

In this tutorial, the learning speed is your choice.

Everything is up to you.

If you are struggling, take a break, or re-read the material.

**Always** make sure you understand **all** the "Try-it-Yourself" examples.

The only way to become a clever programmer is to:

Practice coding! Practice coding! Practice coding!

# Commonly Asked Questions

- How do I get JavaScript?
- Where can I download JavaScript?
- Is JavaScript Free?

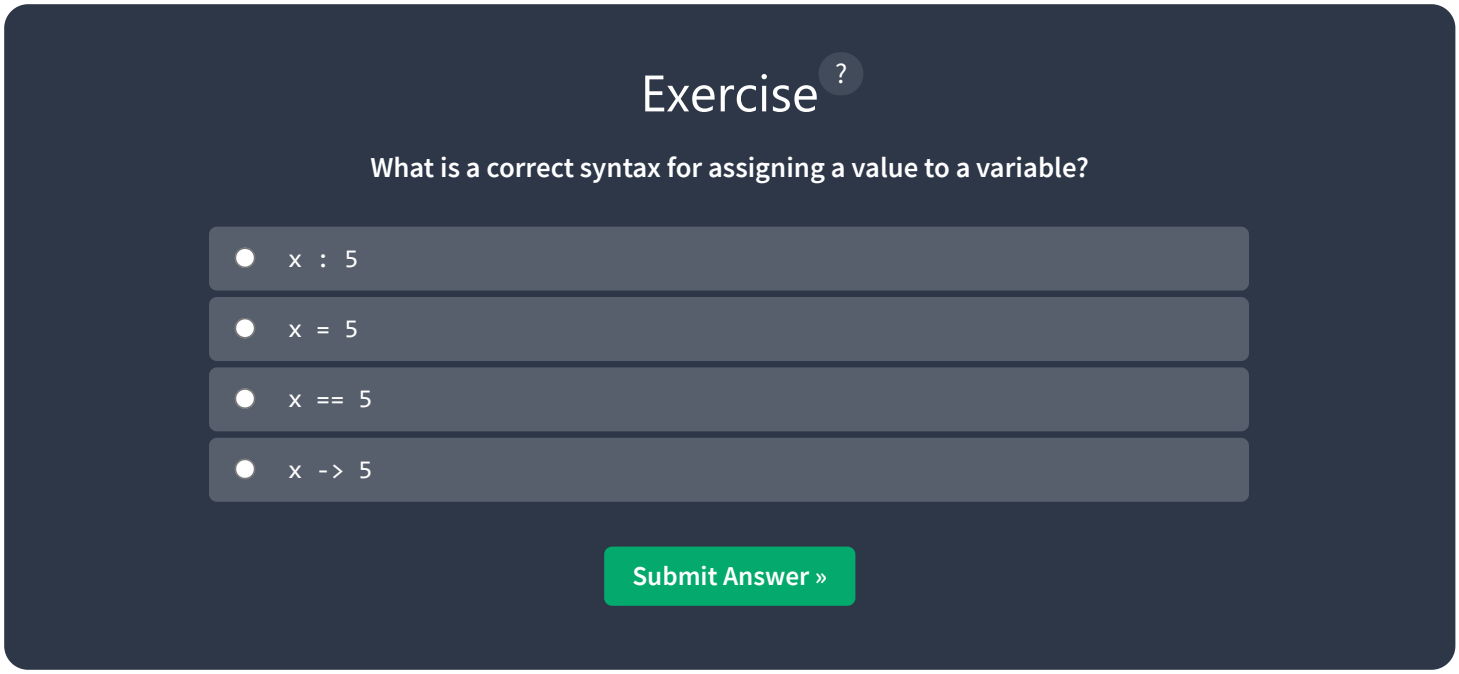
**You don't have to get or download JavaScript.**

**JavaScript is already running in your browser on your computer, on your tablet, and on your smart-phone.**

**JavaScript is free to use for everyone.**

# JavaScript Exercises

Many chapters in this tutorial end with an exercise where you can check your level of knowledge.




See all JavaScript Exercises

# JavaScript Quiz Test

Test your JavaScript skills at W3Schools!

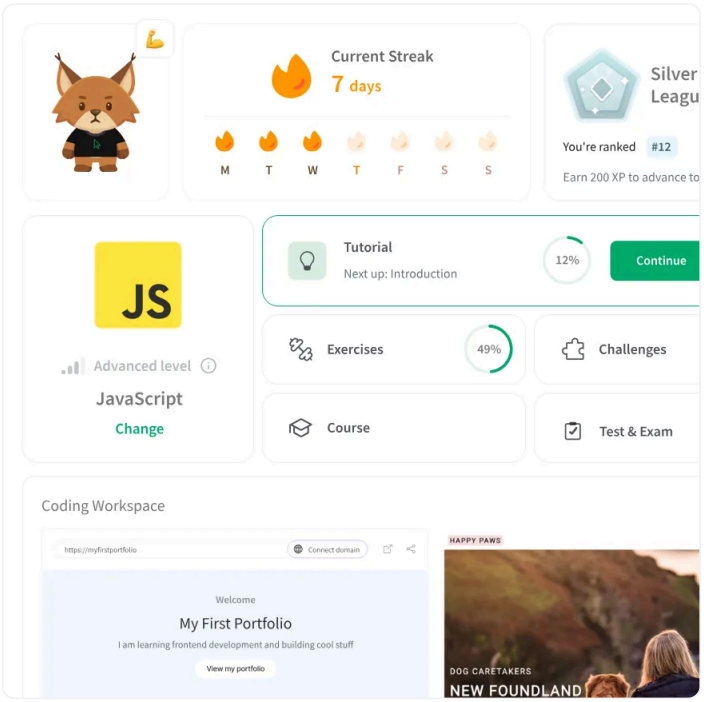
Start JavaScript Quiz!

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### Note

This is an optional feature. You can study at W3Schools without creating an account.

## JavaScript References

W3Schools maintains a complete JavaScript reference, including all HTML and browser objects.

The reference contains examples for all properties, methods and events, and is continuously updated according to the latest web standards.

Complete JavaScript Reference »

JAVASCRIPT CERTIFICATION

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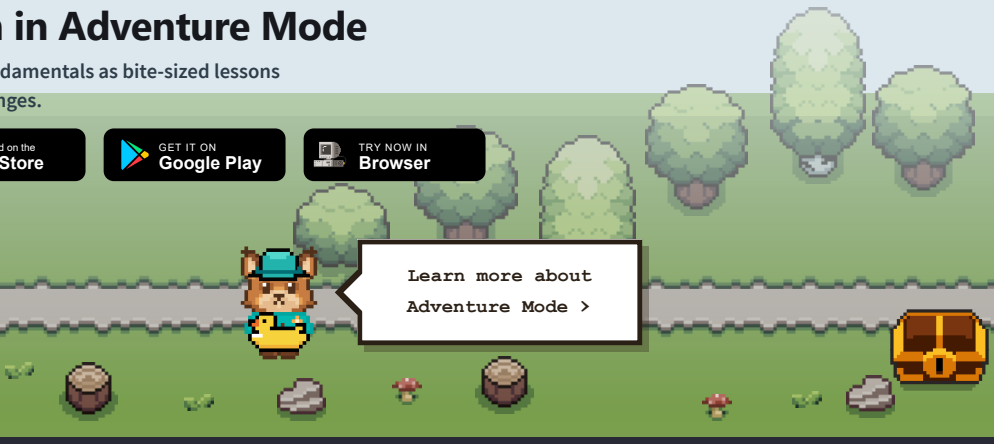
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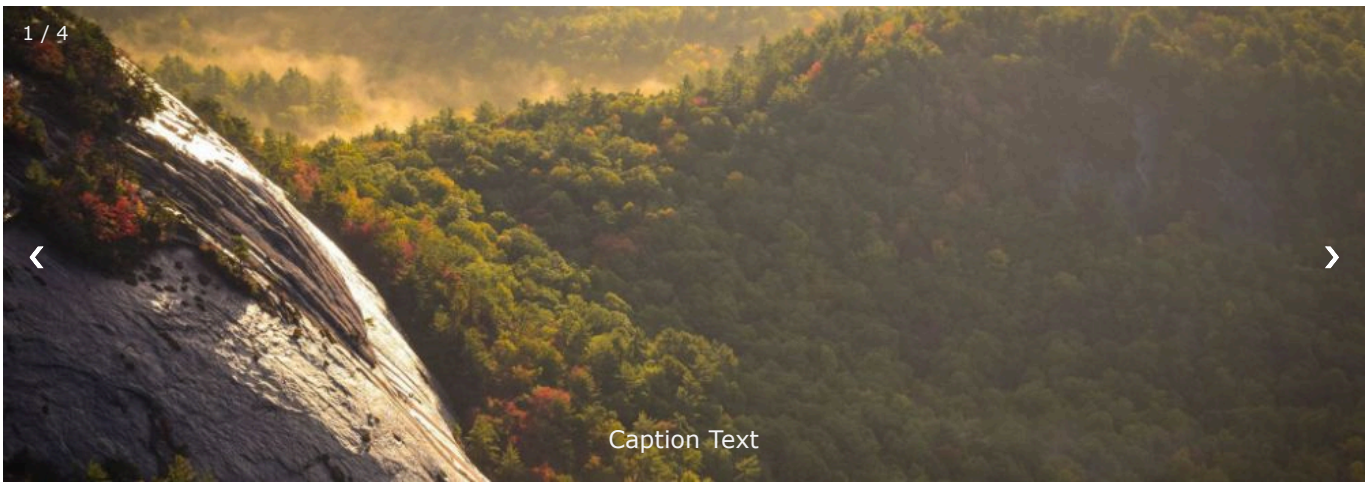
# W3Schools How To

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Code snippets for HTML, CSS and JavaScript.

## Slideshow




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## Login Form

Login

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## Accordion

Section 1 −

Lorem ipsum dolor sit amet, consectetur adipisicing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat.

Section 2 +

Section 3 +

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## Hover Dropdowns

Hover

Learn How To »

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## Click Dropdowns

Click

Learn How To »

## Side Navigation

Sidenav overlay

Sidenav w/opacity

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## Fixed Sidebar

Learn How To »

## Modal Box

Open Modal

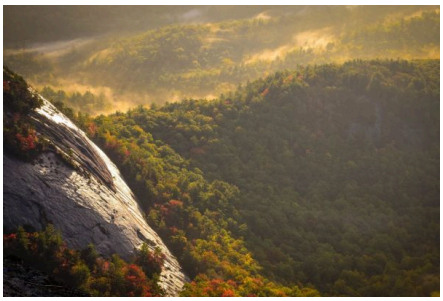
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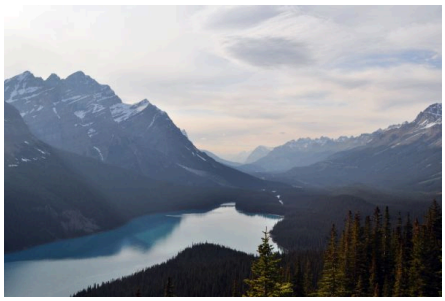
# Modal Image



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# Lightbox





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# Slideshow Gallery






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## Image Hover Overlay



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## Tabs

London | Paris | Tokyo

### London

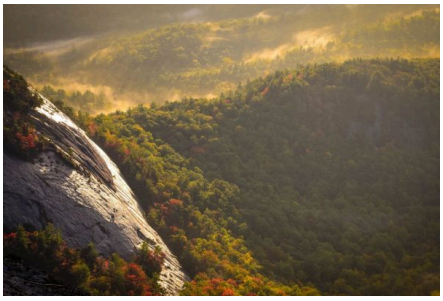
London is the capital city of England.

Learn How To »

## Vertical Tabs

Learn How To »

## Tab Image Gallery








Learn How To »

## Alert Buttons

Success Info Warning Danger Default

Learn How To »

## Outline Buttons

Success Info Warning Danger Default

Learn How To »

## Top Navigation

Home News Contact

Learn How To »

# Responsive Top Navigation

Resize the browser window to see the effect:

Learn How To »

# Fullscreen Navigation

Slide Right Slide Down

Learn How To »

# Login Form in Navbar

Home About Contact Username Password Login

Learn How To »

# Change Background on Scroll

Learn How To »

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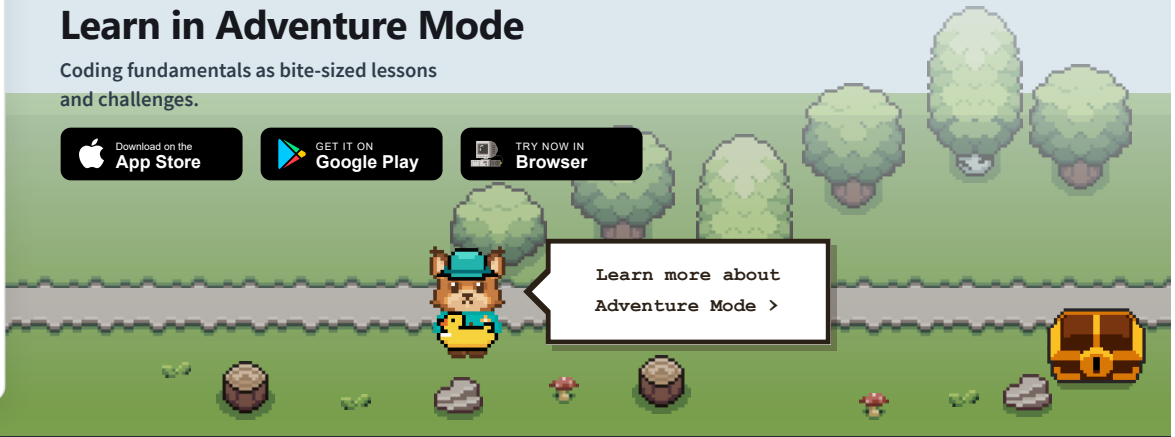
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# SQL Tutorial

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## Learn SQL

SQL is a standard language for storing, manipulating and retrieving data in databases.

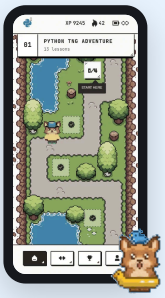
Our SQL tutorial will teach you how to use SQL in MySQL, SQL Server, MS Access, Oracle, PostgreSQL, and more.

Learn SQL now ›

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Learn SQL in bite-sized quests with XP and streaks. Your progress grows for each new adventure! Free on iOS and Android.



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## Examples in Each Chapter

SHARE

With our online SQL editor, you can edit the SQL statements, and click on a button to view the result.

### Example

Get your own SQL Server

```
SELECT * FROM Customers;
```

Try it Yourself »

Click on the "Try it Yourself" button to see how it works.

## SQL Exercises

Many chapters in this tutorial end with an exercise where you can check your level of knowledge.

### Exercise

**Which SQL statement is used to select all records from a table named 'Customers'?**

- SELECT FROM Customers;
- SELECT ALL FROM Customers;
- SELECT * FROM Customers;
- GET ALL FROM Customers;

Submit Answer »

See all SQL Exercises

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## SQL Examples

Learn by examples! This tutorial supplements all explanations with clarifying examples.

See All SQL Examples

## SQL Quiz Test

Test your SQL skills at W3Schools!

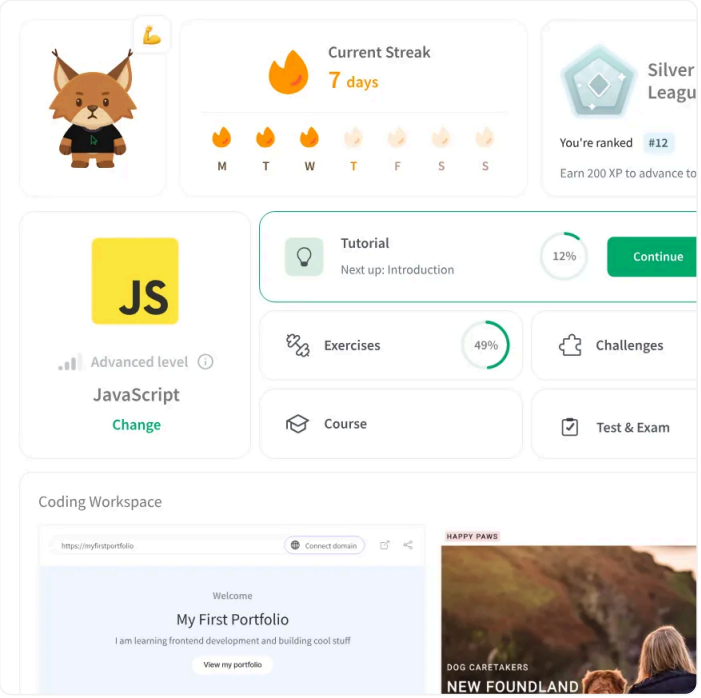
Start SQL Quiz!

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### Note

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## SQL References

At W3Schools you will find a complete reference for keywords and functions:

| | | |
|---|---|---|
| SQL Keyword Reference | MySQL Functions | SQL Server Functions |
| MS Access Functions | SQL Quick Reference | |

## SQL Data Types

Data types and ranges for Microsoft Access, MySQL and SQL Server.

SQL Data Types

SQL CERTIFICATION

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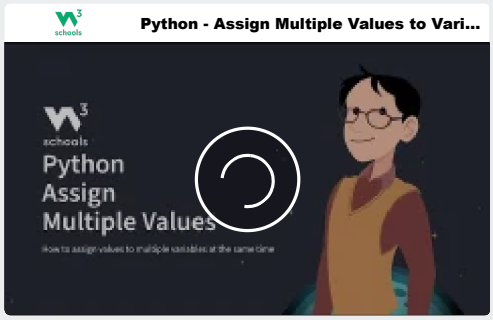

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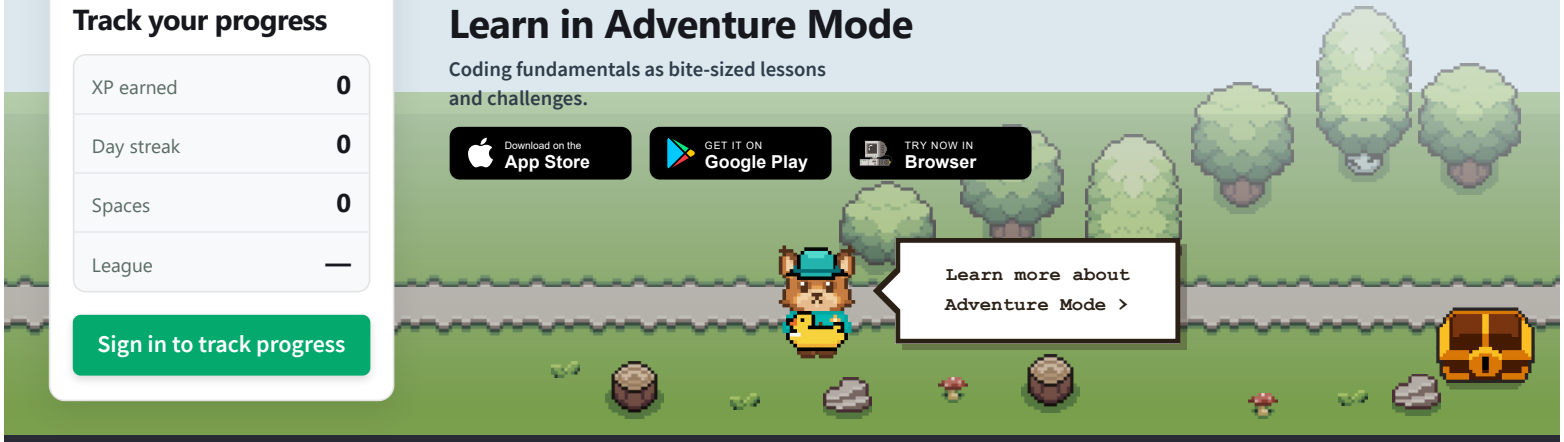
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# Python Tutorial

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## Learn Python

Python is a popular programming language.

Python can be used on a server to create web applications.

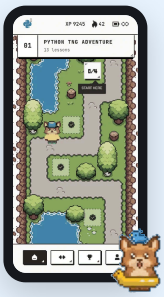
Python is easy to learn - You will enjoy it!

Learn Python now ›

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## Adventure Mode

Learn Python in bite-sized quests with XP and streaks. Your progress grows for each new adventure! Free on iOS and Android.



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## Learning by Examples

SHARE

With our "Try it Yourself" editor, you can edit Python code and view the result.

### Example

Get your own Python Server

```
print("Hello, World!")
```

Try it Yourself »

**Click on the "Try it Yourself" button to see how it works.**

# Python File Handling

In our File Handling section you will learn how to open, read, write, and delete files.

Python File Handling

# Python Database Handling

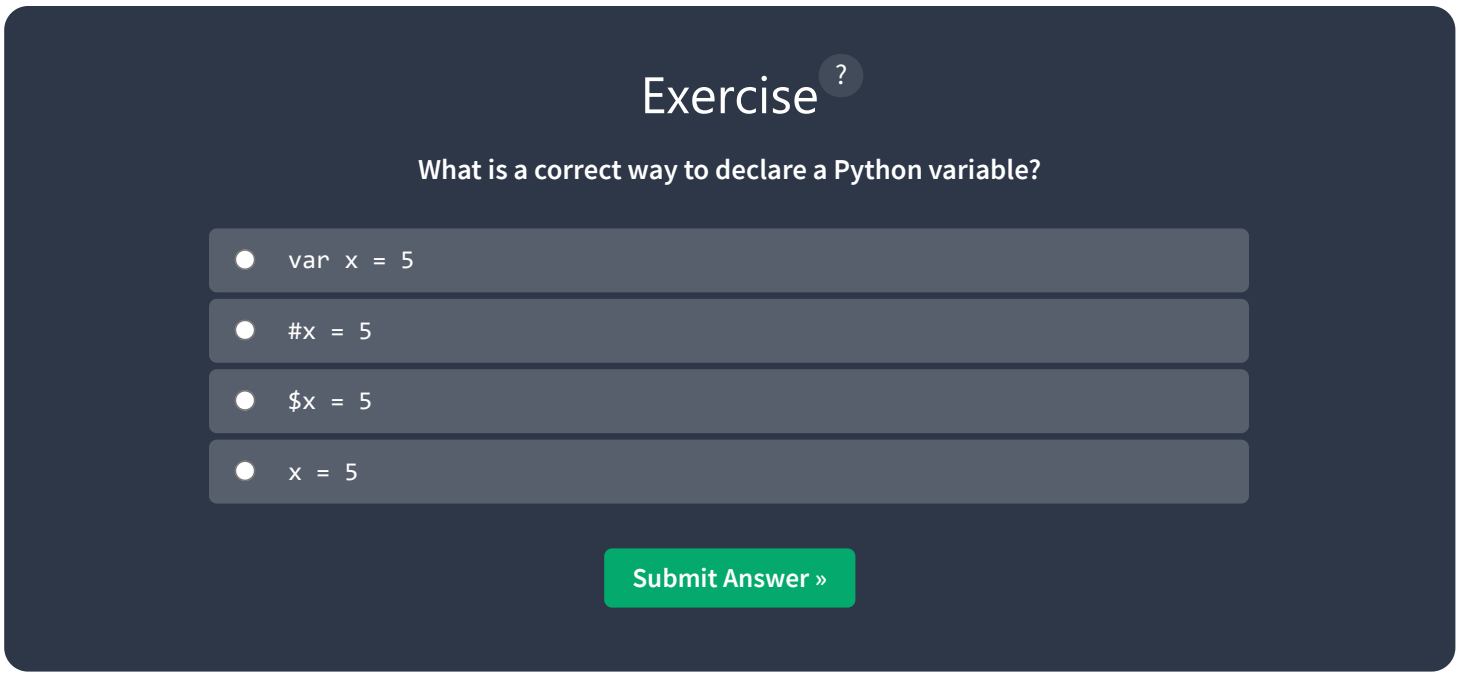
In our database section you will learn how to access and work with MySQL and MongoDB databases:

Python MySQL Tutorial

Python MongoDB Tutorial

# Python Exercises

Many chapters in this tutorial end with an exercise where you can check your level of knowledge.




See all Python Exercises

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# Python Examples

Learn by examples! This tutorial supplements all explanations with clarifying examples.

See All Python Examples

# Python Quiz

Test your Python skills with a quiz.

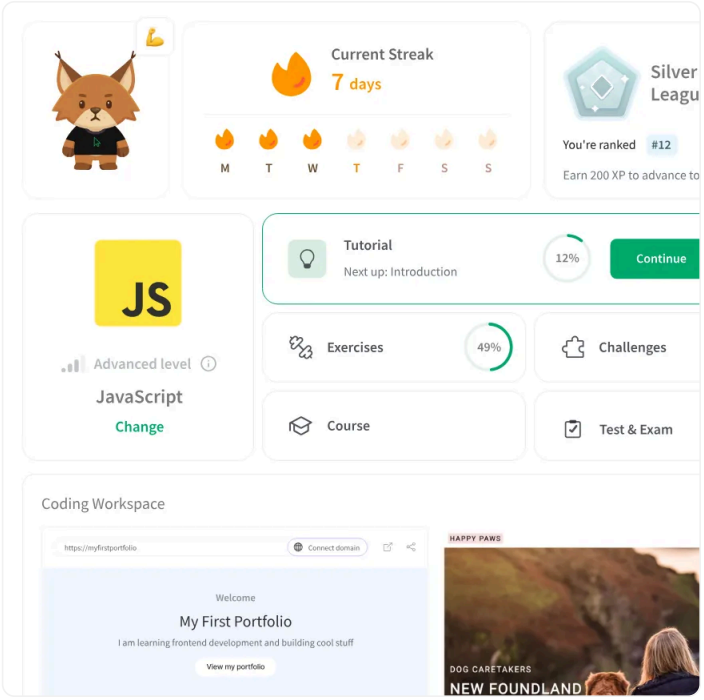
Python Quiz

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# Python Reference

You will also find complete function and method references:

| | | |
|---|---|---|
| Reference Overview | Built-in Functions | String Methods |
| List / Array Methods | Dictionary Methods | Tuple Methods |
| Set Methods | File Methods | Python Keywords |
| Python Exceptions | Python Glossary | Random Module |
| Requests Module | Statistics Module | Math Module |
| CMath Module | | |

# Download Python

Download Python from the official Python web site: https://python.org

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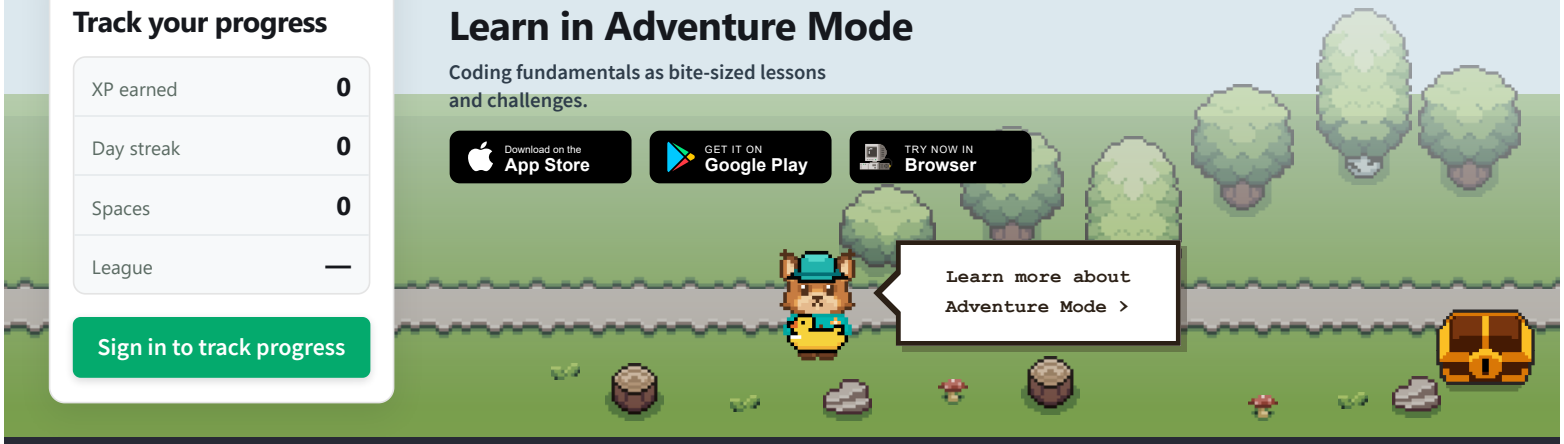
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# W3.CSS Tutorial

‹ Home

Next ›

## Learn W3.CSS

W3.CSS is a CSS framework for developing websites and web apps.

It provides equality for all browsers on all devices.

W3.CSS is free, open source, and uses standard CSS only.

Open Kitchen Sink ›

## Get W3.CSS Certified

Become certified with our W3.CSS exam, includes a professionally curated study kit to guide you from beginner to exam-ready.




Get Certified ›

## See What You Can Build

SHARE

With a few W3.CSS classes you can create cards, responsive layouts, navigation, forms, galleries, dashboards, and complete websites.

# Explore Cities

Discover three great European destinations.

## London

Explore one of the world's great cities.

Explore

## Paris

Discover art, food, and architecture.

Explore

## Rome

Experience thousands of years of history.

Explore

## Example

```
<div class="w3-grid" style="grid-template-columns:repeat(auto-fit,minmax(200px,1fr));gap:16px">

<div class="w3-container w3-card">
  <h2>London</h2>
  <p>Explore one of the world's great cities.</p>
  <button class="w3-button w3-black">Explore</button>
</div>

<div class="w3-container w3-card">
  <h2>Paris</h2>
  <p>Discover art, food, and architecture.</p>
  <button class="w3-button w3-black">Explore</button>
</div>

<div class="w3-container w3-card">
  <h2>Rome</h2>
  <p>Experience thousands of years of history.</p>
  <button class="w3-button w3-black">Explore</button>
</div>

</div>
```

Try it Yourself »

# Why W3.CSS?

## Easy to Learn

W3.CSS uses simple and meaningful class names.

## Responsive

You can create attractive results with very little code.

W3.CSS makes it easy to create layouts that work on desktops, tablets, and phones.

### Modern Layout

W3.CSS supports modern CSS Flexbox and Grid layouts.

### No JavaScript Library

W3.CSS is built with standard CSS and does not depend on any JavaScript library.

## Modern CSS Made Simple

W3.CSS gives you ready-to-use classes for common web design tasks.

### Cards

Create clean content boxes with shadows and spacing.

### Flexbox

Arrange and align content in rows or columns.

### Grid

Build responsive layouts with rows and columns.

You will learn all of these classes, step by step in this tutorial.

## Small Framework. Big Possibilities.

W3.CSS provides the tools needed to build modern websites without a large framework or extra JavaScript libraries.

You can use W3.CSS by itself or combine it with your own CSS or any other CSS.

**W3.CSS is free to use.**

No license is necessary.

## Learn by Building

This tutorial teaches W3.CSS with practical examples.

You will learn how to build real user interfaces instead of only learning individual CSS classes.

1. Learn

2. Try

3. Build

Learn a simple W3.CSS feature.

Edit the examples and see the result.

Combine the classes to build real websites.

Start Learning W3.CSS ❯

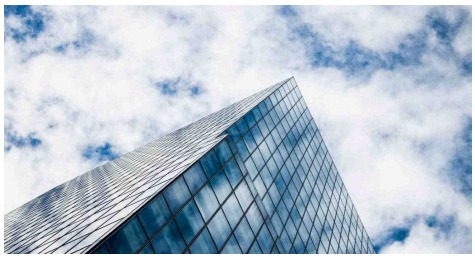
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# W3.CSS Website Templates

W3.CSS can be used to create complete responsive websites.

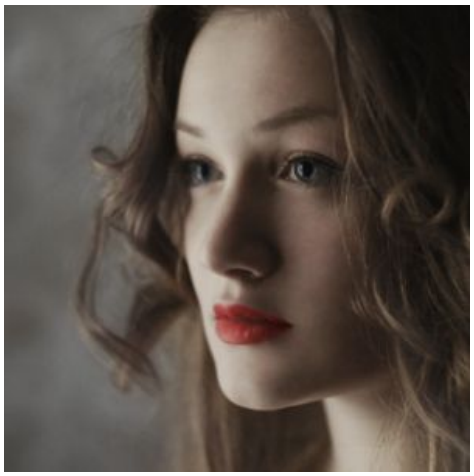
W3Schools provides many free W3.CSS templates that you can edit and use.

## Architect



View Template

## Portfolio



View Template

## Catering




View Template

See All W3.CSS Templates ❯

# W3.CSS Kitchen Sink

Already familiar with CSS frameworks?

See all W3.CSS classes demonstrated.

Open the Kitchen Sink ❯

# W3.CSS Crash Course

Already familiar with HTML and CSS?

Build a complete responsive website in the W3.CSS Crash Course.

Start the Crash Course ❯

# W3.CSS References

Want to study fast?

Our references list all classes and variables available in W3.CSS.

Open W3.CSS References ❯

# Start Learning W3.CSS

W3.CSS is designed to be simple.

In the next chapter, you will see how to start using W3.CSS in your own HTML pages.

Start the Tutorial ❯

# W3.CSS Versions

W3.CSS version 3.01 was released in December 2017.

W3.CSS version 4.01 was released in November 2018.

W3.CSS version 5.01 was released in October 2024.

**W3.CSS version 6.01 was released in October 2026.**

# Note

The W3.CSS style sheet is used in all W3School pages.

The latest versions can be downloaded from W3.CSS Downloads.

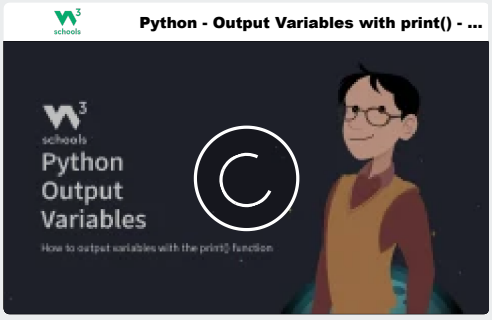
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|---|---|
| XP earned | 0 |
| Day streak | 0 |
| Spaces | 0 |
| League | — |

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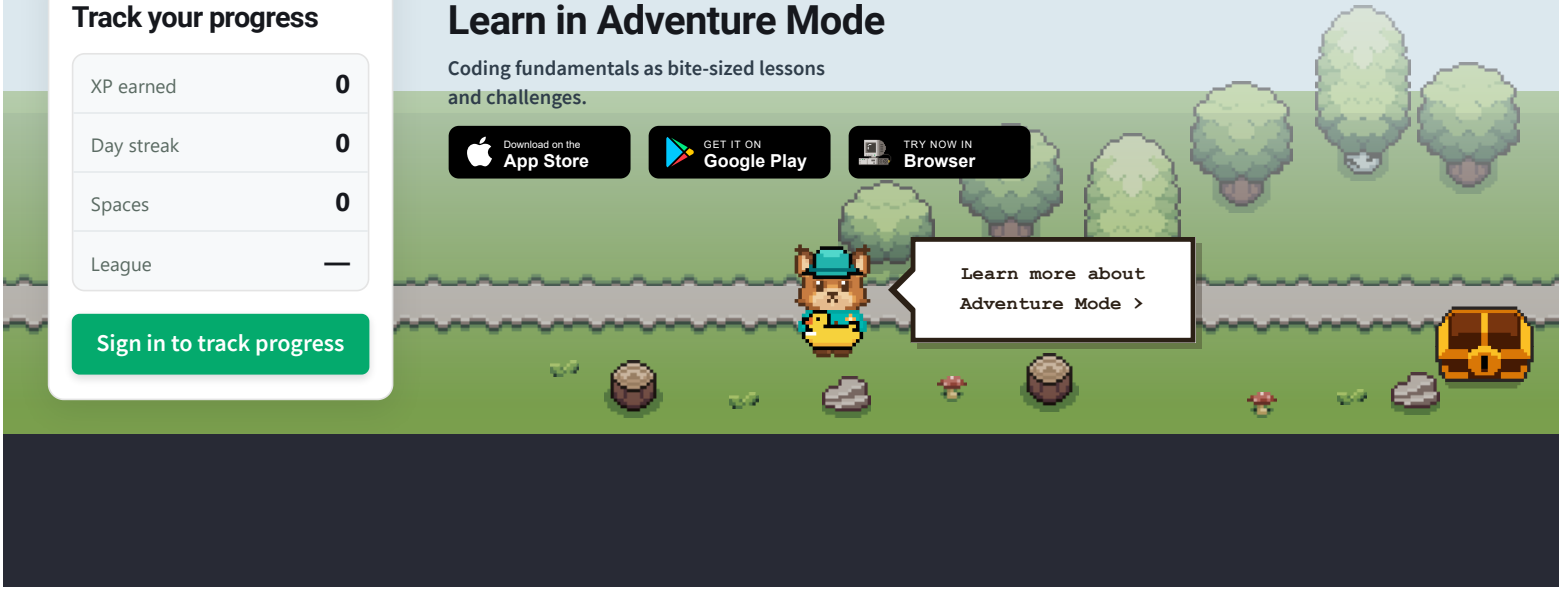
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# Bootstrap Versions

Bootstrap is the world's most famous free CSS framework.
You can choose between the following versions:

## B3

Learn Bootstrap 3 »

Bootstrap 3 is the most stable version of Bootstrap, and it is still supported by the team for critical bugfixes and documentation changes.

## B4

Learn Bootstrap 4 »

Bootstrap 4 is a newer version of Bootstrap; with new components, faster stylesheet and more responsiveness. However, Internet Explorer 9 and down is not supported.

## B5

Learn Bootstrap 5 »

Bootstrap 5 is the **newest** version of Bootstrap; with a smooth overhaul. However, Internet Explorer 11 and down is not supported, and jQuery is replaced with vanilla JavaScript.

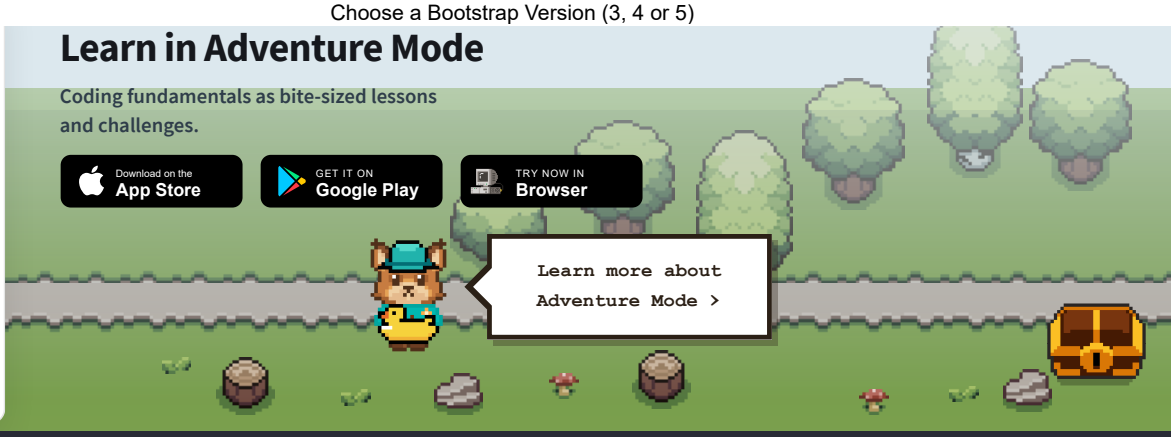
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| | |
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# PHP Tutorial

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## Learn PHP

PHP is a powerful server-side scripting language for making dynamic and interactive Web pages.

PHP is widely-used, open-source and free.

PHP is easy to learn - You will enjoy it!

Learn PHP now ›

## Become PHP Certified

Get certified with our PHP exam, includes a professionally curated study kit to guide you from beginner to exam-ready.




Get Certified ›

Sign in to track your progress

## Easy Learning with "PHP Tryit"

SHARE

With our online "PHP Tryit" editor, you can edit the PHP code, and click on a button to view the result.

### Example

Get your own PHP Server

```
<!DOCTYPE html>
<html>
<body>

<?php
echo 'My first PHP script!';
?>

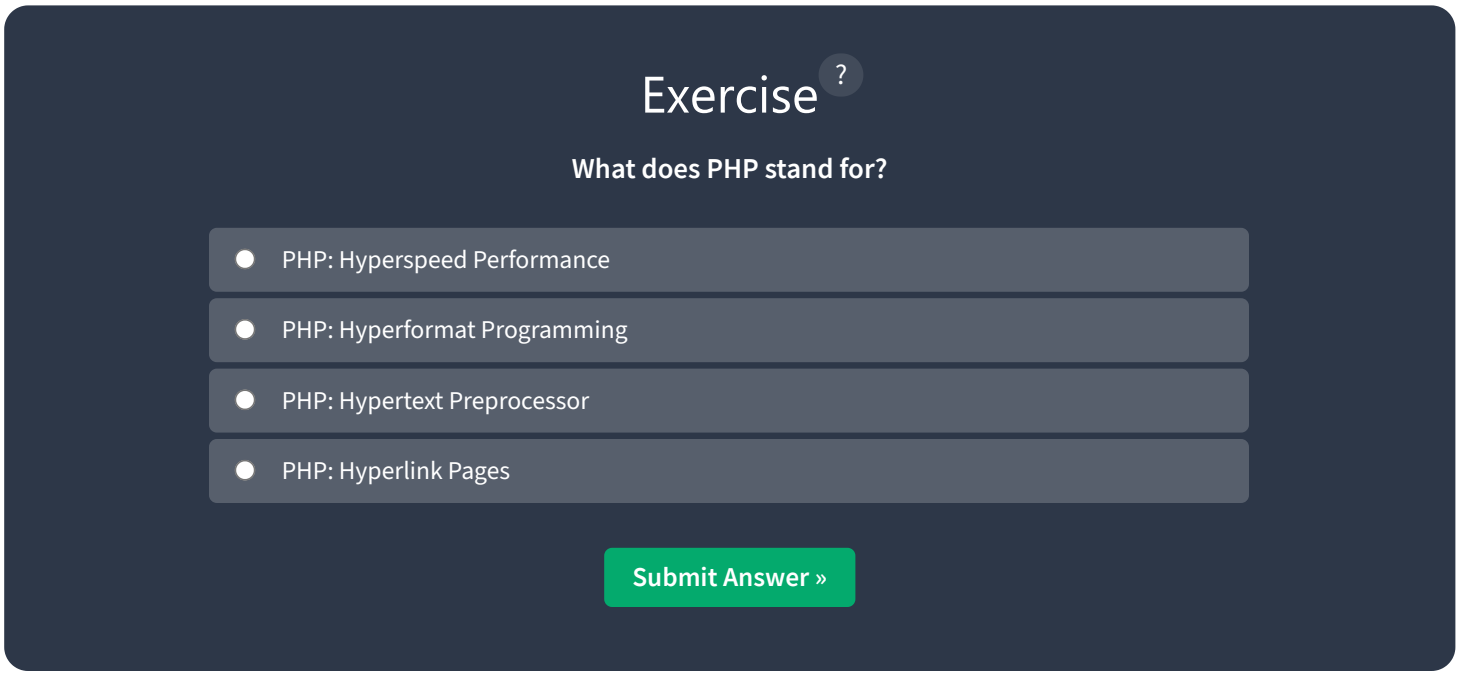
</body>
</html>
```

Try it Yourself »

**Click on the "Try it Yourself" button to see how it works.**

# PHP Exercises

Most chapters in this tutorial end with exercises where you can check your knowledge.




See all PHP Exercises

REMOVE ADS

# PHP Examples

Learn by examples! This tutorial supplements all explanations with clarifying examples.

See All PHP Examples

# PHP Quiz Test

Learn by taking a quiz! This quiz will give you a signal of how much you know, or do not know, about PHP.

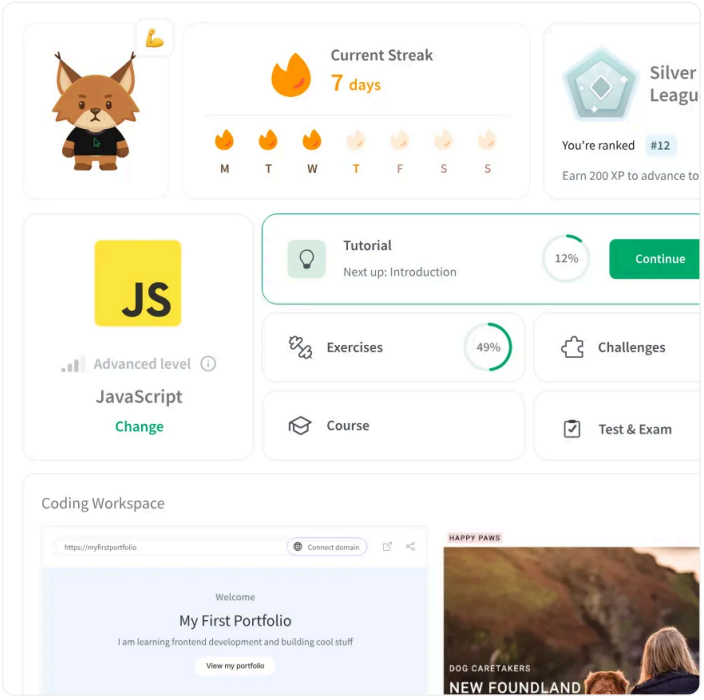
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# PHP References

W3Schools' PHP reference contains different categories of all PHP functions, keywords and constants, along with examples.

Array

Calendar

Date

Directory

Error

Exception

Filesystem

Filter

FTP

JSON

Keywords

Libxml

Mail

Math

Misc

MySQLi

Network

Output Control

RegEx

SimpleXML

Stream

String

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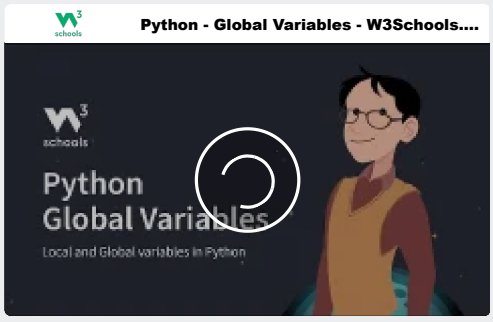
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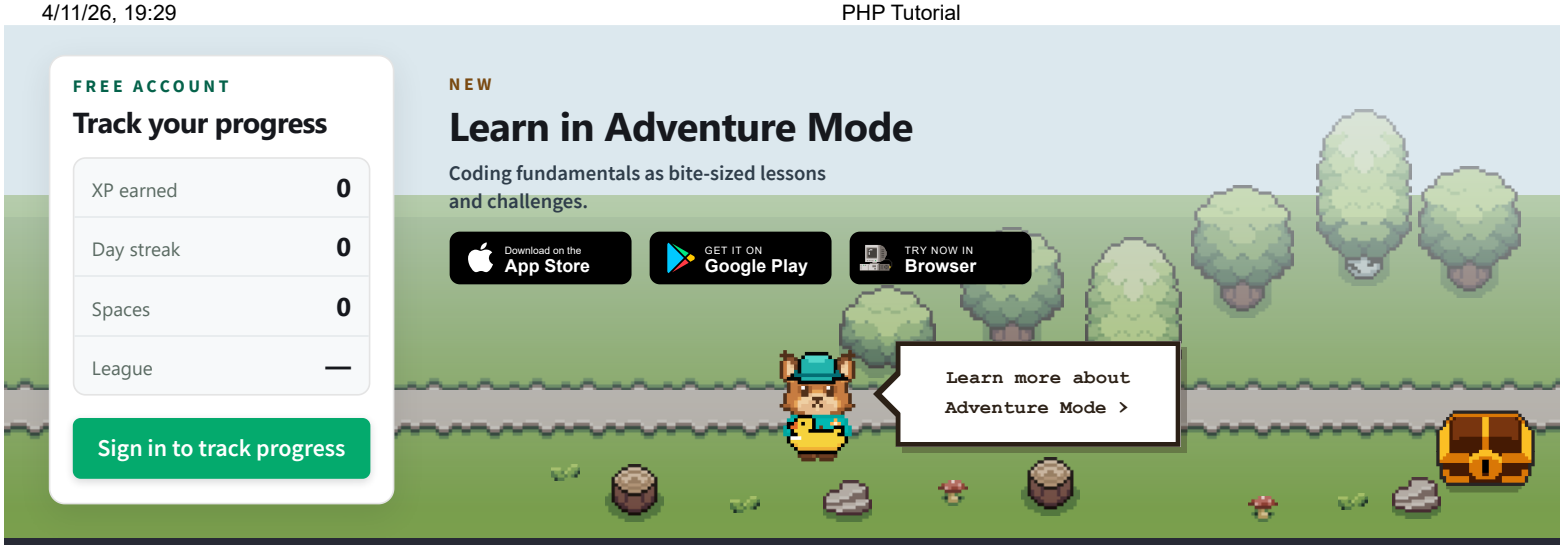
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# Java Tutorial

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## Learn Java

Java is one of the world's most widely used programming languages.

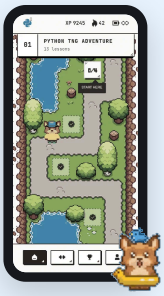
Java is free to use, and runs on all platforms.

Learn Java now ›

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## Examples in Each Chapter

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Our "Try it Yourself" editor makes it easy to learn Java. You can edit Java code and view the result in your browser.

### Example

Get your own Java Server

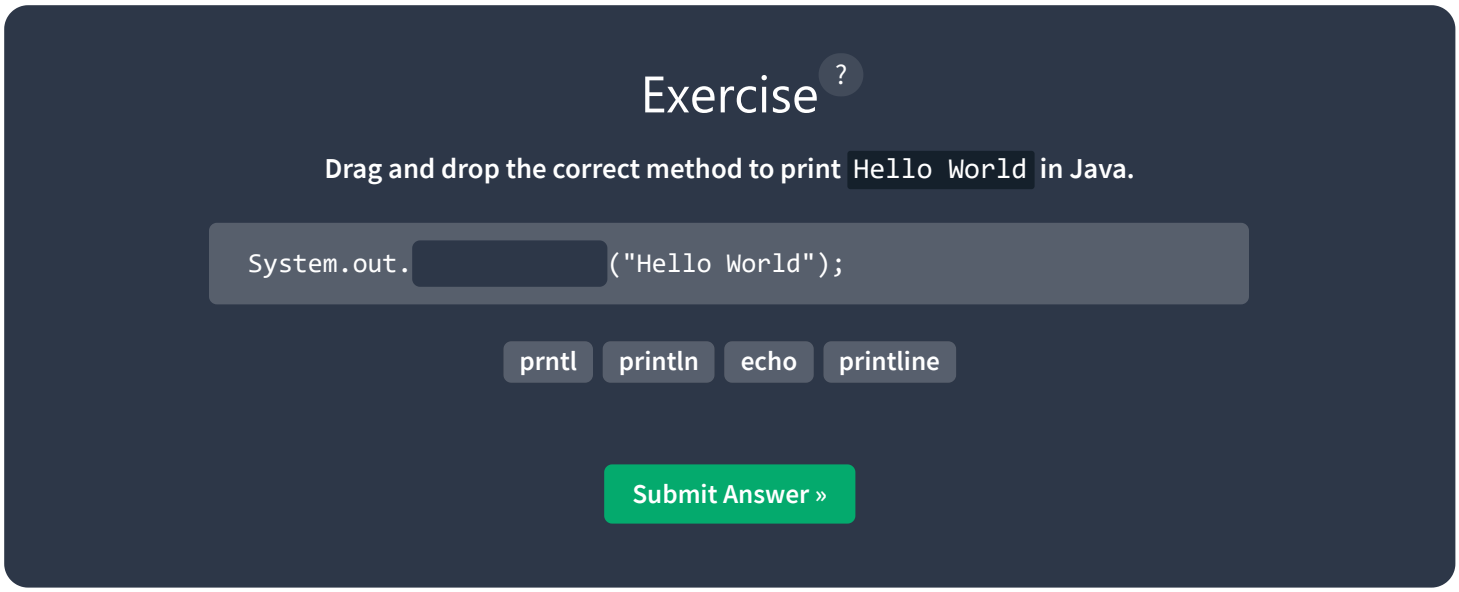
```
public class Main {
  public static void main(String[] args) {
    System.out.println("Hello World");
```

```
  }
}
```

Try it Yourself »

## Exercises

Many chapters in this tutorial end with an exercise where you can check your level of knowledge.




See all Java Exercises

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## What You'll Learn

This tutorial will guide you step by step, with examples and exercises along the way:

- Java syntax and basic structure
- Variables, data types, and operators
- Conditions and loops
- Methods and parameters
- Arrays and strings
- Object-oriented programming
- Classes, objects, inheritance, and polymorphism
- Exception handling
- File handling

- Collections and useful built-in classes

## Java Quiz

Check your understanding with a short quiz and see how well you know Java.

Start Java Quiz

## Learn by Examples

Learn by examples! This tutorial supplements all explanations with clarifying examples.

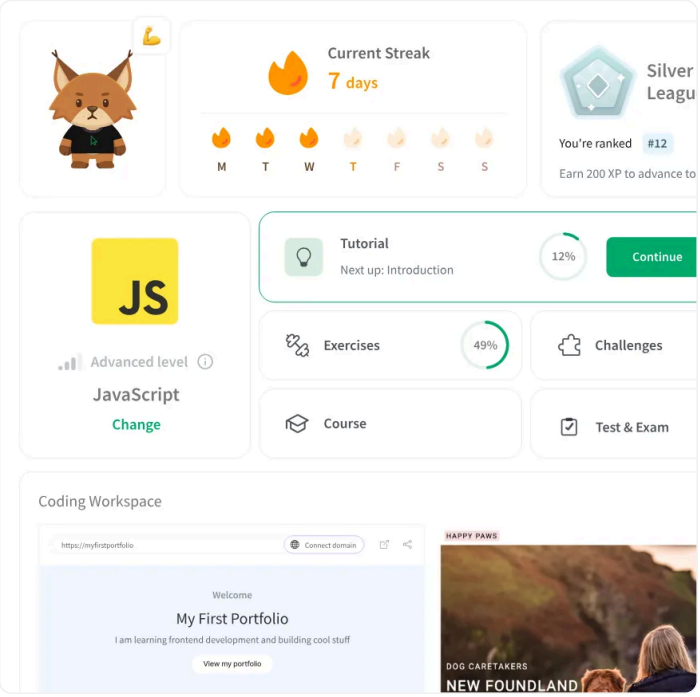
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# Java Reference

You will also find complete keyword and method references:

| | | |
|---|---|---|
| Reference Overview | Java Keywords | String Methods |

Math Methods | Output Methods | Arrays Methods

ArrayList Methods | LinkedList Methods | HashMap Methods

Scanner Methods | File Methods | FileInputStream

FileOutputStream | BufferedReader | BufferedWriter

Iterator Methods | Collections Methods | System Methods

Errors & Exceptions

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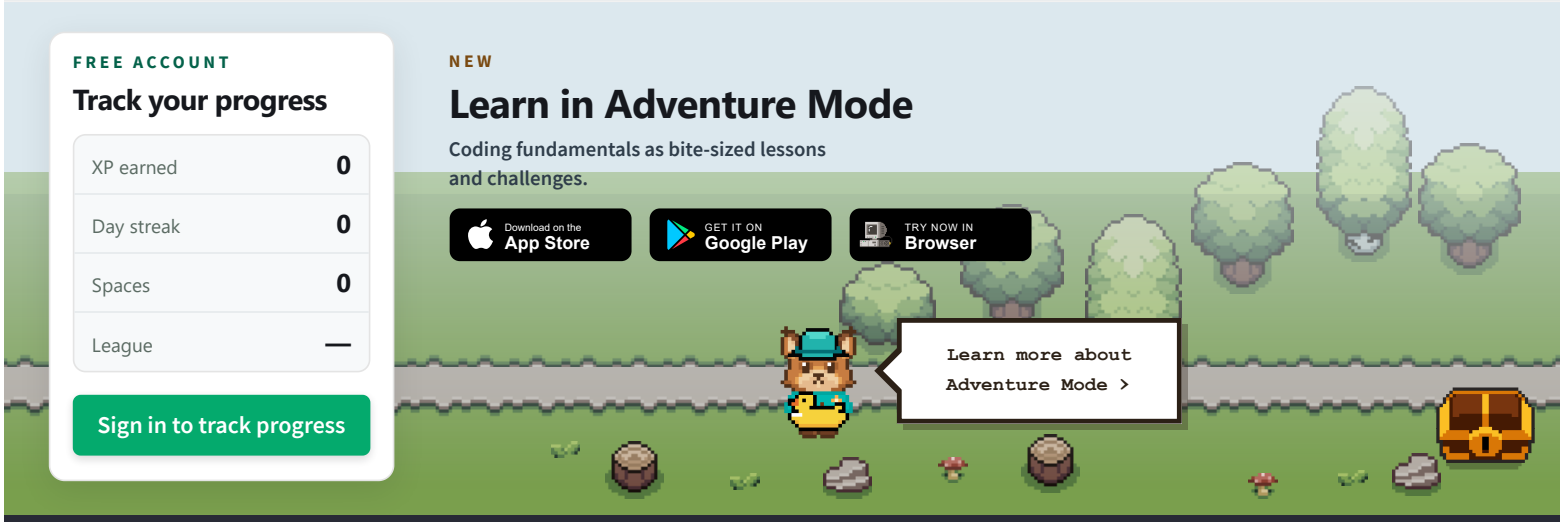


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# C++ Tutorial

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## Learn C++

C++ is a popular programming language.

C++ is used to create computer programs, and is one of the most used languages in game development.

C++ was developed as an extension of C, and both languages have almost the same syntax.

Learn C++ now ❯

## Become C++ Certified

Get certified with our C++ exam, includes a professionally curated study kit to guide you from beginner to exam-ready.




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## Examples in Each Chapter

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Our "Try it Yourself" editor makes it easy to learn C++. You can edit C++ code and view the result in your browser.

### Example

```cpp
#include <iostream>
using namespace std;

int main() {
  cout << "Hello World!";
  return 0;
}
```

Try it Yourself »

We recommend reading this tutorial, in the sequence listed in the left menu.

C++ is an object oriented language and some concepts may be new. Take breaks when needed, and go over the examples as many times as needed.

## C++ Exercises

Many chapters in this tutorial end with an exercise where you can check your level of knowledge.

### Exercise ?

**Drag and drop the correct data type to create a variable that stores an integer value.**

```
          myNum = 15;
cout << myNum;
```

num int double string integer

Submit Answer »

See all C++ Exercises

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# C++ Quiz

Learn by taking a quiz! The quiz will give you a signal of how much you know, or do not know, about C++.

Start C++ Quiz

# Learn by Examples

Learn by examples! This tutorial supplements all explanations with clarifying examples.

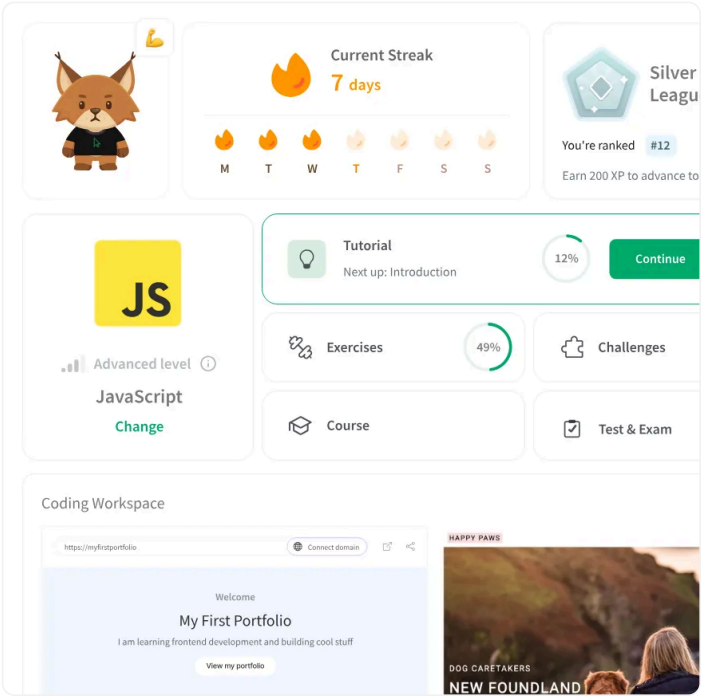
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# C++ Reference

You will also find references for different libraries, keywords and functions:

Reference Overview | C++ Keywords | C++ <iostream> Objects

C++ <fstream> Classes | C++ <cmath> Functions | C++ <string> Functions

C++ <cstring> Functions | C++ <ctime> Functions | C++ <vector> Functions

C++ <algorithm> Functions

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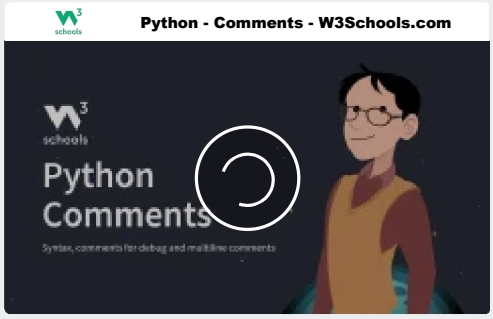
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|---|---|
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## Learn jQuery

jQuery is a JavaScript Library.

jQuery greatly simplifies JavaScript programming.

jQuery is easy to learn.

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## Examples in Each Chapter

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Our "Try it Yourself" editor makes it easy to learn jQuery. You can edit code and view the result in your browser:

### Example

```
$(document).ready(function(){
  $("p").click(function(){
```

```
    $(this).hide();
  });
});
```

Try it Yourself »

**Click on the "Try it Yourself" button to see how it works.**

# jQuery Exercises

## Test Yourself With Exercises

### Exercise:

Use the correct **selector** to hide all <p> elements.

```
$("  ").hide();
```

Submit Answer »

Start the Exercise

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# jQuery Examples

Learn by examples! At W3Schools you will find a lot of jQuery examples to edit and test yourself.

See All jQuery Examples

# jQuery Quiz Test

Test your jQuery skills at W3Schools!

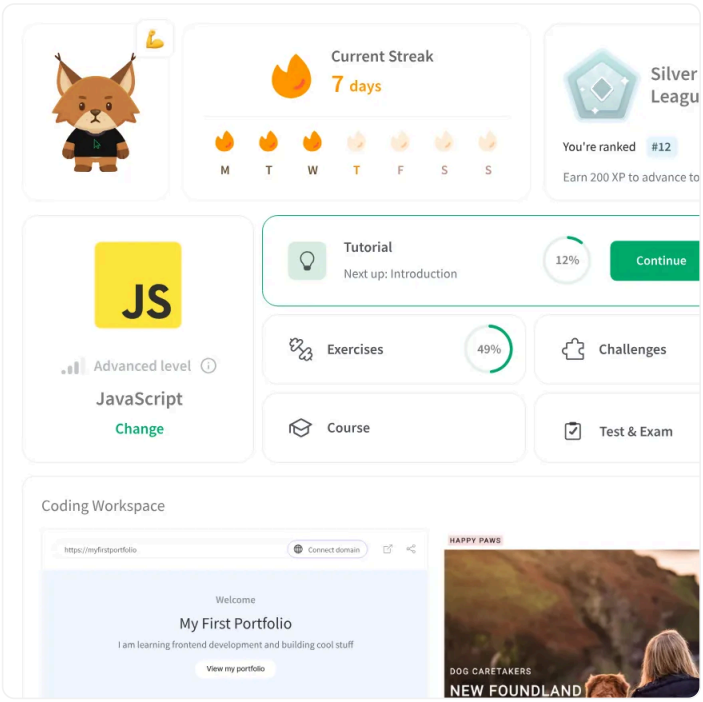
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## jQuery References

At W3Schools you will find a complete reference of all jQuery selectors, methods, properties and events.

jQuery Reference

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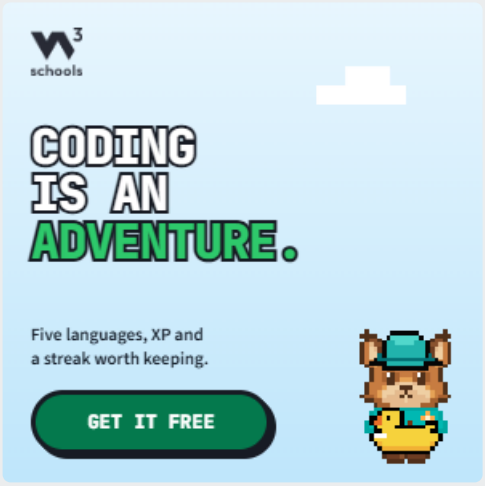
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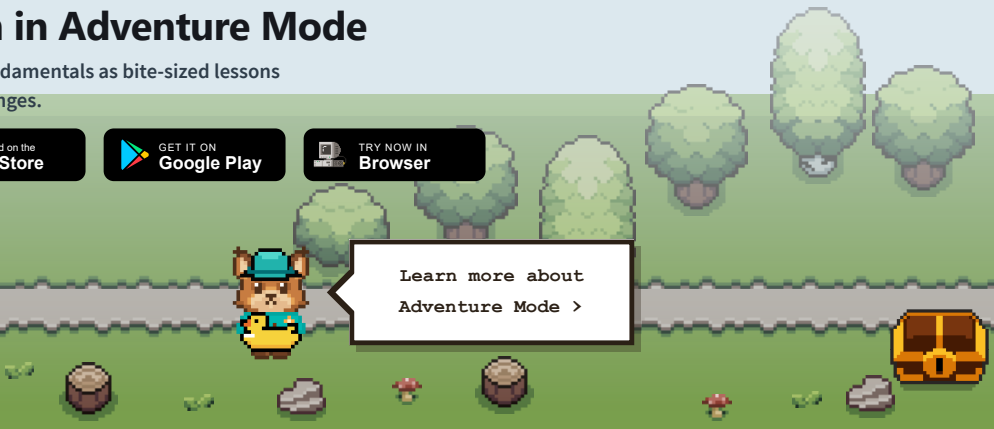
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# HTML Elements Reference

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## HTML Elements Reference

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This page lists all the HTML elements in alphabetical order, with a description of each element.

Search..

| Tag | Description |
|---|---|
| <!--...--> | Defines a comment |
| <!DOCTYPE> | Defines the document type |
| <a> | Defines a hyperlink |
| <abbr> | Defines an abbreviation or an acronym |
| <address> | Defines contact information for the author/owner of a document |
| <area> | Defines an area inside an image map |
| <article> | Defines an article |
| <aside> | Defines content aside from the page content |
| <audio> | Defines embedded sound content |
| <b> | Defines bold text |
| <base> | Specifies the base URL/target for all relative URLs in a document |

| | |
|---|---|
| <bdi> | Isolates a part of text that might be formatted in a different direction from other text outside it |
| <bdo> | Overrides the current text direction |
| <blockquote> | Defines a section that is quoted from another source |
| <body> | Defines the document's body |
| <br> | Defines a single line break |
| <button> | Defines a clickable button |
| <canvas> | Used to draw graphics, on the fly, via scripting (usually JavaScript) |
| <caption> | Defines a table caption |
| <cite> | Defines the title of a work |
| <code> | Defines a piece of computer code |
| <col> | Specifies column properties for each column within a <colgroup> element |
| <colgroup> | Specifies a group of one or more columns in a table for formatting |
| <data> | Adds a machine-readable translation of a given content |
| <datalist> | Specifies a list of pre-defined options for input controls |
| <dd> | Defines a description/value of a term in a description list |
| <del> | Defines text that has been deleted from a document |
| <details> | Defines additional details that the user can view or hide |
| <dfn> | Specifies a term that is going to be defined within the content |
| <dialog> | Defines a dialog box or window |
|  | Defines a section in a document |
| <dl> | Defines a description list |
| <dt> | Defines a term/name in a description list |
|  | Defines emphasized text |
| <embed> | Defines a container for an external application |
| <fieldset> | Groups related elements in a form |
| <figcaption> | Defines a caption for a <figure> element |
| <figure> | Specifies self-contained content |
| <footer> | Defines a footer for a document or section |
| <form> | Defines an HTML form for user input |

| | |
|---|---|
| <h1> to <h6> | Defines HTML headings |
| <head> | Contains metadata/information for the document |
| <header> | Defines a header for a document or section |
| <hgroup> | Defines a header and related content |
| <hr> | Defines a thematic change in the content |
| <html> | Defines the root of an HTML document |
| <i> | Defines a part of text in an alternate voice or mood |
| <iframe> | Defines an inline frame |
| <img> | Defines an image |
| <input> | Defines an input control |
| <ins> | Defines a text that has been inserted into a document |
| <kbd> | Defines keyboard input |
| <label> | Defines a label for an <input> element |
| <legend> | Defines a caption for a <fieldset> element |
| <li> | Defines a list item |
| <link> | Defines the relationship between a document and an external resource (most used to link to style sheets) |
| <main> | Specifies the main content of a document |
| <map> | Defines an image map |
| <mark> | Defines marked/highlighted text |
| <menu> | Defines an unordered list |
| <meta> | Defines metadata about an HTML document |
| <meter> | Defines a scalar measurement within a known range (a gauge) |
| <nav> | Defines navigation links |
| <noscript> | Defines an alternate content for users that do not support client-side scripts |
| <object> | Defines a container for an external application |
| <ol> | Defines an ordered list |
| <optgroup> | Defines a group of related options in a drop-down list |
| <option> | Defines an option in a drop-down list |
| <output> | Defines the result of a calculation |

| | |
|---|---|
| <p> | Defines a paragraph |
| <param> | Defines a parameter for an object |
| <picture> | Defines a container for multiple image resources |
| <pre> | Defines preformatted text |
| <progress> | Represents the progress of a task |
| <q> | Defines a short quotation |
| <rp> | Defines what to show in browsers that do not support ruby annotations |
| <rt> | Defines an explanation/pronunciation of characters (for East Asian typography) |
| <ruby> | Defines a ruby annotation (for East Asian typography) |
| <s> | Defines text that is no longer correct |
| <samp> | Defines sample output from a computer program |
| <script> | Defines a client-side script |
| <search> | Defines a search section |
| <section> | Defines a section in a document |
| <select> | Defines a drop-down list |
| <small> | Defines smaller text |
| <source> | Defines multiple media resources for media elements (<video> and <audio>) |
|  | Defines a section in a document |
|  | Defines important text |
| <style> | Defines style information for a document |
| <sub> | Defines subscripted text |
| <summary> | Defines a visible heading for a <details> element |
| <sup> | Defines superscripted text |
| <svg> | Defines a container for SVG graphics |
| <table> | Defines a table |
| <tbody> | Groups the body content in a table |
| <td> | Defines a cell in a table |
| <template> | Defines a container for content that should be hidden when the page loads |
| <textarea> | Defines a multiline input control (text area) |

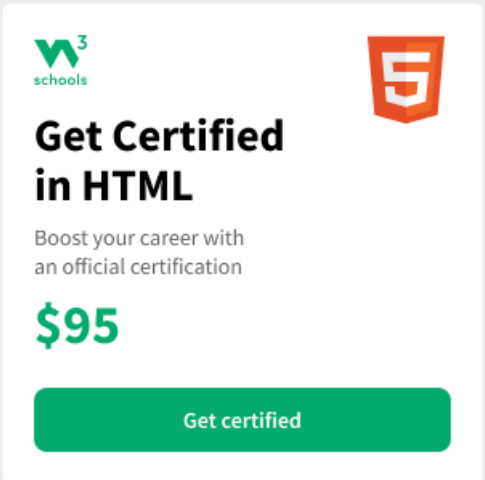
| | |
|---|---|
| <tfoot> | Groups the footer content in a table |
| <th> | Defines a header cell in a table |
| <thead> | Groups the header content in a table |
| <time> | Defines a specific time (or datetime) |
| <title> | Defines a title for the document |
| <tr> | Defines a row in a table |
| <track> | Defines text tracks for media elements (<video> and <audio>) |
| <u> | Defines some text that is unarticulated and styled differently from normal text |
| <ul> | Defines an unordered list |
| <var> | Defines a variable |
| <video> | Defines embedded video content |
| <wbr> | Defines a possible line-break |

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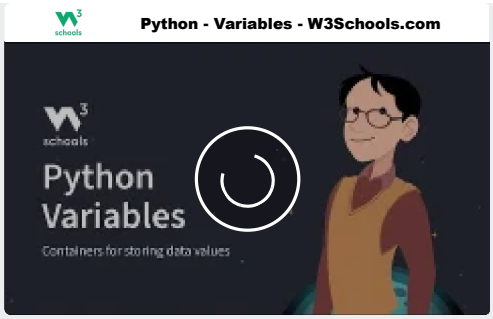
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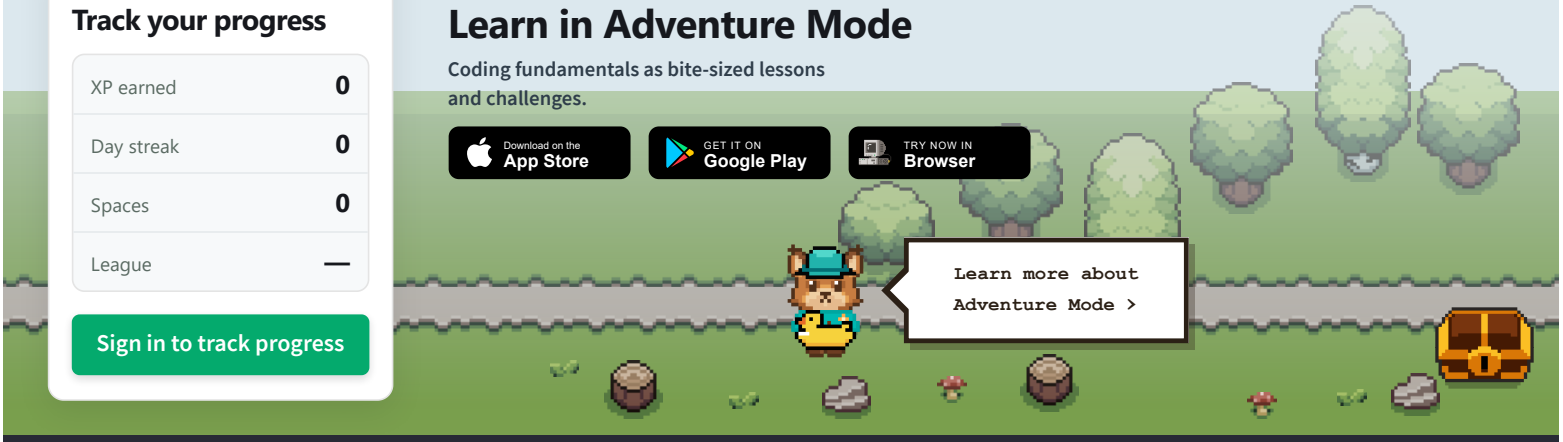
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## CSS Properties Reference

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This page lists all the CSS properties in alphabetical order, with a description of each element.

Search..

## A

| | |
|---|---|
| accent-color | Specifies an accent color for user-interface controls |
| align-content | Specifies the alignment between the lines inside a flexible container when the items do not use all available space |
| align-items | Specifies the alignment for items inside a flexible container |
| align-self | Specifies the alignment for selected items inside a flexible container |
| all | Resets all properties (except unicode-bidi and direction) |
| animation | A shorthand property for all the *animation*-* properties |
| animation-delay | Specifies a delay for the start of an animation |
| animation-direction | Specifies whether an animation should be played forward, backward or in alternate cycles |
| animation-duration | Specifies how long an animation should take to complete one cycle |

| | |
|---|---|
| animation-fill-mode | Specifies a style for the element when the animation is not playing (before it starts, after it ends, or both) |
| animation-iteration-count | Specifies the number of times an animation should be played |
| animation-name | Specifies a name for the @keyframes animation |
| animation-play-state | Specifies whether the animation is running or paused |
| animation-timing-function | Specifies the speed curve of an animation |
| aspect-ratio | Specifies preferred aspect ratio of an element |

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# B

| | |
|---|---|
| backdrop-filter | Defines a graphical effect to the area behind an element |
| backface-visibility | Defines whether or not the back face of an element should be visible when facing the user |
| background | A shorthand property for all the *background-** properties |
| background-attachment | Sets whether a background image scrolls with the rest of the page, or is fixed |
| background-blend-mode | Specifies the blending mode of each background layer (color/image) |
| background-clip | Defines how far the background (color or image) should extend within an element |
| background-color | Specifies the background color of an element |
| background-image | Specifies one or more background images for an element |
| background-origin | Specifies the origin position of a background image |
| background-position | Specifies the position of a background image |
| background-position-x | Specifies the position of a background image on x-axis |
| background-position-y | Specifies the position of a background image on y-axis |
| background-repeat | Sets if/how a background image will be repeated |
| background-size | Specifies the size of the background images |
| block-size | Specifies the size of an element in block direction |
| border | A shorthand property for *border-width, border-style* and *border-color* |
| border-block | A shorthand property for *border-block-width, border-block-style* and *border-block-color* |

| | |
|---|---|
| border-block-color | Sets the color of the borders at start and end in the block direction |
| border-block-end | A shorthand property for *border-block-end-width, border-block-end-style* and *border-block-end-color* |
| border-block-end-color | Sets the color of the border at the end in the block direction |
| border-block-end-style | Sets the style of the border at the end in the block direction |
| border-block-end-width | Sets the width of the border at the end in the block direction |
| border-block-start | A shorthand property for *border-block-start-width, border-block-start-style* and *border-block-start-color* |
| border-block-start-color | Sets the color of the border at the start in the block direction |
| border-block-start-style | Sets the style of the border at the start in the block direction |
| border-block-start-width | Sets the width of the border at the start in the block direction |
| border-block-style | Sets the style of the borders at start and end in the block direction |
| border-block-width | Sets the width of the borders at start and end in the block direction |
| border-bottom | A shorthand property for *border-bottom-width, border-bottom-style* and *border-bottom-color* |
| border-bottom-color | Sets the color of the bottom border |
| border-bottom-left-radius | Defines the radius of the border of the bottom-left corner |
| border-bottom-right-radius | Defines the radius of the border of the bottom-right corner |
| border-bottom-style | Sets the style of the bottom border |
| border-bottom-width | Sets the width of the bottom border |
| border-collapse | Sets whether table borders should collapse into a single border or be separated |
| border-color | Sets the color of the four borders |
| border-end-end-radius | Sets the radius of the corner between the block-end and the inline-end sides of the element |
| border-end-start-radius | Sets the radius of the corner between the block-end and the inline-start sides of the element |
| border-image | A shorthand property for all the *border-image-** properties |
| border-image-outset | Specifies the amount by which the border image area extends beyond the border box |
| border-image-repeat | Specifies whether the border image should be repeated, rounded or stretched |
| border-image-slice | Specifies how to slice the border image |
| border-image-source | Specifies the path to the image to be used as a border |

| | |
|---|---|
| border-image-width | Specifies the width of the border image |
| border-inline | A shorthand property for *border-inline-width, border-inline-style* and *border-inline-color* |
| border-inline-color | Sets the color of the borders at start and end in the inline direction |
| border-inline-end | A shorthand property for *border-inline-end-width, border-inline-end-style* and *border-inline-end-color* |
| border-inline-end-color | Sets the color of the border at the end in the inline direction |
| border-inline-end-style | Sets the style of the border at the end in the inline direction |
| border-inline-end-width | Sets the width of the border at the end in the inline direction |
| border-inline-start | A shorthand property for *border-inline-start-width, border-inline-start-style* and *border-inline-start-color* |
| border-inline-start-color | Sets the color of the border at the start in the inline direction |
| border-inline-start-style | Sets the style of the border at the start in the inline direction |
| border-inline-start-width | Sets the width of the border at the start in the inline direction |
| border-inline-style | Sets the style of the borders at start and end in the inline direction |
| border-inline-width | Sets the width of the borders at start and end in the inline direction |
| border-left | A shorthand property for all the *border-left-** properties |
| border-left-color | Sets the color of the left border |
| border-left-style | Sets the style of the left border |
| border-left-width | Sets the width of the left border |
| border-radius | A shorthand property for the four *border-*-radius* properties |
| border-right | A shorthand property for all the *border-right-** properties |
| border-right-color | Sets the color of the right border |
| border-right-style | Sets the style of the right border |
| border-right-width | Sets the width of the right border |
| border-spacing | Sets the distance between the borders of adjacent cells |
| border-start-end-radius | Sets the radius of the corner between the block-start and the inline-end sides of the element |
| border-start-start-radius | Sets the radius of the corner between the block-start and the inline-start sides of the element |
| border-style | Sets the style of the four borders |
| border-top | A shorthand property for *border-top-width, border-top-style* and *border-top-color* |

| | |
|---|---|
| border-top-color | Sets the color of the top border |
| border-top-left-radius | Defines the radius of the border of the top-left corner |
| border-top-right-radius | Defines the radius of the border of the top-right corner |
| border-top-style | Sets the style of the top border |
| border-top-width | Sets the width of the top border |
| border-width | Sets the width of the four borders |
| bottom | Sets the elements position, from the bottom of its parent element |
| box-decoration-break | Sets the behavior of the background and border of an element at page-break, or, for in-line elements, at line-break. |
| box-reflect | The box-reflect property is used to create a reflection of an element. |
| box-shadow | Attaches one or more shadows to an element |
| box-sizing | Defines how the width and height of an element are calculated: should they include padding and borders, or not |
| break-after | Specifies whether or not a page-, column-, or region-break should occur after the specified element |
| break-before | Specifies whether or not a page-, column-, or region-break should occur before the specified element |
| break-inside | Specifies whether or not a page-, column-, or region-break should occur inside the specified element |

# C

| | |
|---|---|
| caption-side | Specifies the placement of a table caption |
| caret-color | Specifies the color of the cursor (caret) in inputs, textareas, or any element that is editable |
| @charset | Specifies the character encoding used in the style sheet |
| clear | Specifies what should happen with the element that is next to a floating element |
| clip | Deprecated in favor of clip-path. Clips an absolutely positioned element |
| clip-path | Clips an element to a basic shape or to an SVG source |
| color | Sets the color of text |
| color-scheme | Indicates which operating system color scheme an element should render with |
| column-count | Specifies the number of columns an element should be divided into |
| column-fill | Specifies how to fill columns, balanced or not |

| | |
|---|---|
| column-gap | Specifies the gap between the columns |
| column-rule | A shorthand property for all the *column-rule-** properties |
| column-rule-color | Specifies the color of the rule between columns |
| column-rule-style | Specifies the style of the rule between columns |
| column-rule-width | Specifies the width of the rule between columns |
| column-span | Specifies how many columns an element should span across |
| column-width | Specifies the column width |
| columns | A shorthand property for *column-width* and *column-count* |
| @container | Define styles for elements in container, depending on the container's size or style |
| content | Used with the :before and :after pseudo-elements, to insert generated content |
| counter-increment | Increases or decreases the value of one or more CSS counters |
| counter-reset | Creates or resets one or more CSS counters |
| counter-set | Creates or sets one or more CSS counters |
| @counter-style | Lets you define your own counter styles |
| cursor | Specifies the mouse cursor to be displayed when pointing over an element |

# D

| | |
|---|---|
| direction | Specifies the text direction/writing direction |
| display | Specifies how a certain HTML element should be displayed |

# E

| | |
|---|---|
| empty-cells | Specifies whether or not to display borders and background on empty cells in a table |

# F

| | |
|---|---|
| filter | Defines effects (e.g. blurring or color shifting) on an element before the element is displayed |
| flex | A shorthand property for the *flex-grow, flex-shrink*, and the *flex-basis* properties |
| flex-basis | Specifies the initial length of a flexible item |
| flex-direction | Specifies the direction of the flexible items |

| | |
|---|---|
| flex-flow | A shorthand property for the *flex-direction* and the *flex-wrap* properties |
| flex-grow | Specifies how much the item will grow relative to the rest |
| flex-shrink | Specifies how the item will shrink relative to the rest |
| flex-wrap | Specifies whether the flexible items should wrap or not |
| float | Specifies whether an element should float to the left, right, or not at all |
| font | A shorthand property for the *font-style, font-variant, font-weight, font-size/line-height*, and the *font-family* properties |
| @font-face | Specifies a custom font to use to display text |
| font-family | Specifies the font family for text |
| font-feature-settings | Allows control over advanced typographic features in OpenType fonts |
| font-kerning | Controls the usage of the kerning information (how letters are spaced) |
| font-language-override | Controls the usage of language-specific glyphs in a typeface |
| @font-palette-values | Allows you to customize the default values of a font-palette |
| font-size | Specifies the font size of text |
| font-size-adjust | Preserves the readability and size of text when fallback font occurs |
| font-stretch | Selects a normal, condensed, or expanded face from a font family |
| font-style | Specifies the font style for text |
| font-synthesis | Controls which missing typefaces (bold or italic) may be synthesized by the browser |
| font-variant | Specifies whether or not a text should be displayed in a small-caps font |
| font-variant-alternates | Controls the usage of alternate glyphs associated to alternative names defined in @font-feature-values |
| font-variant-caps | Controls the usage of alternate glyphs for capital letters |
| font-variant-east-asian | Controls the usage of alternate glyphs for East Asian scripts (e.g Japanese and Chinese) |
| font-variant-ligatures | Controls which ligatures and contextual forms are used in textual content of the elements it applies to |
| font-variant-numeric | Controls the usage of alternate glyphs for numbers, fractions, and ordinal markers |
| font-variant-position | Controls the usage of alternate glyphs of smaller size positioned as superscript or subscript regarding the baseline of the font |
| font-weight | Specifies the weight of a font |

# G

| | |
|---|---|
| gap | A shorthand property for the *row-gap* and the *column-gap* properties |
| grid | A shorthand property for the *grid-template-rows, grid-template-columns, grid-template-areas, grid-auto-rows, grid-auto-columns*, and the *grid-auto-flow* properties |
| grid-area | Either specifies a name for the grid item, or this property is a shorthand property for the *grid-row-start*, *grid-column-start*, *grid-row-end*, and *grid-column-end* properties |
| grid-auto-columns | Specifies a default column size |
| grid-auto-flow | Specifies how auto-placed items are inserted in the grid |
| grid-auto-rows | Specifies a default row size |
| grid-column | A shorthand property for the *grid-column-start* and the *grid-column-end* properties |
| grid-column-end | Specifies where to end the grid item |
| grid-column-start | Specifies where to start the grid item |
| grid-row | A shorthand property for the *grid-row-start* and the *grid-row-end* properties |
| grid-row-end | Specifies where to end the grid item |
| grid-row-start | Specifies where to start the grid item |
| grid-template | A shorthand property for the *grid-template-rows*, *grid-template-columns* and *grid-areas* properties |
| grid-template-areas | Specifies how to display columns and rows, using named grid items |
| grid-template-columns | Specifies the size of the columns, and how many columns in a grid layout |
| grid-template-rows | Specifies the size of the rows in a grid layout |

# H

| | |
|---|---|
| hanging-punctuation | Specifies whether a punctuation character may be placed outside the line box |
| height | Sets the height of an element |
| hyphens | Sets how to split words to improve the layout of text |
| hyphenate-character | Sets the character used at the end of line, before a hyphenation break |

# I

| | |
|---|---|
| image-rendering | Specifies the type of algorithm to use for image scaling |
| @import | Allows you to import a style sheet into another style sheet |
| initial-letter | Specifies the size of the initial-letter and optionally the number of lines the initial letter should sink (down in the text) |

| | |
|---|---|
| inline-size | Specifies the size of an element in the inline direction |
| inset | Specifies the distance between an element and the parent element |
| inset-block | Specifies the distance between an element and the parent element in the block direction |
| inset-block-end | Specifies the distance between the end of an element and the parent element in the block direction |
| inset-block-start | Specifies the distance between the start of an element and the parent element in the block direction |
| inset-inline | Specifies the distance between an element and the parent element in the inline direction |
| inset-inline-end | Specifies the distance between the end of an element and the parent element in the inline direction |
| inset-inline-start | Specifies the distance between the start of an element and the parent element in the inline direction |
| isolation | Defines whether an element must create a new stacking content |

## J

| | |
|---|---|
| justify-content | Specifies the alignment between the items inside a flexible container when the items do not use all available space |
| justify-items | Is set on the grid container. Specifies the alignment of grid items in the inline direction |
| justify-self | Is set on the grid item. Specifies the alignment of the grid item in the inline direction |

## K

| | |
|---|---|
| @keyframes | Controls the steps in an animation by defining styles for points along the animation sequence |

## L

| | |
|---|---|
| @layer | Controls how the CSS cascade layers evaluates the order of styles |
| left | Specifies the left position of a positioned element |
| letter-spacing | Increases or decreases the space between characters in a text |
| line-break | Specifies how/if to break lines |
| line-height | Sets the line height |

| | |
|---|---|
| list-style | Sets all the properties for a list in one declaration |
| list-style-image | Specifies an image as the list-item marker |
| list-style-position | Specifies the position of the list-item markers (bullet points) |
| list-style-type | Specifies the type of list-item marker |

# M

| | |
|---|---|
| margin | Sets all the margin properties in one declaration |
| margin-block | Specifies the margin in the block direction |
| margin-block-end | Specifies the margin at the end in the block direction |
| margin-block-start | Specifies the margin at the start in the block direction |
| margin-bottom | Sets the bottom margin of an element |
| margin-inline | Specifies the margin in the inline direction |
| margin-inline-end | Specifies the margin at the end in the inline direction |
| margin-inline-start | Specifies the margin at the start in the inline direction |
| margin-left | Sets the left margin of an element |
| margin-right | Sets the right margin of an element |
| margin-top | Sets the top margin of an element |
| marker | Points to a marker that will be drawn on all vertices of an element's path (the first, middle, and last) |
| marker-end | Points to a marker that will be drawn on the last vertex of an element's path |
| marker-mid | Points to a marker that will be drawn on all the middle vertices of an element's path |
| marker-start | Points to a marker that will be drawn on the first vertex of an element's path |
| mask | A shorthand property for *mask-image*, *mask-mode*, *mask-repeat*, *mask-position*, *mask-clip*, *mask-origin*, *mask-size* and *mask-composite* |
| mask-clip | Specifies which area is affected by a mask image |
| mask-composite | Specifies a compositing operation used on the current mask layer with the mask layers below it |
| mask-image | Specifies an image to be used as a mask layer for an element |
| mask-mode | Specifies whether the mask layer image is treated as a luminance mask or as an alpha mask |
| mask-origin | Specifies the origin position (the mask position area) of a mask layer image |

| | |
|---|---|
| mask-position | Sets the starting position of a mask layer image (relative to the mask position area) |
| mask-repeat | Specifies how the mask layer image is repeated |
| mask-size | Specifies the size of a mask layer image |
| mask-type | Specifies whether an SVG <mask> element is treated as a luminance mask or as an alpha mask |
| max-height | Sets the maximum height of an element |
| max-width | Sets the maximum width of an element |
| @media | Sets the style rules for different media types/devices/sizes |
| max-block-size | Sets the maximum size of an element in the block direction |
| max-inline-size | Sets the maximum size of an element in the inline direction |
| min-block-size | Sets the minimum size of an element in the block direction |
| min-inline-size | Sets the minimum size of an element in the inline direction |
| min-height | Sets the minimum height of an element |
| min-width | Sets the minimum width of an element |
| mix-blend-mode | Specifies how an element's content should blend with its direct parent background |

# N

| | |
|---|---|
| @namespace | Defines an XML namespace to be used in the style sheet |

# O

| | |
|---|---|
| object-fit | Specifies how the contents of a replaced element should be fitted to the box established by its used height and width |
| object-position | Specifies the alignment of the replaced element inside its box |
| offset | A shorthand property for the *offset-anchor*, *offset-distance*, *offset-path*, *offset-position*, and the *offset-rotate* properties |
| offset-anchor | Specifies a point on an element that is fixed to the path it is animated along |
| offset-distance | Specifies the position along a path where an animated element is placed |
| offset-path | Specifies the path an element is animated along |
| offset-position | Specifies the initial position of an element along a path |
| offset-rotate | Specifies rotation of an element as it is animated along a path |

| | |
|---|---|
| opacity | Sets the opacity level for an element |
| order | Sets the order of the flexible item, relative to the rest |
| orphans | Sets the minimum number of lines that must be left at the bottom of a page or column |
| outline | A shorthand property for the *outline-width, outline-style*, and the *outline-color* properties |
| outline-color | Sets the color of an outline |
| outline-offset | Offsets an outline, and draws it beyond the border edge |
| outline-style | Sets the style of an outline |
| outline-width | Sets the width of an outline |
| overflow | Specifies what happens if content overflows an element's box |
| overflow-anchor | Specifies whether or not content in viewable area in a scrollable contianer should be pushed down when new content is loaded above |
| overflow-wrap | Specifies whether or not the browser can break lines with long words, if they overflow the container |
| overflow-x | Specifies whether or not to clip the left/right edges of the content, if it overflows the element's content area |
| overflow-y | Specifies whether or not to clip the top/bottom edges of the content, if it overflows the element's content area |
| overscroll-behavior | Specifies whether to have scroll chaining or overscroll affordance in x- and y-directions |
| overscroll-behavior-block | Specifies whether to have scroll chaining or overscroll affordance in the block direction |
| overscroll-behavior-inline | Specifies whether to have scroll chaining or overscroll affordance in the inline direction |
| overscroll-behavior-x | Specifies whether to have scroll chaining or overscroll affordance in x-direction |
| overscroll-behavior-y | Specifies whether to have scroll chaining or overscroll affordance in y-directions |

# P

| | |
|---|---|
| padding | A shorthand property for all the *padding-** properties |
| padding-block | Specifies the padding in the block direction |
| padding-block-end | Specifies the padding at the end in the block direction |
| padding-block-start | Specifies the padding at the start in the block direction |
| padding-bottom | Sets the bottom padding of an element |
| padding-inline | Specifies the padding in the inline direction |

| | |
|---|---|
| padding-inline-end | Specifies the padding at the end in the inline direction |
| padding-inline-start | Specifies the padding at the start in the inline direction |
| padding-left | Sets the left padding of an element |
| padding-right | Sets the right padding of an element |
| padding-top | Sets the top padding of an element |
| @page | Customizes the dimension, orientation, and margins of printed pages |
| page-break-after | Sets the page-break behavior after an element. Replaced by `break-after` property |
| page-break-before | Sets the page-break behavior before an element. Replaced by `break-before` property |
| page-break-inside | Sets the page-break behavior inside an element. Replaced by `break-inside` property |
| paint-order | Sets the order of how an SVG element or text is painted. |
| perspective | Gives a 3D-positioned element some perspective |
| perspective-origin | Defines at which position the user is looking at the 3D-positioned element |
| place-content | Specifies align-content and justify-content property values for flexbox and grid layouts |
| place-items | Specifies align-items and justify-items property values for grid layouts |
| place-self | Specifies align-self and justify-self property values for grid layouts |
| pointer-events | Defines whether or not an element reacts to pointer events |
| position | Specifies the type of positioning method used for an element (static, relative, absolute or fixed) |
| @property | Defines custom CSS properties directly in the stylesheet without having to run any JavaScript |

# Q

| | |
|---|---|
| quotes | Sets the type of quotation marks for embedded quotations |

# R

| | |
|---|---|
| resize | Defines if (and how) an element is resizable by the user |
| right | Specifies the right position of a positioned element |
| rotate | Specifies the rotation of an element |
| row-gap | Specifies the gap between the grid rows |

# S

| | |
|---|---|
| scale | Specifies the size of an element by scaling up or down |
| @scope | Allows you to select elements in specific DOM subtrees and target elements precisely without writing overly-specific selectors |
| scroll-behavior | Specifies whether to smoothly animate the scroll position in a scrollable box, instead of a straight jump |
| scroll-margin | Specifies the margin between the snap position and the container |
| scroll-margin-block | Specifies the margin between the snap position and the container in the block direction |
| scroll-margin-block-end | Specifies the end margin between the snap position and the container in the block direction |
| scroll-margin-block-start | Specifies the start margin between the snap position and the container in the block direction |
| scroll-margin-bottom | Specifies the margin between the snap position on the bottom side and the container |
| scroll-margin-inline | Specifies the margin between the snap position and the container in the inline direction |
| scroll-margin-inline-end | Specifies the end margin between the snap position and the container in the inline direction |
| scroll-margin-inline-start | Specifies the start margin between the snap position and the container in the inline direction |
| scroll-margin-left | Specifies the margin between the snap position on the left side and the container |
| scroll-margin-right | Specifies the margin between the snap position on the right side and the container |
| scroll-margin-top | Specifies the margin between the snap position on the top side and the container |
| scroll-padding | Specifies the distance from the container to the snap position on the child elements |
| scroll-padding-block | Specifies the distance in block direction from the container to the snap position on the child elements |
| scroll-padding-block-end | Specifies the distance in block direction from the end of the container to the snap position on the child elements |
| scroll-padding-block-start | Specifies the distance in block direction from the start of the container to the snap position on the child elements |
| scroll-padding-bottom | Specifies the distance from the bottom of the container to the snap position on the child elements |
| scroll-padding-inline | Specifies the distance in inline direction from the container to the snap position on the child elements |
| scroll-padding-inline-end | Specifies the distance in inline direction from the end of the container to the snap position on the child elements |

| | |
|---|---|
| scroll-padding-inline-start | Specifies the distance in inline direction from the start of the container to the snap position on the child elements |
| scroll-padding-left | Specifies the distance from the left side of the container to the snap position on the child elements |
| scroll-padding-right | Specifies the distance from the right side of the container to the snap position on the child elements |
| scroll-padding-top | Specifies the distance from the top of the container to the snap position on the child elements |
| scroll-snap-align | Specifies where to position elements when the user stops scrolling |
| scroll-snap-stop | Specifies scroll behaviour after fast swipe on trackpad or touch screen |
| scroll-snap-type | Specifies how snap behaviour should be when scrolling |
| scrollbar-color | Specifies the color of the scrollbar of an element |
| shape-outside | Defines a shape for wrapping for the inline content |
| @starting-style | Defines an element's starting styles before the element gets its first style update |
| @supports | Used to test whether a browser supports a CSS feature |

# T

| | |
|---|---|
| tab-size | Specifies the width of a tab character |
| table-layout | Defines the algorithm used to lay out table cells, rows, and columns |
| text-align | Specifies the horizontal alignment of text |
| text-align-last | Describes how the last line of a block or a line right before a forced line break is aligned when text-align is "justify" |
| text-combine-upright | Specifies the combination of multiple characters into the space of a single character |
| text-decoration | Specifies the decoration added to text |
| text-decoration-color | Specifies the color of the text-decoration |
| text-decoration-line | Specifies the type of line in a text-decoration |
| text-decoration-style | Specifies the style of the line in a text decoration |
| text-decoration-thickness | Specifies the thickness of the decoration line |
| text-emphasis | A shorthand property for the *text-emphasis-style* and *text-emphasis-color* properties |
| text-emphasis-color | Specifies the color of emphasis marks |
| text-emphasis-position | Specifies the position of emphasis marks |

| | |
|---|---|
| text-emphasis-style | Specifies the style of emphasis marks |
| text-indent | Specifies the indentation of the first line in a text-block |
| text-justify | Specifies the justification method used when text-align is "justify" |
| text-orientation | Defines the orientation of characters in a line |
| text-overflow | Specifies what should happen when text overflows the containing element |
| text-shadow | Adds shadow to text |
| text-transform | Controls the capitalization of text |
| text-underline-offset | Specifies the offset distance of the underline text decoration |
| text-underline-position | Specifies the position of the underline text decoration |
| top | Specifies the top position of a positioned element |
| transform | Applies a 2D or 3D transformation to an element |
| transform-origin | Allows you to change the position on transformed elements |
| transform-style | Specifies how nested elements are rendered in 3D space |
| transition | A shorthand property for all the *transition-** properties |
| transition-delay | Specifies when the transition effect will start |
| transition-duration | Specifies how many seconds or milliseconds a transition effect takes to complete |
| transition-property | Specifies the name of the CSS property the transition effect is for |
| transition-timing-function | Specifies the speed curve of the transition effect |
| translate | Specifies the position of an element |

# U

| | |
|---|---|
| unicode-bidi | Used together with the direction property to set or return whether the text should be overridden to support multiple languages in the same document |
| user-select | Specifies whether the text of an element can be selected |

# V

| | |
|---|---|
| vertical-align | Sets the vertical alignment of an element |
| visibility | Specifies whether or not an element is visible |

# W

| | |
|---|---|
| white-space | Specifies how white-space inside an element is handled |
| widows | Sets the minimum number of lines that must be left at the top of a page or column |
| width | Sets the width of an element |
| word-break | Specifies how words should break when reaching the end of a line |
| word-spacing | Increases or decreases the space between words in a text |
| word-wrap | Allows long, unbreakable words to be broken and wrap to the next line |
| writing-mode | Specifies whether lines of text are laid out horizontally or vertically |

## Z

| | |
|---|---|
| z-index | Sets the stack order of a positioned element |
| zoom | Specifies the zoom factor for an element. An element can be zoomed in and out |

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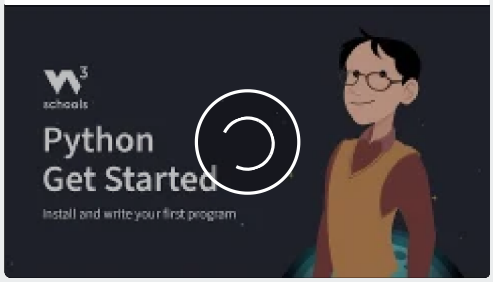
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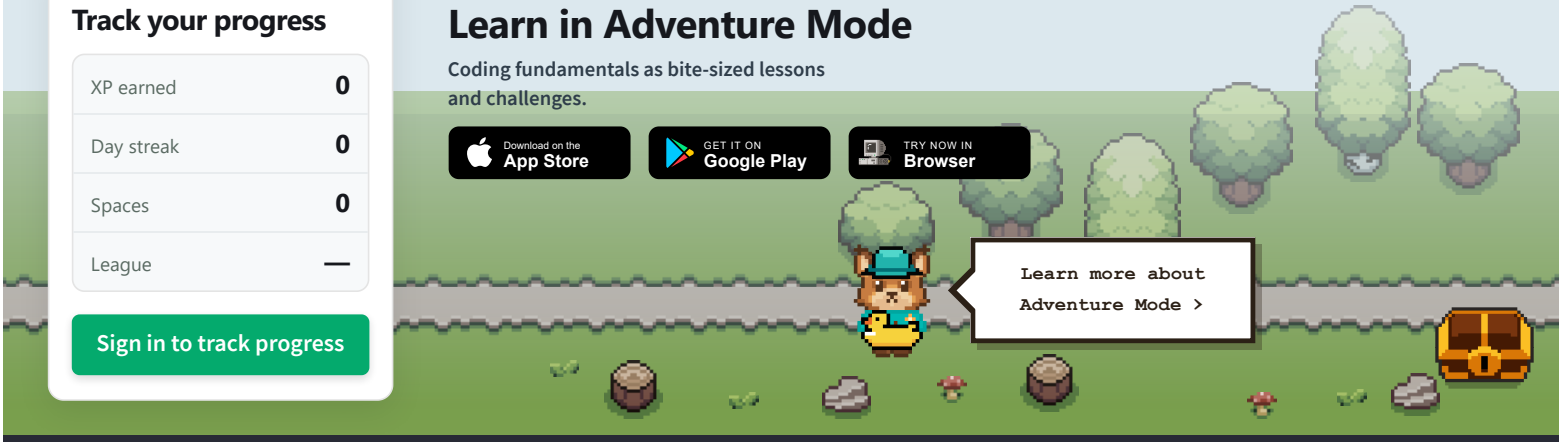
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# Complete JavaScript References

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## JavaScript Primitives

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Primitives | String | Number | Boolean | BigInt | Symbol | undefined | null

## Object Methods and Properties

Array | Boolean | Classes | Date | Error | Function | Global | Iterator | JSON | Map | Math | Number | Object | Operators | Precedence | Promise | Proxy | Reflect | RegExp | Set | Statements | String | Temporal | Typed Array

## Window Objects with Examples

Window | History | Location | Navigator | Screen

## HTML DOM Objects with Examples

Document | Element | Attributes | Events | Event Objects | Collection | Style

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## Web APIs with Examples

Canvas | Console | Fetch | Fullscreen | Geolocation | History | MediaQuery | Storage | Validation

## HTML Elements with Examples

a | abbr | address | area | article | aside | audio | b | base | bdo | blockquote | body | br | button | canvas | caption | cite | code | col | colgroup | datalist | dd | del | details | dfn | dialog | div | dl | dt | em | embed | fieldset | figcaption | figure | footer | form | head | header | h1 - h6 | hr | html | i | iframe | img | ins | input button | input checkbox | input color | input date | input datetime | input datetime-local | input email | input file | input hidden | input image | input month | input number | input password | input radio | input range | input reset | input search | input submit | input text | input time | input url | input week | kbd | label | legend | li | link | map | mark | menu | menuitem | meta | meter | nav | object | ol | optgroup | option | output | p | param | pre | progress | q | s | samp | script | section | select | small | source | span | strong | style | sub | summary | sup | table | tbody | td | tfoot | th | thead | tr | textarea | time | title | track | u | ul | var | video

## Other

CSSStyleDeclaration | JS Conversion

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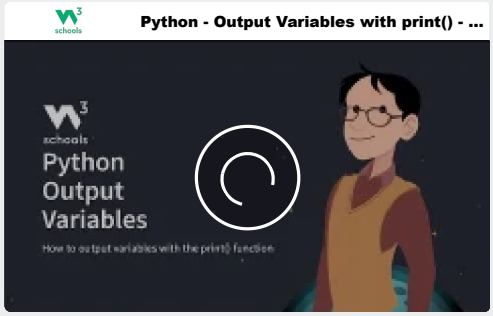
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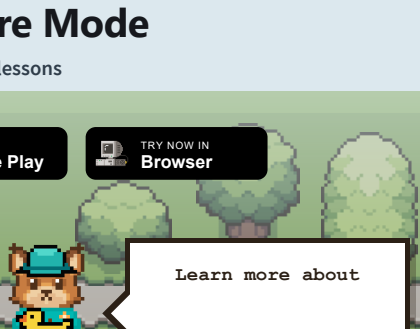


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# SQL Keywords Reference

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## SQL Keywords

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The following table contains all the reserved words in SQL.

| Keyword | Description |
|---|---|
| ADD | Adds a column in an existing table |
| ADD CONSTRAINT | Adds a constraint after a table is already created |
| ALL | Returns true if all of the subquery values meet the condition |
| ALTER | Adds, deletes, or modifies columns in a table, or changes the data type of a column in a table |
| ALTER COLUMN | Changes the data type of a column in a table |
| ALTER TABLE | Adds, deletes, or modifies columns in a table |
| ALTER VIEW | Updates a view |
| AND | Only includes rows where both conditions is true |
| ANY | Returns true if any of the subquery values meet the condition |
| AS | Renames a column or table with an alias |
| ASC | Sorts the result set in ascending order |
| BACKUP DATABASE | Creates a back up of an existing database |

| | |
|---|---|
| BETWEEN | Selects values within a given range |
| CASE | Creates different outputs based on conditions |
| CHECK | A constraint that limits the value that can be placed in a column |
| COLUMN | Changes the data type of a column or deletes a column in a table |
| CONSTRAINT | Adds or deletes a constraint |
| CREATE | Creates a database, index, view, table, or procedure |
| CREATE DATABASE | Creates a new SQL database |
| CREATE INDEX | Creates an index on a table (allows duplicate values) |
| CREATE OR REPLACE VIEW | Updates a view |
| CREATE TABLE | Creates a new table in the database |
| CREATE PROCEDURE | Creates a stored procedure |
| CREATE UNIQUE INDEX | Creates a unique index on a table (no duplicate values) |
| CREATE VIEW | Creates a view based on the result set of a SELECT statement |
| DATABASE | Creates or deletes an SQL database |
| DEFAULT | A constraint that provides a default value for a column |
| DELETE | Deletes rows from a table |
| DESC | Sorts the result set in descending order |
| DISTINCT | Selects only distinct (different) values |
| DROP | Deletes a column, constraint, database, index, table, or view |
| DROP COLUMN | Deletes a column in a table |
| DROP CONSTRAINT | Deletes a UNIQUE, PRIMARY KEY, FOREIGN KEY, or CHECK constraint |
| DROP DATABASE | Deletes an existing SQL database |
| DROP DEFAULT | Deletes a DEFAULT constraint |
| DROP INDEX | Deletes an index in a table |
| DROP TABLE | Deletes an existing table in the database |
| DROP VIEW | Deletes a view |
| EXEC | Executes a stored procedure |
| EXISTS | Tests for the existence of any record in a subquery |
| FOREIGN KEY | A constraint that is a key used to link two tables together |

| | |
|---|---|
| FROM | Specifies which table to select or delete data from |
| FULL OUTER JOIN | Returns all rows when there is a match in either left table or right table |
| GROUP BY | Groups the result set (used with aggregate functions: COUNT, MAX, MIN, SUM, AVG) |
| HAVING | Used instead of WHERE with aggregate functions |
| IN | Check if a specified column's value matches any value within a provided list |
| INDEX | Creates or deletes an index in a table |
| INNER JOIN | Returns rows that have matching values in both tables |
| INSERT INTO | Inserts new rows in a table |
| INSERT INTO SELECT | Copies data from one table into another table |
| IS NULL | Tests for empty values |
| IS NOT NULL | Tests for non-empty values |
| JOIN | Joins tables |
| LEFT JOIN | Returns all rows from the left table, and the matching rows from the right table |
| LIKE | Searches for a specified pattern in a column |
| LIMIT | Specifies the number of records to return in the result set |
| NOT | Only includes rows where a condition is not true |
| NOT NULL | A constraint that enforces a column to not accept NULL values |
| OR | Includes rows where either condition is true |
| ORDER BY | Sorts the result set in ascending or descending order |
| OUTER JOIN | Returns all rows when there is a match in either left table or right table |
| PRIMARY KEY | A constraint that uniquely identifies each record in a database table |
| PROCEDURE | A stored procedure |
| RIGHT JOIN | Returns all rows from the right table, and the matching rows from the left table |
| ROWNUM | Specifies the number of records to return in the result set |
| SELECT | Selects data from a database |
| SELECT DISTINCT | Selects only distinct (different) values |
| SELECT INTO | Copies data from one table into a new table |
| SELECT TOP | Specifies the number of records to return in the result set |
| SET | Specifies which columns and values that should be updated in a table |

| | |
|---|---|
| TABLE | Creates a table, or adds, deletes, or modifies columns in a table, or deletes a table or data inside a table |
| TOP | Specifies the number of records to return in the result set |
| TRUNCATE TABLE | Deletes the data inside a table, but not the table itself |
| UNION | Combines the result set of two or more SELECT statements (only distinct values) |
| UNION ALL | Combines the result set of two or more SELECT statements (allows duplicate values) |
| UNIQUE | A constraint that ensures that all values in a column are unique |
| UPDATE | Updates existing rows in a table |
| VALUES | Specifies the values of an INSERT INTO statement |
| VIEW | Creates, updates, or deletes a view |
| WHERE | Filters a result set to include only records that fulfill a specified condition |

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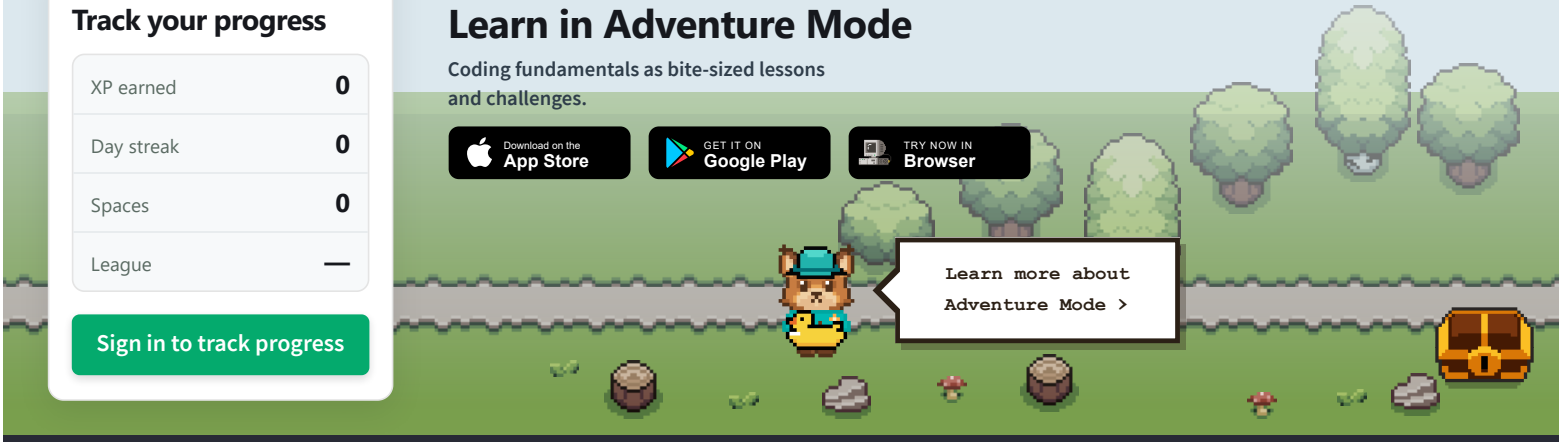
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This section contains a Python reference documentation.

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## Module Reference

Random Module

Requests Module

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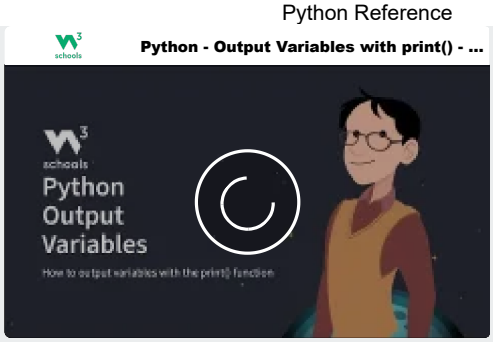
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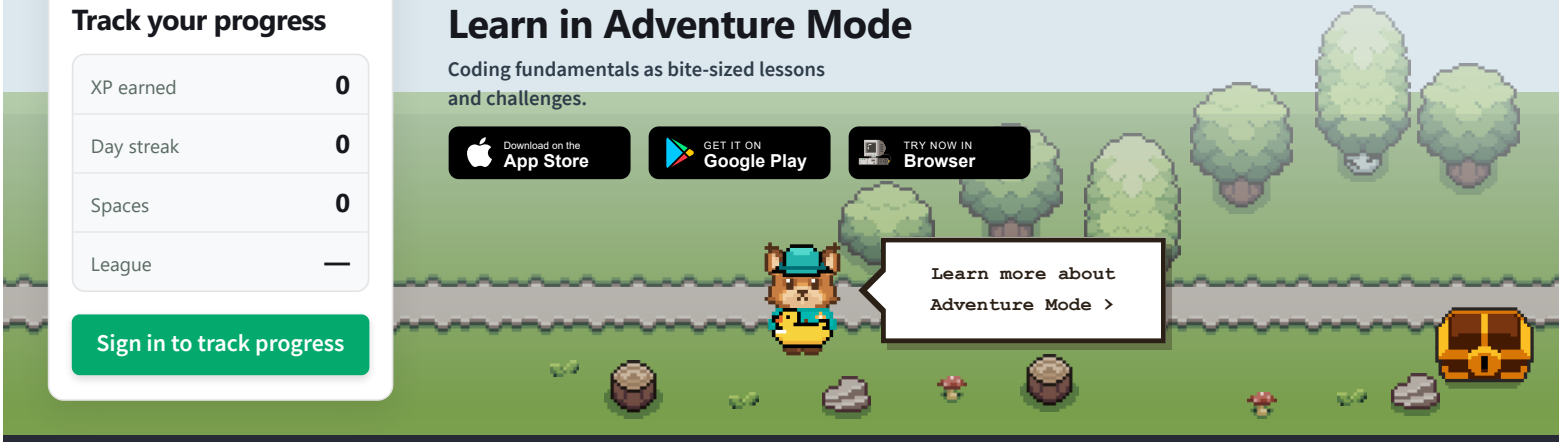
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## W3.CSS Classes

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This reference lists all classes and variables available in W3.CSS.

### Basic Styling

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## Container Classes

With w3-container

## Without w3-container

| Class | Description | |
|---|---|---|
| w3-container | HTML container with 16px left and right padding | Try it |
| | w3-container used to display a header | Try it |
| | w3-container used to display a footer | Try it |
| w3-panel | HTML container with 16px left and right padding and 16px top and bottom margin | Try it |
| | w3-panel used to display a note | Try it |
| | w3-panel used to display a quote | Try it |
| w3-block | Class that can be used to define a full width for any element | Try it |
| w3-code | Code container | Try it |
| w3-codespan | Inline code container (for code snippets) | Try it |
| w3-content | Centers a page container horizontally with a default maximum width of 980px. It does not center text or inline images. | Try it |
| w3-auto | Centers a page container with a maximum width of 1140px above 1205px, 95% above 992px and up to 1205px, and 100% at 992px or less. | Try it |
| w3-stretch | Applies -16px inline margins to extend an element beyond its parent's content edges. | Try it |

# Text Size Classes

Tiny Small Medium Large xLarge xxLarge

| Class | Description | |
|---|---|---|
| w3-tiny | Specifies a font size of 10 pixels | Try it |
| w3-small | Specifies a font size of 12 pixels | Try it |
| w3-medium | Specifies a font size of 15 pixels | Try it |
| w3-large | Specifies a font size of 18 pixels | Try it |
| w3-xlarge | Specifies a font size of 24 pixels | Try it |
| w3-xxlarge | Specifies a font size of 36 pixels | Try it |

| | | |
|---|---|---|
| w3-xxxlarge | Specifies a font size of 48 pixels | Try it |
| w3-jumbo | Specifies a font size of 64 pixels | Try it |
| w3-wide | Adds 4px letter spacing. | Try it |

## Text Style Classes

The w3-serif class changes text to serif.

*The w3-italic class changes text to italic.*

**The w3-bold class changes text to bold.**

| Class | Defines | |
|---|---|---|
| w3-bold | Changes the font to **bold** | Try it |
| w3-text-bold | Changes the font to **bold** (same as w3-bold) | Try it |
| w3-italic | Changes the font to *italic* | Try it |
| w3-text-italic | Changes the font to *italic* (same as w3-italic) | Try it |
| w3-serif | Changes the font to serif | Try it |
| w3-sans-serif | Changes the font to sans-serif | Try it |
| w3-cursive | Changes the font to cursive | Try it |
| w3-monospace | Changes the font to monospace | Try it |

## Text Position and Direction Classes

w3-center

My text.

My label

| | | |
|---|---|---|
| w3-center | Centers text and inline elements inside an element. | Try it |
| w3-text-center | Centers text and inline elements (without !important). | Try it |

| | | |
|---|---|---|
| w3-left-align | Left aligned text | Try it |
| w3-right-align | Right aligned text | Try it |
| w3-justify | Right and left aligned text | Try it |
| w3-rtl | Adjusts text direction for right to left (rtl) layout | Try it |
| w3-ltr | Adjusts text direction for left to right (ltr) layout | Try it |

# Padding Classes

## w3-padding

The w3-padding class adds 8px top and bottom and 16px left and right.

| Class | Defines | |
|---|---|---|
| w3-padding-small | Padding 4px top and bottom, and 8px left and right. | Try it |
| w3-padding | Padding 8px top and bottom, and 16px left and right. | Try it |
| w3-padding-large | Padding 12px top and bottom, and 24px left and right. | Try it |
| w3-padding-16 | Padding 16px top and bottom | Try it |
| w3-padding-24 | Padding 24px top and bottom | Try it |
| w3-padding-32 | Padding 32px top and bottom | Try it |
| w3-padding-48 | Padding 48px top and bottom | Try it |
| w3-padding-64 | Padding 64px top and bottom | Try it |
| w3-padding-top-64 | Padding 64px on top | Try it |
| w3-padding-top-48 | Padding 48px on top | Try it |
| w3-padding-top-32 | Padding 32px on top | Try it |
| w3-padding-top-24 | Padding 24px on top | Try it |

# Margin Classes

The w3-margin-left class adds a 16px left margin to an element.

The w3-margin-right class adds a 16px right margin to an element.

| Class | Description | |
|---|---|---|
| w3-margin | Adds a 16px margin to an element | Try it |
| w3-margin-top | Adds a 16px top margin to an element | Try it |
| w3-margin-right | Adds a 16px right margin to an element | Try it |
| w3-margin-bottom | Adds a 16px bottom margin to an element | Try it |
| w3-margin-left | Adds a 16px left margin to an element | Try it |
| w3-section | Adds a 16px top and bottom margin to an element | Try it |

# Border Classes

## Leftbar

The w3-leftbar class adds a thick left border (bar) to an element.

| Class | Defines | |
|---|---|---|
| w3-border | Borders (top, right, bottom, left) | Try it |
| w3-border-top | Border top | Try it |
| w3-border-right | Border right | Try it |
| w3-border-bottom | Border bottom | Try it |
| w3-border-left | Border left | Try it |
| w3-border-0 | No borders | Try it |
| w3-border-*color* | Border color | Try it |
| w3-bottombar | Thick bottom border (bar) to an element | Try it |
| w3-leftbar | A thick left border (bar) to an element | Try it |
| w3-rightbar | A thick right border (bar) to an element | Try it |
| w3-topbar | A thick top border (bar) to an element | Try it |

| | | |
|---|---|---|
| w3-hover-border-*color* | A hoverable border color | Try it |

## Card Classes

w3-card-2

w3-card-4

| Class | Defines | |
|---|---|---|
| w3-card | A light box shadow. | Try it |
| w3-card-2 | Same as w3-card | Try it |
| w3-card-4 | A stronger box shadow. | Try it |

## Flex Classes

Item 1 Item 2 Item 3

| Class | Defines | |
|---|---|---|
| w3-flex | A flex container for flexible one-dimensional layouts. | Try it |
| w3-flex-column | A flex container with items arranged in a column. | Try it |
| w3-flex-mobile | Arranges flex items in a column on screens 600px wide or smaller when used with w3-flex or any other flex container class. | Try it |
| w3-flex-wrap | A flex container that allows items to wrap to a new line. | Try it |

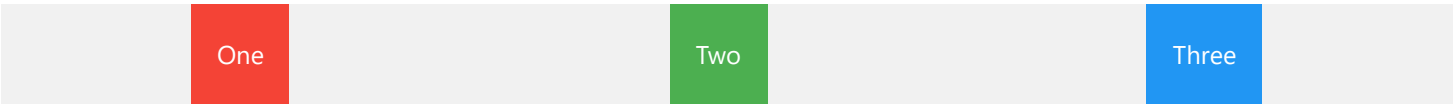
## Flex Item Classes

w3-flex-1 w3-flex-1 w3-flex-auto

| Class | Defines | |
|---|---|---|
| w3-flex-1 | A flex item that grows and shrinks from a zero basis. | Try it |
| w3-flex-auto | A flex item that grows and shrinks from an automatic basis. | Try it |

| | | |
|---|---|---|
| w3-flex-none | A flex item that neither grows nor shrinks. | Try it |
| w3-flex-shrink-0 | A flex item that prevents shrinking. | Try it |
| w3-flex-spacer | An automatic margin at the inline start. In a row flex layout, it pushes this item and following items toward the inline end. | Try it |

# Main Axis Flex Alignment




The main flex axis is horizontal by default, and the cross axis is vertical.

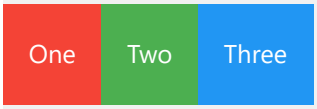
With `w3-flex-column`, the main axis is vertical and the cross axis is horizontal.

These helpers align flex items along the **main axis**:

| Class | Defines | |
|---|---|---|
| w3-flex-start | A flex container that aligns items at **the start**. | Try it |
| w3-flex-center | A flex container that **centers** its items. | Try it |
| w3-flex-end | A flex container that aligns items at **the end**. | Try it |
| w3-flex-around | A flex container with equal space **around** items. | Try it |
| w3-flex-between | A flex container with equal space **between** items. | Try it |
| w3-flex-evenly | A flex container with equal spaces **between** items and **both ends**. | Try it |

# Cross Axis Flex Alignment

These helpers align flex items along the **cross axis**:




| Class | Defines |
|---|---|

| | | |
|---|---|---|
| w3-flex-align-start | A flex container that aligns items at the start of the axis. | Try it |
| w3-flex-align-center | A flex container that centers items. | Try it |
| w3-flex-align-end | A flex container that aligns items at the end of the axis. | Try it |
| w3-flex-align-stretch | A flex container that stretches automatically sized items to fill the cross axis of their flex line. | Try it |

REMOVE ADS

## Basic Grid Classes

1 2 3 4

| Class | Defines | |
|---|---|---|
| w3-grid | A grid container for flexible two-dimensional layouts. | Try it |
| w3-grid-padding | A grid container with a 16px gap between rows and columns. | Try it |
| w3-grid-2 | 2 equal-width columns for all screen sizes. | Try it |
| w3-grid-3 | 3 equal-width columns for all screen sizes. | Try it |
| w3-grid-4 | 4 equal-width columns for all screen sizes. | Try it |

## Responsive Grid Classes








| Class | Defines | |
|---|---|---|
| w3-grid-auto-xsmall | An auto-fitting grid with a 150px minimum column width. | Try it |
| w3-grid-auto-small | An auto-fitting grid with a 200px minimum column width. | Try it |
| w3-grid-auto | An auto-fitting grid with a 250px minimum column width. | Try it |

| | | |
|---|---|---|
| w3-grid-auto-large | An auto-fitting grid with a 300px minimum column width. | Try it |
| w3-grid-auto-xlarge | An auto-fitting grid with a 350px minimum column width. | Try it |
| w3-grid-mobile | Changes a grid to one column on screens 600px wide or smaller when used with any grid container class. | Try it |

## Grid Item Classes

Span 3

4

| Class | Defines | |
|---|---|---|
| w3-grid-span-2 | A grid item that spans 2 columns. | Try it |
| w3-grid-span-3 | A grid item that spans 3 columns. | Try it |
| w3-grid-span-all | A grid item that spans all columns. | Try it |

## Grid and Flexbox Gap Classes

1

2

3

Gap helpers set spacing between rows and columns in Grid and between items and lines in Flexbox. They do not create a layout container.

| Class | Defines | |
|---|---|---|
| w3-gap | A 16px gap between flex or grid items. | Try it |
| w3-gap-small | An 8px gap between flex or grid items. | Try it |
| w3-gap-large | A 24px gap between flex or grid items. | Try it |

## Layout Classes

| Class | Defines | |
|---|---|---|
| w3-cell-row | Container for layout columns (cells). | Try it |
| w3-cell | Layout column (cell). | Try it |
| w3-cell-top | Aligns content at the top of a column (cell). | Try it |

| | | |
|---|---|---|
| w3-cell-middle | Aligns content at the vertical middle of a column (cell). | Try it |
| w3-cell-bottom | Aligns content at the bottom of a column (cell). | Try it |

# Responsive Row Classes

| Class | Defines | |
|---|---|---|
| w3-row | Container for one row of fluid responsive content | Try it |
| w3-row-padding | Row where all columns have a default padding | Try it |
| w3-half | Column that uses half of the row and full width on small screens | Try it |
| w3-third | Column that uses 1/3 of the row and full width on small screens | Try it |
| w3-twothird | Column that uses 2/3 of the row and full width on small screens | Try it |
| w3-quarter | Responsive column that uses 1/4 of the row and full width on small screens | Try it |
| w3-threequarter | Responsive column that uses 3/4 of the row width | Try it |
| w3-col | Responsive column container for HTML content | Try it |
| w3-rest | Occupies the rest of the column width | Try it |
| | | |
| l1 - l12 | Responsive sizes for large screens | Try it |
| m1 - m12 | Responsive sizes for medium screens | Try it |
| s1 - s12 | Responsive sizes for small screens | Try it |
| | | |
| w3-hide-small | Hide content on small screens (less than 601px) | Try it |
| w3-hide-medium | Hide content on medium screens | Try it |
| w3-hide-large | Hide content on large screens (larger than 992px) | Try it |
| | | |
| w3-image | Responsive image | Try it |
| | | |
| w3-mobile | Displays an element as a full-width block on screens 600px wide or smaller | Try it |

# Table Classes

| First Name | Last Name | Points |
|---|---|---|
| Jill | Smith | 50 |
| Eve | Jackson | 94 |
| Adam | Johnson | 67 |

| Class | Defines | |
|---|---|---|
| w3-table | Container for an HTML table | Try it |
| w3-striped | Striped table | Try it |
| w3-border | Bordered table | Try it |
| w3-bordered | Bordered lines | Try it |
| w3-centered | Centers text in table header and data cells. | Try it |
| w3-table-all | Full-width table with borders and alternating row backgrounds. | Try it |
| w3-hoverable | Hoverable table | Try it |
| w3-responsive | Adds horizontal scrolling when content, such as a table, is wider than its container. | Try it |

# Lists and Tags

Jill 4 messages

Eve

Adam New

| Class | Defines | |
|---|---|---|
| w3-badge | Circular badge | Try it |
| w3-tag | Rectangular tag | Try it |
| w3-ul | Styles a list without bullets, with separator lines and padded items | Try it |

# Bar Classes - Navigation

Home Link 1 Link 2 Link 3

| Class | Defines | |
|---|---|---|
| w3-bar | Horizontal bar | Try it |
| w3-bar-block | Vertical bar | Try it |
| w3-bar-item | Provides common style for bar items | Try it |
| w3-sidebar | Side bar | Try it |
| | A side bar can contain all types of content | Try it |
| | A side bar overlaying main content | Try it |
| | A side bar overlaying all main content | Try it |
| | A side bar shifting main content to the right | Try it |
| | A side bar with an overlay background | Try it |
| | A Side bar on the right side | Try it |
| w3-collapse | Used together with w3-sidebar to create a fully automatic responsive side navigation. For this class to work, the page content must be within a "w3-main" class | Try it |
| w3-main | Container for page content when using the w3-collapse class for responsive side navigations | Try it |
| | Fully automatic right-sided responsive side navigation | Try it |

## Dropdown Classes

| | | |
|---|---|---|
| w3-dropdown-click | Clickable dropdown element | Try it |
| w3-dropdown-hover | Shows dropdown content on hover or when keyboard focus is inside the dropdown. | Try it |
| | Hoverable dropdown element (used in w3-bar) | Try it |
| | Hoverable dropdown element (used in w3-bar-block) | Try it |
| | Hoverable dropdown element (used in w3-sidebar) | Try it |
| `w3-dropdown-content` | Hidden, absolutely positioned dropdown content. Hover dropdowns reveal it automatically; click dropdowns require JavaScript. | Learn |

## Button Classes

Button Button Button

| Class | Defines | |
|---|---|---|
| w3-button | Rectangular button with grey background color on hover | Try it |
| w3-btn | Rectangular button with shadows on hover | Try it |
| w3-circle | Can be used to create a circular button | Try it |
| w3-ripple | Rectangular button with reduced opacity when pressed | Try it |
| | Circular floating button with ripple effect | Try it |
| w3-bar | Can be used to group elements (like buttons) in an horizontal bar | Try it |
| w3-block | Class that can be used to define a full width w3-button | Try it |
| | Full width w3-btn | Try it |
| | Full width circular button | Try it |

# Input Classes

## Radio Buttons

Male

Female

Don't know (Disabled)

| Class | Defines | |
|---|---|---|
| w3-input | Input elements | Try it |
| | Input form as a card | Try it |
| | Input elements (top labels) | Try it |
| | Input elements (bottom labels) | Try it |
| w3-check | Checkbox input type | Try it |
| w3-radio | Radio input type | Try it |
| w3-select | Input select element | Try it |
| w3-disabled | Gives an element a disabled appearance and a not-allowed cursor | Try it |

| | | |
|---|---|---|
| w3-animate-input | Animates the width of an input to 100% | Try it |

# Modal Classes

| Class | Defines | |
|---|---|---|
| w3-modal | Modal container | Try it |
| w3-modal-content | Modal pop-up element | Try it |
| w3-tooltip | Container for tooltip text shown on hover or keyboard focus. | Try it |
| w3-text | Tooltip text shown when its tooltip is hovered or contains keyboard focus. | Try it |

# Animation Classes

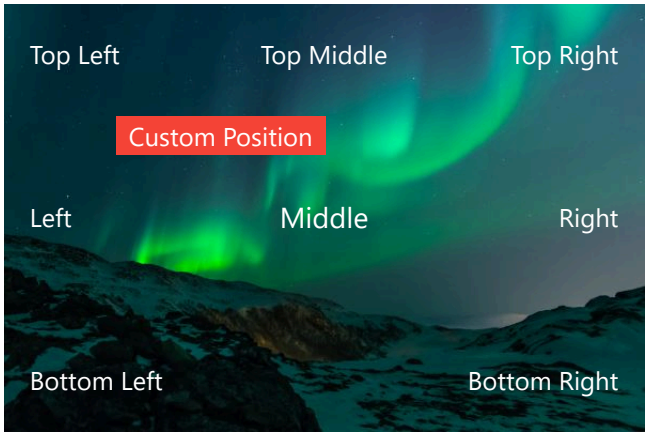
Animation fade in AND out

The w3-animate-fading class fades in and out an element repeatedly.

| Class | Defines | |
|---|---|---|
| w3-animate-top | Animates an element from the top -300px to 0px | Try it |
| w3-animate-left | Animates an element from left -300px to 0px | Try it |
| w3-animate-bottom | Animates an element from the bottom -300px to 0px | Try it |
| w3-animate-right | Animates an element from right -300px to 0px | Try it |
| w3-animate-opacity | Animates an element's opacity from 0 to 1 | Try it |
| w3-animate-zoom | Animates an element from 0 to 100% in size | Try it |
| w3-animate-fading | Animates an element's opacity from 0 to 1 and 1 to 0 (fades in AND out) | Try it |
| w3-spin | Spin an icon 360 degrees | Try it |
| | Spin any element 360 degrees | Try it |
| w3-animate-input | Animates the width of an input field to 100% | Try it |

When the user requests reduced motion, W3.CSS disables the listed animation effects and the input-width transition.

# Display Classes




| Class | Defines | |
|---|---|---|
| w3-center | Centers text and inline elements inside an element. | Try it |
| w3-left | Floats an element to the left (float: left) | Try it |
| w3-right | Floats an element to the right (float: right) | Try it |
| w3-block | Class that can be used to define a full width for any element | Try it |
| w3-circle | Circled content | Try it |
| w3-hide | Hidden content (display:none) | Try it |
| w3-show | Show content (display:block) | Try it |
| w3-show-block | Alias of w3-show (display:block) | Try it |
| w3-show-inline-block | Show content as inline-block (display:inline-block) | Try it |
| w3-top | Fixes an element at the top of the viewport with width:100%. Its horizontal position and maximum width can be customized. | Try it |
| w3-bottom | Fixes an element at the bottom of the viewport with width:100%. Its horizontal position and maximum width can be customized. | Try it |
| w3-display-container | Container for w3-display-*classes* (position: relative) | Try it |
| w3-display-topleft | Displays content at the top left corner of the w3-display-container | Try it |
| w3-display-topright | Displays content at the top right corner of the w3-display-container | Try it |
| w3-display-bottomleft | Displays content at the bottom left corner of the w3-display-container | Try it |
| w3-display-bottomright | Displays content at the bottom right corner of the w3-display-container | Try it |

| | | |
|---|---|---|
| w3-display-left | Displays content to the left (middle left) of the w3-display-container | Try it |
| w3-display-right | Displays content to the right (middle right) of the w3-display-container | Try it |
| w3-display-middle | Displays content in the middle (center) of the w3-display-container | Try it |
| w3-display-topmiddle | Displays content at the top middle of the w3-display-container | Try it |
| w3-display-bottommiddle | Displays content at the bottom middle of the w3-display-container | Try it |
| w3-display-position | Displays content at a specified position in the w3-display-container | Try it |
| w3-display-hover | Shows content when its display container is hovered or contains keyboard focus | Try it |
| w3-clear | Clears floated children using clearfix pseudo-elements | Try it |

# Effect Classes



| Class | Defines | |
|---|---|---|
| w3-opacity | Adds opacity/transparency to an element (opacity: 0.6) | Try it |
| | Add opacity/transparency to text | Try it |
| w3-opacity-off | Turns off opacity/transparency (opacity: 1) | Try it |
| w3-opacity-min | Adds opacity/transparency to an element (opacity: 0.75) | Try it |
| w3-opacity-max | Adds opacity/transparency to an element (opacity: 0.25) | Try it |
| w3-grayscale | Adds a grayscale effect to an element (grayscale: 75%) | Try it |
| w3-greyscale | Same as w3-grayscale | Try it |

| | | |
|---|---|---|
| w3-grayscale-max | Adds a grayscale effect to an element (grayscale: 100%) | Try it |
| w3-greyscale-max | Same as w3-grayscale-max | Try it |
| w3-grayscale-min | Adds a grayscale effect to an element (grayscale: 50%) | Try it |
| w3-greyscale-min | Same-as w3-grayscale-min | Try it |
| w3-sepia-min | Adds a sepia effect to an element (sepia: 50%) | Try it |
| w3-sepia | Adds a sepia effect to an element (sepia: 75%) | Try it |
| w3-sepia-max | Adds a sepia effect to an element (sepia: 100%) | Try it |
| w3-overlay | Creates an overlay effect | Try it |

## Background Color Classes

| | | | |
|---|---|---|---|
| Red<br>w3-red | Pink<br>w3-pink | Purple<br>w3-purple | Deep Purple<br>w3-deep-purple |
| Indigo<br>w3-indigo | Cobalt<br>w3-cobalt | Blue<br>w3-blue | Light Blue<br>w3-light-blue |
| Cyan<br>w3-cyan | Aqua<br>w3-aqua | Teal<br>w3-teal | Emerald<br>w3-emerald |
| Green<br>w3-green | Light Green<br>w3-light-green | Lime<br>w3-lime | Khaki<br>w3-khaki |
| Yellow<br>w3-yellow | Amber<br>w3-amber | Orange<br>w3-orange | Deep Orange<br>w3-deep-orange |
| Blue Gray<br>w3-blue-gray | Blue Grey<br>w3-blue-grey | Olive<br>w3-olive | Taupe<br>w3-taupe |
| Sienna | Brown | Crimson | Sand |

w3-sienna | w3-brown | w3-crimson | w3-sand

| White | Paper | Light Gray | Light Gray |
|---|---|---|---|
| w3-white | w3-paper | w3-light-gray | w3-light-grey |

| Gray | Grey | Dark Gray | Dark Grey |
|---|---|---|---|
| w3-gray | w3-grey | w3-dark-gray | w3-dark-grey |

| Asphalt | Black | Pale Red | Pale Yellow |
|---|---|---|---|
| w3-asphalt | w3-black | w3-pale-red | w3-pale-yellow |

| Pale Green | Pale Blue |
|---|---|
| w3-pale-green | w3-pale-blue |

### Gray and Grey

Where both spellings are listed, they produce the same color.

For example, `w3-gray` and `w3-grey` are identical.

## Semantic Colors

| Danger | Warning | Success | Info |
|---|---|---|---|
| Red often indicates a negative situation. | Orange often indicates a warning. | Green often indicates something positive. | Blue often indicates a neutral information. |
| w3-danger | w3-warning | w3-success | w3-info |

| Note | Transparent |
|---|---|
| Yellow is often used for notifications. | w3-transparent |
| w3-note | |

# Note

The semantic color classes w3-danger, w3-warning, w3-success, w3-info, and w3-note use customizable variable CSS colors.

## Fixed Colors and Dark Mode

Color classes such as `w3-blue`, `w3-text-red`, and `w3-border-green` keep the same colors when dark mode is enabled.

For example, `w3-white` always gives an element a white background, even inside a dark-mode container.

Dark mode changes the default page and component colors using variables:

- `--w3-page-bg` - Default page background.
- `--w3-text-color` - Default text color.
- `--w3-surface` - Component background.
- `--w3-surface-alt` - Alternate background for table stripes and inline code.
- `--w3-codespan-text` - Inline code text color.
- `--w3-border-color` - Default border color.
- `--w3-control-border` - Input and select bottom border color.
- `--w3-border-light` - Table row and list separator color.
- `--w3-badge-bg` - Badge and tag background.
- `--w3-badge-text` - Badge and tag text color.
- `--w3-hover-bg` - Default hover background.
- `--w3-hover-text` - Default hover text color.
- `--w3-danger-bg` - Danger background.
- `--w3-danger-text` - Danger text color.
- `--w3-note-bg` - Note background.
- `--w3-note-text` - Note text color.
- `--w3-info-bg` - Info background.
- `--w3-info-text` - Info text color.
- `--w3-warning-bg` - Warning background.
- `--w3-warning-text` - Warning text color.
- `--w3-success-bg` - Success background.
- `--w3-success-text` - Success text color.
- `--w3-focus-ring` - Keyboard focus outline color.
- `--w3-accent-color` - Checkbox and radio accent color.

## Text Color Classes

| Class | Defines | |
|---|---|---|
| w3-text-red | Text color red | Try it |

| | | |
|---|---|---|
| w3-text-green | Text color green | Try it |
| w3-text-blue | Text color blue | Try it |
| w3-text-yellow | Text color yellow | Try it |
| w3-text-light-grey | Text color light-grey | Try it |
| w3-text-grey | Text color grey | Try it |
| w3-text-dark-grey | Text color dark grey | Try it |
| w3-text-black | Text color black | Try it |
| | Text color white | Try it |
| w3-text-pink | Text color pink | Try it |
| w3-text-purple | Text color purple | Try it |
| w3-text-teal | Text color teal | Try it |
| w3-text-light-green | Text color light green | Try it |
| w3-text-lime | Text color lime | Try it |
| w3-text-deep-purple | Text color deep purple | Try it |
| w3-text-indigo | Text color indigo | Try it |
| w3-text-light-blue | Text color light blue | Try it |
| w3-text-blue-grey | Text color blue grey | Try it |
| w3-text-cyan | Text color cyan | Try it |
| w3-text-aqua | Text color aqua | Try it |
| w3-text-amber | Text color amber | Try it |
| w3-text-orange | Text color orange | Try it |
| w3-text-deep-orange | Text color deep orange | Try it |
| w3-text-sand | Text color sand | Try it |
| w3-text-khaki | Text color khaki | Try it |
| w3-text-brown | Text color brown | Try it |
| w3-text-blue-gray | Same as w3-text-blue-grey | |
| w3-text-dark-gray | Same as w3-text-dark-grey | |

| | |
|---|---|
| w3-text-gray | Same as w3-text-grey |
| w3-text-light-gray | Same as w3-text-light-grey |

# Hover Color Classes

The colors above can also be used as hover background color classes:

Hover

| Class | Defines | |
|---|---|---|
| w3-hover-amber | Applies amber background and its text color on hover | Try it |
| w3-hover-aqua | Applies aqua background and its text color on hover | Try it |
| w3-hover-asphalt | Applies asphalt background and its text color on hover | Try it |
| w3-hover-black | Applies black background and its text color on hover | Try it |
| w3-hover-blue | Applies blue background and its text color on hover | Try it |
| w3-hover-blue-gray | Applies blue-gray background and its text color on hover | Try it |
| w3-hover-blue-grey | Applies blue-grey background and its text color on hover | Try it |
| w3-hover-brown | Applies brown background and its text color on hover | Try it |
| w3-hover-cobalt | Applies cobalt background and its text color on hover | Try it |
| w3-hover-crimson | Applies crimson background and its text color on hover | Try it |
| w3-hover-cyan | Applies cyan background and its text color on hover | Try it |
| w3-hover-dark-gray | Applies dark-gray background and its text color on hover | Try it |
| w3-hover-dark-grey | Applies dark-grey background and its text color on hover | Try it |
| w3-hover-deep-orange | Applies deep-orange background and its text color on hover | Try it |
| w3-hover-deep-purple | Applies deep-purple background and its text color on hover | Try it |
| w3-hover-emerald | Applies emerald background and its text color on hover | Try it |
| w3-hover-gray | Applies gray background and its text color on hover | Try it |
| w3-hover-green | Applies green background and its text color on hover | Try it |
| w3-hover-grey | Applies grey background and its text color on hover | Try it |

| | | |
|---|---|---|
| w3-hover-indigo | Applies indigo background and its text color on hover | Try it |
| w3-hover-khaki | Applies khaki background and its text color on hover | Try it |
| w3-hover-light-blue | Applies light-blue background and its text color on hover | Try it |
| w3-hover-light-gray | Applies light-gray background and its text color on hover | Try it |
| w3-hover-light-green | Applies light-green background and its text color on hover | Try it |
| w3-hover-light-grey | Applies light-grey background and its text color on hover | Try it |
| w3-hover-lime | Applies lime background and its text color on hover | Try it |
| w3-hover-olive | Applies olive background and its text color on hover | Try it |
| w3-hover-orange | Applies orange background and its text color on hover | Try it |
| w3-hover-pale-blue | Applies pale-blue background and its text color on hover | Try it |
| w3-hover-pale-green | Applies pale green background and its text color on hover | Try it |
| w3-hover-pale-red | Applies pale-red background and its text color on hover | Try it |
| w3-hover-pale-yellow | Applies pale-yellow background and its text color on hover | Try it |
| w3-hover-paper | Applies paper background and its text color on hover | Try it |
| w3-hover-pink | Applies pink background and its text color on hover | Try it |
| w3-hover-purple | Applies purple background and its text color on hover | Try it |
| w3-hover-red | Applies red background and its text color on hover | Try it |
| w3-hover-sand | Applies sand background and its text color on hover | Try it |
| w3-hover-sienna | Applies sienna background and its text color on hover | Try it |
| w3-hover-taupe | Applies taupe background and its text color on hover | Try it |
| w3-hover-teal | Applies teal background and its text color on hover | Try it |
| w3-hover-white | Applies white background and its text color on hover | Try it |
| w3-hover-yellow | Applies yellow background and its text color on hover | Try it |

# Hover Text Color Classes

The color above can also be used as hover text color classes:

### Hover

| Class | Defines | |
|---|---|---|
| w3-hover-text-red | Hover text color red | Try it |
| w3-hover-text-green | Hover text color green | Try it |
| w3-hover-text-blue | Hover text color blue | Try it |
| w3-hover-text-yellow | Hover text color yellow | Try it |
| w3-hover-text-amber | Hover text color amber | Try it |
| w3-hover-text-aqua | Hover text color aqua | Try it |
| w3-hover-text-black | Hover text color black | Try it |
| w3-hover-text-blue-gray | Hover text color blue gray | Try it |
| w3-hover-text-blue-grey | Hover text color blue grey | Try it |
| w3-hover-text-brown | Hover text color brown | Try it |
| w3-hover-text-cyan | Hover text color cyan | Try it |
| w3-hover-text-dark-gray | Hover text color dark gray | Try it |
| w3-hover-text-dark-grey | Hover text color dark grey | Try it |
| w3-hover-text-deep-orange | Hover text color deep orange | Try it |
| w3-hover-text-deep-purple | Hover text color deep purple | Try it |
| w3-hover-text-gray | Hover text color gray | Try it |
| w3-hover-text-grey | Hover text color grey | Try it |
| w3-hover-text-indigo | Hover text color indigo | Try it |
| w3-hover-text-khaki | Hover text color khaki | Try it |
| w3-hover-text-light-blue | Hover text color light blue | Try it |
| w3-hover-text-light-gray | Hover text color light gray | Try it |
| w3-hover-text-light-green | Hover text color light green | Try it |
| w3-hover-text-light-grey | Hover text color light grey | Try it |
| w3-hover-text-lime | Hover text color lime | Try it |
| w3-hover-text-orange | Hover text color orange | Try it |
| w3-hover-text-pink | Hover text color pink | Try it |

| | | |
|---|---|---|
| w3-hover-text-purple | Hover text color purple | Try it |
| w3-hover-text-sand | Hover text color sand | Try it |
| w3-hover-text-teal | Hover text color teal | Try it |
| w3-hover-text-white | Hover text color white | Try it |

## Other Hover Classes

| Class | Defines | |
|---|---|---|
| w3-hover-border-*color* | Hover border color | Try it |
| w3-hover-opacity | Adds transparency to an element on hover (opacity: 0.6) | Try it |
| w3-hover-opacity-off | Removes transparency from an element on hover (100% opacity) | Try it |
| w3-hover-shadow | Adds shadow to an element on hover | Try it |
| w3-hover-grayscale | Adds a black and white (100% grayscale) effect to an element | Try it |
| w3-hover-greyscale | Adds a black and white (100% greyscale) effect to an element | Try it |
| w3-hover-sepia | Adds a sepia effect to an element on hover | Try it |
| w3-hover-none | Makes the background transparent and removes the shadow on hover. Does not reset every hover property. | Try it |

## Round Classes



| Class | Defines | |
|---|---|---|
| w3-round | Element rounded (border-radius) 4px | Try it |
| w3-round-small | Element rounded (border-radius) 2px | Try it |
| w3-round-medium | Element rounded (border-radius) 4px | Try it |

| | | |
|---|---|---|
| w3-round-large | Element rounded (border-radius) 8px | Try it |
| w3-round-xlarge | Element rounded (border-radius) 16px | Try it |
| w3-round-xxlarge | Element rounded (border-radius) 32px | Try it |

# Dark Mode Class

| Class | Defines |
|---|---|
| w3-dark-mode | Applies a dark background, light text, and dark-mode colors to a page or section. |

Add `w3-dark-mode` to the `<body>` element for a dark page, or to a container for a dark section.

The class changes selected variables used by components such as tables, forms, and dropdowns.

Explicit color classes such as `w3-blue`, `w3-text-red`, and `w3-border-green` keep their fixed colors.

It also tells the browser to use a dark appearance for built-in controls, such as form controls and scrollbars.

See W3.CSS Dark Mode for examples.

# CSS Variables

Override these variables in your own CSS after loading W3.CSS. You do not need to edit the framework stylesheet.

The table lists the default light-mode values. Dark mode supplies different values for page, surface, border, inline-code, and hover colors.

| Variable | Default Light Value | Controls |
|---|---|---|
| --w3-font-family | Verdana,sans-serif | Default body font family |
| --w3-heading-font-family | "Segoe UI",Arial,sans-serif | Default heading font family |
| --w3-font-size | 15px | Default font size |
| --w3-line-height | 1.5 | Default line height |
| --w3-content-max-width | 980px | Default max-width for w3-content |
| --w3-page-bg | #fff | Default page background |
| --w3-text-color | #000 | Default text color |
| --w3-surface | #fff | Component surface background |
| --w3-surface-alt | #f1f1f1 | Alternate surface background, including table stripes and inline code |
| --w3-codespan-text | #d2143a | Inline code text color |

| | | |
|---|---|---|
| --w3-border-color | #ccc | Default border color |
| --w3-border-light | #ddd | Table row and list separator color |
| --w3-control-border | #808080 | Bottom border of inputs and select elements |
| --w3-hover-bg | #ccc | Default hover background |
| --w3-hover-text | #000 | Default hover text color |
| --w3-badge-bg | #000 | Badge and tag background |
| --w3-badge-text | #fff | Badge and tag text color |
| --w3-focus-ring | #2196F3 | Keyboard focus outline |
| --w3-accent-color | #2196F3 | Checkbox and radio accent color |
| --w3-danger-bg | #dd0000 | Danger background color |
| --w3-danger-text | #fff | Danger text color |
| --w3-note-bg | #fff599 | Note background color |
| --w3-note-text | #000 | Note text color |
| --w3-info-bg | #0a6fc2 | Info background color |
| --w3-info-text | #fff | Info text color |
| --w3-warning-bg | #ffb305 | Warning background color |
| --w3-warning-text | #000 | Warning text color |
| --w3-success-bg | #008a00 | Success background color |
| --w3-success-text | #fff | Success text color |

Override dark-mode colors on `.w3-dark-mode` after loading W3.CSS.

See W3.CSS Variables for examples.

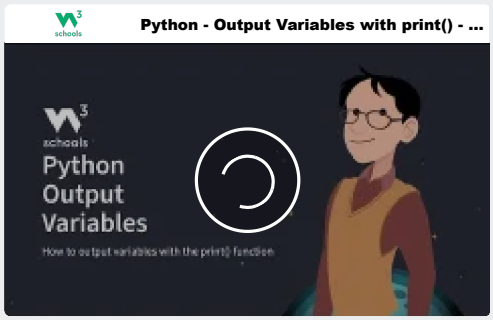
‹ Previous

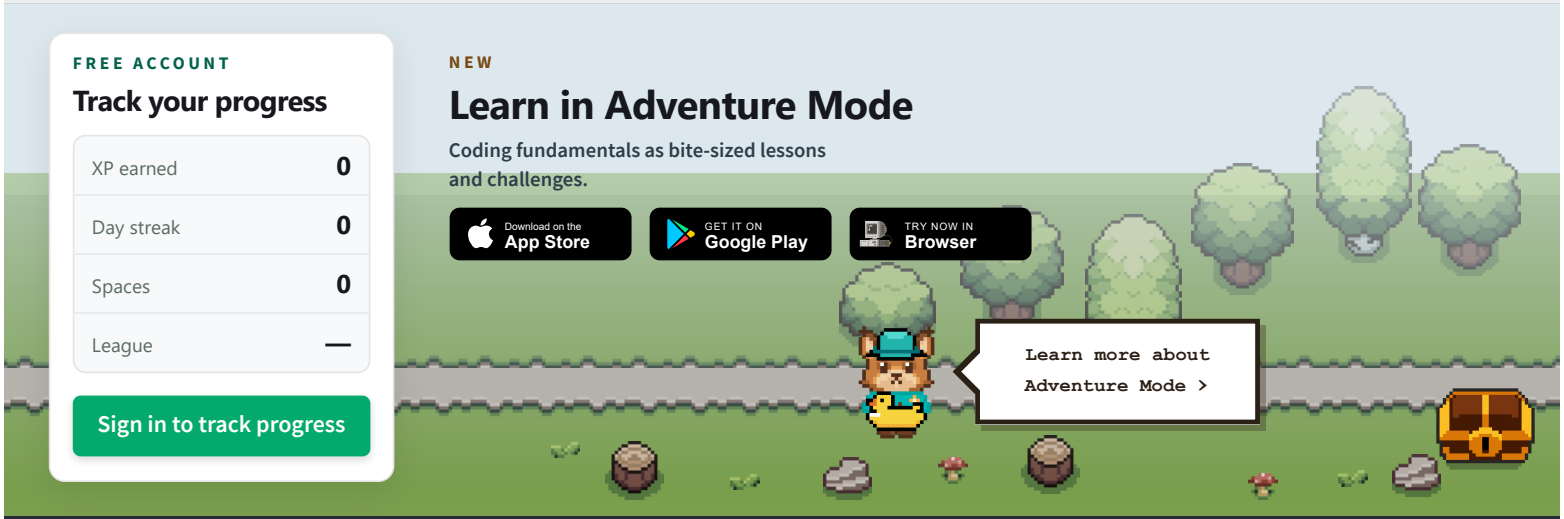
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# Bootstrap Classes Reference

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## Complete List of All Bootstrap Classes

SHARE

Complete list of all Bootstrap classes with description and examples:

Search by class name..

| Class | Description | Example | Category |
|---|---|---|---|
| `.active` | Adds a grey background color to the **table** row ( `<tr>` or table cell ( `<td>` ) (same color used on hover) | Try it | Tables |
| `.active` | Adds a gray background color to the active link in a default **navbar.** Adds a black background and a white color to the current link inside an inverted navbar. | Try it | Navbar |
| `.active` | Adds a blue background color to the active **list item** in a list group | Try it | List Groups |
| `.active` | Adds a blue background color to simulate a "pressed" **button** | Try it | Buttons |
| `.active` | Animates a striped **progress bar** | Try it | Progress Bars |
| `.active` | Adds a blue background color to the active **dropdown item** in a dropdown | Try it | Dropdowns |
| `.active` | Adds a blue background color to the active **pagination** link (to highlight the current page) | Try it | Pagination |

| | | | |
|---|---|---|---|
| `.affix` | The Affix plugin allows an element to become affixed (locked/sticky) to an area on the page. It toggles `position:fixed` on and off | Try it | Affix |
| `.alert` | Creates an alert message box | Try it | Alerts |
| `.alert-danger` | Red alert box. Indicates a dangerous or potentially negative action | Try it | Alerts |
| `.alert-dismissible` | Together with the `.close` class, this class is used to close the alert | Try it | Alerts |
| `.alert-info` | Light-blue alert box. Indicates some information | Try it | Alerts |
| `.alert-link` | Used on links inside alerts to add matching colored links | Try it | Alerts |
| `.alert-success` | Green alert box. Indicates a successful or positive action | Try it | Alerts |
| `.alert-warning` | Yellow alert box. Indicates caution should be taken with this action | Try it | Alerts |
| `.badge` | Creates a circular badge (grey circle - often used as a numerical indicator) | Try it | Badges |
| `.bg-danger` | Adds a red background color to an element. Represents danger or a negative action | Try it | Helpers |
| `.bg-info` | Adds a light-blue background color to an element. Represents some information | Try it | Helpers |
| `.bg-primary` | Adds a blue background color to an element. Represents something important | Try it | Helpers |
| `.bg-success` | Adds a green background color to an element. Indicates success or a positive action | Try it | Helpers |
| `.bg-warning` | Adds a yellow background color to an element. Represents a warning or a negative action | Try it | Helpers |
| `.breadcrumb` | A pagination. Indicates the current page's location within a navigational hierarchy | Try it | Pagination |
| `.btn` | Creates a basic button (gray background and rounded corners) | Try it | Buttons |
| `.btn-block` | Creates a block level button that spans the entire width of the parent element | Try it | Buttons |
| `.btn-danger` | Red button. Indicates danger or a negative action | Try it | Buttons |
| `.btn-default` | Default button. White background and grey border | Try it | Buttons |
| `.btn-group` | Groups buttons together on a single line | Try it | Button Groups |
| `.btn-group-justified` | Makes a group of buttons span the entire width of the screen | Try it | Button Groups |

| | | | |
|---|---|---|---|
| `.btn-group-lg` | Large button group (makes all buttons in a button group larger - increased font-size and padding) | Try it | Button Groups |
| `.btn-group-sm` | Small button group (makes all buttons in a button group smaller) | Try it | Button Groups |
| `.btn-group-xs` | Extra small button group (makes all buttons in a button group extra small) | Try it | Button Groups |
| `.btn-group-vertical` | Makes a button group appear vertically stacked | Try it | Button Groups |
| `.btn-info` | Light-blue button. Represents information | Try it | Buttons |
| `.btn-link` | Makes a button look like a link (get button behavior) | Try it | Buttons |
| `.btn-lg` | Large button | Try it | Buttons |
| `.btn-primary` | Blue button. | Try it | Buttons |
| `.btn-sm` | Small button | Try it | Buttons |
| `.btn-success` | Green button. Indicates success or a positive action | Try it | Buttons |
| `.btn-warning` | Yellow button. Represents warning or a negative action | Try it | Buttons |
| `.btn-xs` | Extra small button | Try it | Buttons |
| `.caption` | Adds a caption text inside a `.thumbnail` | Try it | Images |
| `.caret` | Creates a caret arrow icon ▾, which indicates that the button is a dropdown | Try it | Dropdowns |
| `.carousel` | Creates a carousel (slideshow) | Try it | Carousel |
| `.carousel-caption` | Creates a caption text for each slide in the carousel | Try it | Carousel |
| `.carousel-control` | Container for next and previous links | Try it | Carousel |
| `.carousel-indicators` | Adds little dots/indicators at the bottom of each slide (which indicates how many slides there is in the carousel, and which slide the user are currently viewing) | Try it | Carousel |
| `.carousel-inner` | Container for slide items | Try it | Carousel |
| `.center-block` | Centers any element (Sets an element to `display:block` with `margin-right:auto` and `margin-left:auto`) | Try it | Helpers |
| `.checkbox` | Container for checkboxes | Try it | Inputs |
| `.checkbox-inline` | Makes multiple checkboxes appear on the same line | Try it | Inputs |
| `.clearfix` | Clears floats | Try it | Helpers |
| `.close` | Indicates a close icon | Try it | Helpers |

| | | | |
|---|---|---|---|
| `.col-*-*` | Responsive grid (span 1-12 column). Extra small devices Phones (< 768px), Small devices Tablets (≥768px), Medium devices Desktops (≥992px), Large devices Desktops (≥1200px). Column values can be 1-12. | Try it | Grid |
| `.col-*-offset-*` | Move columns to the right. These classes increase the left margin of a column by * columns | Try it | Grid |
| `.col-*-pull-*` | Changes the order of the grid columns | Try it | Grid |
| `.col-*-push-*` | Changes the order of the grid columns | Try it | Grid |
| `.collapse` | Indicates collapsible content - which can be hidden or shown on demand | Try it | Collapse |
| `.collapse in` | Show the collapsible content by default | Try it | Collapse |
| `.container` | Fixed width container with widths determined by screen sites. Equal margin on the left and right. | Try it | Containers |
| `.container-fluid` | A container that spans the full width of the screen | Try it | Containers |
| `.control-label` | Allows a label to be used for form validation | Try it | Forms |
| `.danger` | Adds a red background to the table row ( `<tr>` or table cell ( `<td>` ). Indicates a dangerous or potentially negative action | Try it | Tables |
| `.disabled` | Disables a **button** (adds opacity and a "no-parking-sign" icon on hover) | Try it | Buttons |
| `.disabled` | Disables a **dropdown** item (adds a grey text color and a "no-parking-sign" icon on hover) | Try it | Dropdowns |
| `.disabled` | Disables a **pagination** link (cannot be clicked - adds a grey text color and a "no-parking-sign" icon on hover) | Try it | Pagination |
| `.disabled` | Disables a **list** item in a list group (cannot be clicked - adds a grey background color and a "no-parking-sign" icon on hover) | Try it | List Groups |
| `.divider` | Used to separate links in the dropdown menu with a thin horizontal border | Try it | Dropdowns |
| `.dl-horizontal` | Lines up the terms `<dt>` and descriptions `<dd>` in `<dl>` elements side-by-side. Starts off like default `<dl>` s, but when the browser window expands, it will line up side-by-side | Try it | Typography |
| `.dropdown` | Creates a toggleable menu that allows the user to choose one value from a predefined list | Try it | Dropdowns |
| `.dropdown-header` | Used to add headers inside the dropdown menu | Try it | Dropdowns |
| `.dropdown-menu` | Adds the default styles for the dropdown menu container | Try it | Dropdowns |
| `.dropdown-menu-right` | Right-aligns a dropdown menu | Try it | Dropdowns |

| | | | |
|---|---|---|---|
| `.dropdown-toggle` | Used on the button that should hide and show (toggle) the dropdown menu | Try it | Dropdowns |
| `.dropup` | Indicates a dropup menu (upwards instead of downwards) | Try it | Dropdowns |
| `.embed-responsive` | Container for embedded content. Makes videos or slideshows scale properly on any device | Try it | Images |
| `.embed-responsive-16by9` | Container for embedded content. Creates an 16:9 aspect ratio embedded content | Try it | Images |
| `.embed-responsive-4by3` | Container for embedded content. Creates an 4:3 aspect ratio embedded content | Try it | Images |
| `.embed-responsive-item` | Used inside `.embed-responsive`. Scales the video nicely to the parent element | Try it | Images |
| `.fade` | Adds a fading effect when closing an alert box | Try it | Alerts |
| `.form-control` | Used on input, textarea, and select elements to span the entire width of the page and make them responsive | Try it | Forms |
| `.form-control-feedback` | Form validation class | Try it | Inputs 2 |
| `.form-control-static` | Adds plain text next to a form label within a horizontal form | Try it | Inputs 2 |
| `.form-group` | Container for form input and label | Try it | Forms |
| `.form-inline` | Makes a <form> left-aligned with inline-block controls (This only applies to forms within viewports that are at least 768px wide) | Try it | Forms |
| `.form-horizontal` | Aligns labels and groups of form controls in a horizontal layout | Try it | Forms |
| `.glyphicon` | Creates an icon. Bootstrap provides 260 free glyphicons from the Glyphicons Halflings set | Try it | Glyphicons |
| `.has-danger` | Adds a red color to the label and a red border to the input, as well as an error icon inside the input (used together with `.has-feedback`) | Try it | Forms |
| `.has-feedback` | Adds feedback icons for inputs (checkmark, warning and error signs) | Try it | Forms |
| `.has-success` | Adds a green color to the label and a green border to the input, as well as a checkmark icon inside the input (used together with `.has-feedback`) | Try it | Forms |
| `.has-warning` | Adds a yellow/orange color to the label and a yellow/orange border to the input, as well as a checkmark icon inside the input (used together with `.has-feedback`) | Try it | Forms |
| `.help-block` | A block of help text that breaks onto a new line and may extend beyond one line. | Try it | Input Sizing |

| | | | |
|---|---|---|---|
| `.hidden` | Forces an element to be hidden ( `display:none` ) | Try it | Helpers |
| `.hidden-*` | Hides content depending on screen size | Try it | Helpers |
| `.hide` | Deprecated. Use `.hidden` instead | Try it | Helpers |
| `.h1 - .h6` | Makes an element look like a heading of the chosen class (h1-h6) | Try it | Typography |
| `.icon-bar` | Used in the navbar to create a hamburger menu (three horizontal bars) | Try it | Navbar |
| `.icon-next` | Unicode icon (arrow pointing right), used in carousels. This is often replaced with a glyphicon | Try it | Carousel |
| `.icon-prev` | Unicode icon (arrow pointing left), used in carousels. This is often replaced with a glyphicon | Try it | Carousel |
| `.img-circle` | Shapes an image to a circle (not supported in IE8 and earlier) | Try it | Images |
| `.img-responsive` | Makes an image responsive | Try it | Images |
| `.img-rounded` | Adds rounded corners to an image | Try it | Images |
| `.img-thumbnail` | Shapes an image to a thumbnail (borders) | Try it | Images |
| `.in` | Fades in tabs | Try it | Tabs |
| `.info` | Adds a light-blue background to the table row ( `<tr>` or table cell ( `<td>` ). Indicates a neutral informative change or action | Try it | Tables |
| `.initialism` | Displays the text inside an `<abbr>` element in a slightly smaller font size | Try it | Typography |
| `.input-group` | Container to enhance an input by adding an icon, text or a button in front or behind it as a "help text" | Try it | Inputs |
| `.input-group-lg` | Large input group | Try it | Inputs |
| `.input-group-sm` | Small input group | Try it | Inputs |
| `.input-group-addon` | Together with the `.input-group` class, this class makes it possible to add an icon or help text next to the input field | Try it | Inputs |
| `.input-group-btn` | Together with the `.input-group` class, this class attaches a button next to an input. Often used as a search bar | Try it | Inputs |
| `.input-lg` | Large input field | Try it | Input Sizing |
| `.input-sm` | Small input field | Try it | Input Sizing |
| `.invisible` | Makes an element invisible ( `visibility:hidden` ). **Note:** Even though the element is invisible, it will take up space on the page | Try it | Helpers |
| `.item` | Class added to each carousel item. May be text or images | Try it | Carousel |

| | | | |
|---|---|---|---|
| `.jumbotron` | Creates a padded grey box with rounded corners that enlarges the font sizes of the text inside it. Creates a big box for calling extra attention to some special content or information | Try it | Jumbotron |
| `.label` | Adds a grey rounded box to an element. Provides additional information about something (e.g. "New") | Try it | Labels |
| `.label-danger` | Red label | Try it | Labels |
| `.label-info` | Light-blue label | Try it | Labels |
| `.label-success` | Green label | Try it | Labels |
| `.label-warning` | Yellow label | Try it | Labels |
| `.lead` | Increase the font size and line height of a paragraph | Try it | Typography |
| `.left` | Used to identify the left carousel control | Try it | Carousel |
| `.list-group` | Creates a bordered list group for `<li>` elements | Try it | List Group |
| `.list-group-item` | Added to each `<li>` element in the list group | Try it | List Group |
| `.list-group-item-heading` | Creates a list group heading (used on other elements besides `<li>`) | Try it | List Group |
| `.list-group-item-text` | Used for item text inside the list group (used on other elements besides `<li>`) | Try it | List Group |
| `.list-group-item-danger` | Red background color for a list item in a list group | Try it | List Group |
| `.list-group-item-info` | Light-blue background color for a list item in a list group | Try it | List Group |
| `.list-group-item-success` | Green background color for a list item in a list group | Try it | List Group |
| `.list-group-item-warning` | Yellow background color for a list item in a list group | Try it | List Group |
| `.list-inline` | Places all list items on a single line (horizontal menu) | Try it | Tabs |
| `.list-unstyled` | Removes all default list-style (bullets, left margin, etc.) styling from a `<ul>` or `<ol>` list | Try it | Typography |
| `.mark` | Highlights text: Highlighted text | Try it | Typography |
| `.media` | Aligns media objects (like images or videos - often used for comments in a blog post etc) | Try it | Media Objects |
| `.media-body` | Text that should appear next to a media object | Try it | Media Objects |

| | | | |
|---|---|---|---|
| `.media-heading` | Creates a heading inside the media object | Try it | Media Objects |
| `.media-list` | Nested media lists | Try it | Media Objects |
| `.media-object` | Indicates a media object (image or video) | Try it | Media Objects |
| `.modal` | Identifies the content as a modal and brings focus to it | Try it | Modals |
| `.modal-body` | Defines the style for the body of the modal. Add any HTML markup here (p, img, etc) | Try it | Modals |
| `.modal-content` | Styles the modal (border, background-color, etc). Inside this, add the modal's header, body and footer, if needed | Try it | Modals |
| `.modal-dialog` | Sets the proper width and margin of the modal | Try it | Modals |
| `.modal-footer` | The footer of the modal (often contains an action button and a close button) | Try it | Modals |
| `.modal-header` | The header of the modal (often contains a title and a close button) | Try it | Modals |
| `.modal-lg` | Large modal (wider than default) | Try it | Modals |
| `.modal-open` | Used on the `<body>` element to prevent page scrolling (`overflow:hidden`) | Try it | Modals |
| `.modal-sm` | Small modal (less width) | Try it | Modals |
| `.modal-title` | The title of the modal | Try it | Modals |
| `.nav nav-tabs` | Indicates a tabbed menu | Try it | Tabs |
| `.nav nav-pills` | Indicates a pill menu | Try it | Tabs |
| `.nav .navbar-nav` | Used on a `<ul>` container that contains the list items with links inside a navigation bar | Try it | Navbar |
| `.nav-justified` | Centers tabs/pills. Note that on screens smaller than 768px the items are stacked (content will remain centered) | Try it | Tabs |
| `.nav-stacked` | Vertically stack tabs or pills | Try it | Tabs |
| `.nav-tabs` | Creates a tabbed menu | Try it | Tabs |
| `.navbar` | Creates a navigation bar | Try it | Navbar |
| `.navbar-brand` | Added to a link or a header element inside the navbar to represent a logo or a header | Try it | Navbar |
| `.navbar-btn` | Vertically aligns a button inside a navbar | Try it | Navbar |
| `.navbar-collapse` | Collapses the navbar (hidden and replaced with a menu/hamburger icon on mobile phones and small tablets) | Try it | Navbar |

| | | | |
|---|---|---|---|
| `.navbar-default` | Creates a default navigation bar (light-grey background color) | Try it | Navbar |
| `.navbar-fixed-bottom` | Makes the navbar stay at the bottom of the screen (sticky/fixed) | Try it | Navbar |
| `.navbar-fixed-top` | Makes the navbar stay at the top of the screen (sticky/fixed) | Try it | Navbar |
| `.navbar-form` | Added to form elements inside the navbar to vertically center them (proper padding) | Try it | Navbar |
| `.navbar-header` | Added to a container element that contains the link/element that represent a logo or a header | Try it | Navbar |
| `.navbar-inverse` | Creates a black navigation bar (instead of light-grey) | Try it | Navbar |
| `.navbar-left` | Aligns nav links, forms, buttons, or text, in the navbar to the left | Try it | Navbar |
| `.navbar-link` | Styles an element to look like a link inside the navbar (anchors get proper padding and an underline on hover, while other elements like p or span gets a default hover effect - white color in an inversed navbar and a black color in a default navbar) | Try it | Navbar |
| `.navbar-nav` | Used on a `<ul>` container that contains the list items with links inside a navigation bar | Try it | Navbar |
| `.navbar-right` | Aligns nav links, forms, buttons, or text in the navbar to the right. | Try it | Navbar |
| `.navbar-static-top` | Removes left, top and right borders (rounded corners) from the navbar (default navbar has a gray border and a 4px border-radius by default) | Try it | Navbar |
| `.navbar-text` | Vertical align any elements inside the navbar that are not links (ensures proper padding) | Try it | Navbar |
| `.navbar-toggle` | Styles the button that should open the navbar on small screens. Often used together with three `.icon-bar` classes to indicate a toggleable menu icon (hamburger/bars) | Try it | Navbar |
| `.next` | Used in the carousel control to identity the next control | Try it | Carousel |
| `.next` | Used to align pager buttons to the right side of the page (next button) | Try it | Pager |
| `.page-header` | Adds a horizontal line under the heading (+ adds some extra space around the element) | Try it | Page Header |
| `.pager` | Creates previous/next buttons (used on `<ul>` elements) | Try it | Pager |
| `.pagination` | Creates a pagination (Useful when you have a web site with lots of pages. Used on `<ul>` elements) | Try it | Pagination |
| `.pagination-lg` | Large pagination (each pagination link gets a font-size of 18px. Default is 14px) | Try it | Pagination |

| | | | |
|---|---|---|---|
| `.pagination-sm` | Small pagination (each pagination link gets a font-size of 12px. Default is 14px) | Try it | Pagination |
| `.panel` | Creates a bordered box with some padding around its content | Try it | Panels |
| `.panel-body` | Container for content inside the panel | Try it | Panels |
| `.panel-collapse` | Collapsible panel (toggle between hiding and showing panel(s)) | Try it | Collapse |
| `.panel-danger` | Red panel. Indicates danger | Try it | Panels |
| `.panel-info` | Light-blue panel. Indicates information | Try it | Panels |
| `.panel-success` | Green panel. Indicates success | Try it | Panels |
| `.panel-warning` | Yellow panel. Indicates warning | Try it | Panels |
| `.panel-footer` | Creates a panel footer (light background color) | Try it | Panels |
| `.panel-group` | Used to group many panels together. This removes the bottom margin below each panel | Try it | Panels |
| `.panel-heading` | Creates a panel header (light background color) | Try it | Panels |
| `.panel-title` | Used inside a `.panel-heading` to adjust the styling of the text (removes margins and adds a font-size of 16px) | Try it | Panels |
| `.popover` | Popup-box that appears when the user clicks on an element | Try it | Popover |
| `.pre-scrollable` | Makes a `<pre>` element scrollable ( `max-height` of 350px and provide a y-axis scrollbar) | Try it | Helpers |
| `.prev` | Used in carousels to indicate a "previous" link | Try it | Carousel |
| `.previous` | Used to align pager buttons to the left side of the page (previous button) | Try it | Pager |
| `.progress` | Container for progress bars | Try it | Progress Bars |
| `.progress-bar` | Creates a progress bar | Try it | Progress Bars |
| `.progress-bar-danger` | Red progress bar. Indicates danger | Try it | Progress Bars |
| `.progress-bar-info` | Light-blue progress bar. Indicates information | Try it | Progress Bars |
| `.progress-bar-striped` | Creates a striped progress bar | Try it | Progress Bars |
| `.progress-bar-success` | Green progress bar. Indicates success | Try it | Progress Bars |
| `.progress-bar-warning` | Yellow progress bar. Indicates warning | Try it | Progress Bars |

| | | | |
|---|---|---|---|
| `.pull-left` | Float an element to the left | Try it | Helpers |
| `.pull-right` | Float an element to the right | Try it | Helpers |
| `.right` | Used to identify the right carousel control | Try it | Carousel |
| `.row` | Container for responsive columns | Try it | Grid |
| `.row-no-gutters` | Removes the gutters from a row and its columns | Try it | Grid |
| `.show` | Shows an element ( `display:block)` | Try it | Helpers |
| `.small` | Creates a lighter, secondary text in any heading | Try it | Typography |
| `.sr-only` | Hides an element on all devices except for screen readers | Try it | Helpers |
| `.sr-only-focusable` | Hides an element on all devices except for screen readers | Try it | Helpers |
| `.success` | Adds a green background color to a table row ( `<tr>` or table cell ( `<td>` ). Indicates success or a positive action | Try it | Tables |
| `.tab-content` | Used together with `.tab-pane` to creates toggleable/dynamic tabs/pills | Try it | Tabs |
| `.tab-pane` | Used together with `.tab-content` to creates toggleable/dynamic tabs/pills | Try it | Tabs |
| `.table` | Adds basic styling to a table (padding, bottom borders, etc) | Try it | Tables |
| `.table-bordered` | Adds borders on all sides of the table and cells | Try it | Tables |
| `.table-condensed` | Makes a table more compact by cutting cell padding in half | Try it | Tables |
| `.table-hover` | Creates a hoverable table (adds a grey background color on table rows on hover) | Try it | Tables |
| `.table-responsive` | Makes a table responsive (adds a horizontal scrollbar when needed) | Try it | Tables |
| `.text-capitalize` | Indicates capitalized text | Try it | Typography |
| `.text-center` | Center-aligns text | Try it | Typography |
| `.text-danger` | Red text color. Indicates danger | Try it | Typography |
| `.text-hide` | Hides text (helps replace an element's text content with a background image) | Try it | Typography |
| `.text-info` | Light-blue text color. Indicates information | Try it | Typography |
| `.text-justify` | Indicates justified text | Try it | Typography |
| `.text-left` | Aligns the text to the left | Try it | Typography |

| | | | |
|---|---|---|---|
| `.text-lowercase` | Changes text to lowercase | Try it | Typography |
| `.text-muted` | Grey text color | Try it | Typography |
| `.text-nowrap` | Prevents the text from wrapping | Try it | Typography |
| `.text-primary` | Blue text color | Try it | Typography |
| `.text-right` | Aligns text to the right | Try it | Typography |
| `.text-success` | Green text color. Indicates success | Try it | Typography |
| `.text-uppercase` | Makes text uppercase | Try it | Typography |
| `.text-warning` | Yellow/orange text color. Indicates warning | Try it | Typography |
| `.thumbnail` | Adds a border around an element (often images or videos) to make it look like a thumbnail | Try it | Images |
| `.tooltip` | Popup-box that appears when the user moves the mouse pointer over an element | Try it | Tooltip |
| `.visible-*` | Deprecated as of v3.2.0. Used to show and/or hide content by device. **Note:** Use `.hidden-*` instead | Try it | Helpers |
| `.visible-print-block` | Displays the element ( `display:block` ) in print (pre)view | | Helpers |
| `.visible-print-inline` | Displays the element ( `display:inline` ) in print (pre)view | | Helpers |
| `.visible-print-inline-block` | Displays the element ( `display:inline-block` ) in print (pre)view | | Helpers |
| `.hidden-print` | Hides the element ( `display:none` ) in print (pre)view | | Helpers |
| `.warning` | Adds a yellow background color to the table row ( `<tr>` or table cell ( `<td>` ). Indicates a warning | Try it | Tables |
| `.well` | Adds a rounded border around an element with a gray background color and some padding | Try it | Wells |
| `.well-lg` | Large well (more padding) | Try it | Wells |
| `.well-sm` | Small well (less padding) | Try it | Wells |

The table above shows all available Bootstrap 3 classes.

**Tip:** To see the full list of all **Bootstrap 4** classes, go to our All Bootstrap 4 CSS Classes Reference.

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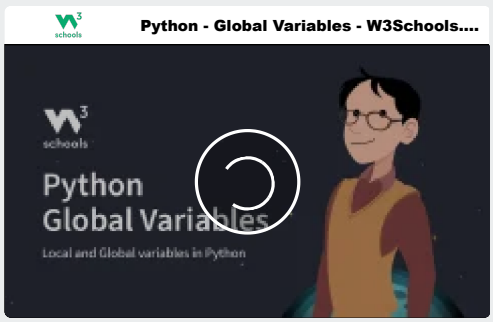
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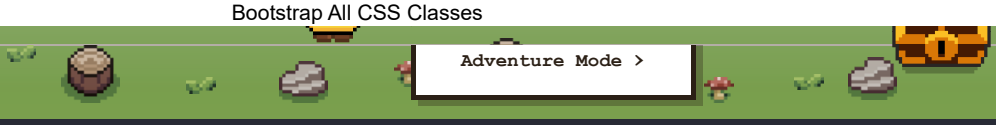
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# PHP Reference

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## PHP Reference

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This section contains a complete PHP reference documentation.

The PHP reference contains different categories of all PHP functions, keywords and constants, along with examples.

- Array
- Calendar
- Date
- Directory
- Error
- Exception
- Filesystem
- Filter

FTP

JSON

Keywords

Libxml

Mail

Math

Misc

MySQLi

Network

Output Control

RegEx

SimpleXML

Stream

String

Var Handling

XML Parser

Zip

Timezones

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# HTML Color Names

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## Color Names Supported by All Browsers

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All modern browsers support the following 140 color names (click on a color name, or a hex value, to view the color as the background-color along with different text colors):

Click here to see the 140 colors sorted by HEX Value

| | |
|---|---|
| **AliceBlue**<br>#F0F8FF<br>Color Mixer Color Picker | **AntiqueWhite**<br>#FAEBD7<br>Color Mixer Color Picker |
| **Aqua**<br>#00FFFF<br>Color Mixer Color Picker | **Aquamarine**<br>#7FFFD4<br>Color Mixer Color Picker |
| **Azure**<br>#F0FFFF<br>Color Mixer Color Picker | **Beige**<br>#F5F5DC<br>Color Mixer Color Picker |
| **Bisque**<br>#FFE4C4 | **Black**<br>#000000 |

Color Mixer Color Picker

Color Mixer Color Picker

BlanchedAlmond
#FFEBCD
Color Mixer Color Picker

Blue
#0000FF
Color Mixer Color Picker

BlueViolet
#8A2BE2
Color Mixer Color Picker

Brown
#A52A2A
Color Mixer Color Picker

BurlyWood
#DEB887
Color Mixer Color Picker

CadetBlue
#5F9EA0
Color Mixer Color Picker

Chartreuse
#7FFF00
Color Mixer Color Picker

Chocolate
#D2691E
Color Mixer Color Picker

Coral
#FF7F50
Color Mixer Color Picker

CornflowerBlue
#6495ED
Color Mixer Color Picker

Cornsilk
#FFF8DC
Color Mixer Color Picker

Crimson
#DC143C
Color Mixer Color Picker

Cyan
#00FFFF
Color Mixer Color Picker

DarkBlue
#00008B
Color Mixer Color Picker

DarkCyan
#008B8B
Color Mixer Color Picker

DarkGoldenRod
#B8860B
Color Mixer Color Picker

DarkGray

DarkGrey

#A9A9A9
Color Mixer Color Picker

#A9A9A9
Color Mixer Color Picker

DarkGreen
#006400
Color Mixer Color Picker

DarkKhaki
#BDB76B
Color Mixer Color Picker

DarkMagenta
#8B008B
Color Mixer Color Picker

DarkOliveGreen
#556B2F
Color Mixer Color Picker

DarkOrange
#FF8C00
Color Mixer Color Picker

DarkOrchid
#9932CC
Color Mixer Color Picker

DarkRed
#8B0000
Color Mixer Color Picker

DarkSalmon
#E9967A
Color Mixer Color Picker

DarkSeaGreen
#8FBC8F
Color Mixer Color Picker

DarkSlateBlue
#483D8B
Color Mixer Color Picker

DarkSlateGray
#2F4F4F
Color Mixer Color Picker

DarkSlateGrey
#2F4F4F
Color Mixer Color Picker

DarkTurquoise
#00CED1
Color Mixer Color Picker

DarkViolet
#9400D3
Color Mixer Color Picker

DeepPink
#FF1493
Color Mixer Color Picker

DeepSkyBlue
#00BFFF
Color Mixer Color Picker

DimGray
#696969
Color Mixer Color Picker

DimGrey
#696969
Color Mixer Color Picker

DodgerBlue
#1E90FF
Color Mixer Color Picker

FireBrick
#B22222
Color Mixer Color Picker

FloralWhite
#FFFAF0
Color Mixer Color Picker

ForestGreen
#228B22
Color Mixer Color Picker

Fuchsia
#FF00FF
Color Mixer Color Picker

Gainsboro
#DCDCDC
Color Mixer Color Picker

GhostWhite
#F8F8FF
Color Mixer Color Picker

Gold
#FFD700
Color Mixer Color Picker

GoldenRod
#DAA520
Color Mixer Color Picker

Gray
#808080
Color Mixer Color Picker

Grey
#808080
Color Mixer Color Picker

Green
#008000
Color Mixer Color Picker

GreenYellow
#ADFF2F
Color Mixer Color Picker

HoneyDew
#F0FFF0
Color Mixer Color Picker

HotPink
#FF69B4

IndianRed
#CD5C5C

Color Mixer Color Picker

Color Mixer Color Picker

**Indigo**
#4B0082
Color Mixer Color Picker

**Ivory**
#FFFFF0
Color Mixer Color Picker

**Khaki**
#F0E68C
Color Mixer Color Picker

**Lavender**
#E6E6FA
Color Mixer Color Picker

**LavenderBlush**
#FFF0F5
Color Mixer Color Picker

**LawnGreen**
#7CFC00
Color Mixer Color Picker

**LemonChiffon**
#FFFACD
Color Mixer Color Picker

**LightBlue**
#ADD8E6
Color Mixer Color Picker

**LightCoral**
#F08080
Color Mixer Color Picker

**LightCyan**
#E0FFFF
Color Mixer Color Picker

**LightGoldenRodYellow**
#FAFAD2
Color Mixer Color Picker

**LightGray**
#D3D3D3
Color Mixer Color Picker

**LightGrey**
#D3D3D3
Color Mixer Color Picker

**LightGreen**
#90EE90
Color Mixer Color Picker

**LightPink**
#FFB6C1
Color Mixer Color Picker

**LightSalmon**
#FFA07A
Color Mixer Color Picker

**LightSeaGreen**

**LightSkyBlue**

#20B2AA
Color Mixer Color Picker

#87CEFA
Color Mixer Color Picker

LightSlateGray
#778899
Color Mixer Color Picker

LightSlateGrey
#778899
Color Mixer Color Picker

LightSteelBlue
#B0C4DE
Color Mixer Color Picker

LightYellow
#FFFFE0
Color Mixer Color Picker

Lime
#00FF00
Color Mixer Color Picker

LimeGreen
#32CD32
Color Mixer Color Picker

Linen
#FAF0E6
Color Mixer Color Picker

Magenta
#FF00FF
Color Mixer Color Picker

Maroon
#800000
Color Mixer Color Picker

MediumAquaMarine
#66CDAA
Color Mixer Color Picker

MediumBlue
#0000CD
Color Mixer Color Picker

MediumOrchid
#BA55D3
Color Mixer Color Picker

MediumPurple
#9370DB
Color Mixer Color Picker

MediumSeaGreen
#3CB371
Color Mixer Color Picker

MediumSlateBlue
#7B68EE
Color Mixer Color Picker

MediumSpringGreen
#00FA9A
Color Mixer Color Picker

MediumTurquoise
#48D1CC
Color Mixer Color Picker

MediumVioletRed
#C71585
Color Mixer Color Picker

MidnightBlue
#191970
Color Mixer Color Picker

MintCream
#F5FFFA
Color Mixer Color Picker

MistyRose
#FFE4E1
Color Mixer Color Picker

Moccasin
#FFE4B5
Color Mixer Color Picker

NavajoWhite
#FFDEAD
Color Mixer Color Picker

Navy
#000080
Color Mixer Color Picker

OldLace
#FDF5E6
Color Mixer Color Picker

Olive
#808000
Color Mixer Color Picker

OliveDrab
#6B8E23
Color Mixer Color Picker

Orange
#FFA500
Color Mixer Color Picker

OrangeRed
#FF4500
Color Mixer Color Picker

Orchid
#DA70D6
Color Mixer Color Picker

PaleGoldenRod
#EEE8AA
Color Mixer Color Picker

PaleGreen
#98FB98
Color Mixer Color Picker

PaleTurquoise
#AFEEEE

PaleVioletRed
#DB7093

Color Mixer Color Picker

Color Mixer Color Picker

PapayaWhip
#FFEFD5
Color Mixer Color Picker

PeachPuff
#FFDAB9
Color Mixer Color Picker

Peru
#CD853F
Color Mixer Color Picker

Pink
#FFC0CB
Color Mixer Color Picker

Plum
#DDA0DD
Color Mixer Color Picker

PowderBlue
#B0E0E6
Color Mixer Color Picker

Purple
#800080
Color Mixer Color Picker

RebeccaPurple
#663399
Color Mixer Color Picker

Red
#FF0000
Color Mixer Color Picker

RosyBrown
#BC8F8F
Color Mixer Color Picker

RoyalBlue
#4169E1
Color Mixer Color Picker

SaddleBrown
#8B4513
Color Mixer Color Picker

Salmon
#FA8072
Color Mixer Color Picker

SandyBrown
#F4A460
Color Mixer Color Picker

SeaGreen
#2E8B57
Color Mixer Color Picker

SeaShell
#FFF5EE
Color Mixer Color Picker

Sienna

Silver

#A0522D
Color Mixer Color Picker

#C0C0C0
Color Mixer Color Picker

SkyBlue
#87CEEB
Color Mixer Color Picker

SlateBlue
#6A5ACD
Color Mixer Color Picker

SlateGray
#708090
Color Mixer Color Picker

SlateGrey
#708090
Color Mixer Color Picker

Snow
#FFFAFA
Color Mixer Color Picker

SpringGreen
#00FF7F
Color Mixer Color Picker

SteelBlue
#4682B4
Color Mixer Color Picker

Tan
#D2B48C
Color Mixer Color Picker

Teal
#008080
Color Mixer Color Picker

Thistle
#D8BFD8
Color Mixer Color Picker

Tomato
#FF6347
Color Mixer Color Picker

Turquoise
#40E0D0
Color Mixer Color Picker

Violet
#EE82EE
Color Mixer Color Picker

Wheat
#F5DEB3
Color Mixer Color Picker

White
#FFFFFF
Color Mixer Color Picker

WhiteSmoke
#F5F5F5
Color Mixer Color Picker

| Yellow | YellowGreen |
| --- | --- |
| #FFFF00 | #9ACD32 |
| Color Mixer Color Picker | Color Mixer Color Picker |

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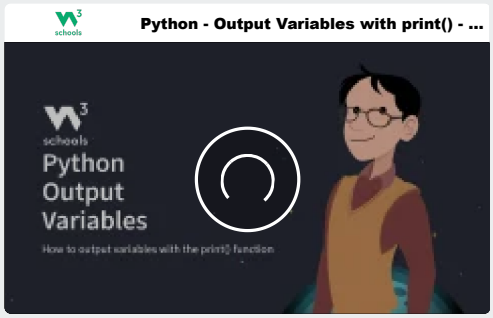
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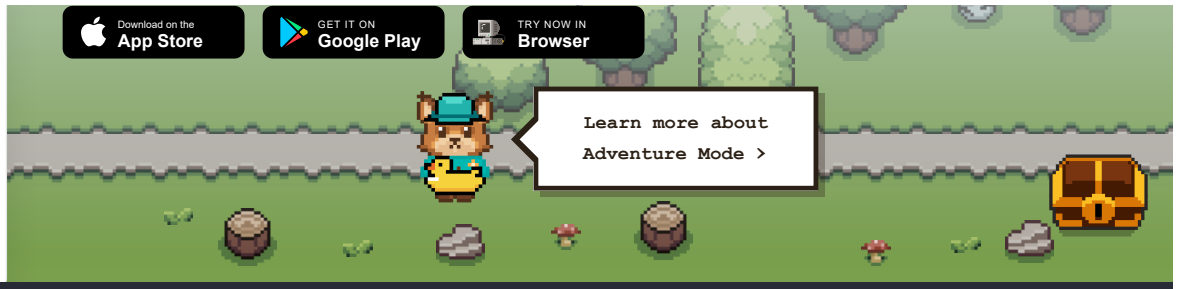
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# Java Keywords

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## Java Reserved Keywords

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Java has a set of keywords that are reserved words that cannot be used as variables, methods, classes, or any other identifiers:

| Keyword | Description |
|---|---|
| abstract | A non-access modifier. Used for classes and methods: An abstract class cannot be used to create objects (to access it, it must be inherited from another class). An abstract method can only be used in an abstract class, and it does not have a body. The body is provided by the subclass (inherited from) |
| assert | For debugging |
| boolean | A data type that can only store true or false values |
| break | Breaks out of a loop or a switch block |
| byte | A data type that can store whole numbers from -128 and 127 |
| case | Marks a block of code in switch statements |
| catch | Catches exceptions generated by try statements |
| char | A data type that is used to store a single character |
| class | Defines a class |
| continue | Continues to the next iteration of a loop |
| const | Defines a constant. Not in use - use final instead |

| | |
|---|---|
| default | Specifies the default block of code in a switch statement |
| do | Used together with while to create a do-while loop |
| double | A data type that can store fractional numbers from 1.7e−308 to 1.7e+308 |
| else | Used in conditional statements |
| enum | Declares an enumerated (unchangeable) type |
| exports | Exports a package with a module. New in Java 9 |
| extends | Extends a class (indicates that a class is inherited from another class) |
| final | A non-access modifier used for classes, attributes and methods, which makes them non-changeable (impossible to inherit or override) |
| finally | Used with exceptions, a block of code that will be executed no matter if there is an exception or not |
| float | A data type that can store fractional numbers from 3.4e−038 to 3.4e+038 |
| for | Create a for loop |
| goto | Not in use, and has no function |
| if | Makes a conditional statement |
| implements | Implements an interface |
| import | Used to import a package, class or interface |
| instanceof | Checks whether an object is an instance of a specific class or an interface |
| int | A data type that can store whole numbers from -2147483648 to 2147483647 |
| interface | Used to declare a special type of class that only contains abstract methods |
| long | A data type that can store whole numbers from -9223372036854775808 to 9223372036854775808 |
| module | Declares a module. New in Java 9 |
| native | Specifies that a method is not implemented in the same Java source file (but in another language) |
| new | Creates new objects |
| package | Declares a package |
| private | An access modifier used for attributes, methods and constructors, making them only accessible within the declared class |
| protected | An access modifier used for attributes, methods and constructors, making them accessible in the same package and subclasses |
| public | An access modifier used for classes, attributes, methods and constructors, making them accessible by any other class |
| requires | Specifies required libraries inside a module. New in Java 9 |

| | |
|---|---|
| return | Finished the execution of a method, and can be used to return a value from a method |
| short | A data type that can store whole numbers from -32768 to 32767 |
| static | A non-access modifier used for methods and attributes. Static methods/attributes can be accessed without creating an object of a class |
| strictfp | Obsolete. Restrict the precision and rounding of floating point calculations |
| super | Refers to superclass (parent) objects |
| switch | Selects one of many code blocks to be executed |
| synchronized | A non-access modifier, which specifies that methods can only be accessed by one thread at a time |
| this | Refers to the current object in a method or constructor |
| throw | Creates a custom error |
| throws | Indicates what exceptions may be thrown by a method |
| transient | Used to ignore an attribute when serializing an object |
| try | Creates a try...catch statement |
| var | Declares a variable. New in Java 10 |
| void | Specifies that a method should not have a return value |
| volatile | Indicates that an attribute is not cached thread-locally, and is always read from the "main memory" |
| while | Creates a while loop |

**Note:** `true`, `false`, and `null` are not keywords, but they are literals and reserved words that cannot be used as identifiers.

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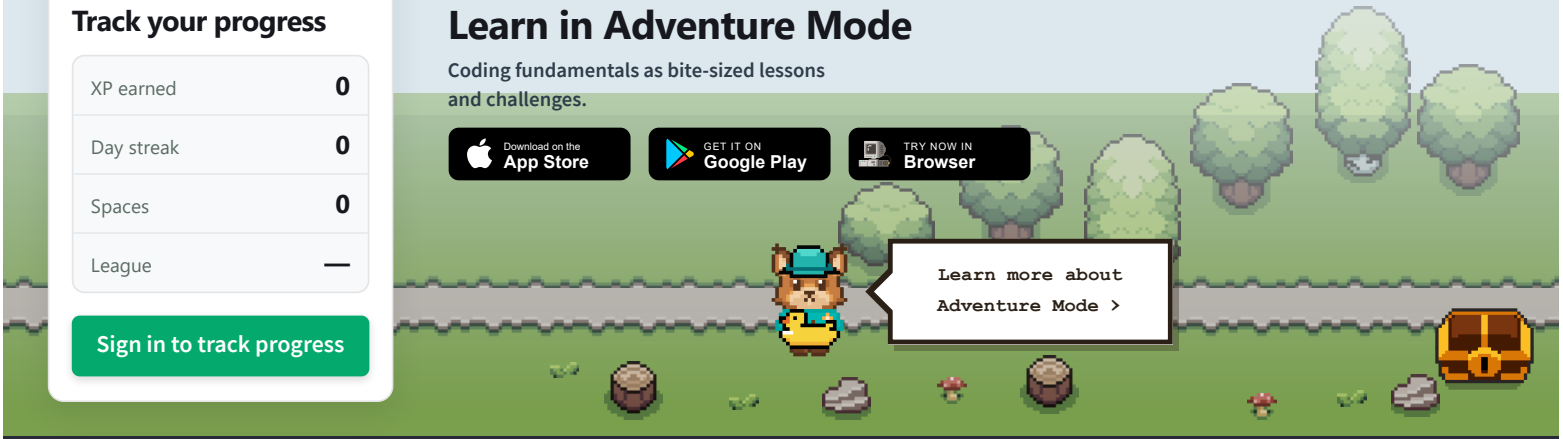
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# AngularJS References

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## AngularJS Directives

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| Directive | Description |
|---|---|
| ng-app | Defines the root element of an application. |
| ng-bind | Binds the content of an HTML element to application data. |
| ng-bind-html | Binds the innerHTML of an HTML element to application data, and also removes dangerous code from the HTML string. |
| ng-bind-template | Specifies that the text content should be replaced with a template. |
| ng-blur | Specifies a behavior on blur events. |
| ng-change | Specifies an expression to evaluate when content is being changed by the user. |
| ng-checked | Specifies if an element is checked or not. |
| ng-class | Specifies CSS classes on HTML elements. |
| ng-class-even | Same as ng-class, but will only take effect on even rows. |
| ng-class-odd | Same as ng-class, but will only take effect on odd rows. |
| ng-click | Specifies an expression to evaluate when an element is being clicked. |
| ng-cloak | Prevents flickering when your application is being loaded. |
| ng-controller | Defines the controller object for an application. |

| | |
|---|---|
| ng-copy | Specifies a behavior on copy events. |
| ng-csp | Changes the content security policy. |
| ng-cut | Specifies a behavior on cut events. |
| ng-dblclick | Specifies a behavior on double-click events. |
| ng-disabled | Specifies if an element is disabled or not. |
| ng-focus | Specifies a behavior on focus events. |
| ng-form | Specifies an HTML form to inherit controls from. |
| ng-hide | Hides or shows HTML elements. |
| ng-href | Specifies a url for the <a> element. |
| ng-if | Removes the HTML element if a condition is false. |
| ng-include | Includes HTML in an application. |
| ng-init | Defines initial values for an application. |
| ng-jq | Specifies that the application must use a library, like jQuery. |
| ng-keydown | Specifies a behavior on keydown events. |
| ng-keypress | Specifies a behavior on keypress events. |
| ng-keyup | Specifies a behavior on keyup events. |
| ng-list | Converts text into a list (array). |
| ng-maxlength | Specifies the maximum number of characters allowed in the input field. |
| ng-minlength | Specifies the minimum number of characters allowed in the input field. |
| ng-model | Binds the value of HTML controls to application data. |
| ng-model-options | Specifies how updates in the model are done. |
| ng-mousedown | Specifies a behavior on mousedown events. |
| ng-mouseenter | Specifies a behavior on mouseenter events. |
| ng-mouseleave | Specifies a behavior on mouseleave events. |
| ng-mousemove | Specifies a behavior on mousemove events. |
| ng-mouseover | Specifies a behavior on mouseover events. |
| ng-mouseup | Specifies a behavior on mouseup events. |
| ng-non-bindable | Specifies that no data binding can happen in this element, or its children. |
| ng-open | Specifies the open attribute of an element. |

| | |
|---|---|
| ng-options | Specifies <options> in a <select> list. |
| ng-paste | Specifies a behavior on paste events. |
| ng-pluralize | Specifies a message to display according to en-us localization rules. |
| ng-readonly | Specifies the readonly attribute of an element. |
| ng-repeat | Defines a template for each data in a collection. |
| ng-required | Specifies the required attribute of an element. |
| ng-selected | Specifies the selected attribute of an element. |
| ng-show | Shows or hides HTML elements. |
| ng-src | Specifies the src attribute for the <img> element. |
| ng-srcset | Specifies the srcset attribute for the <img> element. |
| ng-style | Specifies the style attribute for an element. |
| ng-submit | Specifies expressions to run on onsubmit events. |
| ng-switch | Specifies a condition that will be used to show/hide child elements. |
| ng-transclude | Specifies a point to insert transcluded elements. |
| ng-value | Specifies the value of an input element. |

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# AngularJS Directives on HTML Elements

AngularJS modifies the default behavior of some HTML elements.

| Element | Description |
|---|---|
| a | AngularJS modifies the <a> element's default behaviors. |
| form | AngularJS modifies the <form> element's default behaviors. |
| input | AngularJS modifies the <input> element's default behaviors. |
| script | AngularJS modifies the <script> element's default behaviors. |
| select | AngularJS modifies the <select> element's default behaviors. |
| textarea | AngularJS modifies the <textarea> element's default behaviors. |

# AngularJS Filters

| Filter | Description |
|---|---|
| currency | Format a number to a currency format. |
| date | Format a date to a specified format. |
| filter | Select a subset of items from an array. |
| json | Format an object to a JSON string. |
| limitTo | Limits an array, or a string, into a specified number of elements/characters. |
| lowercase | Format a string to lower case. |
| number | Format a number to a string. |
| orderBy | Orders an array by an expression. |
| uppercase | Format a string to upper case. |

Filters are explained in Angular Filters.

# AngularJS Validation Properties

- $dirty
- $invalid
- $error

Validation is explained in Angular Validation.

# AngularJS Global API

## Converting

| API | Description |
|---|---|
| angular.lowercase() | Converts a string to lowercase |
| angular.uppercase() | Converts a string to uppercase |
| angular.copy() | Creates a deep copy of an object or an array |
| angular.forEach() | Executes a function for each element in an object or array |

## Comparing

| API | Description |
|---|---|
| angular.isArray() | Returns true if the reference is an array |
| angular.isDate() | Returns true if the reference is a date |
| angular.isDefined() | Returns true if the reference is defined |
| angular.isElement() | Returns true if the reference is a DOM element |
| angular.isFunction() | Returns true if the reference is a function |
| angular.isNumber() | Returns true if the reference is a number |
| angular.isObject() | Returns true if the reference is an object |
| angular.isString() | Returns true if the reference is a string |
| angular.isUndefined() | Returns true if the reference is undefined |
| angular.equals() | Returns true if two references are equal |

## JSON

| API | Description |
|---|---|
| angular.fromJson() | Takes a JSON string and returns an JavaScript object |
| angular.toJson() | Takes a JavaScript object and returns a JSON string |

## Basic

| API | Description |
|---|---|
| angular.bootstrap() | Starts AngularJS manually |
| angular.element() | Wraps an HTML element as an jQuery element |
| angular.module() | Creates, registers, or retrieves an AngularJS module |

The Global API is explained in Angular API.

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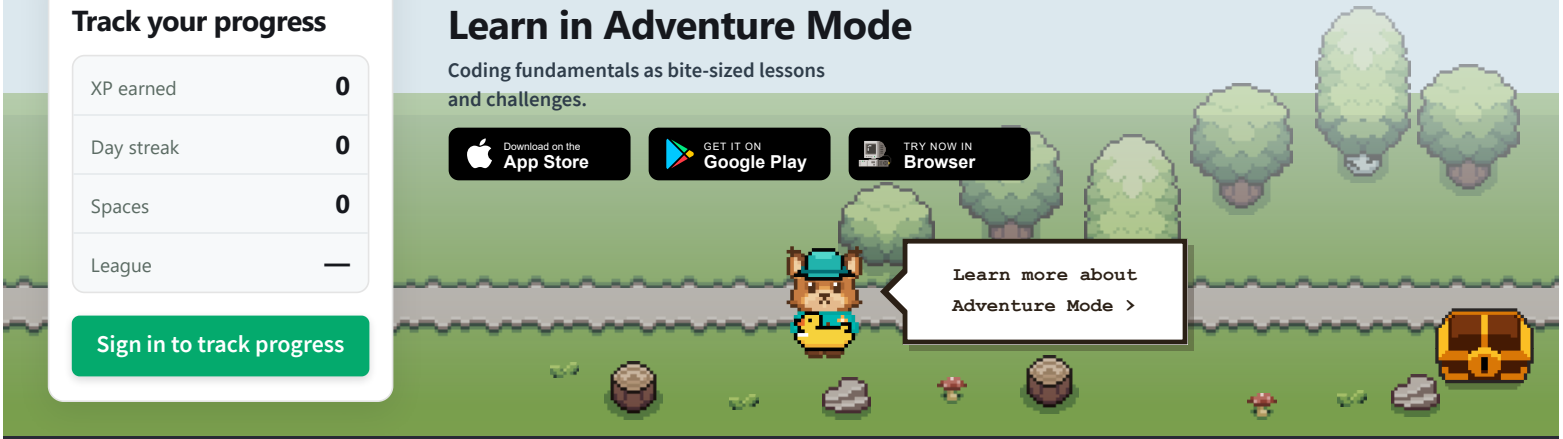
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# jQuery Reference

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This section contains a complete jQuery reference documentation.

## jQuery Reference

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The jQuery reference contains a list of all jQuery selectors, methods, properties and events, along with examples.

- Selectors
- Events
- Effects
- HTML/CSS
- Traversing
- AJAX
- Misc

Properties

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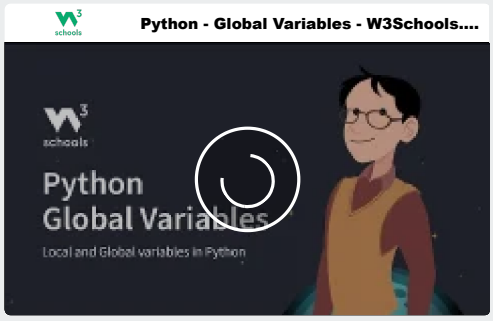
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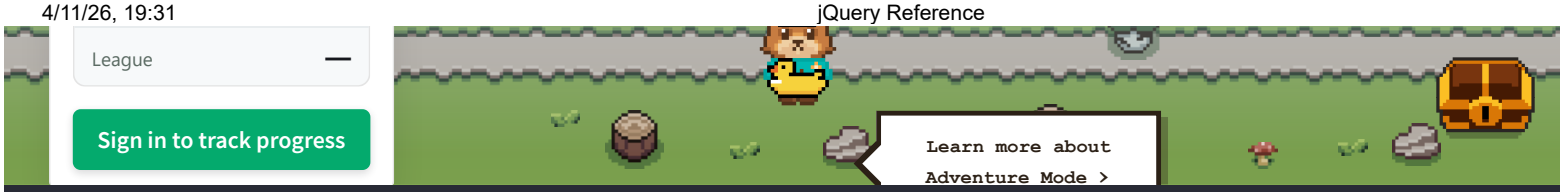



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# HTML Examples

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## HTML Examples

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This page covers all the HTML examples from the HTML tutorial. Each HTML example opens in our Try-it editor, showing the source code and the result.

### HTML Basic

- HTML document
- HTML headings
- HTML paragraphs
- HTML links
- HTML images
- HTML buttons
- HTML lists

Examples explained

### HTML Attributes

- The title attribute
- The href attribute

- The width and height attributes
- The alt attribute
- Attribute without quotes
- Attribute without quotes does not work

**Examples explained**

## HTML Headings

- HTML headings
- HTML horizontal rules
- HTML head

**Examples explained**

## HTML Paragraphs

- HTML paragraphs
- More HTML paragraphs
- The use of line breaks in HTML
- Poem problems (some problems with HTML formatting)
- How to control the line breaks and spaces with the <pre> tag

**Examples explained**

## HTML Styles

- HTML styles
- HTML background color
- HTML text color
- HTML text font
- HTML text size
- HTML text alignment

Examples explained

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## HTML Text Formatting

Bold formatting using the <b> element

Strong formatting using the  element

Italic formatting using the <i> element

Emphasized formatting using the  element

Small formatting using the <small> element

Marked formatting using the <mark> element

Marked deleted using the <del> element

Marked inserted using the <ins> element

Marked deleted and inserted using <del> and <ins>

Subscript formatting using the <sub> element

Superscript formatting using the <sup> element

Examples explained

## HTML Quotations and Citations

Formatting short quotations with the <q> element.

Formatting quoted sections with the <blockquote> element.

Formatting document author/owner information with the <address> element

Formatting abbreviations and acronyms the <abbr> element

Formatting work title with the <cite> element

Formatting text direction with the <bdo> element

Examples explained

## HTML Comments

Hidden comments

Conditional comments

Comments for debugging

Examples explained

## HTML CSS

HTML with inline CSS

HTML with internal CSS

HTML with external CSS

HTML with CSS fonts

HTML with CSS using the id attribute

HTML with CSS using the class attribute

HTML and CSS borders

HTML and CSS padding

HTML and CSS margin

HTML and CSS full demo

Examples explained

## HTML Links

Linking, using an absolute URL

Linking, using a relative URL

Changing the color of links

Removing the underline from links

Changing the target of a link

An image as a link

Creating a bookmark link

A link that breaks out of a frame

A mailto link

A mailto link with subject

Examples explained

## HTML Images

An image

An image height and width using attributes

An image height and width using CSS

An image height and width using both

An image in another folder

An image with a broken link

An image on another server

Using an image as a link

A moving image

An image map with clickable regions

A floating image

Examples explained

## HTML Tables

Basic HTML tables

A table with borders

A table with collapsed borders

A table with cell padding

A table with headings

A table with left-aligned headings

Horizontal/Vertical table headings

A table with a caption

Table cells that span more than one column

Table cells that span more than one row

A table with cell spacing

A table with HTML tags inside

Tables with different style using id I

Tables with different style using id II

Tables with different style using class I

Tables with different style using class II

Examples explained

## HTML Lists

An unordered list (default)

An unordered list with disc bullets

An unordered list with circle bullets

An unordered list with square bullets

An unordered list without bullets

An ordered list (default)

An ordered list with numbers

An ordered list with letters

An ordered list with lowercase letters

An ordered list with roman numbers

An ordered list with lowercase roman numbers

A description list

A nested list I

A nested list II

A horizontal list

A horizontal list menu

Examples explained

## HTML Block and inline elements

The  element

The  element

Styling a  element

Styling a  element

Examples explained

## HTML Div Element

 elements come with linebreaks

 as a container

Center align a  element

Multiple  elements

Floating  elements

Position  elements with display:inline-block

Position  elements with display:flex

Position  elements with display:grid

Examples explained

## HTML Classes

Style all elements with a specified class name

Access elements with a specified class name, with JavaScript

Multiple classes

Same class, different tag

Examples explained

## HTML Id

Style an element with a specific id

Difference between class and id

Access an element with a specific id, with JavaScript

Examples explained

## HTML Layout

- Layout using float
- Layout using flexbox
- Layout using flexbox 2
- Layout using flexbox 3

Examples explained

## HTML IFrame

- Inline frame (a frame inside an HTML page)

Examples explained

## HTML head Elements

- A valid HTML document with no <html> <body, and <head>
- A valid HTML document with no <head> element
- The <title> element defines the document title
- The <style> element contains style information
- The <link> element defines a relationship to an external resource
- The <meta> element defines special meta information
- The <script> element defines client-side JavaScripts
- The <base> element defines the base URL for all URLs

Examples explained

## HTML Scripts

- Insert a script
- Use of the <noscript> tag

Examples explained

## HTML Computercode Elements

- Keyboard input formatting using the <kbd> element
- Computer output formatting using the <samp> element
- Programming code formatting using the <code> element
- Programming code formatting preserving whitespace and line-breaks
- Variable formatting using the <var> element

Examples explained

## HTML Forms

- Form with text input
- Form with radio button input
- Form with text fields and a submit button
- Form with a text fields without a name attribute
- Grouping Form Data

Examples explained

## HTML Form Elements

- A simple drop-down list
- A drop-down list with a pre-selected value
- A textarea (a multi-line text input field)
- An input button
- Using the <datalist> Element
- Using the <output> Element

Examples explained

## HTML Input Types

Input type text

Input type password

Input type radio

Input type checkbox

Input type button

Input type number - with restrictions

Input type number - with steps

Input type date - with date picker

Input type date - with restrictions

Input type color - with color picker

Input type range

Input type month

Input type week

Input type time

Input type datetime-local

Input type email

Input type search

Input type tel

Input type url

**Examples explained**

## HTML Input Attributes

The autocomplete attribute

The novalidate attribute

The autofocus_attribute

The form attribute

The formaction attribute

The formenctype attribute

The formmethod attribute

The formnovalidate attribute

The formtarget attribute

The height and width attributes

The list attribute

The min and max attributes

The multiple attribute

The pattern attribute

The placeholder attribute

The required attribute

The step attribute

**Examples explained**

## HTML Canvas Graphics

Draw on the canvas with JavaScript

Draw a line with lineTo()

Draw a circle with arc()

Draw a text with fillText()

Draw a text with strokeText()

Draw a linear gradient

Draw a circular gradient

Draw an image with drawImage()

**Examples explained**

## HTML SVG Graphics

SVG Circle

SVG Rectangle

SVG Rounded Rectangle

SVG Star

SVG Logo

Examples explained

## HTML Media

Play Bunny

Play bear video with controls

Play bear video with autoplay

Play Horse sound with controls

Examples explained

## HTML Geolocation

Get geolocation coordinates

Handle geolocation errors

Get geolocation and watch the position

Examples explained

## HTML Local Storage

Store a name permanently

Store a counter permanently

Store a counter for one session

Examples explained

## HTML Media

Play a video file

Play an audio file in HTML

Play a YouTube video in HTML

Examples explained

# More HTML Examples

- HTML drag and drop
- HTML web workers
- HTML server sent events

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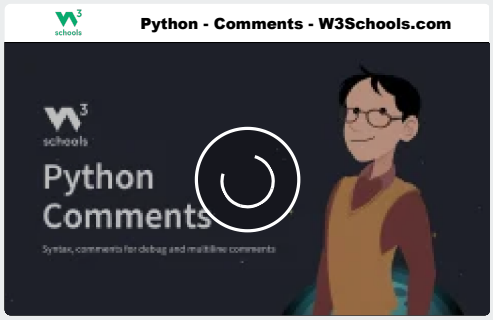
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## CSS Examples

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This page covers all the CSS examples from the CSS tutorial. Each CSS example opens in our Try-it editor, showing the source code and the result.

### CSS Syntax

CSS syntax

CSS syntax explained

### CSS Selectors

The element selector

The id selector

The class selector (for all elements)

The class selector (for only <p> elements)

An HTML element that refer to two classes

The universal selector

Grouping selectors

CSS selectors explained

## CSS How To / Where To

- External CSS
- Internal CSS
- Inline CSS
- Multiple style sheets
- Cascading order

CSS how to explained

## CSS Comments

- Single-line comment
- Multi-line comment

CSS comments explained

## CSS Colors

- Set the background color of an element
- Set the text color
- Set the border color
- Set different color values
- Set RGB values
- Set HEX values
- Set HSL values

CSS colors explained

## CSS Backgrounds

- Set the background color of a page
- Set the background color of different elements

Set an image as the background of a page

How to repeat a background image only horizontally

How to position a background image

A fixed background image (this image will not scroll with the rest of the page)

All the background properties in one declaration

Advanced background example

**CSS background properties explained**

## CSS Borders

Set the width of the four borders

Set the width of the top border

Set the width of the bottom border

Set the width of the left border

Set the width of the right border

Set the style of the four borders

Set the style of the top border

Set the style of the bottom border

Set the style of the left border

Set the style of the right border

Set the color of the four borders

Set the color of the top border

Set the color of the bottom border

Set the color of the left border

Set the color of the right border

All the border properties in one declaration

Add rounded borders to an element

Set different borders on each side

All the top border properties in one declaration

All the bottom border properties in one declaration

All the left border properties in one declaration

All the right border properties in one declaration

**CSS border properties explained**

## CSS Margins

Specify different margins for each side of an element

Use shorthand margin property with four values

Use shorthand margin property with three values

Use shorthand margin property with two values

Use shorthand margin property with one value

Set margin to auto to center the element within its container

Let the left margin be inherited from the parent element

Margin collapse

**Margin properties explained**

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## CSS Padding

Specify different padding for each side of an element

Use shorthand padding property with four values

Use shorthand padding property with three values

Use shorthand padding property with two values

Use shorthand padding property with one value

Padding and element width (without box-sizing)

Padding and element width (with box-sizing)

Set padding-left of an element

Set padding-right of an element

Set padding-top of an element

Set padding-bottom of an element

**CSS padding properties explained**

## CSS Height/Width

Set the height and width of an element

Set max-width of an element

Set the height and width of different elements

Set the height and width of an image using percent

Set min-width and max-width of an element

Set min-height and max-height of an element

**CSS height/width properties explained**

## CSS Box Model

Demonstrating the box model

Calculate the total width

**CSS box model explained**

## CSS Outline

Draw a line around an element (outline)

Set the style of an outline

Set the color of an outline

Set the width of an outline

Use the shorthand outline property

Outline with rounded corners

Using outline-offset

Using outline-offset with negative value

**CSS outline properties explained**

## CSS Text

- Set the text color of different elements
- Set the text color and background color of different elements
- Align the text
- Using text-align: justify
- Using text-align-last
- Vertical alignment of an image inside text
- Set the text direction of an element
- Using text-decoration-line
- Using text-decoration-color
- Using text-decoration-style
- Using text-decoration-thickness
- Using the text-decoration shorthand property
- Using text-decoration: none
- Using text-transform
- Using text-indent
- Using letter-spacing
- Using line-height
- Using word-spacing
- Using white-space
- Add a text shadow for an element
- Add a colored text shadow for an element
- Add a colored text shadow with blur effect
- Text shadow on white text
- Text shadow with red neon glow
- Text shadow with red and blue neon glow
- Another text shadow effect

**CSS text properties explained**

## CSS Fonts

- Set different fonts for a text
- Using fallback font
- Using the font-style property
- Using the font-weight property
- Set the font-variant property
- Set font size in px
- Set font size in em
- Set font size in rem
- Set font size in vw
- Using the font shorthand property

**Font properties explained**

## CSS Icons

- Font Awesome icons
- Bootstrap icons
- Google icons

**CSS icons explained**

## CSS Links

- Style a link
- Style a link depending on state
- Style a link with text-decoration
- Style a link with background-color
- Create a link button
- Create a link button with borders
- Add other styles to links
- Different types of cursors

CSS link properties explained

## CSS Lists

- All the different list item markers in lists
- Set an image as the list-item marker
- Position the list-item marker
- Remove default list settings
- All list properties in one declaration
- Style lists with colors
- Full-width bordered list

CSS list properties explained

## CSS Tables

- Specify a black border for table, th, and td elements
- Specify a green border for table, th, and td elements
- Using border-collapse
- Use padding in table
- Using border-spacing
- Using border-spacing
- Single border around the table
- Full-width table
- Half-width table
- Fixed-width table
- Auto width table
- Set height of table headers
- Center-align the content of td elements
- Left-align the content in th
- Vertical alignment of content
- Horizontal dividers for a table

- Highlight table rows on mouse over
- Zebra-striped table
- Colored table header
- A responsive table
- Using caption-side
- Create a fancy table

**CSS table properties explained**

## CSS Display

- Using visibility:hidden
- Using display:none
- Difference between display:none and visiblity: hidden
- All different display values
- How to display a block-level element as an inline element
- How to display an inline element as a block-level element
- How to use CSS and JavaScript to show a hidden element
- How to use CSS and JavaScript to toggle between show and hide

**CSS display properties explained**

## CSS Positioning

- Using position: static
- Using position: relative
- Using position: fixed
- Using position: absolute
- Using position: sticky
- Position image text (top left corner)
- Position image text (top right corner)
- Position image text (bottom left corner)

Position image text (bottom right corner)

Position image text (centered)

Using positioning and z-index

Using z-index to set stack order

Using no z-index

**CSS positioning properties explained**

## CSS Overflow

Using the overflow property

Using overflow: visible

Using overflow: hidden

Using overflow: scroll

Using overflow: auto

Using overflow-x and overflow-y.

**CSS overflow properties explained**

## CSS Floating

A simple use of the float property

An image with border and margins that floats to the right in a paragraph

An image with a caption that floats to the right

Let the first letter of a paragraph float to the left

Turning off float (using the clear property)

Turning off float (using the "clearfix" hack)

Create floating boxes

Create side-by-side images

Create equal-height boxes (with flexbox)

Creating a horizontal menu

**CSS float properties explained**

## CSS Inline-block

- Show the differences between inline, inline-block and block
- Using inline-block to create navigation links

**CSS inline-block explained**

## CSS Aligning Elements

- Center align with margin
- Center aligning text
- Center an image
- Center align with flexbox
- Center align with grid
- Left/Right aligning with position
- Left/Right aligning with float
- Center vertically and horizontally with position

**CSS align properties explained**

## CSS Combinators

- Descendant selector
- Child selector
- Adjacent Sibling selector
- General Sibling selector

**CSS combinators explained**

## CSS Pseudo-classes

- Display links in different colors depending on state
- Add other styles to hyperlinks
- Using :hover on div

Use of :focus

Combine pseudo-classes and HTML classes

Use :hover to show tooltip

:first-child - style all first child p elements

:first-child - style the first em element in all p elements

:first-child - style all em elements in all first child p elements

Use of :lang

**CSS pseudo-classes explained**

## CSS Pseudo-elements

Make the first letter special in a text

Make the first line special in a text

Make the first letter and first line special

Use :before to insert some content before an element

Use :after to insert some content after an element

Use :marker to style list item markers

Use :selection to style user-selected text

Use :backdrop to style the viewbox behind a dialog box

**CSS pseudo-elements explained**

## CSS Opacity

Creating transparent images

Creating transparent images - mouseover effect

Reversed mouseover effect for transparent images

Transparent boxes with opacity

Opacity vs background RGBA

Text in transparent box

**CSS image opacity explained**

## CSS Navigation Bars

- Vertical navbar - fully styled
- Horizontal navbar - fully styled
- Horizontal navbar using float
- Horizontal navbar using flex
- Vertical navbar - full-height and fixed
- Horizontal navbar - fixed top
- Sticky navigation bar
- Responsive top navigation
- Responsive side navigation

**CSS navigation bars explained**

## CSS Dropdowns

- Dropdown text
- Dropdown menu
- Right-aligned dropdown menu
- Dropdown image
- Dropdown navigation bar

**CSS dropdowns explained**

## CSS Image Gallery

- Image gallery
- Responsive Image gallery

**CSS image gallery explained**

## CSS Image Sprites

- An image sprite

An image sprite - a navigation list

An image sprite with hover effect

**CSS image sprites explained**

## CSS Attribute Selectors

Selects all <a> elements with a target attribute

Selects all <a> elements with a target="_blank" attribute

Selects all elements with a title attribute that contains a space-separated list of words, one of which is "flower"

Selects all elements with a class attribute value that begins with "top" (must be whole word)

Selects all elements with a class attribute value that begins with "test"

Selects all elements with a class attribute value that ends with "test"

Selects all elements with a class attribute value that contains "te"

**CSS attribute selectors explained**

## CSS Forms

Full-width input field

Padded input field

Bordered input field

Bottom bordered input field

Colored input fields

Focused input fields

Focused input fields 2

Input with icon/image

Animated search input

Styling textareas

Styling select menus

Styling input buttons

Responsive form

CSS forms explained

## CSS Counters

- Create a counter
- Decrease a counter by 1
- Increase a counter by 2
- Using two counters
- Using the counters() function

CSS counters explained

## CSS Website Layout

- Website layout - header
- Website layout - Navbar
- Website layout - 3 column to 1 column
- Website layout - Basic footer
- Website layout - Fixed footer
- Responsive Website Example

CSS website layout explained

## CSS Rounded Corners

- Add rounded corners to elements
- Round each corner separately
- Create elliptical corners

CSS rounded corners explained

## CSS Border Images

- Create an image border around an element, using the round keyword
- Create an image border around an element, using the stretch keyword

Image border - Different slice values

CSS border images explained

## CSS Backgrounds

Add multiple background images for an element

Specify the size of a background image

Scale a background image using "contain" and "cover"

Define sizes of multiple background images

Full-size background image

Background hero image

Use background-origin to specify where the background image is positioned

Use background-clip with a background image

Use background-clip with a background color

CSS backgrounds explained

## CSS Colors

Different RGBA colors

Different HSLA colors

Different colors with opacity

CSS colors explained

## CSS Color Keywords

Using the transparent keyword

Using the currentcolor keyword

Using the currentcolor keyword

Using the currentcolor keyword

Using the inherit keyword

CSS color keywords explained

## CSS Gradients

- Linear Gradient - top to bottom
- Linear Gradient - bottom to top
- Linear Gradient - left to right
- Linear Gradient - diagonal
- Linear Gradient - with a specified angle
- Linear Gradient - with multiple color stops
- Linear Gradient - color of a rainbow + text
- Linear Gradient - with transparency
- Linear Gradient - a repeating linear gradient
- Radial Gradient - evenly spaced color stops
- Radial Gradient - differently spaced color stops
- Radial Gradient - set shape
- Radial Gradient - different size keywords
- Radial Gradient - a repeating radial gradient
- Conic Gradient - Three colors
- Conic Gradient - Five colors
- Conic Gradient - Three colors and degrees
- Conic Gradient - Pie chart
- Conic Gradient - Pie chart 2
- Conic Gradient - With a from angle
- Conic Gradient - With a center position
- Conic Gradient - Repeating a conic gradient
- Conic Gradient - Repeating a conic gradient 2

CSS gradients explained

## CSS Shadow Effects

Simple shadow effect

Add a color to the shadow

Add a blur effect to the shadow

White text with black shadow

A red neon glow shadow

A red and blue neon glow shadow

White text with black, blue, and darkblue shadow

Border around text

Add a simple box-shadow to an element

Add color to box-shadow

Add color and blur effect to box-shadow

Add color, blur, and spread effect to box-shadow

Using the inset keyword in box-shadow

Add multiple shadows for an element

Create paper-like cards (text)

Create paper-like cards (polaroid images)

**CSS shadow effects explained**

## CSS Text Effects

Specify how hidden, overflowed content should be signaled to the user

How to display the overflowed content when hover over the element

Allow long words to be able to be broken and wrap onto the next line

Specify whether lines of text are laid out horizontally or vertically

**CSS text explained**

## CSS Custom Fonts

Use your "own" fonts in @font-face rule

Use your "own" fonts in @font-face rule (bold)

CSS custom fonts explained

## CSS 2D Transforms

translate() - move an element from its current position

rotate() - rotate an element clockwise

rotate() - rotate an element counter-clockwise

scale() - increase an element

scale() - decrease an element

skewX() - skews an element along the X-axis

skewY() - skews an element along the Y-axis

skew() - skews an element along the X and Y-axis

matrix() - rotate, scale, move, and skew an element

CSS 2D transforms explained

## CSS 3D Transforms

rotateX() - rotate an element around its X-axis at a given degree

rotateY() - rotate an element around its Y-axis at a given degree

rotateZ() - rotate an element around its Z-axis at a given degree

CSS 3D transforms explained

## CSS Transitions

Transition - change width of an element

Transition - change width and height of an element

Specify different speed curves for a transition

Specify a delay for a transition effect

Add a transformation to a transition effect

Specify all transition properties in one shorthand property

CSS transitions explained

## CSS Animations

- Bind an animation to an element
- Animation - change background-color of an element
- Animation - change background-color and position of an element
- Delay an animation
- Run animation 3 times before it stops
- Run animation for ever
- Run animation in reverse direction
- Run animation in alternate cycles
- Speed curves for animations
- Animation shorthand property

**CSS animations explained**

## CSS Tooltips

- Right tooltip
- Left tooltip
- Top tooltip
- Bottom tooltip
- Tooltip with arrow
- Animated tooltip

**CSS tooltips explained**

## CSS Style Images

- Rounded image
- Circled image
- Thumbnail image
- Thumbnail image as link
- Responsive image

Polaroid images

Image with opacity

Image text (top left corner)

Image text (top right corner)

Image text (bottom left corner)

Image text (bottom right corner)

Image text (centered)

Image hover - Fade in text

Image hover - Fade in a box

Image hover - Slide in from top

Image hover - Slide in from bottom

Image hover - Slide in from left

Image hover - Slide in from right

Flip an image on hover

Responsive Image Gallery

**CSS styling images explained**

## CSS Image Modal

Responsive modal image gallery

**CSS image modal explained**

## CSS Image Centering

Center image horizontally with margin: auto

Center image horizontally with display: flex

Center image both vertically and horizonatlly with display: flex

Center image both vertically and horizonatlly with display: grid

**CSS image centering explained**

## CSS Image Filters

- Filter blur()
- Filter brightness()
- Filter contrast()
- Filter drop-shadow()
- Filter grayscale()
- Filter hue-rotate()
- Filter invert()
- Filter opacity()
- Filter saturate()
- Filter sepia()

**CSS image filters explained**

## CSS Image Shapes

- Clip an image to a circle
- Clip an image to a circle and position the center of circle to the right
- Circle() and shape-outside
- Clip an image to an ellipse
- Clip an image to an ellipse and position the center of ellipse to the right
- Ellipse() and shape-outside
- Clip an image to a polygon

**CSS image shapes explained**

## CSS Object-fit

- Use object-fit: cover
- Use object-fit: contain
- Use object-fit: fill
- Use object-fit: none

Use object-fit: scale-down

**CSS object-fit explained**

## CSS Object-position

Position the image so that the great old building is in center

Position the image so that the famous Eiffel Tower is in center

**CSS object-position explained**

## CSS Masking

Using the mask-image property

Omit the mask-repeat property

Position the mask image

A linear gradient mask layer

A linear-gradient along with text masking

A radial gradient mask layer (circle)

A radial gradient mask layer (ellipse)

A conic gradient mask layer

An SVG mask layer (circle)

An SVG mask layer (ellipse)

An SVG mask layer (triangle)

An SVG mask layer (star)

An SVG mask layer (multiple circles)

**CSS masking explained**

## CSS Buttons

Basic CSS buttons

Button colors

Button sizes

- Rounded buttons
- Colored button borders
- Hoverable buttons
- Shadow buttons
- Disabled buttons
- Button width
- Button groups
- Bordered button group
- Animated Button (Hover Effect)
- Animated Button (Ripple Effect)
- Animated Button (Pressed Effect)

**CSS buttons explained**

## CSS Pagination

- A Simple pagination
- Pagination with an active class
- Pagination with a disabled class
- Pagination with hover effect
- Pagination with transition effect on hover
- Breadcrumb Pagination

**CSS pagination explained**

## CSS Multiple Columns

- Create multiple columns
- Specify the gap between columns
- Specify the style of the rule between columns
- Specify the width of the rule between columns
- Specify the color of the rule between columns

Specify the width, style and color of the rule between columns

Specify how many columns an element should span across

Specify a suggested, optimal width for the columns

**CSS multiple columns explained**

## CSS User Interface

Let a user resize the width of an element

Let a user resize the height of an element

Let a user resize both the width and height of an element

Add space between an outline and the border of an element

**CSS user interface explained**

## CSS Variables

Using the var() function

Override global variable with local variable

Add new local variable

Get and change variable with JavaScript

Change variables in media queries

**CSS variables explained**

## CSS @property Rule

Simple use of @property

Override the custom property

Type checking and fallback value of @property

Inherit value (fale) of @property

Inherit value (true) of @property

Use @property to animate a gradient

CSS @property explained

## CSS Box Sizing

Width of elements without box-sizing

Width of elements with box-sizing

Form elements + box-sizing

CSS box sizing explained

## CSS Media Queries

Change the background-color to lightgreen if the viewport is 480px wide or wider

Show a menu that will float to the left of the page if the viewport is 480px wide or wider

CSS media queries explained

## CSS Media Queries - More Examples

Set different background colors depending on screen width

Responsive columns using flexbox

Responsive navigation menu

Hide elements with media queries

Responsive font size

Change background-color if screen orientation is landscape

Respect user preferences with media query

Responsive image gallery

Responsive website

CSS media queries examples explained

## CSS Flexbox

Flexbox with three flex items

flex-direction - row

flex-direction - column

flex-direction - row-reverse

flex-direction - column-reverse

flex-wrap - nowrap

flex-wrap - wrap

flex-wrap - wrap-reverse

flex-flow

justify-content - center

justify-content - flex-start

justify-content - flex-end

justify-content - space-around

justify-content - space-between

justify-content - space-evenly

align-items - center

align-items - flex-start

align-items - flex-end

align-items - stretch

align-items - baseline

align-content - center

align-content - flex-start

align-content - flex-end

align-content - space-between

align-content - space-around

align-content - space-evenly

Perfect centering

**CSS flexbox explained**

## CSS Flex Items

Order the flex items

flex-grow property

flex-shrink property

flex-basis property

flex property

flex-basis property

align-self - center

align-self - flex-start and flex-end

**CSS flex items explained**

## CSS Flexbox Responsive

Responsive flexbox

Responsive flexbox 2

Responsive image gallery with flexbox

Create a responsive website with flexbox

**CSS flexbox responsive explained**

## CSS Grid Intro

Grid layout

Grid container and grid items

**CSS grid intro explained**

## CSS Grid Container

Grid container

display: grid

display: inline-grid

**CSS grid container explained**

## CSS Grid Tracks

grid-template-columns: auto

grid-template-columns mixed

grid-template-columns: fr unit

grid-template-columns fr unit 2

grid-template-columns: repeat()

grid-template-columns: minmax()

The grid-template-rows property

**CSS grid tracks explained**

## CSS Grid Gaps

column-gap property

row-gap property

gap property - One value

gap property - Two values

**CSS grid gaps explained**

## CSS Grid Align

justify-content: space-evenly

justify-content: space-around

justify-content: space-between

justify-content: center

justify-content: start

justify-content: end

align-content: center

align-content: space-evenly

align-content: space-around

align-content: space-between

align-content: start

align-content: end

place-content: center

place-content: end space-between

**CSS grid align explained**

## CSS Grid Spanning Items

A five items grid layout

The grid-column-start and grid-column-end properties

The grid-column property

The grid-row-start and grid-row-end properties

The grid-row property

**CSS grid item explained**

## CSS Grid Naming Items

Using grid-area to name grid items 1

Using grid-area to name grid items 2

Using grid-area to name grid items 3

Using grid-area to name grid items 4

**CSS grid naming items explained**

## CSS Grid Aligning Items

The justify-self property

The align-self property

**CSS grid align items explained**

## CSS Grid Order Items

Set the visual order of the grid items

CSS grid order items explained

## CSS 12-column Grid Layout

A responsive 12-column Grid layout

CSS 12-column grid layout explained

## CSS @supports Rule

@supports - fallback styling for flex

@supports - fallback styling for grid

CSS @supports rule explained

## CSS Responsive Webdesign

Responsive grid view

Responsive grid view with breakpoints

Typical breakpoints

Media query orientation: portrait / landscape

Hide elements on different screen sizes

Change font size on different screen sizes

Respect user preferences

Responsive image: width property

Responsive image: max-width property

Responsive image on responsive webpage

Background image and background-size: contain

Background image and background-size: 100% 100%

Background image and background-size: cover

Change background image depending on browser width

Change background image with HTML picture element

Responsive video: width property

| |
|---|
| Responsive video: max-width property |
| Responsive video on responsive webpage |
| Responsive framework: W3.CSS |
| Responsive framework: Bootstrap |

CSS responsive webdesign explained

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# JavaScript Examples

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## What can JavaScript do?

SHARE

- JavaScript can change HTML content
- JavaScript can change HTML attributes
- JavaScript can change CSS style
- JavaScript can hide HTML elements
- JavaScript can show hidden HTML elements

Examples Explained

## Where to Insert JavaScript

- JavaScript in <head>
- JavaScript in <body>
- JavaScript in an external file
- JavaScript in an external url
- JavaScript in an external folder

Where to Explained

# JavaScript Output

- Writing into the HTML output
- Writing into an HTML element
- Writing into an window alert box
- Writing into the browser console

**Output Explained**

# JavaScript Syntax

- JavaScript statements
- JavaScript numbers
- JavaScript strings
- JavaScript variables
- JavaScript operators
- JavaScript assignment
- JavaScript expressions (using constants)
- JavaScript expressions (using strings)
- JavaScript expressions (using variables)
- JavaScript keywords
- JavaScript comments
- JavaScript is case sensitive

**Syntax Explained**

REMOVE ADS

# JavaScript Statements

- JavaScript statements are commands to the browser
- JavaScript code is a sequence of statements

JavaScript statements are separated with semicolon

Multiple statement on one line is allowed

JavaScript statements can be grouped together in code blocks

You can break a code line after an operator or a comma.

**Statements Explained**

## JavaScript Comments

Single line comments

Single line comments at the end of a line

Multiple lines comments

Single line comment to prevent execution

Multiple lines comment to prevent execution

**Comments Explained**

## JavaScript Variables

JavaScript variables

JavaScript variables as algebra

JavaScript numbers and strings

JavaScript var keyword.

Declaring many variables in one statement

Declaring many variables multiline

A variable without a value returns the value undefined

Re-declaring a variable will not destroy the value

Adding JavaScript numbers

Adding JavaScript strings

Adding strings and numbers

**Variables Explained**

# JavaScript Arithmetic

| |
|---|
| The addition (+) operator |
| The subtraction (-) operator |
| The multiplication (*) operator |
| The division (/) operator |
| The modulus (%) operator |
| The increment (++) operator |
| The decrement (--) operator |

Arithmetic Explained

# JavaScript Assignment

| |
|---|
| The = assignment operator |
| The += assignment operator |
| The -= assignment operator |
| The *= assignment operator |
| The /= assignment operator |
| The %= assignment operator |

Assignment Explained

# JavaScript String Concatenation

| |
|---|
| Adding two strings together using the concatenating (+) operator |
| Adding two strings together with a space in the first string |
| Adding two strings together with a space in between |
| Adding two strings together using the += operator |
| Adding strings and numbers |

Concatenation Explained

# JavaScript Data Types

- Declare (create) strings
- Declare (create) numbers
- Declare (create) an array
- Declare (create) an object
- Find the type of a variable
- Adding two numbers and a string
- Adding a string and two numbers
- An undefined variable
- An empty variable

**Data types Explained**

# JavaScript Objects

- Create a JavaScript variable
- Create a JavaScript object
- Create a person object (single line)
- Create a person object (multiple lines)
- Access object properties using .property
- Access object properties using [property]
- Access a function property as a method
- Access a function property as a property

**Objects Explained**

# JavaScript Functions

- A simple function
- A function with an argument
- A function with an argument 2

A function that returns a value

A function that converts Fahrenheit to Celsius

A function call without ()

**Functions Explained**

## JavaScript Events

An onclick event changes an HTML element

An onclick event changes its own element

An onclick event calls a function

**Events Explained**

## JavaScript Strings

Strings can be written with single or double quotes.

Show some string examples

Backslash before quotes accepts quotes as quotes.

Find the length of a string

You can break text string with a backslash.

You cannot break code with a backslash.

Find the position of the first occurrence of a text in a string - indexOf()

Search for a text in a string and return the text if found - match()

Replace characters in a string - replace()

Convert string to upper case - toUpperCase()

Convert string to lower case - toLowerCase()

Split a string into an array - split()

**Strings Explained**

## JavaScript Numbers

| |
|---|
| Numbers can be written with or without decimals |
| Extra large or extra small numbers can be written with exponent notation |
| Number are considered accurate only up to 15 digits |
| Floating point arithmetic is not always 100% accurate |
| But it helps to multiply and divide by 10 |
| Adding two numbers results in a new number |
| Adding two numeric strings results in a concatenated string |
| Adding a number and a numeric string also results in a concatenated string |
| Adding a numeric string and a number also results in a concatenated string |
| Common mistake when adding strings and numbers 1 |
| Common mistake when adding strings and numbers 2 |
| JavaScript will try to convert strings to numbers when dividing |
| JavaScript will try to convert strings to numbers when multiplying |
| JavaScript will try to convert strings to numbers when subtracting |
| JavaScript will NOT convert strings to numbers when adding |
| A number divided by a string is NaN (Not a Number) |
| A number divided by a numeric string is a number |
| The global JavaScript function isNaN() returns if a value is a number |
| Using NaN in a mathematical operation will always return NaN |
| Using NaN in a mathematical string operation will concatenate NaN |
| NaN (Not a Number) is a number (Yes! typeof NaN returns number) |
| Infinity is returned if you calculate a number outside the largest possible number |
| Division by zero also generates Infinity |
| Infinity is a number (typeof Infinity returns number) |
| Constants, preceded by 0x, are interpreted as hexadecimal |
| The toString() method can output numbers as hex, octal, and binary |
| Numbers can be objects |
| Numbers and objects cannot be safely compared |
| Objects and objects cannot be safely compared |

Numbers Explained

# JavaScript Number Methods

| |
|---|
| The toString() method converts a number to a string |
| The valueOf() method returns a number as a number |
| The toExponential() returns a number with exponential notation |
| The toFixed() method rounds a number to a number of digits |
| The toPrecision() method a number written with a specified length |
| The global method Number() converts variables to numbers |
| The global method Number() can even convert dates to numbers |
| The global method parseInt() converts strings to numbers |
| The global method parseFloat() converts strings to numbers |
| MAX_VALUE returns the largest possible number in JavaScript |
| MIN_VALUE returns the smallest possible number in JavaScript |
| POSITIVE_INFINITY represents infinity |
| POSITIVE_INFINITY is returned on overflow |
| NEGATIVE_INFINITY represents negative infinity |
| NEGATIVE_INFINITY is returned on overflow |
| NaN Represents "Not-a-Number" |
| Arithmetic performed on a string will result in NaN |
| Using a Number property on a variable will return undefined |

Numbers Methods Explained

# JavaScript Maths

| |
|---|
| Math.PI returns the value of PI |
| Math.round(x) returns the rounded value of x |
| Math.pow(x, y) returns the value of x to the power of y |
| Math.sqrt(x) returns the square root of x |

Math.abs(x) returns the absolute (positive) value of x

Math.ceil(x) returns the value of x rounded up

Math.floor(x) returns the value of x rounded down

Math.sin(x) returns the sin of the angle x (given in radians)

Math.cos(x) returns the cosin of the angle x (given in radians)

Math.max() return the number with the highest value from a list of arguments

Math.min() to return the number with the lowest value from a list of arguments

Converting Celsius to Fahrenheit

**Maths Explained**

# JavaScript Random

Math.random() returns a random number between 0 (included) and 1 (excluded)

How to return a random integer between 0 and 9 (both included)

How to return a random integer between 0 and 10 (both included)

How to return a random integer between 0 and 99 (both included)

How to return a random integer between 0 and 100 (both included)

How to return a random integer between 1 and 10 (both included)

How to return a random integer between 1 and 100 (both included)

How to return a random integer between x (included) and y (excluded)

How to return a random integer between x and y (both included)

**Random Explained**

# JavaScript Dates

Use Date() to display today's date and time

Use getFullYear() display the year

Use getTime() to calculate the number of milliseconds since 1970

Use setFullYear() to set a specific date

Use toUTCString() to convert today's date (according to UTC) to a string

Use getDay() to display the weekday as a number

Use getDay() and an array to display the weekday as a name

Display a clock

**Dates Explained**

# JavaScript Arrays

Create an array I

Create an array II

Access an array element

Change an array element

Access a full array

Find the length of an array

Loop through an array

Add an element to an array

Add undefined "holes" to an array

How to recognize an array I

How to recognize an array II

**Arrays Explained**

# JavaScript Array Methods

Add an element to an array

Remove the last element of an array - pop()

Join all elements of an array into a string - join()

Join two arrays - concat()

Join three arrays - concat()

Add an element to position 2 in an array - splice()

Convert an array to a string - toString()

Add new elements to the beginning of an array - unshift()

Remove the first element of an array - shift()

Select elements from an array - slice()

**Array Methods Explained**

# JavaScript Array Sort

Sort an array in ascending order

Sort an array in descending order

Sort an array of numbers ascending

Sort an array of numbers descending

Sort numbers (alphabetically or numerically)

Sort array numbers in random order

Find the lowest number in an array

Find the highest number in an array

Find the lowest number in an array using Math.min()

Find the highest number in an array using Math.max()

Using a "homemade" myArrayMin method

Using a "homemade" myArrayMax method

Sort objects by numeric properties

Sort objects by string properties

**Array Sort Explained**

# JavaScript Array Iteration

Array.forEach()

Array.map()

Array.filter()

Array.reduce()

Array.reduceRight()

Array.every()

Array.some()

Array.indexOf()

Array.lastIndexOf()

Array.find()

Array.findIndex()

**Array Iteration Explained**

# JavaScript Type Conversion

Display the typeof all variable types

Display the constructor of all variable types

Convert a number to a string using String()

Convert a number to a string using toString()

Find out if a variable is an array

Find out if a variable is a date

**Type Conversion Explained**

# JavaScript Booleans

Display the value of Boolean(10 > 9)

Display the value of 10 > 9

Everything with a real value is true

The Boolean value of zero is false

The Boolean value of minus zero is false

The Boolean value of an empty string is false

The Boolean value of undefined is false

The Boolean value of null is false

The Boolean value of false is false

The Boolean value of NaN is false

Booleans Explained

## JavaScript Comparisons

- Assign 5 to x, and display the value of (x == 8)
- Assign 5 to x, and display the value of (x == 5)
- Assign 5 to x, and display the value of (x === 5)
- Assign 5 to x, and display the value of (x === "5")
- Assign 5 to x, and display the value of (x != 8)
- Assign 5 to x, and display the value of (x !== 5)
- Assign 5 to x, and display the value of (x !== "5")
- Assign 5 to x, and display the value of (x > 8)
- Assign 5 to x, and display the value of (x < 8)
- Assign 5 to x, and display the value of (x >= 8)
- Assign 5 to x, and display the value of (x <= 8)

Comparisons Explained

## JavaScript Conditionals

- The if statement
- The else statement
- The else if statement
- Random link
- Switch statement

Conditionals Explained

## JavaScript Loops

- For loop
- Looping an Array

Looping through HTML headers

While loop

Do While loop

Break a loop

Break and continue a loop

Use a for...in statement to loop through the elements of an object

**Loops Explained**

## JavaScript Error Handling

The try...catch statement

The try...catch statement with a confirm box

The onerror event

**Errors Explained**

## JavaScript Regular Expressions

Search for an expression in a string

Search for an expression and replace it

**Regular Expressions Explained**

## JavaScript Objects

Creating a JavaScript variable

Creating a JavaScript object

Creating a JavaScript object (single line)

Creating a JavaScript object (multiple lines)

Creating a JavaScript object using new

Creating JavaScript objects using a constructor

Creating built-in JavaScript objects

The best way to create JavaScript variables

JavaScript objects are mutable

**Objects Explained**

## JavaScript Object Properties

Accessing properties using .property

Accessing properties using [property]

Accessing properties using for in

Adding new properties to existing objects

Deleting properties from objects

**Object Properties Explained**

## JSON Objects

Accessing properties using .property

Accessing properties using [property]

Looping through properties

Looping through property values

Access nested JSON objects

Modify values using the dot notation

Modify values using the bracket notation

Delete object properties

**JSON Object Properties Explained**

## JSON Arrays

Accessing array values

Looping through an array using for-in

Looping through an array using for

Access nested JSON arrays

Modify array values

Delete array items

**JSON Arrays Explained**

## JSON Parse

Use JSON parse

Using JSON parse in an AJAX example

Using JSON parse on an array

Parsing dates

Parsing dates using the reviver function

Parsing functions

**JSON Parse Explained**

## JSON Stringify

Use JSON stringify

Using JSON stringify on an array

Stringify dates

Stringify functions

Stringify functions using the toString() method

**JSON Stringify Explained**

## JSON PHP

Get JSON from a php file

Get JSON array from php

Get JSON from a database

Loop through the result from a database

Send JSON using POST method

JSON PHP Explained

## JSON HTML

Make an HTML table based on JSON data

Make a dynamic HTML Table

Make an HTML drop down list based on JSON data

JSON HTML Explained

## JSON JSONP

Simple JSONP example

Create a dynamic script tag

JSONP example with dynamic result

JSONP example with a callback function

JSON JSONP Explained

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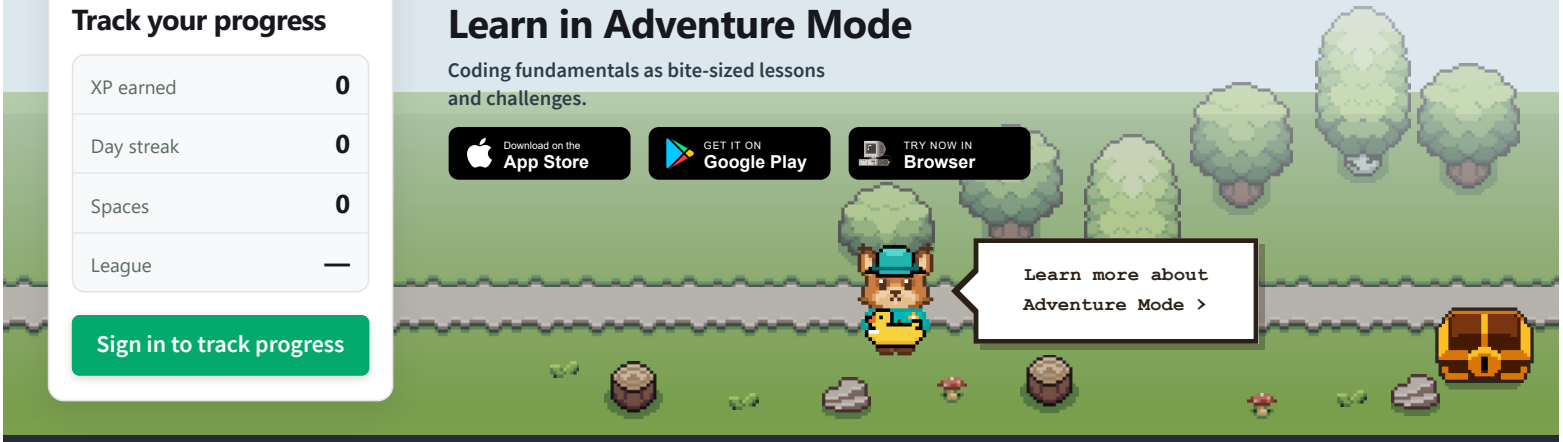
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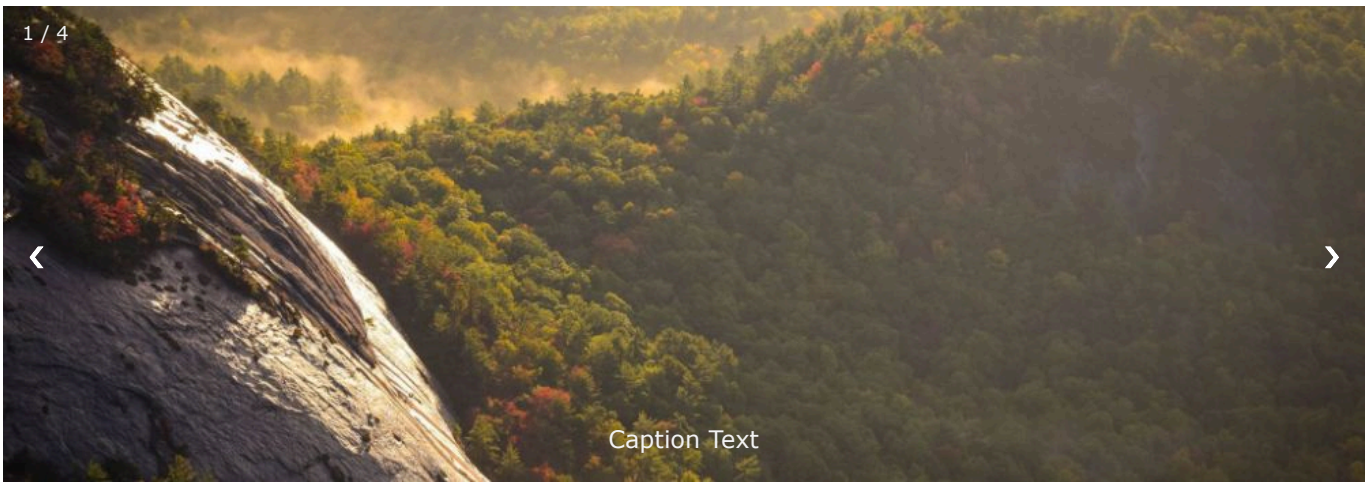
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Code snippets for HTML, CSS and JavaScript.

## Slideshow




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## Login Form

Login

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## Accordion

Section 1 −

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Section 2 +

Section 3 +

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## Hover Dropdowns

Hover

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## Click Dropdowns

Click

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## Side Navigation

Sidenav overlay

Sidenav w/opacity

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## Fixed Sidebar

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## Modal Box

Open Modal

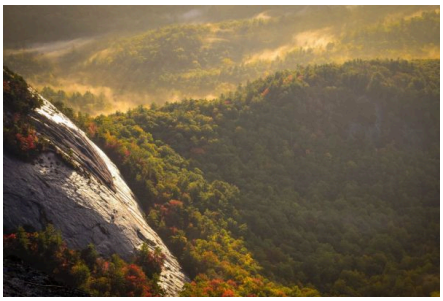
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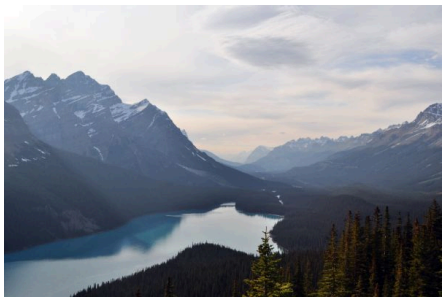
# Modal Image



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# Lightbox





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# Slideshow Gallery






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## Image Hover Overlay



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## Tabs

London Paris Tokyo

### London

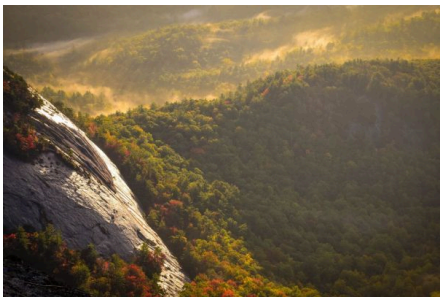
London is the capital city of England.

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## Vertical Tabs

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## Tab Image Gallery








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## Alert Buttons

Success Info Warning Danger Default

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## Outline Buttons

Success Info Warning Danger Default

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## Top Navigation

Home News Contact

Learn How To »

# Responsive Top Navigation

Resize the browser window to see the effect:

Learn How To »

# Fullscreen Navigation

Slide Right Slide Down

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# Login Form in Navbar

Home About Contact Username Password Login

Learn How To »

# Change Background on Scroll

Learn How To »

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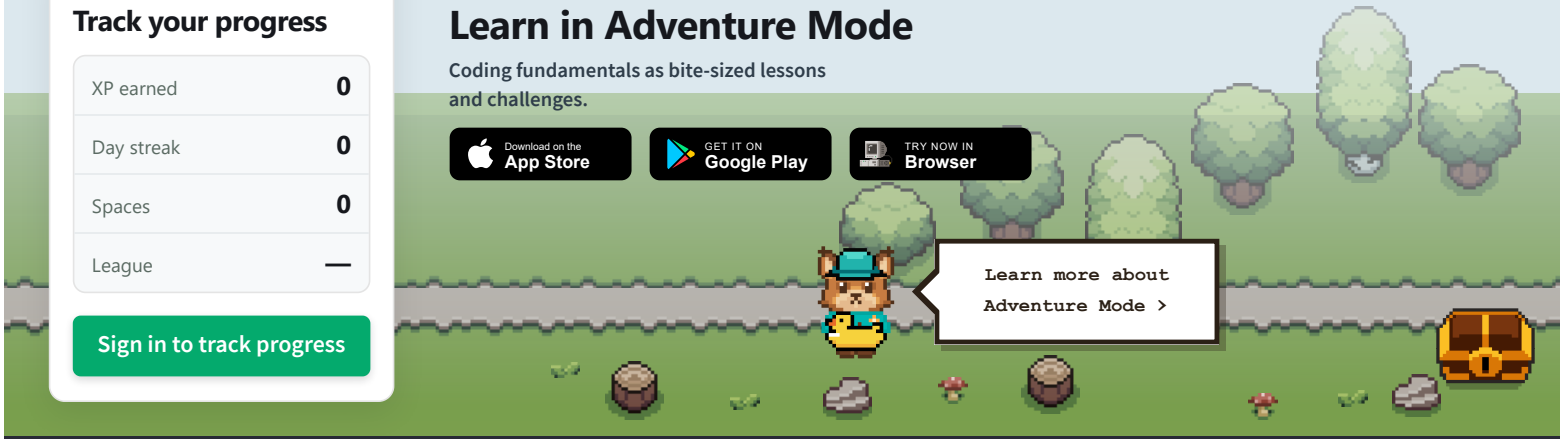

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# SQL Examples

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## SQL Examples

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This page covers all the SQL examples from the SQL tutorial. Each example opens in our SQL editor, showing the SQL command and the result.

## SQL Syntax

Select all the records from a specific table ("Customers")

Example Explained

## SQL SELECT

SELECT Column

SELECT *

Examples Explained

## SQL SELECT DISTINCT

SELECT DISTINCT

Examples Explained

## SQL WHERE

WHERE Clause

Text Fields vs. Numeric Fields

Examples Explained

## SQL AND, OR and NOT Operators

AND

OR

NOT

Combining AND, OR and NOT

Examples Explained

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## SQL ORDER BY

ORDER BY

ORDER BY DESC

ORDER BY Several Columns

Examples Explained

## SQL NULL Values

IS NULL Operator

IS NOT NULL Operator

Examples Explained

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## SQL SELECT TOP

| |
|---|
| SELECT TOP |
| LIMIT |
| SELECT TOP PERCENT |
| SELECT TOP and add a WHERE Clause |

Examples Explained

## SQL MIN() and MAX()

| |
|---|
| MIN() |
| MAX() |

Examples Explained

## SQL COUNT, AVG() and SUM()

| |
|---|
| COUNT() |
| AVG() |
| SUM() |

Examples Explained

## SQL LIKE

LIKE - select all table rows starting with "a"

LIKE - select all table rows ending with "a"

LIKE - select all table rows that have "or" in any position

LIKE - select all table rows that have "r" in the second position

LIKE - select all table rows that starts with "a" and ends with "o"

LIKE - select all table rows that does NOT start with "a"

**Examples Explained**

# SQL Wildcards

Using the % Wildcard

Using the _ Wildcard

Using the [charlist] Wildcard

Using the [!charlist] Wildcard

**Examples Explained**

# SQL IN

IN

NOT IN

**Examples Explained**

# SQL BETWEEN

BETWEEN

NOT BETWEEN

BETWEEN with IN

BETWEEN Text Values

NOT BETWEEN Text Values

**Examples Explained**

# SQL Aliases

- Alias for Columns
- Two Aliases
- Alias for Tables

**Examples Explained**

# SQL Joins

- INNER JOIN
- LEFT JOIN
- RIGHT JOIN
- Self JOIN

**Examples Explained**

# SQL UNION

- UNION
- UNION ALL
- UNION With WHERE
- UNION ALL With WHERE

**Examples Explained**

# SQL GROUP BY

- GROUP BY
- GROUP BY and ORDER BY
- GROUP BY With JOIN

**Examples Explained**

# SQL HAVING

HAVING and GROUP BY

HAVING and ORDER BY

**Examples Explained**

# SQL EXISTS

EXISTS

**Example Explained**

# SQL ANY and ALL

ANY

ALL

**Examples Explained**

# SQL CASE

CASE 1

CASE 2

**Examples Explained**

# SQL Comments

Single Line Comments

Single Line Comments At The End Of a Line

Multi-line Comments

**Examples Explained**

# SQL Database

SQL Database tutorials can be found here:

- SQL Create DB
- SQL Drop DB
- SQL Backup DB
- SQL Create Table
- SQL Drop Table
- SQL Alter Table
- SQL Constraints
- SQL Not Null
- SQL Unique
- SQL Primary Key
- SQL Foreign Key
- SQL Check
- SQL Default
- SQL Index
- SQL Auto Increment
- SQL Dates
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# Python Examples

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## Python Syntax

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Print "Hello World"

Comments in Python

Docstrings

Syntax Explained

## Python Variables

Create a variable

Output both text and a variable

Add a variable to another variable

Variables Explained

## Python Numbers

Verify the type of an object

Create integers

Create floating point numbers

Create scientific numbers with an "e" to indicate the power of 10

Create complex numbers

**Numbers Explained**

# Python Casting

Casting - Integers

Casting - Floats

Casting - Strings

**Casting Explained**

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# Python Strings

Get the character at position 1 of a string

Substring. Get the characters from position 2 to position 5 (not included)

Remove whitespace from the beginning or at the end of a string

Return the length of a string

Convert a string to lower case

Convert a string to upper case

Replace a string with another string

Split a string into substrings

**Strings Explained**

# Python Operators

Addition operator

Subtraction operator

Multiplication operator

Division operator

Modulus operator

Assignment operator

Operators Explained

# Python Lists

Create a list

Access list items

Change the value of a list item

Loop through a list

Check if a list item exists

Get the length of a list

Add an item to the end of a list

Add an item at a specified index

Remove an item

Remove the last item

Remove an item at a specified index

Empty a list

Using the list() constructor to make a list

Lists Explained

# Python Tuples

Create a tuple

Access tuple items

Change tuple values

Loop through a tuple

Check if a tuple item exists

Get the length of a tuple

Delete a tuple

Using the tuple() constructor to create a tuple

**Tuples Explained**

# Python Sets

Create a set

Loop through a set

Check if an item exists

Add an item to a set

Add multiple items to a set

Get the length of a set

Remove an item in a set

Remove an item in a set by using the discard() method

Remove the last item in a set by using the pop() method

Empty a set

Delete a set

Using the set() constructor to create a set

**Sets Explained**

# Python Dictionaries

Create a dictionary

Access the items of a dictionary

Change the value of a specific item in a dictionary

Print all key names in a dictionary, one by one

Print all values in a dictionary, one by one

Using the values() function to return values of a dictionary

Loop through both keys an values, by using the items() function

Check if a key exists

Get the length of a dictionary

Add an item to a dictionary

Remove an item from a dictionary

Empty a dictionary

Using the dict() constructor to create a dictionary

**Dictionaries Explained**

# Python If ... Else

The if statement

The elif statement

The else statement

Short hand if

Short hand if ... else

The and keyword

The or keyword

**If ... Else Explained**

# Python While Loop

The while loop

Using the break statement in a while loop

Using the continue statement in a while loop

**While Loop Explained**

# Python For Loop

The for loop

Loop through a string

Using the break statement in a for loop

Using the continue statement in a for loop

Using the range() function in a for loop

Else in for loop

Nested for loop

**For Loop Explained**

# Python Functions

Create and call a function

Function parameters

Default parameter value

Let a function return a value

Recursion

**Functions Explained**

# Python Lambda

A lambda function that adds 10 to the number passed in as an argument

A lambda function that multiplies argument a with argument b

A lambda function that sums argument a, b, and c

**Lambda Explained**

# Python Arrays

Create an array

Access the elements of an array

Change the value of an array element

Get the length of an array

Loop through all elements of an array

Add an element to an array

Remove an element from an array

Arrays Explained

# Python Classes and Objects

Create a class

Create an object

The __init__() Function

Create object methods

The self parameter

Modify object properties

Delete object properties

Delete an object

Classes/Objects Explained

# Python Iterators

Return an iterator from a tuple

Return an iterator from a string

Loop through an iterator

Create an iterator

Stop iteration

Iterators Explained

# Python Modules

Use a module

Variables in module

Re-naming a module

Built-in modules

Using the dir() function

Import from module

**Modules Explained**

# Python Dates

Import the datetime module and display the current date

Return the year and name of weekday

Create a date object

The strftime() Method

**Dates Explained**

# Python Math

Find the lowest and highest value in an iterable

Return the absolute value of a number

Return the value of x to the power of y ($x^y$)

Return the square root of a number

Round a number upwards and downwards to its nearest integer

Return the value of PI

**Math Explained**

# Python JSON

Convert from JSON to Python

Convert from Python to JSON

Convert Python objects into JSON strings

Convert a Python object containing all the legal data types

Use the indent parameter to define the numbers of indents

Use the separators parameter to change the default separator

Use the sort_keys parameter to specify if the result should be sorted or not

**JSON Explained**

# Python RegEx

Search a string to see if it starts with "The" and ends with "Spain"

Using the findall() function

Using the search() function

Using the split() function

Using the sub() function

**RegEx Explained**

# Python PIP

Using a package

**PIP Explained**

# Python Try Except

When an error occurs, print a message

Many exceptions

Use the else keyword to define a block of code to be executed if no errors were raised

Use the finally block to execute code regardless if the try block raises an error or not

**Try Except Explained**

# Python File Handling

Read a file

Read only parts of a file

Read one line of a file

Loop through the lines of a file to read the whole file, line by line

**File Handling Explained**

# Python MySQL

Create a connection to a database

Create a database in MySQL

Check if a database exist

Create a table

Check if a table exist

Create primary key when creating a table

Insert a record in a table

Insert multiple rows

Get inserted ID

Select all records from a table

Select only some of the columns in a table

Use the fetchone() method to fetch only one row in a table

Select with a filter

Wildcards characters

Prevent SQL injection

Sort the result of a table alphabetically

Sort the result in a descending order (reverse alphabetically)

Delete records from an existing table

Prevent SQL injection

Delete an existing table

Delete a table if it exist

Update existing records in a table

Prevent SQL injection

Limit the number of records returned from a query

Combine rows from two or more tables, based on a related column between them

LEFT JOIN

RIGHT JOIN

MySQL Explained

# Python MongoDB

Create a database

Check if a database exist

Create a collection

Check if a collection exist

Insert into collection

Return the id field

Insert multiple documents

Insert multiple documents with specified IDs

Find the first document in the selection

Find all documents in the selection

Find only some fields

Filter the result

Advanced query

Filter with regular expressions

Sort the result alphabetically

Sort the result descending (reverse alphabetically)

Delete document

Delete many documents

Delete all documents in a collection

Delete a collection

Update a document

Update many/all documents

Limit the result

MongoDB Explained

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# W3.CSS Examples

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## W3.CSS Colors

SHARE

| |
|---|
| Background Colors |
| Text Colors |
| Hover Colors |

Examples explained

## W3.CSS Containers

| |
|---|
| Containers |
| Containers with color |
| Container header using  |
| Container header using <header> |
| Container footer using  |
| Container footer using <footer> |
| Container <article> and <section> |
| Container with  elements |

Container with semantic elements

Container padding

Container with headers and paragraphs

Examples explained

# W3.CSS Borders

Borders

Colored borders

Rounded borders

Thick borders

Hoverable borders

Examples explained

# W3.CSS Panels

Panels

Panel notes

Panel quotes

Panel alerts

Panel cards

Panel round

Hide/close a panel

Show/Open a panel

Examples explained

REMOVE ADS

# W3.CSS Cards

Cards

Colored cards

Card content

Photo cards

Hoverable cards

Avatar card with buttons

Avatar card with full-width button

Card Widgets

Examples explained

# W3.CSS Fonts

Headings

Font-size classes

Override the W3.CSS defaults

Change the default page font

How to use a font class

How to use external fonts

External google font: lobster

Font effects

Font effects 2

Examples explained

# W3.CSS Text

Text alignment

Center elements

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Text shadow

Text opacity

Text special effects

Examples explained

## W3.CSS Round

Round classes

Circle class

Text inside a circle

Circle inside a circle

Circle inside a circle 2

Examples explained

## W3.CSS Padding

Padding number classes

Padding size classes

Examples explained

## W3.CSS Margins

Margin on all sides

Margin top

Margin bottom

Margin left

Margin right

Margin sections

Examples explained

## W3.CSS Display

- Display container
- Display container with padding
- Display container for image
- Displaying a flag
- Floating classes
- Hide and show classes
- Toggle hide and show

Examples explained

## W3.CSS Buttons

- Buttons
- Button colors
- Hover colors
- Button shapes
- Button sizes
- Button borders
- Buttons with text effects
- Buttons with wide text effects
- Padded buttons
- Left and right buttons
- Full-width buttons
- Disabled buttons
- Button groups on the same line
- Button bars
- Buttons with ripple effects

Examples explained

## W3.CSS Notes

Notes

Grey with rounded borders

Pale blue with left and right bar

Pale red with left bar

Pale yellow with border

Pale green with border and bottom bar

Yellow with top and bottom bar

Examples explained

# W3.CSS Quotes

Quotes

Large quotes

Blockquotes

Custom quotes 1

Custom quotes 2

Custom quotes 3

Black quote

Quotes as cards

Quotes as cards 2

Examples explained

# W3.CSS Alerts

Basic alerts

Closable alerts

Rounded alerts

Alert cards

Examples explained

# W3.CSS Tables

| |
|---|
| Basic table |
| Bordered table (horizontal dividers) |
| Striped table |
| Bordered striped table |
| Borders around the table |
| Table all (combines stripes, borders, etc) |
| Flipping the stripes |
| Centered table |
| Table with a colored heading |
| Table colors |
| Hoverable table (grey color) |
| Specific hoverable table color |
| Table card |
| Responsive table |
| Tiny table |
| Small table |
| Large table |
| xLarge table |
| xxLarge table |
| xxxLarge table |
| Jumbo table |

Examples explained

# W3.CSS Lists

| |
|---|
| Basic list |
| Bordered list |
| List header |

List card

Centered list

Colored list

Colored list item

Hoverable list (grey color)

Specific hoverable list color

Closable list

Padded list

Avatar list

Tiny list

Small list

Large list

xLarge list

xxLarge list

xxxLarge list

Jumbo list

**Examples explained**

# W3.CSS Images

Rounded image

Circled image

Bordered image

Image card

Image text

Responsive image

- Responsive image with max width
- Image opacity
- Image grayscale
- Image sepia
- Image hover effects
- Image opacity off
- Photo Album

Examples explained

## W3.CSS Inputs

- Top labels
- Bottom labels
- Input cards
- Colored labels
- Bordered inputs
- Rounded borders
- Borderless input
- Colored inputs
- Hoverable inputs
- Animated inputs
- Checkboxes
- Radio buttons
- Select menu
- Bordered select menu
- Form elements in a three-column grid
- Two-column layout with labels

Examples explained

# W3.CSS Badges

- Badges
- Colored Badges
- Badges in buttons
- Badges in lists I
- Badges in lists II
- Badges in tables
- Large badges
- UTF-8 badges

Examples explained

# W3.CSS Tags, Labels, and Signs

- Tags and labels
- Colored tags
- Large tags
- Tags as letters
- Tags in a row
- Tag as a sign
- Road signs
- Large signs
- Large signs 2
- Rounded signs
- Rotated tags
- Spinning tags

Examples explained

# W3.CSS Icons

Font awesome icons

Google material design icons

Bootstrap icons

Examples explained

## W3.CSS Responsive

The w3-half class

The w3-third class

The w3-twothird class

The w3-quarter class

The w3-threequarter class

Nested rows (w3-half inside w3-half)

Columns using w3-rest

Columns using percent

The w3-content class

Difference between w3-row and w3-row-padding

Examples explained

## W3.CSS Animate

Top animation

Bottom animation

Left animation

Right animation

Fading animation

Fading infinite animation

Zoom animation

Spin animation

Fade all

Examples explained

## W3.CSS Dropdowns

- Hoverable dropdown menu
- Hoverable dropdown
- Dropdown in navbar
- Clickable dropdown menu
- Image dropdown
- Card dropdown
- Animated dropdown
- Right-aligned dropdown

Examples explained

## W3.CSS Accordions

- Basic accordion
- Accordion buttons
- Active accordion
- Accordion width
- Accordion links
- Accordion card with lists
- Accordion and dropdown in sidebar
- Animated accordion

Examples explained

## W3.CSS Navigation

- Basic navigation
- Colored bar

Bordered bar

Active link in bar

Hoverable bar links

Right-aligned links

Bar font-size

Bar padding

Bar width

Bar icons

Bar with input

Bar with dropdown

Bar with an active dropdown and icon

Bar with clickable dropdown

Top bar

Bottom bar

Collapsible bar

**Examples explained**

# W3.CSS Sidebar

Basic sidebar (always displayed)

Collapsible sidebar

Sidebar hiding a part of the page content

Sidebar hiding all page content

Sidebar shift content to the right

Sidebar color

Bordered sidebar

Bottom bordered sidebar (dividers)

Sidebar card

Hoverable sidebar links (background color)

- Hoverable sidebar links (text color)
- Sidebar size
- Sidebar with icons (icon bar)
- Sidebar with dropdown
- Sidebar with accordion
- Animated sidebar
- Sidebar with overlay effect
- Sidebar with content

Examples explained

## W3.CSS Tabs

- Basic tab
- Active/current tab
- Vertical tab
- Animated tab content
- Tabbed image gallery
- Tabs in a grid

Examples explained

## W3.CSS Pagination

- Basic pagination
- Pagination arrows
- Active pagination link
- Pagination hover color
- Pagination size
- Bordered pagination
- Rounded bordered pagination
- Centered pagination

Next/previous pagination arrows

Examples explained

## W3.CSS Progress Bars

Basic progress bar

Progress bar width

Progress bar sizes

Progress bar colors

Rounded progress bars

Progress bar labels

Dynamic progress bar

Dynamic progress bar with centered label

Dynamic progress bar with left-aligned label

Dynamic progress bar with label placed outside

Examples explained

## W3.CSS Slideshow

Manual slideshow with images

Automatic slideshow with images

Automatic HTML slides

Slideshow captions

Slideshow indicators with numbers and prev/next buttons

Slideshow indicators with icons and bullets

Images as indicators

Animated slides

Automatic slideshow with infinite fading

Examples explained

# W3.CSS Modal

| |
|---|
| Basic modal (dialog box/popup window) |
| Modal with containers |
| Modal card |
| Animated modal |
| Fade in modal |
| Modal image |
| Modal image gallery |
| Modal login form |
| Modal tab |
| How to close the modal |
| Modal lightbox (modal image gallery) |

Examples explained

# W3.CSS Tooltips

| |
|---|
| Inline tooltip text |
| Inline tooltip tag |
| Tooltip text before a hoverable image |
| Tooltip text after a hoverable image |
| Absolute tooltip |
| Colored tooltip |
| Rounded tooltip |
| Small tooltip |
| Large tooltip |

Examples explained

# W3.CSS Responsive Grid

Fluid grid demonstration

Two equal columns

Two unequal columns

Three equal columns

Three unequal columns

Six equal columns

Mixed: Mobile and Laptops

Mixed: Mobile, Tablets and Laptops

Difference between w3-row and w3-row-padding

Columns using w3-rest

Columns using percent

**Examples explained**

# W3.CSS Code

Displaying examples

Displaying code

Displaying colored HTML

Displaying colored CSS

Displaying colored JavaScript code

Displaying inline code (w3-codespan)

**Examples explained**

# W3.CSS Filters

Filter Tables

Filter Lists

Filter Dropdowns (click)

Filter Dropdowns (hover)

Examples explained

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Build skills.
Get hired.

$95 per paid course
one-time payment

Exam and certificate included.

Get started

REMOVE ADS




FREE ACCOUNT

Track your progress

| | |
|---|---|
| XP earned | 0 |
| Day streak | 0 |
| Spaces | 0 |
| League | — |

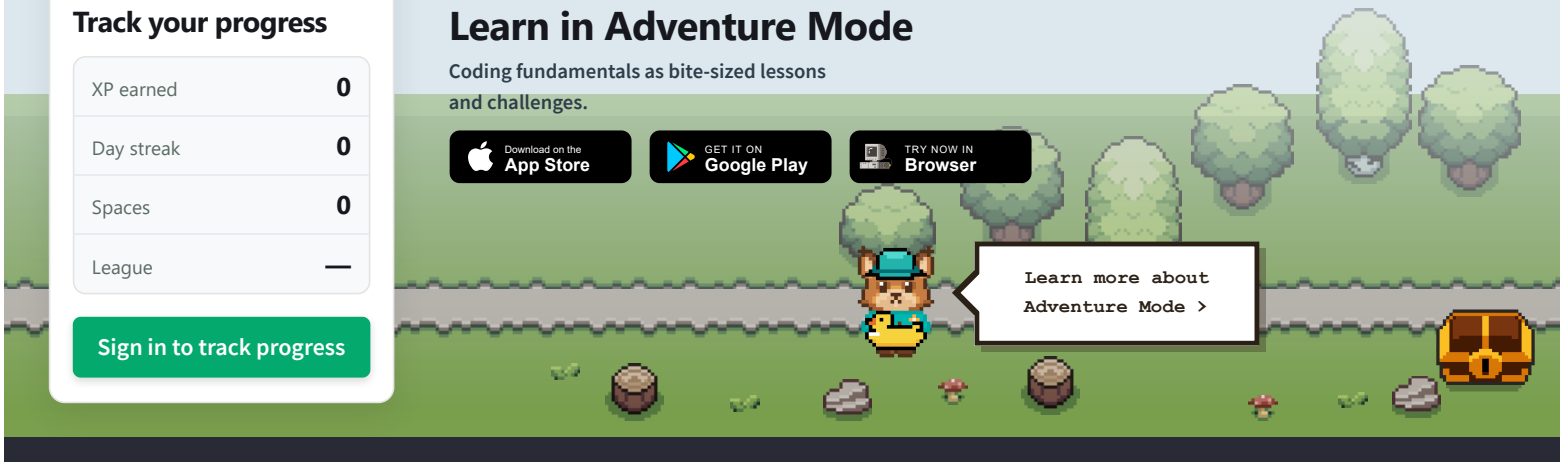
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# Bootstrap Examples

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## Bootstrap Get Started

SHARE

- Container
- Fluid container

Examples explained

## Bootstrap Grid Basic

- Three equal columns
- Two unequal columns

Examples explained

## Bootstrap Typography

- Headings <h1> to <h6>
- The <small> element
- The <mark> element

- The <abbr> element
- The <blockquote> element
- Reversed blockquote
- The <dl> element
- The <code> element
- The <kbd> element
- The <pre> element
- Contextual text colors
- Contextual background colors

Examples explained

## Bootstrap Tables

- Basic table
- Striped rows
- Bordered table
- Hover rows
- Condensed table
- Contextual classes
- Responsive tables

Examples explained

REMOVE ADS

## Bootstrap Images

- Rounded image
- Circle image
- Thumbnail image
- Responsive images

Image gallery

Responsive embeds

Examples explained

## Bootstrap Jumbotron

Jumbotron inside container

Jumbotron outside container

Creating a page header

Examples explained

## Bootstrap Wells

Basic well

Well size

Examples explained

## Bootstrap Alerts

Alerts

Closing alerts

Animated alerts

Examples explained

## Bootstrap Buttons

Button styles

Button tags

Button sizes

Block level buttons

Active/disabled buttons

Examples explained

## Bootstrap Button Groups

Button group

Button group sizes

Vertical button group

Justified button group

Justified button group 2

Nesting button groups and dropdown menus

Split buttons

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## Bootstrap Glyphicons

Glyphicons

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## Bootstrap Badges and Labels

Badges

Badge inside a button

Labels

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Examples explained

## Bootstrap Progress Bars

Progress bar

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Striped progress bar

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Stacked progress bar

Examples explained

## Bootstrap Pagination

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Examples explained

## Bootstrap Pager

Pager

Align buttons

Examples explained

## Bootstrap List Groups

List group

List group with badges

List group with links

Active list item

Disabled list item

Contextual classes

Custom content

Examples explained

## Bootstrap Panels

Panels

Panel heading

Panel footer

Panel group

Contextual panel classes

Examples explained

## Bootstrap Dropdowns

Dropdown

Dropdown divider

Dropdown header

Disabled dropdown item

Right-aligned dropdown menu

Dropup

Dropdown accessibility

Examples explained

## Bootstrap Collapsible

Collapsible button

Collapsible link

Collapsible "in" (show by default)

Collapsible panel

Collapsible list group

Accordion

Examples explained

## Bootstrap Tabs and Pills

Inline list

Tabs

Tabs with dropdown menu

Pills

Vertical pills

Vertical pills in a row

Pills with dropdown menu

Centered tabs and pills

Toggleable/dynamic tabs

Toggleable/dynamic pills

Examples explained

## Bootstrap Navbar

Navbar

Inverted navbar

Fixed top navbar

Fixed bottom navbar

Navbar with dropdown

Right-aligned navbar

Collapsible navbar

Examples explained

## Bootstrap Forms

Vertical form

Inline form

Inline form without labels

Horizontal form

Examples explained

## Bootstrap Inputs

Input

Textarea

Checkbox

Inline checkbox

Radio buttons

Inline radio buttons

Select menu

Examples explained

## Bootstrap Inputs 2

Static control

Control states

Inline form: control states

Examples explained

## Bootstrap Input Sizing

Input sizing

Input sizing horizontal label

Input width (columns)

Input help text

Examples explained

## Bootstrap Media Objects

- Left and right-aligned media object
- Top, middle and bottom-aligned media object
- Nested media object
- Nested media object II

Examples explained

## Bootstrap Carousel

- Carousel
- Caption slides

Examples explained

## Bootstrap Modal

- Modal
- Small modal
- Large modal

Examples explained

## Bootstrap Tooltip

- Tooltip
- Positioning tooltips

Examples explained

## Bootstrap Popover

Popover

Positioning popovers

Dismissable popover

Hoverable popover

Examples explained

## Bootstrap Scrollspy

Horizontal scrollspy

Vertical scrollspy

Examples explained

## Bootstrap Affix

Horizontal affix

Vertical affix

Scrollspy and affix (horizontal)

Scrollspy and affix (vertical)

Examples explained

## Bootstrap Filters

Filter table

Filter list

Filter dropdown

Filter anything

Examples explained

## Bootstrap Grid: Stacked to Horizontal

Stacked to horizontal grid

Stacked to horizontal grid with fluid container

Examples explained

## Bootstrap Grid: Small Devices

Small grid

Small grid

Examples explained

## Bootstrap Grid: Medium Devices

Medium grid

Medium grid 2

Examples explained

## Bootstrap Grid: Large Devices

Medium grid

Medium grid 2

Examples explained

## Bootstrap Grid Examples

Three equal columns

Three unequal columns

Two unequal columns

Two columns with two nested columns

Mixed columns: mobile and desktop

Mixed columns: mobile, tablet and desktop

Clear floats

Offsetting columns

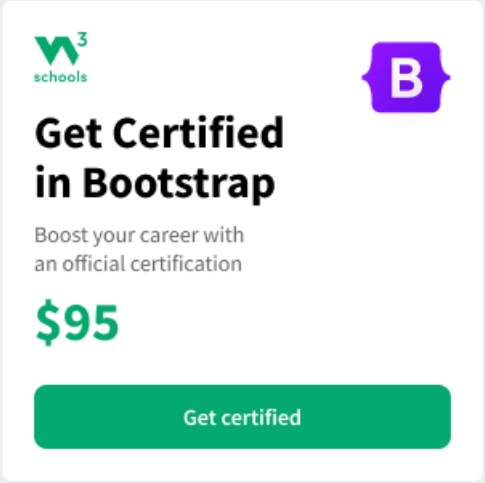
Push and Pull - Change column order

Examples explained

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Track your progress

XP earned 0

Day streak 0

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# PHP Examples

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## PHP Examples

SHARE

This page covers all the PHP examples from the PHP tutorial. Each PHP example opens in our Try-it editor, showing the source code and the result.

## PHP Install

Check PHP version

Install explained

## PHP Syntax

Write text to the output using PHP

Keywords, classes, functions, and user-defined functions ARE NOT case-sensitive

Variable names ARE case-sensitive

Syntax explained

## PHP Comments

Single-line comment before code line

Single-line comment at the end of code line

Use comment to prevent code from being executed

Multi-line comment

Multi-line comment to ignore code

**Comments explained**

# PHP Variables

Create variables

Output some text and the value of a variable

Output the sum of two variables

Assign the same value to multiple variables in one line

Use var_dump() to return the data type of variables

Global scope (variable outside function)

Local scope (variable inside function)

Static scope (let a local variable not be deleted after execution of function)

Use the global keyword to access a global variable from within a function

Use the $GLOBALS[] array to access a global variable from within a function

**Variables explained**

REMOVE ADS

# PHP Echo and Print

Echo - Output text

Echo - Output variables

Echo - Single or double quotes?

Print - Output text

Print - Output variables

Print - Single or double quotes?

**Echo and Print explained**

# PHP Data Types

PHP string

PHP integer

PHP float

PHP boolean

PHP array

PHP object

PHP NULL

**Data Types explained**

# PHP Strings

strlen() - Return the length of a string

str_word_count() - Count the number of words in a string

str_contains() - Check if a string contains a specific substring

strpos() - Search for a specific text within a string

strtoupper() - Returns a string in upper case

strtolower() - Returns a string in lower case

str_replace() - Replaces some characters with some other characters in a string

strrev() - Reverse a string

trim() - Removes any whitespace from the beginning/end of a string

explode() - Splits a string into an array

substr() - Extract a part of a string (slice a string)

Use escape character

**Strings explained**

# PHP Numbers

is_int() - Check if variable is of type integer

is_float() - Check if variable is of type float

is_infinite() - Check if a numeric value is infinite

is_nan() - Check if a value is "Not A Number" (NAN)

is_numeric() - Check if a variable is a number or a numeric string

is_numeric() - Check if a variable is a number or a numeric string

intval() - Return the integer value of a variable

**Numbers explained**

# PHP Casting

Cast to String

Cast to Integer

Cast to Float

Cast to Boolean

Cast to Array

Cast to Object

Cast to NULL

**Casting explained**

# PHP Math

pi() - Return value of PI

min(), max() - Find the lowest and highest value in a list of arguments

abs() - Return the absolute (positive) value of a number

sqrt() - Return the square root of a number

round() - Round a floating-point number to its nearest integer

rand() - Generate a random number

rand() - Generate a random number between 10 and 100

Math explained

## PHP Constants

define() - Create a constant

Declare a constant inside a function, and output it outside the function

const keyword - Create a constant

Create Array constants with define() or const

Constants explained

## PHP Magic Constants

The __CLASS__ constant

The __DIR__ constant

The __FILE__ constant

The __FUNCTION__ constant

The __FUNCTION__ constant

The __METHOD__ constant

The __NAMESPACE__ constant

The __TRAIT__ constant

The ClassName::class constant

Magic Constants explained

## PHP Operators

Arithmetic operator: Addition (+)

Arithmetic operator: Subtraction (-)

Arithmetic operator: Multiplication (*)

Arithmetic operator: Division (/)

Arithmetic operator: Modulus (%)

Arithmetic operator: Exponentiation (**)

Assignment operator: x = y

Assignment operator: x += y

Assignment operator: x -= y

Assignment operator: x *= y

Assignment operator: x /= y

Assignment operator: x %= y

Comparison operator: Equal (==)

Comparison operator: Identical (===)

Comparison operator: Not equal (!=)

Comparison operator: Not equal (<>)

Comparison operator: Not identical (!==)

Comparison operator: Greater than (>)

Comparison operator: Less than (<)

Comparison operator: Greater than or equal (>=)

Comparison operator: Less than or equal (<=)

Comparison operator: Spaceship (<=>)

Increment operator: ++$x

Increment operator: $x++

Decrement operator: --$x

Decrement operator: $x--

Logical operator: and

Logical operator: or

Logical operator: xor

Logical operator: && (and)

Logical operator: || (or)

Logical operator: not

String operator: Concatenation of $txt1 and $txt2

String operator: Appends $txt2 to $txt1

Array operator: Union (+)

Array operator: Equality (==)

Array operator: Identity (===)

Array operator: Inequality (!=)

Array operator: Inequality (<>)

Array operator: Non-identity (!==)

Conditional assignment operator: Ternary (?:)

Conditional assignment: Null coalescing (??)

**Operators explained**

# PHP If Statements

The if statement executes some code only if the specified condition is true

The if...else statement executes some code if a condition is true, and another code if condition is false

The if...elseif...else statement executes different codes for more than two conditions

Shorthand if

Shorthand if...else

Nested if

**If Statements explained**

# PHP Switch Statement

The switch statement performs different actions based on different conditions

The default keyword specifies the code to run if there is no case match

Combining cases

**Switch Statement explained**

# PHP Match Expression

The match expression performs different actions based on different conditions

Match multiple values for the same code block

**Match Expression explained**

## PHP Loops

while loop - Loops through a block of code as long as the specified condition is true

do...while loop - Loops through a block of code at least once, and then repeats the loop as long as the specified condition is true

Another do...while loop

for loop - Loops through a block of code a specified number of times

foreach loop - Loops through a block of code for each element in an array or each property in an object

**Loops explained**

## PHP Functions

Create and call a function

Function with one parameter

Function with two parameters

Function with default parameter value

Function with return value

Passing arguments by reference

Variable number of parameters

**Functions explained**

## PHP Arrays

Create and display an indexed array

Access a specific array item with index number

Change value of array item

Loop through an indexed array

Create and display an associative array

Access a specific array item with key name

Change value of array item

Loop through an associative array

Add more array items to the end of an indexed array

Add more array items to the end of an associative array

array_push() - Add one or more array items to the end of an array

array_unshift() - Add one or more array items to the beginning of an array

array_splice() - Remove a portion of an array and replace it with new items

array_merge() - Merge two or more arrays

array_pop() - Remove the last item of an array

array_shift() - Remove the first item of an array

sort() - Sorts an indexed array in ascending order

rsort() - Sorts an indexed array in descending order

asort() - Sorts an associative array in ascending order (value)

arsort() - Sorts an associative array in descending order (value)

ksort() - Sorts an associative array in ascending order (key)

krsort() - Sorts an associative array in descending order (key)

Create a multidimensional array

Loop through a multidimensional array (for loop)

Loop through a multidimensional array (foreach loop)

**Arrays explained**

# PHP Superglobals

$GLOBAL - Used to access global variables from anywhere in the script

$_SERVER - Holds information about headers, paths, and script locations

$_REQUEST - Contains data from submitted forms, URL query strings, and HTTP cookies

$_POST - Contains an array of variables received via the HTTP POST method

$_GET - Contains an array of variables received via the HTTP GET method

**Superglobals explained**

# PHP Regular Expressions

Regex Modifier: i

Regex Modifier: m

Regex Modifier: [abc]

Regex Modifier: [^abc]

Regex Modifier: [a-z]

Regex Modifier: [A-z]

Regex Modifier: [A-Z]

Regex Modifier: [123]

Regex Modifier: [0-5]

Regex Modifier: [0-9]

Regex Modifier: |

Regex Modifier: .

Regex Modifier: ^

Regex Modifier: $

Regex Modifier: \d

Regex Modifier: \D

Regex Modifier: \s

Regex Modifier: \S

Regex Modifier: \w

Regex Modifier: \W

Regex Modifier: \b

Regex Modifier: \u

Regex Modifier: n+

Regex Modifier: n{3}

Regex Modifier: n{2, 5}

Regex Modifier: n{3,}

**Regular Expressions explained**

# PHP Regular Expression Functions

preg_match() - Returns 1 if the pattern was found in the string, otherwise 0

preg_match_all() - Returns how many matches were found for a pattern in a string

preg_replace() - Replaces all matches of the pattern in a string with another string

preg_split() - Splits a string into an array using matches of a regular expression as separators

preg_grep() - Returns an array containing only elements from the input that match the given pattern

**Regular Exp Functions explained**

# PHP Form Validation

PHP Form Validation

**Form Validation explained**

# PHP Date and Time

date() Format today's date in several ways

Automatically update the copyright year

Output the current time (server time)

Set timezone, then output current time

Return default timezone

mktime() - Create a date and time from a number of parameters

Format a timestamp to a readable date and time

strtotime() - Convert strings to dates

Output the dates for the next six Saturdays

Date and Time explained

## PHP Include Files

Use include to include "footer.php" in a page

Use include to include "menu.php" in a page

Use include to include "vars.php" in a page

Use include to include a non-existing file

Use require to include a non-existing file

Include Files explained

## PHP File Handling

readfile() - Read a file and write it to the output buffer

File Handling explained

## PHP File Open/Read/Close

Use fopen(), fread(), and fclose() to open, read, and close a file

Use fgets() to read a single line from a file

Use feof() to read through a file, line by line, until end-of-file is reached

Use fgetc() to read a single character from a file

File Open/Read/Close explained

## PHP Sessions

Start a new session, and set session variables

Get session variable values

Get all session variable values

Modify a session variable

Destroy a session

Sessions explained

## PHP Filters

- filter_list() - List all supported filter names and ids
- Sanitize and validate an email
- Sanitize and validate a URL
- Validate an integer
- Validate an integer that is 0
- Validate an IP address
- Validate an integer within a range
- Validate IPv6 address
- Validate URL - Must contain querystring
- Sanitize string - Remove characters with ASCII value > 127

Filters explained

## PHP Callback Functions

- Use a named function as a callback
- Use an anonymous function as a callback
- Callbacks in user defined functions

Callback Functions explained

## PHP and JSON

- Encode an associative array into a JSON object
- Encode an indexed array into a JSON object
- Decode JSON data into a PHP object
- Decode JSON data into an associative array
- Access the values from a PHP object

Access the values from an associative array

Loop through the values of a PHP object

Loop through the values of an associative array

**PHP and JSON explained**

# PHP Exceptions

Throw an exception without catching it

Display a message when an exception is thrown, and continue script

Display a message when an exception is thrown, and then indicate that the process is complete

Output information about an exception that was thrown

**PHP Exceptions explained**

# PHP Object-Oriented Programming (OOP)

Create a class and two objects

The instanceof keyword checks if an object belongs to a specific class

__construct() - Create a constructor

__destruct() - Create a destructor

Access modifier: Public

Access modifier: Private

Access modifier: Protected

Inheritance

A constant can be accessed fom outside the class

A constant can be accessed fom inside the class

Abstract class

Abstract method with argument

Define and use an Interface

Use an Interface 2

Define and use a Trait

Use multiple Traits

Declare a Static method

Declare a Static property

PHP OOP explained

# PHP MySQL Database

Select data with MySQLi (Object-oriented)

Select data with MySQLi (Procedural)

Put result in an HTML table (MySQLi Object-oriented)

Select and filter data with MySQLi (Object-oriented)

Select and filter data with MySQLi (Procedural)

Select and order by with MySQLi (Object-oriented)

Select and order by with MySQLi (Procedural)

MySQL Database explained

# PHP SimpleXML Parser

simplexml_load_string() - Read XML data from a string

simplexml_load_file() - Read XML data from a file

Get node values

Get node values of specific elements

Get node values - loop

Get attribute values

Get attribute values - loop

SimpleXML Parser explained

# PHP XML Expat Parser

Initialize an XML Expat parser, define some handlers, then parse an XML file

XML Expat Parser explained

# PHP and AJAX

AJAX and PHP

AJAX and MySQL

AJAX explained

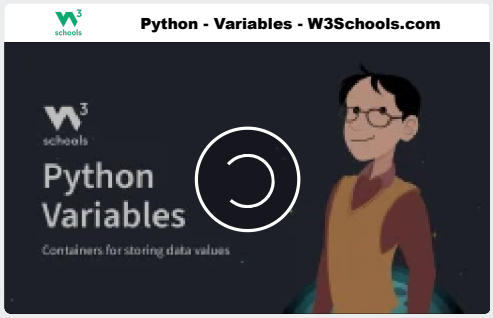
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# Java Examples

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## Java Syntax

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Create a simple "Hello World" program

Syntax Explained

## Java Comments

Single-line comment before a line of code

Single-line comment at the end of a line of code

Multi-line comment

Comments Explained

## Java Variables

Create a string variable

Create an integer variable

Create a variable without assigning the value, and assign the value later

Overwrite an existing variable value

Create a final variable (unchangeable and read-only)

Combine text and a variable on display

Add a variable to another variable

Declare many variables of the same type with a comma-separated list

**Variables Explained**

## Java Data Types

A demonstration of different data types in Java

Create a byte type

Create a short type

Create an int type

Create a long type

Create a float type

Create a double type

Create a boolean type

Create a char type

Create a String type

**Data Types Explained**

## Java Type Casting

Widening Casting

Narrowing Casting

**Type Casting Explained**

## Java Operators

Addition operator

Subtraction operator

Multiplication operator

Division operator

Modulus operator

Increment operator

Decrement operator

Assignment operator

Comparison operator

Logical operator

**Operators Explained**

# Java Strings

Create a string

Quotes inside a string

Find the length of a string

Using methods to convert strings to uppercase and lowercase

Finding a string in a string

String concatenation

If you add a number and a string, the result will be a string concatenation

**Strings Explained**

# Java Math

Math.max(x,y) - return the highest value of x and y

Math.min(x,y) - return the lowest value of x and y

Math.sqrt(x) - return the square root of x

Math.abs(x) - return the absolute (positive) value of x

Math.random() - return a random number between 0 and 1

**Math Explained**

# Java Booleans

- Create a boolean type
- Find out if an expression is true or false
- Use the "equal to" operator to evaluate a boolean expression

**Booleans Explained**

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# Java If...Else (Conditions)

- The if statement
- The else statement
- The else if statement

**If...Else Explained**

# Java Switch

- The switch statement
- The switch statement with a default keyword

**Switch Explained**

# Java Loops

- While loop
- Do while loop
- For loop
- For-each loop
- Break a loop
- Continue a loop

Loops Explained

## Java Arrays

- Create and access an array
- Change an array element
- Find the length of an array
- Access an array
- Loop through an array
- Loop through an array with for-each
- Multidimensional array
- Loop through a multidimensional array

Arrays Explained

## Java Methods

- Create and call a method
- Call a method multiple times
- Method with parameters
- Method with multiple parameters
- Return value
- A method with if...else
- Method overloading
- Recursion
- Recursive function with a halting condition

Methods Explained

## Java Classes and Objects

- Create a class and an object of a class

Create multiple objects of a class

Accessing class attributes (variables)

Modify attributes

Override existing attribute values

Multiple attributes

Create a class method

Method with parameters

Access class methods with an object

Create a class constructor

Constructor with parameters

**Classes and Objects Explained**

# Java Modifiers

Public class

Default class

Public attributes

Private attributes

Default attributes

Protected attributes

Final class

Abstract class with abstract methods

**Modifiers Explained**

# Java Encapsulation

Java encapsulation (getters and setters)

**Encapsulation Explained**

# Java Packages

Import a class from the Java API

Import a package from the Java API

Create a package

**Packages Explained**

# Java Inheritance

Java inheritance (extends)

Java polymorphism

**Inheritance Explained**

# Java Inner Classes

Inner class

Private inner class (error when trying to access it from an outside class)

Static inner class

Access outer class from inner class

**Inner Classes Explained**

# Java Abstraction & Interfaces

Java abstract classes and methods

Java interface (implements)

Multiple interfaces

**Interfaces Explained**

# Java Enums

Create an enum with a group of constants

An enum inside a class

An enum in a switch statement

Loop through an enum

**Enums Explained**

## Java User Input

Read user input

Read user input of various types

**User Input Explained**

## Java Dates

Display current date

Display current time

Display current date and time

Formatting date and time

**Dates Explained**

## Java ArrayList

Create an ArrayList

Access an item in an ArrayList

Remove an item from an ArrayList

Remove all items from an ArrayList

Get the size of an ArrayList

Loop through an ArrayList

Loop through an ArrayList with for-each

Create an ArrayList that should store numbers (integers)

**ArrayList Explained**

# Java HashMap

- Create a HashMap
- Access an item in a HashMap
- Remove an item from a HashMap
- Remove all items from a HashMap
- Get the size of a HashMap
- Loop through a HashMap
- Create a HashMap that should store String keys and Integer values

**HashMap Explained**

# Java HashSet

- Create a HashSet and add items to it
- Check if an item exists in a HashSet
- Remove an item from a HashSet
- Remove all items from a HashSet
- Get the size of a HashSet
- Loop through a HashSet
- Create a HashSet that should store Integer values

**HashSet Explained**

# Java Iterator

- Getting an iterator
- Looping throug a collection
- Removing items from a collection

**Iterator Explained**

# Java Wrapper Classes

Create an ArrayList that should store numbers (integers)

Create wrapper objects

Using wrapper methods

Convert wrapper objects to strings

**Wrapper Classes Explained**

# Java Exceptions (Try...Catch)

The try...catch statement

The finally statement

The throw statement

**Exceptions Explained**

# Java Regular Expressions

Search for the word "w3schools" in a sentence

**Regular Expressions Explained**

# Java Threads

Running a thread by extending the thread class

Running a thread by implementing the Runnable interface

Concurrency problems

Avoid concurrency problems

**Threads Explained**

# Java Files

Create a file

| |
|---|
| Write to a file |
| Read a file |
| Get file information |
| Delete a file |

Files Explained

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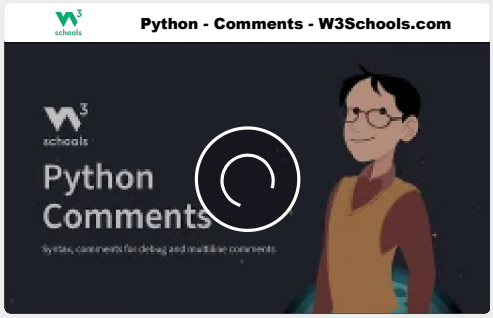
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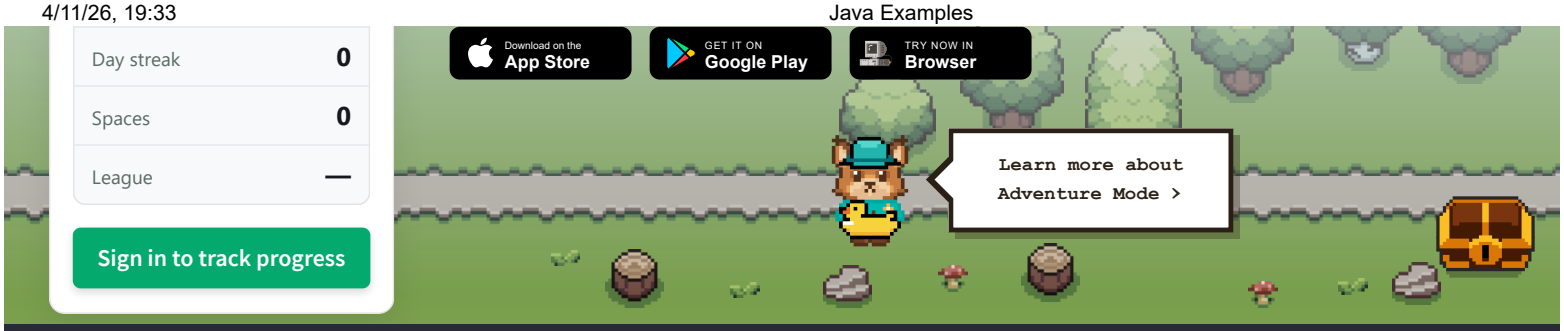
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## XML Examples

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This page covers all the examples from the XML tutorial.

### Viewing XML Files

View a simple XML file (note.xml)
View the same XML file with an error
View an XML CD catalog
View an XML plant catalog
View an XML food menu

**Examples explained**

### XML and CSS

View an XML CD catalog
View the corresponding CSS file
Display the CD catalog formatted with the CSS file

**Examples explained**

### XML and XSLT

View an XML food menu
Display the food menu styled with an XSLT style sheet

Examples explained

## Parsing XML and the XML DOM

View a simple XML file (note.xml)
Parse an XML string
Parse the XML file

Examples explained

## XML Output From a Server

See how ASP can return XML
See how PHP can return XML
View XML output from a database

Examples explained

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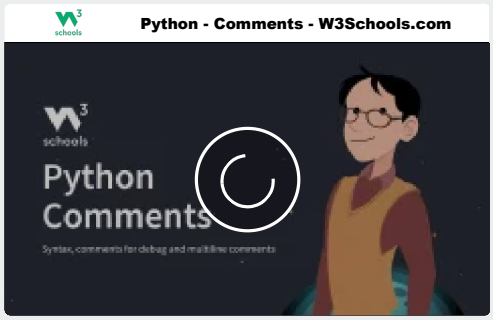
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# jQuery Examples

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## jQuery Selectors

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| |
|---|
| $("p").hide() - Hide all <p> elements |
| $("#test").hide() - Hide the element with id="test" |
| $(".test").hide() - Hide all elements with class="test" |
| $(this).hide() - Hide the current HTML element |

Selectors explained

## jQuery Events

| |
|---|
| jQuery click() |
| jQuery dblclick() |
| jQuery mouseenter() |
| jQuery mouseleave() |

jQuery mousedown()

jQuery mouseup()

jQuery hover()

jQuery focus() and blur()

Events explained

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# jQuery Hide/Show

jQuery hide()

jQuery hide() and show()

jQuery toggle()

jQuery hide()

Hide and Show explained

# jQuery Fade

jQuery fadeIn()

jQuery fadeOut()

jQuery fadeToggle()

jQuery fadeTo()

Fade explained

# jQuery Slide

jQuery slideDown()

jQuery slideUp()

jQuery slideToggle()

Slide explained

## jQuery Animate

- jQuery animate()
- jQuery animate() - manipulate multiple CSS properties
- jQuery animate() - using relative values
- jQuery animate() - using pre-defined values
- jQuery animate() - several calls after each other

Animate explained

## jQuery Stop Animations

- jQuery stop() sliding
- jQuery stop() animation (with parameters)

Stop Animations explained

## jQuery HTML Get Content and Attributes

- jQuery text() and html() - Get content
- jQuery val() - Get content
- jQuery attr() - Get attribute value

Get Content and Attributes explained

## jQuery HTML Set Content and Attributes

- jQuery text(), html(), and val() - Set content
- jQuery text() and html() - Set content with a callback function
- jQuery attr() - Set attribute value
- jQuery attr() - Set multiple attribute values
- jQuery attr() - Set attribute value with a callback function

Set Content and Attributes explained

## jQuery HTML Add Elements/Content

- jQuery append()
- jQuery prepend()
- jQuery append() - Insert several new elements
- jQuery after() and before()
- jQuery after() - Insert several new elements

Add Elements/Content explained

## jQuery HTML Remove Elements/Content

- jQuery remove()
- jQuery remove() - with a parameter
- jQuery empty()

Remove Elements/Content explained

## jQuery Get and Set CSS Classes

- jQuery addClass()
- jQuery addClass() - Multiple classes
- jQuery removeClass()
- jQuery toggleClass()

Get and Set CSS explained

## jQuery css() Method

- jQuery css() - return CSS property
- jQuery css() - set CSS property
- jQuery css() - set CSS properties

The css() method explained

## jQuery Dimensions

- jQuery - return width() and height()
- jQuery - return innerWidth() and innerHeight()
- jQuery - return outerWidth() and outerHeight()
- jQuery - return outerWidth(true) and outerHeight(true)
- jQuery - return width() and height() of document and window
- jQuery - set width() and height()

Dimensions explained

## jQuery Traversing Ancestors

- jQuery parent()
- jQuery parents()
- jQuery parentsUntil()

Traversing Ancestors explained

## jQuery Traversing Descendants

- jQuery children()
- jQuery find()

Traversing Descendants explained

## jQuery Traversing Siblings

- jQuery siblings()
- jQuery next()
- jQuery nextAll()
- jQuery nextUntil()

Traversing Siblings explained

## jQuery Traversing Filtering

- jQuery first()
- jQuery last()
- jQuery eq()
- jQuery filter()
- jQuery not()

Traversing Filtering explained

## jQuery Filters

- Filter table
- Filter list
- Filter anything

Filters explained

## jQuery AJAX load() Method

- jQuery load()
- jQuery load()
- jQuery load() - with callback

AJAX load() explained

## jQuery AJAX get() and post() Methods

- jQuery get()
- jQuery post()

AJAX get() and post() explained

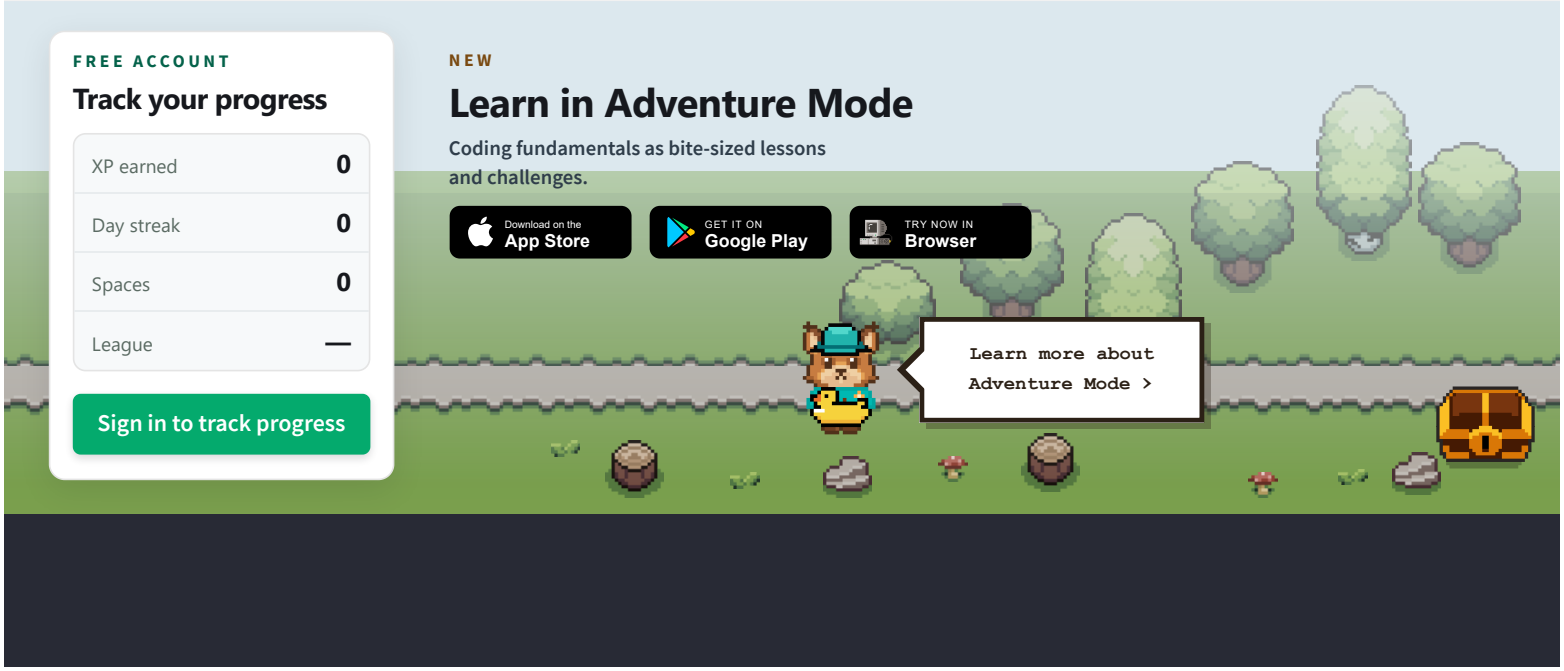
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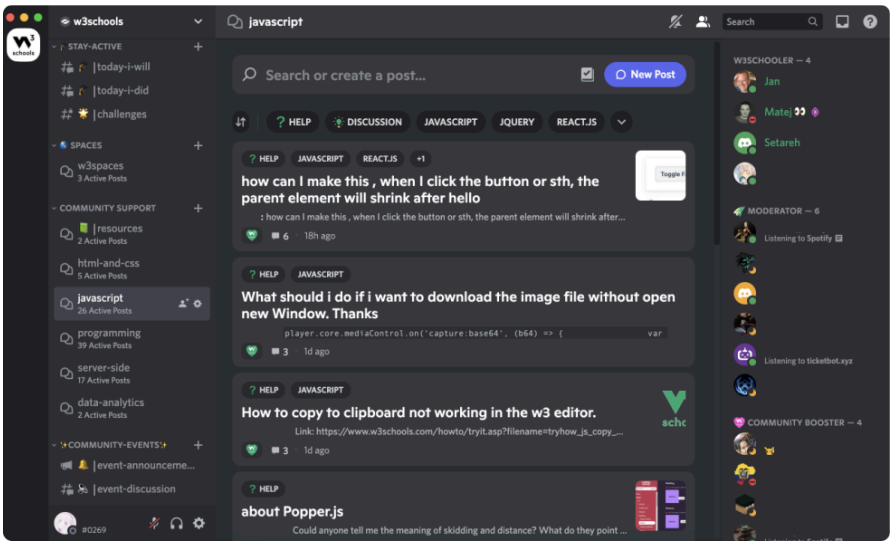
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This is how ChatGPT describes W3Schools:

*"W3Schools is a popular online platform that provides web development tutorials and references for various web technologies. It serves as a comprehensive resource for individuals who are learning or looking to enhance their skills in HTML, CSS, JavaScript, and other related technologies.*

*The website offers a wide range of educational materials, including beginner-friendly tutorials, interactive examples, and hands-on exercises. These resources are designed to help users understand the fundamentals of web development and gain practical experience. Users can learn about topics such as web design, front-end development, back-end development, database management, and more.*

*One of the notable features of W3Schools is its "Try it Yourself" editor, which allows users to experiment with code directly within their web browser. This interactive environment enables learners to test out different code snippets, modify them, and see the results in real-time, providing an effective way to practice and reinforce their understanding.*

*In addition to tutorials, W3Schools also provides comprehensive references for various web technologies, which serve as handy guides for developers when they need quick information or syntax references. The platform covers a wide range of subjects, including HTML elements, CSS properties, JavaScript functions, SQL queries, and more.*

*Overall, W3Schools is a well-regarded resource that has been utilized by numerous individuals seeking to learn web development concepts or refresh their knowledge. However, it's worth noting that while W3Schools is a valuable learning platform, it is not affiliated with the official World Wide Web Consortium (W3C), the international standards organization for the web."*

# You Can Help

Many people work very hard to ensure w3schools remains useful, updated, and interesting.

If you find an error, or a broken link, please tell us about it.

You can send us a message using the "CONTACT US" link at the bottom of any page.

# Add a Link to Us

If you want others to discover W3schools, please add a link to us.

To add a simple text link, insert the following HTML code on your site:

```
<a href="https://www.w3schools.com/">W3Schools.com</a>
```

If you want to link with a banner, please choose from: Banners and Buttons

# W3Schools Demographics

| Origin | Percent |
|---|---|
| Asia | 38 % |
| Americas | 30 % |
| Europe | 27 % |
| Africa | 3 % |
| Oceania | 2 % |

Source: Google Analytics

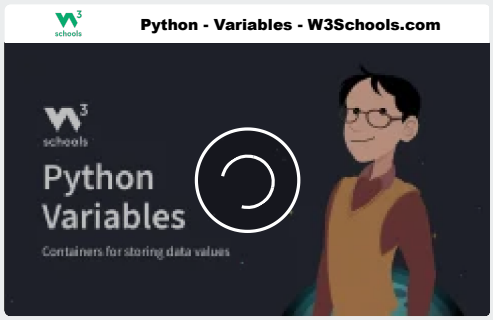
‹ Home

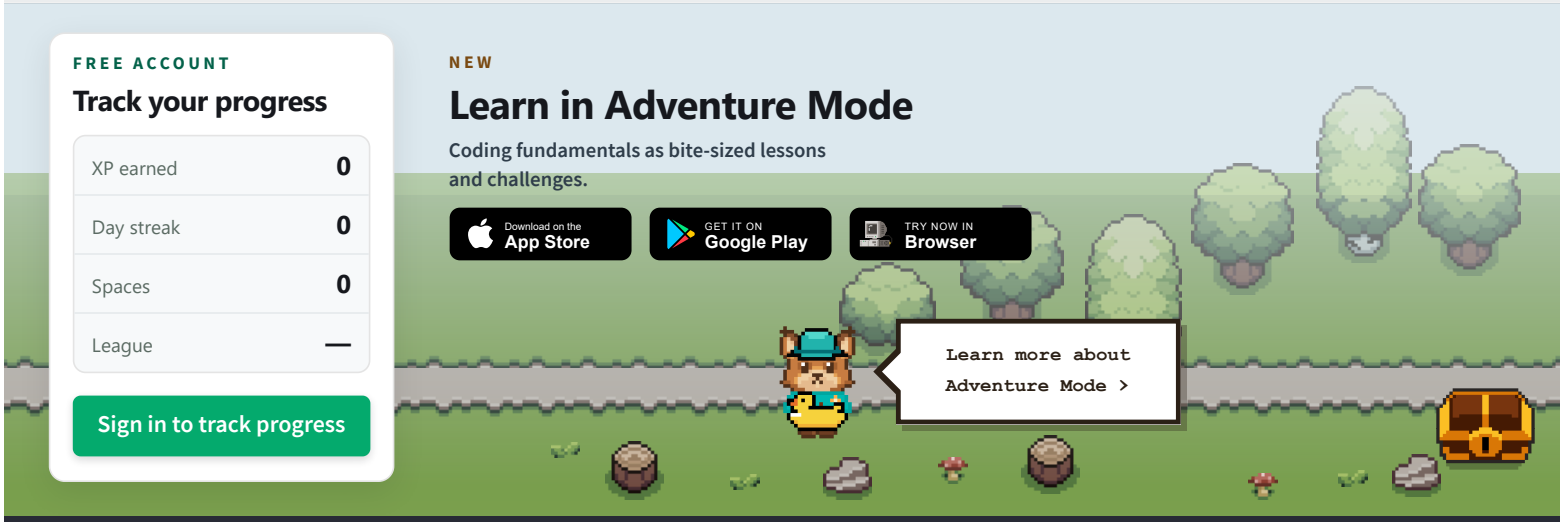
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## Top Tutorials

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Java Tutorial
C++ Tutorial
jQuery Tutorial

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jQuery Examples

## Get Certified

HTML Certificate
CSS Certificate
JavaScript Certificate
Front End Certificate
SQL Certificate
Python Certificate
PHP Certificate
jQuery Certificate
Java Certificate
C++ Certificate
C# Certificate
XML Certificate





W3Schools is optimized for learning and training. Examples might be simplified to improve reading and learning. Tutorials, references, and examples are constantly reviewed to avoid errors, but we cannot warrant full correctness of all content. While using W3Schools, you agree to have read and accepted our terms of use, cookies and privacy policy.

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# Terms of Service

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## Introduction

Welcome to W3Schools. By visiting W3Schools.com ("Website") or using our products and services ("Services"), you agree to the terms and conditions (hereafter referred to as the "Terms"). The Terms apply to everyone using the Website, including without limitation, visitors, vendors, customers, merchants, and content contributors.

Please read the Terms carefully before using the Website or the Services. By accessing any part of the Website, you agree to be bound by the Terms. If you don't agree with the Terms, you shouldn't access the Website or use our Services.

## Overview

The Website is operated and managed by Refsnes Data AS ("Refsnes") and W3Schools Network AS ("Network"), collectively referred to as "W3Schools". Refsnes provides the Website including all content, information, and related tools. Network operates certain solutions (such as Spaces, Campus (previously Academy), Courses, Challenges, and My Learning) and a billing system. The two entities Refsnes and Network are separately responsible for the Services they provide, manage and operate.

Any new features or tools added to the Website shall be subject to the Terms. You can review the most current version of the Terms at any time on this page. We reserve the right to update, change, or replace parts of the Terms by posting updates or changes at the Website. Please check this page periodically for changes. Your continued use of or access to the Website after posting any changes constitutes acceptance of such changes.

## Use of the Website and the Services

Our Service is generally not intended for users under the age of 16. We do not knowingly collect personal information from anyone under the age of 13 in ways not allowed by the Children's Online Privacy Protection Act (COPPA). If you are a parent or

legal guardian and believe that your child has shared personal information in a way that is not allowed by law, please contact us at help@w3schools.com. We will look into it and delete the information as required by law.

When a school uses our Services in an educational setting, it is the school's responsibility to ensure it has the legal right to share student information with us. This includes getting parental consent where the law requires it, such as under COPPA. By using our services on behalf of students, the school confirms that it has obtained or holds all the necessary permissions.

Some parts of the Services require you to create an account. When you register, we collect information about you, like your name and email address. Your account information and how we use your data are detailed in our Privacy Policy.

You agree not to use our products for illegal or unauthorized purposes. You must comply with all laws, rules, and regulations, including copyright laws.

Do not transmit any viruses, worms, or harmful code through the Services.

If you breach our Terms, we may terminate your access to the Services immediately. At our sole discretion, we reserve the right to remove or restrict your account and any content you've created, such as websites or other user-generated data.

# Intellectual Property Rights and Fair Use

All content provided by W3Schools, including text, graphics, logos, images, and software, is owned by W3Schools. Any use beyond what is specifically allowed under the Terms is forbidden, ref. the Norwegian "Lov om opphavsrett til åndsverk m.v."

## Fair Use

You can still use W3Schools copyright material under fair use for:

- Research
- News reporting
- Citations
- Commentary

Fair use also includes:

- Linking to W3Schools
- Including references to W3Schools in search engines

# Using W3Schools in Teaching

Fair use allows the use of copyrighted material in teaching, provided it meets the following guidelines:

## Permitted Use

- Copying examples and code snippets for non-profit teaching or research
- Copying small portions suitable for classroom instruction

## Not Permitted Use

- Copying for profitable or commercial use

- Mass or verbatim copying
- Copying large portions of material

# Privacy and Cookies

W3Schools collects, uses, and processes personal data as described in the Privacy Policy, which explains the legal basis we rely on for each purpose. Where we rely on your consent, such as for analytics and personalized advertising, we ask for it separately through our consent tool rather than treating your use of the Website as agreement. You can change or withdraw that consent at any time, and using the Website does not oblige you to give it.

Users have certain rights regarding their data. For more information on these rights and how to exercise them, please see our Privacy Policy.

We use cookies and similar tracking technologies to improve user experience. For more information on how we use cookies, please see our Cookie Policy.

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# Products and Services

To keep our offerings current, product descriptions and prices are subject to change at any time without prior notice. Without warning, we can discontinue the supply of any Services if our rules and guidelines or applicable laws are violated.

We seek to maintain high standards and to provide accurate information. However, we cannot guarantee that every product, service, or element of information will meet your expectations.

# Modifications to the Service and Prices

By ordering a subscription, the price for such subscription will remain firm for the period the subscription is purchased for. We reserve the right to adjust prices for any subsequent period(s).

We are not liable to you or any third party for any loss suffered due to modifications to the Website, price adjustments, suspension or discontinuation of the Service.

# Accuracy and Completeness of Information

We do our best to provide accurate and current information. However, we cannot guarantee that everything on the Website is complete or always up to date. Please do not rely solely on the material at the Website when making important decisions. You are using the information on the Website at your own risk.

Some of the content you find on the Website may be historical. Since historical information may become incorrect we provide at the Website such information for reference only. We can change the contents of the Website at any time without creating a liability for W3Schools. It's your responsibility to keep track of any changes.

If you find any errors, please let us know at help@w3schools.com so we can correct them.

# Refund Policy

## Exams and Courses:

You can withdraw from a course or exam within 14 days of receiving your initial access (marked by the date we send you the access email). If you have already used any exam attempts, the purchase is final and not eligible for a refund. For questions or issues related to the product you have purchased, please contact us at hello@w3schools.com.

## Plans and Subscriptions:

We do not provide refunds for plans or subscriptions. You may cancel your subscription anytime, but no refunds will be issued for the period you paid for.

# Mobile Apps

These Terms also apply when you use our mobile apps, such as the W3Schools Adventure app.

## In-App Purchases and Subscriptions:

- All payments in our apps are handled by the Apple App Store or Google Play, not by W3Schools
- Subscriptions renew automatically unless you cancel at least 24 hours before the current period ends
- You can manage or cancel your subscription in your App Store or Google Play account settings
- Deleting an app does not cancel your subscription
- Refunds for in-app purchases are handled by Apple or Google, following their own refund policies

We use RevenueCat to manage in-app subscriptions. See our Privacy Policy for more information about how your data is handled.

# Domains

We provide a domain purchasing service that seamlessly integrates with the W3Schools web hosting and development platform ("Spaces"). It allows you to search, select, and purchase a domain that automatically links to your Spaces.

When you buy a domain, you become its sole owner and must follow all related terms and conditions.

W3Schools works with third-party providers, such as Amazon and its Route 53 services, for domain registration and distribution. You can review their terms here:

- Amazon Route 53 Domain Registration Agreement
- Amazon Registrar Policy Page
- Gandi Contract Page

As the distributor, W3Schools handles purchasing and billing. If you have any complaints, claims, or concerns about these third-party products, please contact us at support@w3schools.com.

The mutual agreement for a domain renews automatically each year. If you don't want to continue with a domain purchase from us, you must cancel the renewal before the end of the current agreement. See the Billing and Account Information section for more billing information.

We are not liable for any harm, errors, or omissions caused by the domain registry administrator's actions, including domain registration or renewal requests. Before purchasing a domain, please review all associated third-party policies and practices to ensure you understand their terms.

# Billing and Account Information

We reserve the right to refuse any order placed with us. At our sole discretion, we can limit or cancel the quantities purchased by any individual, household, or for any specific order. These restrictions also apply to orders made under the same customer account, using the same payment method, or associated with the same billing or shipping address.

If we need to change or cancel an order, we will email you using the email address you provided at the time of purchase.

Please know that you agree to provide current, complete, and accurate purchase and account information for all orders made through our website. You must keep your account details updated at all times, including your email address and payment method, so we can process your transactions and contact you if needed.

# Optional Tools and Third-Party Links

We sometimes provide access to third-party tools we do not develop, monitor, or control. These tools are offered "as is" and "as available," without warranties, endorsements, or promises. We are not responsible for any issues or damages that arise from your use of these optional third-party tools.

If you choose to use these tools, you do so at your own risk. Before you proceed with any such tools, please ensure you understand and agree to the terms set by the third-party provider.

The Website may include links to external websites or services we do not operate or control. While we try to avoid linking to harmful or offensive content, we cannot guarantee the accuracy or quality of third-party materials, products, or services. We are not liable for any harm or damages you experience with third-party websites or services.

Before you engage in any transaction with a third party, review their policies and practices to ensure you are comfortable with them. If you have complaints, claims, concerns, or questions about third-party products or services, please direct them to the appropriate third party.

# Errors, Inaccuracies, and Omissions

Occasionally, the information on the Website or the Service contains typographical errors, inaccuracies, or omissions related to product descriptions, pricing, promotions, offers, product shipping charges, transit times, or availability. We reserve the right to correct these errors, inaccuracies, or omissions and to change or update information or cancel orders if the information on the Website or the Services is inaccurate. We can do this at any time without prior notice, including after you have placed an order.

# Prohibited Use

In addition to other restrictions described in these Terms, the following activities are forbidden when using the Website or its content:

a. Engaging in any unlawful activity
b. Encouraging others to engage in unlawful activities
c. Violating international, national, regional, or local laws, regulations, or rules
d. Infringing upon our intellectual property rights or those of others
e. Harassing, abusing, insulting, harming, defaming, slandering, disparaging, intimidating, or discriminating against any individual or group, including but not limited to those defined by gender, sexual orientation, religion, ethnicity, race, age, national origin, or disability
f. Providing false or misleading information
g. Uploading or transmitting viruses, malicious code, or any harmful elements designed to affect the functionality or operation of the Service, any related website, other websites, or the broader Internet
h. Collecting or tracking personal information about others without authorization
i. Engaging in spamming, phishing, pharming, pretexting, spidering, crawling, or scraping
j. Using the Website or its content for offensive or unethical purposes
k. Interfering with or attempting to bypass the security features of the Service, related websites, other websites, or the Internet in general

We reserve the right to terminate your use of the Website or the Services if you engage in prohibited activities.

# Disclaimer of Warranties and Limitation of Liability

We do not guarantee that using the Service will be uninterrupted, timely, secure, or error-free.

We do not guarantee that results obtained using the Service will be accurate or reliable.

You create and own your user-generated content and are fully responsible for it. W3Schools is not liable for any errors or faults in your user-generated content.

By using the Service, you accept that you do so at your own risk. Unless otherwise stated, we offer the Website and the Services "as is" and "as available" without express or implied warranties or guarantees.

Under no circumstances shall W3Schools, its directors, officers, employees, affiliates, agents, contractors, interns, suppliers, service providers, or licensors be held liable for any injury, loss, claim, or any direct, indirect, incidental, punitive, special, or consequential damages of any kind.

# Indemnification

By using our Services, you agree that W3Schools and its related parties (owners, subsidiaries, affiliates, partners, officers, directors, agents, contractors, licensors, service providers, subcontractors, suppliers, interns, and employees) are not responsible for any third-party claims or lawsuits that come from your violation of these Terms, any referenced documents, or any laws or rights of others. In addition, W3Schools reserves the right to claim from you a reasonable attorney's fee incurred by W3Schools and/or its Associates in relation to your breach.

# Entire Agreement

If we fail to exercise or enforce any right or provision of these Terms, this does not mean we give up that right or provision.

These Terms and any policies or rules on the Website make up the entire agreement between you and W3Schools. They govern your use of the Service and replace any previous agreements or discussions, whether written or spoken between you and W3Schools.

Any uncertainty in interpreting these Terms should not be used to disadvantage the party responsible for drafting them.

## Governing Law and Jurisdiction

These Terms and any separate agreements whereby we provide you with the Services shall be governed by and construed in accordance with the laws of Norway unless otherwise agreed to and accepted by W3Schools. In that case, it will only apply to the signing entity.

You acknowledge that these Terms form a contract between you and W3Schools, even if they are presented electronically and not physically signed. This contract governs your use of the Services.

## Changes to the Terms

You can review this page's most up-to-date version of these Terms anytime.

We reserve the right to update, change, or replace any part of these Terms by posting updates on our website. It's your responsibility to check our website regularly for changes.

If you continue to use or access the Website or the Service after we post changes, you thereby accept those changes.

## Contact Information

If you have any questions or concerns about these Terms, please contact us at help@w3schools.com

Last updated: 2026.09.16

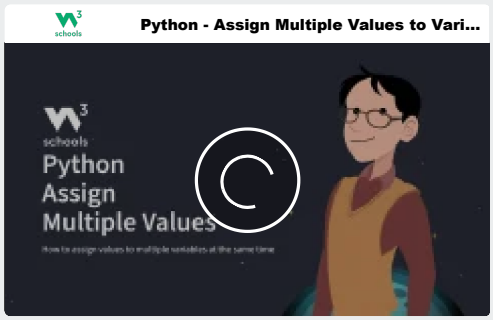
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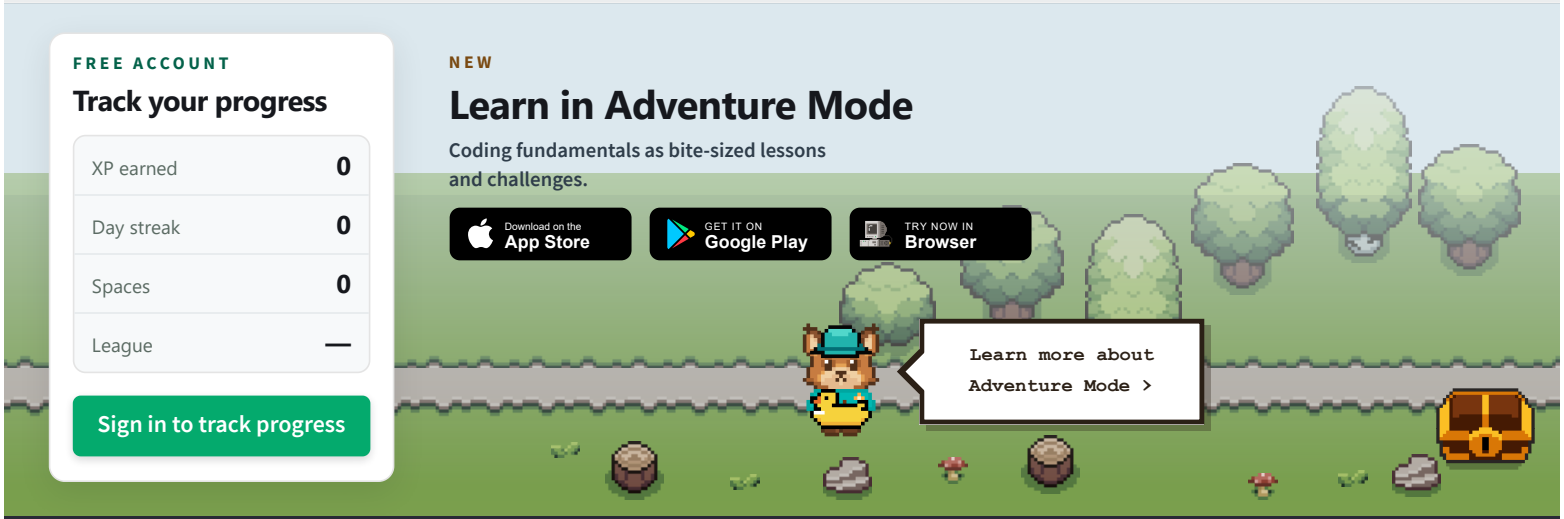
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W3.CSS Tutorial
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PHP Tutorial
Java Tutorial
C++ Tutorial
jQuery Tutorial

## Top References

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Java Certificate
C++ Certificate
C# Certificate
XML Certificate



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# Privacy Policy

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## Introduction

This Privacy Policy explains how we collect, use, and share your personal information when you use www.w3schools.com (the "Website") and or our product and or our services (the "Services").

We categorize our users as either "Customers" or "Free Users":

- **Customers** are users who have entered into an agreement with us or our resellers, either alone or with others, to access ad-free paid Services
- **Free Users** use the Website and the Services for free

When we refer to "Users" without specifying Customers or Free Users, we mean both groups.

This Privacy Policy is here to tell you what we collect and what we do with it, rather than to be agreed to. Where the law requires your consent before we process something, such as analytics and personalized advertising, we ask for it separately through our consent tool, and you can change or withdraw it at any time.

## About W3Schools

W3Schools is an online platform for learning web development and programming skills, offering tutorials, courses, certifications and various subscription plans. Our educational resources and interactive tools enable users to learn programming languages, build projects, track their progress and practice coding through exercises and quizzes. We fund the Website by showing adverts to our Free Users as you browse the Website.

1. Visit the Website
2. Use the Services
3. Interact with us in any way, such as by email requests, receiving support or by filling out a form (collectively "Support")

# Information We Collect

We collect information about you based on how you interact with us, your choices, the products and features you use, your location, in accordance with applicable laws. This information comes either directly from you or automatically through your use of the Services.

## Information That You Provide Directly

We collect the following information when you use the Website and or the Services.

- **Contact Information:** When you register, inquire about, or purchase our Services, participate in our marketing activities, or seek support, we collect your contact details. This includes your full name, email address, phone number, and location.
- **Account Information:** To create, update, or manage your account on the Website, we collect information like your email address, full name and phone number.
- **Transactional Information:** We gather payment and transaction details related to your use of the Services, such as encrypted payment card data and billing address.
- **Support and Troubleshooting Information:** If you contact us for Support, we collect information about your preferences and any further details you provide. This includes messages, chat content and attachments, which help us assist you and comply with legal obligations.
- **Voluntary Information:** Any other information you choose to provide, such as profile updates or feedback on your experience with the Services. By providing this information, you confirm that you own or have permission to share it.

## Information from Sub-Processors

We receive information about you from our sub-processors who support our business operations. We use this information only for the purposes described in this policy.

## Information We Automatically Collect

When you use the Website and or the Services, we automatically collect certain information about you, your device, and how you interact with us as follows:

- **Telemetry Information.** We collect statistical and aggregated information about how you use the Website and or the Services.
- **Usage Information.** We track how you interact with the Website and or the Services. This includes clicks, pages viewed, searches, the web browser used, page response times, errors, date and time stamps, request details (like speed and frequency), the site you came from ("referring page"), the site you visit after ours ("exit page"), and the time spent on the Website.
- **Device Information.** We collect details about your device and how it interacts with the Website and or the Services. This includes browser type and settings, device type, screen size, operating system, model, language preferences, internet service provider, mobile network, push notification tokens, system activity, hardware settings, application version, and unique device identifiers like device type ID.
- **Service-Generated Information.** We use data generated by your use of the Services, such as log files, diagnostics, capacity, and usage information, to determine storage needs, performance data from our servers, data settings, and system configurations, including your chosen privacy and security settings.
- **Cookies.** We use cookies and similar tracking technologies to collect information from your browser, depending on your settings when using the Website and or the Services. As explained in our Cookie Policy, you can choose to delete or not accept our cookies.

Sometimes, in compliance with applicable laws or our customer agreements, we combine the information we collect from different sources. For example, we might merge usage data with your feedback to improve the Services or inform you about relevant products.

# How We Use Information

We use your information to operate, improve, and maintain the Website and or the Services. Here are the ways we may use your information:

- **Service Operation:** We use your information to run and manage the Website and or the Services effectively. This includes operating, delivering, monitoring, and maintaining the Services, as well as troubleshooting and performing system maintenance and upgrades.
- **Product Development and Improvement:** Your information helps us enhance the Services and develop new features. In addition we use your information to improve functionality, quality, and user experience.
- **Support:** We use your information to provide customer assistance and technical support. This includes responding to your requests and inquiries to ensure a smooth experience with the Services.
- **Account Administration:** We use your information to create and manage your account, complete transactions and send information such as purchase confirmations, receipts, and invoices.
- **Marketing and Promotional Activities:** We use your information to personalize and enhance our marketing efforts. This includes developing and improving our marketing activities, analyzing trends and usage to provide content or features that match your interests and preferences and informing you about promotions, offers, updates, and new products or services that might interest you. We may send you newsletters, promotional emails, and other marketing communications tailored to your interactions with the Services.
- **Security:** We use your information to protect you and the Services by seeking to detect and to investigate, prevent and or respond to potential threats, fraudulent transactions, unauthorized access and other malicious or illegal activities.
- **Legal, Safety, and Compliance:** We use your information to comply with applicable laws and regulations. This involves meeting legal obligations, responding to court orders, and ensuring compliance with our terms and policies.

## For Free Users Only:

We use your personal information to provide you with targeted advertisements or marketing communications that we believe may interest you. These ads are managed by partners who specialize in advertising across multiple websites. If you consent to personalized advertising, these partners create and read persistent identifiers stored on your device. Most of these identifiers are not built from anything you told us: they identify your browser rather than you by name. They are not anonymous, though, because they are designed to recognize the same device over time and across other websites, so that ads can be personalized for you, so that you do not see the same ads too often, and so that advertisers can be told which ads were effective.

If you are signed in to a W3Schools account and have consented to personalized advertising, one of our partners also uses an identifier derived from your email address. Your email address is converted into a coded value inside your browser before anything is shared, so the partner does not receive the address itself. That coded value is still personal data: it is stable, and it can be used to recognize the same person across different websites and devices, rather than only within one browser. We do not do this for accounts that do not receive advertising, including paid plans and classroom accounts created through W3Schools Campus (previously Academy).

You can withdraw your consent at any time using the Manage Consent control below, and doing so stops these identifiers from being created or read. For more information about targeted advertising, visit the Network Advertising Initiative's ("NAI") educational page at http://www.networkadvertising.org/understanding-online-advertising/how-does-it-work

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# How We Share Your Information

We share your information to provide the Website and or the Services, comply with legal obligations, and as part of our regular business activities. We only share your information with those who need it to ensure that proper protections and agreements are in place.

- **Sub-processors.** To deliver the Website, the Services, and the Support, we share your information with our sub-processors (Overview). We only use sub-processors who have adequate contractual obligations in their standard agreements.They act based on our instructions and in line with our Privacy Policy, along with strict confidentiality and security measures. This includes:
  - **Cloud Services:** For data hosting, storage, and content delivery
  - **Billing and Payments:** To manage and process transactions
  - **Customer Support:** To address and resolve inquiries and support requests
  - **Security Monitoring:** To protect against fraud or illegal or harmful activities
  - **Domain Registration:** To register domains and manage related services for you
  - **Analytics:** To understand the performance and impact of our sites and services
- **For legal reasons.** We will share your information if we believe it´s necessary to:
  - Comply with applicable laws, regulations, legal processes, or governmental requests
  - Enforce our agreements, Terms of Service and this Privacy Policy
  - Detect, prevent, or address fraud, security or technical issues
  - Protect the rights, property or safety of W3Schools, our Users, or the public
- **Business transfers.** Suppose W3Schools is involved in a merger, acquisition, or sale of assets. In that case, we will continue to ensure the confidentiality of your information.

## For Free Users Only:

We share your information with partners who specialize in advertising across multiple websites. This sharing allows us to personalize your experience and provide you with targeted advertisements. These partners do not all have the same role, and the difference affects how you exercise your rights.

**Companies that act on our instructions.** Some partners handle data only as we direct them to, under contracts that limit them to that purpose. They are sub-processors, in the same sense as the companies listed in the appendix to this policy:

- Actirise: Privacy Policy – sells and serves advertising space on our behalf
- Viously: Privacy Policy – delivers video advertising on our behalf

- LiveRamp: Privacy Policy – creates the identifier derived from your email address described above. For users in Norway, the rest of the EEA, the United Kingdom and Switzerland, LiveRamp acts as our data processor. Your email address is deleted as soon as the identifier has been created, the identifier itself expires after 30 days, and this processing takes place within the EU and the UK. It applies only if you are signed in, have consented to personalized advertising, and are on an account that receives advertising.

**Companies that decide for themselves.** The identifier and the advertising request are then passed on to a wider advertising market, including supply-side and demand-side platforms and identity providers such as ID5 and Criteo. These companies are not acting on our instructions. They decide independently how the data they receive is used, which makes them separate controllers of it. You have rights in relation to that data against them directly, as well as against us, and each of them publishes its own privacy policy. Google takes part in more than one role depending on the service involved, and describes its practices in its own policy:

- Google: Privacy Policy

The complete and current list of advertising companies, together with controls to allow or refuse each of them individually, is available in our consent tool below.

Opting Out of Targeted Advertising:

- W3Schools:
  Manage Consent
- Digital Advertising Alliance: http://optout.aboutads.info/
- Your Online Choice: https://www.youronlinechoices.com.

**Note:** Opting out will not turn off ads; it will only stop them from being tailored to your interests.

# Your Privacy Rights

You have choices and rights regarding how we collect and use your information. Here's a summary of those options and how to exercise them:

- **Opt-out of Marketing Communications.** You can opt out of receiving marketing emails by clicking the "Unsubscribe" link in any marketing email or by contacting us directly. Please note that if you are a Customer, you will still receive transactional communications related to the Website and or the Services.
- **Account Preferences.** You can update or correct certain account information at any time by logging into your account or contacting our support team. If you wish to delete your account, please follow the instructions provided in our documentation. We will retain certain information as described in the How We Retain Your Information section.
- **Data Deletion.** If you wish to delete your account and personal data, follow these steps. Go to your W3Schools Profile and click on 'Delete My Account' at the bottom of the page. Your account and related data will be deleted within 30 days. Be aware that deletion is permanent, content cannot be transferred to a new account, and no refunds will be issued for past purchases.
- **Managing Cookies and Tracking Technologies.** You can manage your cookie preferences as described in the Cookie Policy.
- **Managing Vendor Consent** You may update and customise your Vendor Consent preferences at any time by clicking here:
  Manage Consent

Depending on your location, you may have additional rights concerning your personal information. These rights vary by jurisdiction. To learn more about your specific rights, please see the Jurisdiction-Specific Information.

# How We Secure Information

We use appropriate administrative, technical and physical safeguards to protect your information from unauthorized access, use, alteration, destruction, loss or disclosure. We also require our sub-processors, who handle your information on our behalf, to follow industry-standard security measures and meet their privacy and security obligations. However, we are not responsible for how our sub-processors handle privacy and security beyond the information we provide to or receive from them.

While we strive to protect your information, no security measures are entirely foolproof. If you have any questions about the security of the Website and the Services, please contact us using the information in the Contact Us section

# How We Retain Your Information

We retain your information for the minimum period necessary to fulfill our legal and contractual obligations, develop the Website and or the Services, resolve disputes, enforce our rights, and for legitimate business purposes such as tax or accounting requirements, as described in this policy and recommended by industry standards. When we no longer have a legitimate business need to process your information, we delete or anonymize it.

# Third-Party Services

The Website and the Services may contain links to or integrations with other websites or applications not operated or controlled by W3Schools. While we strive to avoid offensive or harmful content on these links, we are not responsible for the content, accuracy, or opinions expressed in these third-party services. We recommend you review each third-party service's terms and privacy policies before use to ensure they meet your compliance, privacy, and security requirements.

# Mobile Apps

This Privacy Policy also applies when you use our mobile apps, such as the W3Schools Adventure app.

When you use our apps, we collect the same types of information as described in this policy. This includes account information, usage information, and device information.

In-app purchases and subscriptions are handled by the Apple App Store or Google Play. We never see or store your full payment card details.

We use RevenueCat, a sub-processor, to manage in-app subscriptions. RevenueCat receives limited information, such as your user ID, device information, transaction ID, and subscription status. You can read the RevenueCat Privacy Policy for more information.

# How We Transfer Information

When you access or use the Website and or the Services or provide your information, you consent to process your information in the United States or Europe, where we operate. W3Schools may store, process, and transmit your information outside your country of residence to provide the Website and the Services. These countries may have different data protection laws. They might not offer the same level of protection as your country or the country where you initially provided the information. As applicable laws

require, whenever we transfer your information, we take appropriate steps to protect it, including using standard contractual clauses or other legal mechanisms.

Our data privacy practices follow the EU-U.S. Data Privacy Framework (DPF). This means we comply with strict requirements for handling, storing, and sharing your information per the framework. You can read more about the framework here

# Minimum Age For Use Of Our Site

Our Service is generally not intended for users under the age of 16. We do not knowingly collect personal information from anyone under the age of 13 in ways not allowed by the Children's Online Privacy Protection Act (COPPA). If you are a parent or legal guardian and believe that your child has shared personal information in a way that is not allowed by law, please contact us at help@w3schools.com. We will look into it and delete the information as required by law.

Accounts that do not receive advertising, including paid plans and classroom accounts created through W3Schools Campus (previously Academy), are excluded from advertising entirely. For these accounts we do not load our advertising partners, no advertising identifiers are created, and their activity is not used for ad personalization. We still collect basic analytics for these accounts, as described under How We Use Information.

# Changes To This Privacy Policy

We periodically review and update this Privacy Policy to reflect new Services or changes to our practices. You can see when this policy was last updated by referring to the "Last Update" date at the bottom of the policy. We encourage you to review the policy whenever you interact with us to stay informed about our privacy practices and how you can help protect your privacy.

# Contact us

If you have any questions about this Privacy Policy, please contact us at help@w3schools.com

# Jurisdiction-Specific Information

Depending on your location, certain data privacy rights and regulations may apply to you. Please refer to the sections below that are relevant to your jurisdiction:

## European Economic Area (EEA) and the United Kingdom (UK)

This section applies to individuals located in the European Economic Area (EEA) and the United Kingdom (UK).

### Legal Bases for Processing Personal Data

W3Schools processes your personal data only when we have a valid legal basis to do so. These include:

- **Your Consent:** When you have given us clear permission for specific purposes
- **Contractual Necessity:** When the processing is required to fulfill a contract with you or to take steps before entering into a contract

- **Legitimate Interests:** When processing is necessary for our legitimate business purposes, as long as your rights and freedoms are not overridden
- **Legal Obligations:** When we must process your data to comply with applicable laws

### International Data Transfer

If your data is collected in Europe and stored outside of the European Economic Area (EEA), we ensure that your personal data is transferred with appropriate safeguards in place, such as Standard Contractual Clauses (SCCs)

### Your Rights

If you are located in the European Union (EU) or Switzerland, you have the following rights regarding the personal data we hold about you:

- **Right of Access:** You have the right to request access to the personal data we process about you
- **Right to Rectification:** You can request that we correct any inaccurate or incomplete personal data without undue delay
- **Right to Erasure:** You can request the deletion of your personal data in certain circumstances, such as when it is no longer needed for the purposes for which it was collected
- **Right to Restriction:** You can ask us to limit the processing of your personal data in specific situations, such as when you contest its accuracy, allowing us time to verify its correctness
- **Right to Data Portability:** You can request a copy of your personal data in a structured, commonly used, and machine-readable format to transfer it to another organization
- **Right to Object:** You have the right to object to the processing of your personal data for purposes based on our legitimate interests or for direct marketing

If you wish to exercise any of these rights, please contact us using the details provided at the end of this Privacy Policy. We will respond to your request within 30 days. To process your request, we need to verify your identity. In some cases, applicable laws may require or allow us to decline your request. If this happens, we will explain the reason, subject to legal restrictions.

We retain personal data as needed to fulfill the purposes for which it was collected. This includes retaining information for our legitimate interests, such as complying with legal obligations, resolving disputes, preventing fraud, and enforcing agreements, even after a data subject request.

If your personal data was processed by us on behalf of a business customer and you wish to exercise your rights, please contact that customer directly. If you prefer to contact us, include the name of the customer on whose behalf we processed your data. We will forward your request to them and support them to the extent required by applicable law in responding to your request.

## United States

This section applies to individuals in the U.S. and supplements our Privacy Notice to comply with U.S. Data Privacy Laws, including the California Consumer Privacy Act (CCPA).

### Data Collection, Use, and Sharing

We collect and process personal information in order to provide the Website and the Services for our Users. The types of data we collect include:

- **Personal Identifiers:** Name, email address, and IP address
- **Customer Records:** Billing address, subscription details, and payment information (processed securely and in encrypted form)
- **Commercial Information:** Information about your purchases, subscriptions, and engagement with our services

- **Site Activity:** Data about your interactions with the Website, such as pages visited, tutorials accessed, and performance metrics
- **Geolocation Data:** Approximate location derived from your IP address

For detailed information about what we collect and the sources of this data, see the Information We Collect section. We collect and use this information to provide and improve our services, process transactions, ensure security, and for other business purposes outlined in the How We Use Information section.

We disclose the following categories of information for business or commercial purposes: personal identifiers, commercial information, site activity, geolocation data, and inferred insights. We work with trusted partners, including service providers, payment processors, and advertising platforms, to operate the Website and manage the Services. For further details, see the How We Share Your Information section.

## Your Privacy Rights

Under U.S. Data Privacy Laws, you have the following rights regarding the personal information we collect:

- **Right of Access:** You can request access to and receive a summary of the personal information we have collected about you
- **Right to Know:** You can request details about the categories and specific pieces of personal information we have collected in the last 12 months
- **Right to Rectify or Update:** You can ask us to correct inaccurate or incomplete personal information. We recommend updating your information in your Account Settings first, if possible
- **Right to Delete or Limit:** You can request that we delete your personal data or limit how we process it, in line with legal requirements
- **Right to Object:** You can object to the processing of your personal data, as permitted by applicable laws
- **Right to Withdraw Consent:** If you previously gave consent for us to process your data, you can withdraw it at any time
- **Right to Request Information:** You can ask for details about how we collect, share, and disclose your personal information from the last 12 months
- **Right to Opt-Out of the Sale of Information:** To opt out of the sale of your personal data
- **Right to Opt-Out of Sharing for Advertising:** To opt out of sharing your data for targeted advertising
- **Right to Non-Discrimination:** Exercising your privacy rights will not affect the services we provide to you
- **Right to Data Portability:** You can request a copy of your personal information in a format that is easy to use and transfer to another organization, when feasible

## Exercising Your Privacy Rights

If you want to exercise any of these rights, contact us using the information in the Contact Us section. Clearly state which right you are exercising and provide details about your request. We will confirm receipt of your request within 10 days. To process your request, we will require information to verify your identity. If we cannot verify your identity, we will not be able to fulfill your request to access or delete your data.

To verify your identity, we require you to provide certain identifying information related to your account (i.e., user ID) or details about your recent interactions with us. We cannot process your request unless we verify your identity and confirm that the personal information relates to you or the individual you are acting on behalf of.

If you wish to authorize an agent to make a request on your behalf, you must provide a signed written permission authorizing them to act for you. Requests submitted by an agent without proper authorization will be denied.

We process verified requests within 45 days. However, certain requests cannot be fulfilled if they conflict with:

- Federal, state, or local laws;
- Regulatory investigations;

- Subpoenas; or
- Legal claims or defenses.

# Appendix: Sub-Processors

| Company | Location | Service |
|---|---|---|
| Actirise | EU | Ad-Service |
| Airwallex | EU | Payment processing |
| Amazon | USA | Cloud Infrastructure for the Website and the Services |
| Brevo | EU | Market activities |
| Google, Inc | USA | Site analytics (web) and customer interactions (email) |
| LiveRamp | EU / UK | Advertising identifier derived from the email address of signed-in users who have consented to personalized advertising |
| Paypal | USA | Payment processing |
| Pipedrive | EU | CRM System |
| RevenueCat | USA | In-app subscription management (mobile apps) |
| Revinova | USA | LMS platform - (Specific for AWS courses) |
| Shopify | USA | E-commerce store |
| Stripe | USA | Payment processing |
| Viously | EU | Ad-Service |
| Zendesk | EU | Customer support |

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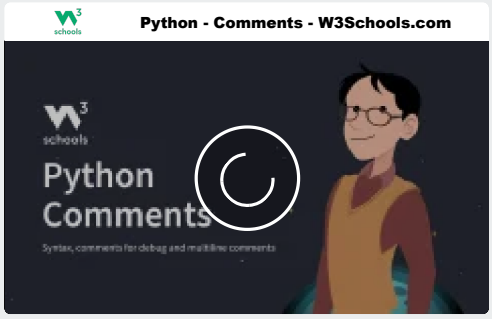
✓ Shareable credentials

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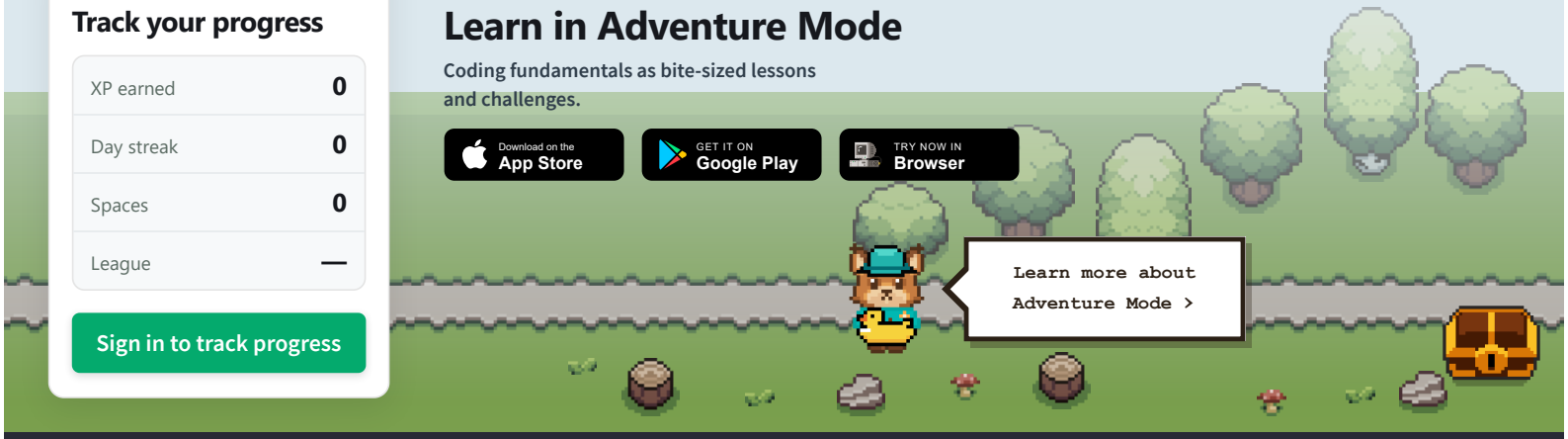
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C# Certificate
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# Terms of Service

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## Introduction

Welcome to W3Schools. By visiting W3Schools.com ("Website") or using our products and services ("Services"), you agree to the terms and conditions (hereafter referred to as the "Terms"). The Terms apply to everyone using the Website, including without limitation, visitors, vendors, customers, merchants, and content contributors.

Please read the Terms carefully before using the Website or the Services. By accessing any part of the Website, you agree to be bound by the Terms. If you don't agree with the Terms, you shouldn't access the Website or use our Services.

## Overview

The Website is operated and managed by Refsnes Data AS ("Refsnes") and W3Schools Network AS ("Network"), collectively referred to as "W3Schools". Refsnes provides the Website including all content, information, and related tools. Network operates certain solutions (such as Spaces, Campus (previously Academy), Courses, Challenges, and My Learning) and a billing system. The two entities Refsnes and Network are separately responsible for the Services they provide, manage and operate.

Any new features or tools added to the Website shall be subject to the Terms. You can review the most current version of the Terms at any time on this page. We reserve the right to update, change, or replace parts of the Terms by posting updates or changes at the Website. Please check this page periodically for changes. Your continued use of or access to the Website after posting any changes constitutes acceptance of such changes.

## Use of the Website and the Services

Our Service is generally not intended for users under the age of 16. We do not knowingly collect personal information from anyone under the age of 13 in ways not allowed by the Children's Online Privacy Protection Act (COPPA). If you are a parent or

legal guardian and believe that your child has shared personal information in a way that is not allowed by law, please contact us at help@w3schools.com. We will look into it and delete the information as required by law.

When a school uses our Services in an educational setting, it is the school's responsibility to ensure it has the legal right to share student information with us. This includes getting parental consent where the law requires it, such as under COPPA. By using our services on behalf of students, the school confirms that it has obtained or holds all the necessary permissions.

Some parts of the Services require you to create an account. When you register, we collect information about you, like your name and email address. Your account information and how we use your data are detailed in our Privacy Policy.

You agree not to use our products for illegal or unauthorized purposes. You must comply with all laws, rules, and regulations, including copyright laws.

Do not transmit any viruses, worms, or harmful code through the Services.

If you breach our Terms, we may terminate your access to the Services immediately. At our sole discretion, we reserve the right to remove or restrict your account and any content you've created, such as websites or other user-generated data.

# Intellectual Property Rights and Fair Use

All content provided by W3Schools, including text, graphics, logos, images, and software, is owned by W3Schools. Any use beyond what is specifically allowed under the Terms is forbidden, ref. the Norwegian "Lov om opphavsrett til åndsverk m.v."

## Fair Use

You can still use W3Schools copyright material under fair use for:

- Research
- News reporting
- Citations
- Commentary

Fair use also includes:

- Linking to W3Schools
- Including references to W3Schools in search engines

# Using W3Schools in Teaching

Fair use allows the use of copyrighted material in teaching, provided it meets the following guidelines:

## Permitted Use

- Copying examples and code snippets for non-profit teaching or research
- Copying small portions suitable for classroom instruction

## Not Permitted Use

- Copying for profitable or commercial use

- Mass or verbatim copying
- Copying large portions of material

# Privacy and Cookies

W3Schools collects, uses, and processes personal data as described in the Privacy Policy, which explains the legal basis we rely on for each purpose. Where we rely on your consent, such as for analytics and personalized advertising, we ask for it separately through our consent tool rather than treating your use of the Website as agreement. You can change or withdraw that consent at any time, and using the Website does not oblige you to give it.

Users have certain rights regarding their data. For more information on these rights and how to exercise them, please see our Privacy Policy.

We use cookies and similar tracking technologies to improve user experience. For more information on how we use cookies, please see our Cookie Policy.

REMOVE ADS

# Products and Services

To keep our offerings current, product descriptions and prices are subject to change at any time without prior notice. Without warning, we can discontinue the supply of any Services if our rules and guidelines or applicable laws are violated.

We seek to maintain high standards and to provide accurate information. However, we cannot guarantee that every product, service, or element of information will meet your expectations.

# Modifications to the Service and Prices

By ordering a subscription, the price for such subscription will remain firm for the period the subscription is purchased for. We reserve the right to adjust prices for any subsequent period(s).

We are not liable to you or any third party for any loss suffered due to modifications to the Website, price adjustments, suspension or discontinuation of the Service.

# Accuracy and Completeness of Information

We do our best to provide accurate and current information. However, we cannot guarantee that everything on the Website is complete or always up to date. Please do not rely solely on the material at the Website when making important decisions. You are using the information on the Website at your own risk.

Some of the content you find on the Website may be historical. Since historical information may become incorrect we provide at the Website such information for reference only. We can change the contents of the Website at any time without creating a liability for W3Schools. It's your responsibility to keep track of any changes.

If you find any errors, please let us know at help@w3schools.com so we can correct them.

# Refund Policy

## Exams and Courses:

You can withdraw from a course or exam within 14 days of receiving your initial access (marked by the date we send you the access email). If you have already used any exam attempts, the purchase is final and not eligible for a refund. For questions or issues related to the product you have purchased, please contact us at hello@w3schools.com.

## Plans and Subscriptions:

We do not provide refunds for plans or subscriptions. You may cancel your subscription anytime, but no refunds will be issued for the period you paid for.

# Mobile Apps

These Terms also apply when you use our mobile apps, such as the W3Schools Adventure app.

## In-App Purchases and Subscriptions:

- All payments in our apps are handled by the Apple App Store or Google Play, not by W3Schools
- Subscriptions renew automatically unless you cancel at least 24 hours before the current period ends
- You can manage or cancel your subscription in your App Store or Google Play account settings
- Deleting an app does not cancel your subscription
- Refunds for in-app purchases are handled by Apple or Google, following their own refund policies

We use RevenueCat to manage in-app subscriptions. See our Privacy Policy for more information about how your data is handled.

# Domains

We provide a domain purchasing service that seamlessly integrates with the W3Schools web hosting and development platform ("Spaces"). It allows you to search, select, and purchase a domain that automatically links to your Spaces.

When you buy a domain, you become its sole owner and must follow all related terms and conditions.

W3Schools works with third-party providers, such as Amazon and its Route 53 services, for domain registration and distribution. You can review their terms here:

- Amazon Route 53 Domain Registration Agreement
- Amazon Registrar Policy Page
- Gandi Contract Page

As the distributor, W3Schools handles purchasing and billing. If you have any complaints, claims, or concerns about these third-party products, please contact us at support@w3schools.com.

The mutual agreement for a domain renews automatically each year. If you don't want to continue with a domain purchase from us, you must cancel the renewal before the end of the current agreement. See the Billing and Account Information section for more billing information.

We are not liable for any harm, errors, or omissions caused by the domain registry administrator's actions, including domain registration or renewal requests. Before purchasing a domain, please review all associated third-party policies and practices to ensure you understand their terms.

# Billing and Account Information

We reserve the right to refuse any order placed with us. At our sole discretion, we can limit or cancel the quantities purchased by any individual, household, or for any specific order. These restrictions also apply to orders made under the same customer account, using the same payment method, or associated with the same billing or shipping address.

If we need to change or cancel an order, we will email you using the email address you provided at the time of purchase.

Please know that you agree to provide current, complete, and accurate purchase and account information for all orders made through our website. You must keep your account details updated at all times, including your email address and payment method, so we can process your transactions and contact you if needed.

# Optional Tools and Third-Party Links

We sometimes provide access to third-party tools we do not develop, monitor, or control. These tools are offered "as is" and "as available," without warranties, endorsements, or promises. We are not responsible for any issues or damages that arise from your use of these optional third-party tools.

If you choose to use these tools, you do so at your own risk. Before you proceed with any such tools, please ensure you understand and agree to the terms set by the third-party provider.

The Website may include links to external websites or services we do not operate or control. While we try to avoid linking to harmful or offensive content, we cannot guarantee the accuracy or quality of third-party materials, products, or services. We are not liable for any harm or damages you experience with third-party websites or services.

Before you engage in any transaction with a third party, review their policies and practices to ensure you are comfortable with them. If you have complaints, claims, concerns, or questions about third-party products or services, please direct them to the appropriate third party.

# Errors, Inaccuracies, and Omissions

Occasionally, the information on the Website or the Service contains typographical errors, inaccuracies, or omissions related to product descriptions, pricing, promotions, offers, product shipping charges, transit times, or availability. We reserve the right to correct these errors, inaccuracies, or omissions and to change or update information or cancel orders if the information on the Website or the Services is inaccurate. We can do this at any time without prior notice, including after you have placed an order.

# Prohibited Use

In addition to other restrictions described in these Terms, the following activities are forbidden when using the Website or its content:

a. Engaging in any unlawful activity
b. Encouraging others to engage in unlawful activities
c. Violating international, national, regional, or local laws, regulations, or rules
d. Infringing upon our intellectual property rights or those of others
e. Harassing, abusing, insulting, harming, defaming, slandering, disparaging, intimidating, or discriminating against any individual or group, including but not limited to those defined by gender, sexual orientation, religion, ethnicity, race, age, national origin, or disability
f. Providing false or misleading information
g. Uploading or transmitting viruses, malicious code, or any harmful elements designed to affect the functionality or operation of the Service, any related website, other websites, or the broader Internet
h. Collecting or tracking personal information about others without authorization
i. Engaging in spamming, phishing, pharming, pretexting, spidering, crawling, or scraping
j. Using the Website or its content for offensive or unethical purposes
k. Interfering with or attempting to bypass the security features of the Service, related websites, other websites, or the Internet in general

We reserve the right to terminate your use of the Website or the Services if you engage in prohibited activities.

# Disclaimer of Warranties and Limitation of Liability

We do not guarantee that using the Service will be uninterrupted, timely, secure, or error-free.

We do not guarantee that results obtained using the Service will be accurate or reliable.

You create and own your user-generated content and are fully responsible for it. W3Schools is not liable for any errors or faults in your user-generated content.

By using the Service, you accept that you do so at your own risk. Unless otherwise stated, we offer the Website and the Services "as is" and "as available" without express or implied warranties or guarantees.

Under no circumstances shall W3Schools, its directors, officers, employees, affiliates, agents, contractors, interns, suppliers, service providers, or licensors be held liable for any injury, loss, claim, or any direct, indirect, incidental, punitive, special, or consequential damages of any kind.

# Indemnification

By using our Services, you agree that W3Schools and its related parties (owners, subsidiaries, affiliates, partners, officers, directors, agents, contractors, licensors, service providers, subcontractors, suppliers, interns, and employees) are not responsible for any third-party claims or lawsuits that come from your violation of these Terms, any referenced documents, or any laws or rights of others. In addition, W3Schools reserves the right to claim from you a reasonable attorney's fee incurred by W3Schools and/or its Associates in relation to your breach.

# Entire Agreement

If we fail to exercise or enforce any right or provision of these Terms, this does not mean we give up that right or provision.

These Terms and any policies or rules on the Website make up the entire agreement between you and W3Schools. They govern your use of the Service and replace any previous agreements or discussions, whether written or spoken between you and W3Schools.

Any uncertainty in interpreting these Terms should not be used to disadvantage the party responsible for drafting them.

## Governing Law and Jurisdiction

These Terms and any separate agreements whereby we provide you with the Services shall be governed by and construed in accordance with the laws of Norway unless otherwise agreed to and accepted by W3Schools. In that case, it will only apply to the signing entity.

You acknowledge that these Terms form a contract between you and W3Schools, even if they are presented electronically and not physically signed. This contract governs your use of the Services.

## Changes to the Terms

You can review this page's most up-to-date version of these Terms anytime.

We reserve the right to update, change, or replace any part of these Terms by posting updates on our website. It's your responsibility to check our website regularly for changes.

If you continue to use or access the Website or the Service after we post changes, you thereby accept those changes.

## Contact Information

If you have any questions or concerns about these Terms, please contact us at help@w3schools.com

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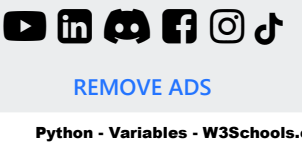
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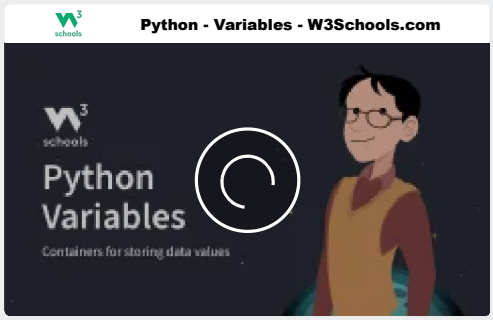
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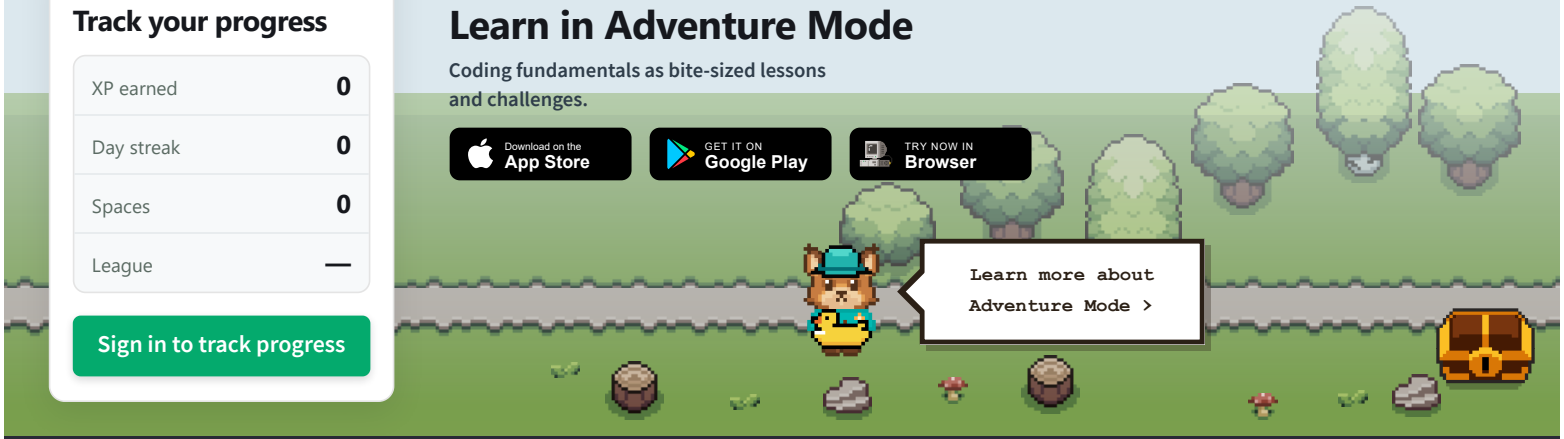
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# W3.CSS Tutorial

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## Learn W3.CSS

W3.CSS is a CSS framework for developing websites and web apps.

It provides equality for all browsers on all devices.

W3.CSS is free, open source, and uses standard CSS only.

Open Kitchen Sink ›

## Get W3.CSS Certified

Become certified with our W3.CSS exam, includes a professionally curated study kit to guide you from beginner to exam-ready.




Get Certified ›

## See What You Can Build

SHARE

With a few W3.CSS classes you can create cards, responsive layouts, navigation, forms, galleries, dashboards, and complete websites.

# Explore Cities

Discover three great European destinations.

## London

Explore one of the world's great cities.

Explore

## Paris

Discover art, food, and architecture.

Explore

## Rome

Experience thousands of years of history.

Explore

## Example

```
<div class="w3-grid" style="grid-template-columns:repeat(auto-fit,minmax(200px,1fr));gap:16px">

<div class="w3-container w3-card">
  <h2>London</h2>
  <p>Explore one of the world's great cities.</p>
  <button class="w3-button w3-black">Explore</button>
</div>

<div class="w3-container w3-card">
  <h2>Paris</h2>
  <p>Discover art, food, and architecture.</p>
  <button class="w3-button w3-black">Explore</button>
</div>

<div class="w3-container w3-card">
  <h2>Rome</h2>
  <p>Experience thousands of years of history.</p>
  <button class="w3-button w3-black">Explore</button>
</div>

</div>
```

Try it Yourself »

# Why W3.CSS?

## Easy to Learn

W3.CSS uses simple and meaningful class names.

## Responsive

You can create attractive results with very little code.

W3.CSS makes it easy to create layouts that work on desktops, tablets, and phones.

### Modern Layout

W3.CSS supports modern CSS Flexbox and Grid layouts.

### No JavaScript Library

W3.CSS is built with standard CSS and does not depend on any JavaScript library.

## Modern CSS Made Simple

W3.CSS gives you ready-to-use classes for common web design tasks.

### Cards

Create clean content boxes with shadows and spacing.

### Flexbox

Arrange and align content in rows or columns.

### Grid

Build responsive layouts with rows and columns.

You will learn all of these classes, step by step in this tutorial.

## Small Framework. Big Possibilities.

W3.CSS provides the tools needed to build modern websites without a large framework or extra JavaScript libraries.

You can use W3.CSS by itself or combine it with your own CSS or any other CSS.

**W3.CSS is free to use.**

No license is necessary.

## Learn by Building

This tutorial teaches W3.CSS with practical examples.

You will learn how to build real user interfaces instead of only learning individual CSS classes.

1. Learn

2. Try

3. Build

Learn a simple W3.CSS feature.

Edit the examples and see the result.

Combine the classes to build real websites.

Start Learning W3.CSS ❯

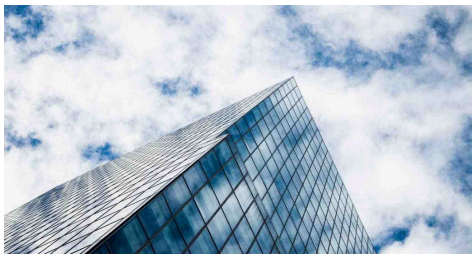
REMOVE ADS

# W3.CSS Website Templates

W3.CSS can be used to create complete responsive websites.

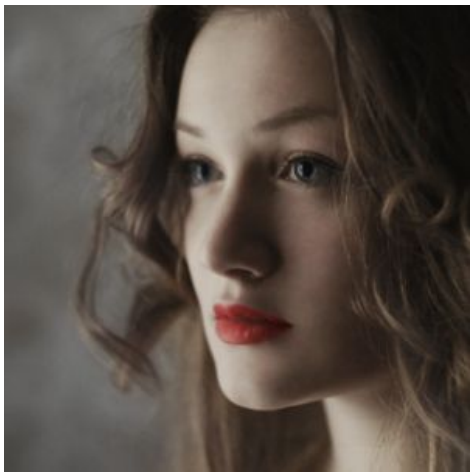
W3Schools provides many free W3.CSS templates that you can edit and use.

## Architect



View Template

## Portfolio



View Template

## Catering




View Template

See All W3.CSS Templates ❯

# W3.CSS Kitchen Sink

Already familiar with CSS frameworks?

See all W3.CSS classes demonstrated.

Open the Kitchen Sink ❯

# W3.CSS Crash Course

Already familiar with HTML and CSS?

Build a complete responsive website in the W3.CSS Crash Course.

Start the Crash Course ❯

# W3.CSS References

Want to study fast?

Our references list all classes and variables available in W3.CSS.

Open W3.CSS References ❯

# Start Learning W3.CSS

W3.CSS is designed to be simple.

In the next chapter, you will see how to start using W3.CSS in your own HTML pages.

Start the Tutorial ❯

# W3.CSS Versions

W3.CSS version 3.01 was released in December 2017.

W3.CSS version 4.01 was released in November 2018.

W3.CSS version 5.01 was released in October 2024.

**W3.CSS version 6.01 was released in October 2026.**

# Note

The W3.CSS style sheet is used in all W3School pages.

The latest versions can be downloaded from W3.CSS Downloads.

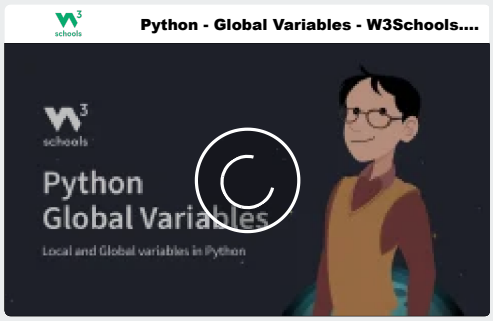
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| XP earned | 0 |
| Day streak | 0 |
| Spaces | 0 |
| League | — |

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